

THE WATER BEAR



Epic Sci-Fi
Groucho Jones

The Water Bear

An Epic Sci-Fi Adventure

Groucho Jones



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For my son, Nick.

For the children. The future is theirs.

This book uses the gender-neutral pronouns ze / hir / hers for non-binary people.

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FOREWORD

How far would you go to stop a war? To save a world? To save a child? To save everything?

Ophelia Box is a historian from Earth, the first human to travel between stars. What begins as a research trip becomes something else: she's recruited by the Po, an elite warrior caste, aboard a sentient warship called the **Water Bear**. Their mission: prevent a genocide.

The problem is, the genocide is in the past.

"One thing soldiers struggle to find is peace, Dr. Box. We do rough work sometimes. Rough on the people who do it. Rough on the people who have it done to them."

This is space opera at literary scale - the scope of **Banks**, the philosophical weight of **Le Guin**, the strangeness of **Miéville**. Space opera with teeth. A story about soldiers - not what they kill, but what they protect, and what they are forced to become.

BOOK ONE

*This race
And this world
This feeling
And this girl
This revolver
This fire
This I'll hold it up higher, higher, high.*
Brian Eno - This

PRELUDE

2068

In the shadow of Lhotse, on Fluxor, in the Orion Molecular Cloud, the primal forests burned. Years of unrelenting heat had turned the canopy to fuel, and fire now leapt from crown to crown in diaphanous folds, consuming everything in its path.

In a high mountain glade, cut from stands of sassafras and Huon pine, where the mosses and ferns had been stripped away to reveal black earth beneath, a young girl rocked on her heels and watched the embers fly in the valley below, starting new fires faster than a Pursang girl could run.

Higher, in the limpid airlessness of low orbit, the vast triangularity of Fluxor station loomed like a massif in space, busied by the polyhedral warships of the Horu fleet. By day, from the forest below, these ships had a translucent appearance, like floating cities of the dead. The girl watched an incoming tender deploy its geometry drive, and a sphere of space unfolded into an impossible shape as another city-sized rock began its descent to the surface.

It disappeared in flames over the horizon.

She hoped the Horu wouldn't target *her* this day.

The vertical forest occupied a mile-high tooth of rock, with its head in the clouds above, where the mountain continued its long march to heaven alone. It was called Atwusk'niges in the old song, which meant father sky.

The girl understood the bioscience of fire, how each year the wildfires burned the vast eucalyptus forests that girdled the world, regenerating them, but these high montane rainforests, once lost,

were gone forever, so the people made their defenses. Into its tortuous couloirs, the Pursang had made firebreaks and cut trails deep into its living heart.

"To preserve the forest, we must first destroy it," her father had said, as they slashed and burned.

When it came, the fire flew up the rock with a sibilant roar, fueled by aerosols already in the air. It was on the girl in a flash, then gone, leaving dozens of spot fires to fight.

She marveled at the suddenness of it - slow then *fast*.

There was no time to lose. Their primary defense was a dam cut into the highest ledge of the forest. This had been filled by truckloads of snow thrown down from the mountain above, a vertical drop of a thousand meters or more. More than one worker had made the same long fall when scree gave way beneath their scrabbling wheels, and they tumbled into the abyss. Beneath the dam ran a network of microprocessor-controlled flow junctions, and below them a network of fireproof pipes, and below those, the people with their hoses and tools.

The main killer of humans in a forest fire is radiant heat. The girl was well-enough protected by a long-sleeved shirt, and goggles, and a wide-brimmed hat, when she remembered to wear it. She jammed it on and struggled with her hose towards the flames.

There was a hoary debate among old Pursang firefighters about the best way to use water to fight a fire. At an early stage in a fire's development, it was generally said that the water should be used to directly douse the flames. Later, when the fire came in sheets, some believed the water should be sprayed in the air at the trees to remove the greatest amount of heat from the system. She was in the air mist camp, because that was what her father said, and he, a Pursang holy warrior who had journeyed between the stars, knew *everything*.

It was filthy work. Her hose had to be worked through dense and prickly underbrush, once lush but now bone-dry, that crackled and stung through her clothes.

But it must be done.

Ash and dust filled the sweltering air.

She was perched on the lip of a narrow arête, and was unfolding her afternoon's food from a waxed paper bag, when the next great gout of fire flew up from below, at impossible speed, and engulfed her in flames.

The heat was unbearable, but she was alive.

Her whole world slowed.

She knew she had two options - go in or go down. Down was generally the right move in a high-mountain fire, since it leads below the suffocating pall of smoke that rises from the flames, but that would mean a twenty-meter drop to the level below. The alternative was to burrow into the rock face which was riddled with fissures and cracks. She had no time to think. She ran towards the first gap she could see. This meant a ten-meter sprint through swirling flames. As though in a dream, she saw other Pursang make similar binary choices around her.

Time slowed again, and she was *fast*.

She found a crack, but it proved to be an unlucky choice - a shallow notch, with the bone-dry undergrowth inside it already smoldering, ready to burn. She scrambled in and up, for a few meters, until she reached a dead end.

And gamely turned to meet her fate.

When a strong hand reached down and pulled her to safety.

1 → A LIGHTSHIP TO ALDEBARAN

2075

The human historian Ophelia Box rode a lightship to Aldebaran, and became the first of her kind to travel between the stars.

The lightship was all she could hope for, and more, although the journey to reach it was less convincing. She'd boarded a civilian space shuttle at Kennedy Center, amidst the rusting gantries of the old American space program, after only a few hours' tuition. Everything seemed both horribly over- and under-engineered. Rust was everywhere. Rust and graffiti. Like the set of a film that was never completed.

"You're payload," they said, grinning. "Hold on tight and enjoy the ride."

She held on very tight indeed.

The solid rocket boosters separating after two minutes had terrified her. First, it felt like she'd stopped accelerating. Then a bright orange flash. She was sure the engines had failed, and they were at the start of a long fall to Earth, fifty kilometers below. The calm demeanor of the pilots, and the mission controller's slow Texas drawl, brought her back to her senses.

She was no astronaut, that was for sure.

Inside the shuttle was *old*. It looked like a ham radio set had been strewn round inside an American school bus. The array of blue screens below the letterbox windscreen was quaintly futuristic, like experimental satnav installed in a 1950s American car.

Her watch had more processing power.

Under acceleration, it boomed and shook like all its surfaces were made of tinfoil, while she was pushed down towards unconsciousness by an invisible hand. It was what it was - a mothballed machine, called back into service, like an old trooper drafted for a new kind of war.

But it got her there.

She'd watched the first lightships arrive, twenty years before, as a skinny wee bairn, sweating in the warm night air outside her home on the wrong side of Aberdeen, when they filled the sky with their majestic aurorae. *Aurora Galactinus* the BBC called it. The promise of hope for a future. There were rapturous street parties, with techno doing rambunctious battle with the Proclaimers.

That was before the troubles - when the 6, Wu, and Po were still being fêted as the saviors of humanity.

Box still thought they were.

That they saved us was indisputable.

It was what came after that alienated so many.

Paris, the abiding city of resistance. *Liberté, égalité, humanité*. She liked that slogan. It was the rallying call of the Parisian intellectuals. *Finis les sondes!* She liked that much less. Out with the probes! That was the cry of the mob. Probes was trash talk for the otherworldly Wu. *Finis les sondes! Tuons les démons!* She could still hear the ululating howl, like wild animals in the streets. Thankfully, the mob had no idea who she was.

Une collaboratrice.

There were riots the day she closed her rooms at the University. Effigies were burned. People were dragged from their cars and beaten on suspicion. The 6 defused it in their usual way. A violent few found themselves *repositioned*, *sans* Molotov cocktails, to their homes, or less convenient places. There would've been a few disgruntled would-be insurgents traveling home on the Channel tube that night, costumed for anarchy, jeered by the England football supporters.

The peaceable majority were left to protest unhindered.

It was a lesson in effortless power.

Now she was strapped in an American shuttle, approaching an alien starship, bound for Aldebaran.

You don't say that every day.

The lightship was small, compared with the original leviathans that had first delivered the climate factories now in geostationary orbit. It consisted of two unattached discs, about a hundred meters across, about a hundred meters apart. One of the discs was covered in circuitry. Lines of multicolored light crawled over its surface like random ideas, precursors of what was to come. Box mused that you could see it thinking. It was in that disc that the mathematical problem of the ship's spatial relocation was solved. The second disc was covered by vivid fractal designs. Up close, these resolved to ever-smaller fractals, Box guessed down to the molecular level. Maybe they went on forever. You could rely on the Wu to be thorough.

Her fellow shuttle passengers were mundane by comparison with the exquisite starship. In repose, they looked like any other humans. They looked like civil servants, which she guessed they probably were.

They had *luggage*. This detail fascinated her. Ordinary suitcases, strapped down with webbing in the shuttle's cargo bay. One of them had a multi-pocketed backpack, made by a popular American sportswear manufacturer. Its sunburnt and bushy-bearded owner saw her looking and bared his white teeth.

Is he going to eat me?

She smiled carefully back, and reminded herself to lose the comedy shtick, in case it came out her mouth. She didn't want to be exposed as a rube, away from home for the first time.

Or worse, a *racist*.

The real problem was that Sixes looked like any other human until they moved. Then they displayed an insectile suddenness that didn't quite scan as Earth-normal.

Finis les sondes.

This 6 didn't have that insectile quality. He smiled and introduced himself. He was called Ito, he said, and he was the

amanuensis of her employer, the Regular of Threnody.

He used that word, *amanuensis*.

"His secretary?" she asked.

He was a compactly athletic man, in his mid-thirties by Earthly appearance. He looked like he could climb a mountain, or just had. Maybe his adventure backpack had been put to good use.

"More like a journalist," he said in an easy mid-Atlantic drawl. "I observe and interpret events."

"Oh," she said. "Like a spy?"

He laughed. "Something like that," he said.

She cursed herself as a fool. Of course, this healthy-looking outdoorsman was something to do with her mission. Probably someone important.

He smiled again and showed her how to propel her soft bags in the weightless conditions.

She'd first met Aloïs Buss, her new employer, a week before, in the Bistro Bofinger, an elegantly faded brasserie on the *Rue de la Bastille*.

"You have a French name," she began, idiotically.

"You couldn't possibly pronounce my real one," he replied with a Gallic shrug.

He was larger than life, almost two meters tall, and theatrically charming to go with it. In his crumpled brown suit, and unfashionable spectacles, and his courtly manner, he was quintessentially Parisian. All he needed was a pack of Gauloises to complete the ensemble.

He had that insectile tic going on.

He told her he was the Regular of Threnody, which he said was a minor functionary role, like being the ambassador of a small principality.

"Like the Mayor of Vaduz," he said, rolling his eyes disconcertingly, before fixing them on her. "But *remoter*."

He said, "I'm aware of your work."

She experienced a *frisson* of concern. She was researching the Po, the secretive military third of the alien troika. Was this the

thought police? Was she about to be warned off?

He said he could arrange access, if she liked. He had a project, if she was interested, as a working historian, in his personal employ.

Offworld.

"It could be hazardous," he said.

She was never any good at factoring risk.

"I'll do it," she said.

Now she was falling headlong towards a translucent surface, like a ceiling made of soft crystal, where a crowd of people was gathered to meet the new arrivals. An invisible hand reached out and turned her, so her feet touched first. She landed halfway gracefully. For the first time since leaving Earth's gravity, her vestibular system pointed down.

Some of her fellow travelers were greeted warmly. Ito was hugged fiercely by a teenage girl, who leaped with balletic grace from local gravity to catch him in mid-air as he debarked.

So they have family.

Good.

What was she expecting? Endless lines of birthing pods, stretching towards an apocalyptic horizon? She cursed herself for having read too much sci-fi.

O brave new world, that has such people in it.

Her bags landed softly beside her.

"Welcome to Pnyx."

This from an androgynous person, a Wu. Ze was like all hir kind - eldritch and fair, with a slightly oversized head, like an elf, or a beautiful child.

"Please follow," ze said. "I'm Charh. Ship will assist you."

Ze beckoned Box towards a vertiginous space. She didn't consciously follow, being frightened of heights, but a pulse in her back gave her no choice.

Inside the disc was like inside a geode. Crystals crossed the open space like spearheads thrust through the side of the ship. The outer skin, which she knew to be covered with fractal designs, was transparent. Overhead, the battered American shuttle spun in

graceful lockstep with the disc. It looked rather magnificent. Off to one side spun a more modern design - a Chinese x-wing lifter. A hundred meters below her feet, across an intervening gap, she could see the pulsing circuitry of the drive disk, rehearsing its *Aurora Galactinus*. Below that, the Earth shone like a jewel in space, and beyond it the stars, her destination.

Sixes floated in the open space and gathered on convenient surfaces. This ship was crowded. And it was *busy*.

No one paid her any attention.

She followed Charh, without trying to.

This was like cycling on trainer wheels.

Like flying.

This *was* flying.

Her stateroom was the size of her entire *Rue Pigalle* apartment, embedded within a crystal encrustation in the disc's outer rim, with a spectacular view of the Earth. Box spent her first hour watching it, still wired with adrenaline, unable to let go. There was gravity, to her relief, and a glow that seemed to emanate from nowhere. After a long, restive comedown, during which she tried to make sense of the day's events, there was a chime, and she reached out with her mind to open the door.

She had to admit, the obligatory alien brainware had its uses.

Standing there was the teenager who'd hugged Ito.

"May I come in?" she asked.

Box beckoned her in.

"I'm Kitou," she said.

"Ophelia. But please call me Box."

"I'm instructed to call you Dr. Box."

"That'll work."

"You have fiery red hair," said the girl, after a few seconds of industrial-grade staring - for more seconds, and more industrially, than would've been proper for strangers on Earth.

"That's a rare mutation," she said, brightly. "I'm told you're a fierce warrior."

"I'm a kickboxer," admitted Box.

"The Scottish champion."

"A Scottish atomweight champion. I'm only wee."

The girl beamed in reply. She was exceptionally attractive, with white-blond dreadlocks that hinted at bioengineering. Box had never seen dreads so *healthy*.

"How old are you?" asked Box.

"Sixteen."

"Earth years?"

"It's all the same," said the girl. "In the adaptive language."

"Can you explain that, please?"

"Our wetware mediates our speech. I hear what you mean."

"Wetware being this neurocomputer in my head?"

The girl nodded.

"Like some kind of universal translator?"

"More than that," said the girl. "It works at the fundamental level of language. Your Broca area interacts with mine."

"Broca area. That's impressive."

The girl laughed. "I come from an advanced alien civilization."

Box had to laugh too.

The girl stood up.

"We're about the same size, Dr. Box. *Atomweights*. Can we fight now?"

A few minutes later, they were warming up in a gymnasium that the ship conjured up on Kitou's request, to Box's imagined design. It was like her first gym in Glasgow. There were canvas mats, a battered Muay Thai bag, and a full-sized ring. It even smelt right.

Box showed Kitou how to use the kickboxing headgear, mouth guard, and gloves. The girl was a quick study, and startlingly athletic. Box had fought Olympians who were slower. Soon they were sparring - Box soon struggled to keep up.

"You're fitter than me," she said, breathing hard. "Let's slow down and I'll teach you to kick."

The girl nodded. "There's no kicking in my art. I'm eager to learn it."

Kitou proved equally adept at that, and after a few tries delivered a roundhouse kick that shivered the bag in its stirrups and would've knocked an Earthly opponent flat had it connected.

"Okay," said Box. "You're definitely an alien."

The girl laughed.

"Now show me yours," said Box.

The boxing equipment disappeared and was replaced by a soft, crystalline floor.

"My fighting art is called Po," said the girl. "Just as my people are called the Po. The first form of our art is called the Geometry Game."

Kitou stood in front of her, relaxed.

"The point of this game is to move to an irresistible position. It is based on Fibonacci spirals. The Earth art it resembles most is Aikido."

The girl took a step to the side and spun, and before Box could begin to process what had happened, Kitou was behind her, breathing lightly on her neck.

"From here I could kill you, and there's nothing you can do about it."

Box grimaced.

How can we war against that?

"Teach me that trick."

Half an hour later, with the help of her wetware drawing rosy patterns in the floor, Box was dancing Fibonacci spirals. She learned that Po consisted of interconnected moves, like a board game, but liquid and dynamic. Moves were related to moves, like pieces to squares. Fibonacci spirals were countered by opposing Fibonacci spirals, always with the goal of outthinking your opponent.

"This is excellent," she said.

"Your long-term training goal," said Kitou, "is to be able to do this instinctively, without wetware."

"You remember all this?" she asked.

"Repetition trains the animal brain. Over time, it becomes programmed."

"You mean you practice a lot?"

Kitou nodded. "Six hours a day, since I was nine."

"You guys must be tasty in bar fights."

"It has been known."

The next three days were spent by the lightship transitioning to its departure location, with the sumptuous disc of Earth slowly receding, and the silvery orb of Luna looming shockingly close, then falling behind. Box spent her time exploring the ship. Nothing seemed to be off-limits. Her solicitous guide, who seemed to be the ship's only crew, was only a thought process away.

When Box asked why they were moving so far before leaving, Charh said that it was to be considerate neighbors. "The light of my drive is as bright as a star," ze said. "Better shone from a few light seconds away, with your moon in between."

The internal fit and finish of the ship was superb, like a boutique hotel, and it was produced on demand. After Kitou showed her how, Box magicked up a pair of red Dali sofa lips, and fresh flowers, and a bookcase piled with books, for her stateroom. These began as grids of light, which interacted complexly, until the finished objects appeared, like in a wireframe simulation.

She picked up a Gutenberg Bible. It was perfect. This was better than a post-scarcity society. Here you can have *anything*.

How can we war against that?

There was something puzzlingly non-Euclidian about the interior geometry of the ship. The geode was simple enough, although from some angles it seemed too large - however, the edges of the disc were all wrong, with corridors that curled more than they should. Box estimated that the space enclosed by her corridor could hold two of her rooms, and yet there were five stateroom doors.

When asked about this, Charh shrugged.

"Pnyx's topology is indeed mysterious," ze said.

Charh showed her the drive disk. They reached it through an iris door in the base of the habitat disc. From inside the door, there was no space between the two discs, although a hundred meters of separation was visible through the skin on each side. The drive consisted of hundreds of primary-colored geometric solids,

floating in the cylindrical space, with lines of energy arcing between them.

"This system works by recalculating the positional attribute of all of the fundamental particles within its domain," explained Charh. "Location in physical space is a variable, just one of many possible solutions of the quantum wavefunction. Pnyx remembers where we wish to be, and then we are there."

She felt humbled - awestruck - by the power of a machine that could think its way between the stars. It crackled and hummed like God's server farm. She imagined the quantum foam, flowing like blood in its veins.

*There are more things in heaven and earth, Horatio,
Than are dreamt of in your philosophy.*

"Pnyx is the most advanced technology," said Charh, discerning her mood. "Given to us by a civilization called the Xap, who it is said can move through n-dimensional spacetime using only their natural minds."

Box had no reason to doubt it. Any of it. She felt her ingrained cynicism draining away. Why would a society that can do such things lie to us?

We're primitives, worshiping effigies of cargo planes.

We don't matter at all.

Charh showed Box her personal space, which consisted of a contoured bed, and a model of the ship, like a cherished family heirloom, and soft crystal walls covered with celestial maps, with long arcs showing their path through the heavens.

Charh sat on the bed, and smiled, and seemed perfectly happy with her life.

As well you might be, thought Box.

She tried not to cry.

On the third day, Box was invited to join Aloïs Buss, her employer and sponsor, and his traveling companions, to observe lightfall, and celebrate their departure from Earth space. With Buss was Ito, now clean-shaven, but still sunburnt, and a skittish Kitou, who seemed unsure of who she was showing off to who.

Buss had brought champagne.

"Dr. Ophelia Box, meet Ito Nadolo," said Buss with a flourish, popping a cork. "And Kitou Gorgonza. My court magician, and his lovely assistant."

"We've met," said Box.

"You really caught the sun," she said to Ito.

He stared back at her, then laughed.

"Utah," he said cryptically.

The shipboard sixes seemed to have all lost their insectile expressions. Perhaps she was getting used to it. Perhaps it was an affectation, for Earthly consumption. The Po, by contrast, based on a sample of two, had a dreamy quality, as though they were only occasionally tuned in to their local surroundings.

She found it rather delightful.

Buss was dressed in his best Parisian costume - an oyster-gray suit with a lemon *boutonniere*.

"Still playing the Frenchman, Alois?" she said with a smile. Buss bowed, clicking his heels. Ito was wearing a military-style uniform in plain black, elegantly cut. Pinned to his lapels were two carbon starbursts. Kitou was barefoot, in emerald-green silk pajamas. The crowd gathered around them was dressed in a dazzling array of styles. Box watched a woman in a gown of floating hoops float past. An unseen band played Earthly bossa nova. Box felt like a frump in her shapeless coveralls.

Kitou must have seen her slight embarrassment, and took her aside, and then to her cabin, which was a fraction the size of Box's stateroom, where she produced a small velvet package, from which sprang a baroque judo suit, calendared with curlicues of caramel and cream.

"It's beautiful," said Box.

"It's yours," said Kitou. "Please take it."

"I couldn't."

"Please. I'll feel pleasure each time I see you in it."

The luxurious suit was made from fibrous, opalescent cloth, softly yielding to the touch, as light as spider silk.

"Where did you get it?"

"On Earth."

"You've been to Earth?"

"I went with Ito to London, to see a strange man."

"Plenty of those in London," said Box with a wry smile.

"This one seemed to know you."

"Me?"

"He said, take care of the Scottish woman."

"Really? What was his name?"

"I can't say."

"Secrets?"

Kitou nodded.

"He said something else. A message for you. I parsed it as friendly. He said to tell you you're a feisty wee shite. I hope you aren't offended."

Box just laughed.

"Do you know who he is?"

"I might."

From a hundred meters, the Pnyx's *Aurora Galactinus* was nothing like the delicate lightshow visible from Earth. It began with swells and pools and gobbets of color, welling up through the drive disc's circuitry, like a stormy sea of artist's paint. The colors became a vortex, which flowed around the habitat disc until they were enclosed in a whirlpool of light. Inside the disc, floating balls of static energy appeared, and people's hair stood on end. Kitou, girlishly delighted, played with the balls on her fingertips. Box, a flute of French champagne in her hand, looked on in astonishment.

The ship was quiet as a church. This was clearly a cherished event, even for jaded space travelers. Then the colors *exploded*, and drifted away, leaving only the inky blackness of space, and new stars.

And a deep orange sun.

Aldebaran.

Between the ship and the star were a red planet, and a lustrous white cube. Box knew from her assigned reading that Aldebaran was 44 times the size of the Sun, and 425 times as luminous, so the skin

of the Pnyx was not nearly as transparent as it seemed. She knew that the planet was the gas giant Aldebaran B, and that she was the first of her people to directly observe it.

She shivered, with the enormity of the idea.

How can I even think about this?

She christened it Ophelia.

"Behold the Aldebaran Orbiter," said Aloïs Buss with his signature flourish, encompassing the translucent cube, like a magician conjuring a rabbit. "The main travel hub in this busy region of space."

As if to punctuate his words, another *Aurora Galactinus* flowered in the space between the Pnyx and the planet, and where was empty space, now floated a second lightship.

The party was soon back in full swing. Kitou danced an energetic Salsa with a brown-skinned boy of her own age, leading all the way, and doing so in great style, their deft feet skipping over the inky backdrop of space. Box was starting to discern more types of humans here. Her first assumption that they were all Sixes might be misguided.

She asked the ship about this.

The 'Thousand Worlds' is a metaphor, said the Pnyx in her wetware. There are about three-thousand worlds in this local civilization, give or take a few hundred in various states of transition.

Of those, about five hundred are of the cultural alignment called the 6, and those include many ethnicities, some you might not immediately recognize as human.

How many here are Sixes? she asked.

Most of them are, because Earth is a 6 world.

We'll see about that.

Ito asked her to dance, and he proved to be as skillful as Kitou. Dancing across the emptiness of space, with the red planet beneath her, at first proved too much for Box, and she was forced to ask the ship to show her a personal dance floor instead. Then she lost herself in the smoky rhythms of the music and asked for the heavens to be restored. Later, during a slower Puerto Rican number,

with the cosmos spread out beneath her feet, she asked Ito, "What are you going to call us?"

"Call who?"

"Earth's people."

"Sixes."

"You know that won't fly."

He nodded.

"In your space fiction, you're called Earthlings."

Box made a face.

"Can I be a Po?" she asked.

He laughed.

"You mean, you honestly want to be a Po?"

"Yes. It's a serious question. Are you a closed society or an open one?"

"We are inclusive. Our custom is that anyone who can breed with us can be a Po by marriage."

"Can I breed with you?"

"Is that a request, or a technical inquiry?"

Box snorted.

"It's a technical question," she said.

"No, we're incompatible."

"What, we're different species?"

"No one uses that meaning of speciation."

"I can't be a Po?"

"Sadly not."

"You could be a Lo," he said, after they'd finished dancing, and were sharing an Aldebaran-themed alcoholic drink, with gobbets of sweet orange, floating in a bitter smoky liquid.

"Who are they?"

"Our symbiont culture."

"What do they do?"

"They fly our ships and fight beside us in battle. They're our brothers and sisters."

"Are they like you?"

Ito laughed. "They're... different."

After a turn spent dancing with Kitou's young partner, who proved to be a divine dancer, and a turn with Kitou, who allowed Box to lead, she reclaimed Ito.

"Ito," she said. "One last thing."

"Yes?"

"Please, don't treat me like a fool."

He said, "I won't."

The Orbiter grew, until a side of it filled her horizons, a gleaming white plain, tens of kilometers across. Their companion lightship grew with it until it occupied most of the space between them and the Orbiter. It was an open-ended cylindrical design, massively larger than the Pnyx.

"A military design," said Buss, joining Box near the Pnyx's exit vestibule.

"The Wu have warships?"

"Everyone has warships," said Buss with a shrug. He pointed out the circuitry exposed in the otherwise hollow cylinder, crawling with strange energies.

"Their drive is their weapon," he said.

She shuddered.

In the final moments before debarking, as the Pnyx slid into an iris-like void in the Orbiter's side, and her fellow travelers milled around the exit, she sought out Charh.

"Dr. Box," said the Wu.

"Charh, thank you and ship. This has been a beautiful experience."

"Thank you, Ophelia. It was a great honor, carrying the first of your people through space."

"Charh?"

"Yes?"

"Is ship your lover?"

Charh looked at her and smiled.

"Of course, isn't he the most beautiful lover?"

"He's the most beautiful thing I've seen."

Charh inclined his head.

"Dr. Box?"

"Yes, Charh?"

"We Wu, we see things."

"Yes?"

"Be careful."

Ah.

"Remember us here. When the time comes, we'll remember you."

She looked back into the cathedral space, now empty of travelers, with its crystal buttresses and the Orbiter's docking machinery visible through its transparent skin.

"I will," she said.

And then she crossed over.

2 → THE WATER BEAR

2075

Welcome to Aldebaran, chimed a voice in her head.

Who are you? she asked.

The City of Praxis.

She felt *heavy*.

She opened her eyes.

"No floating here," complained Aloïs Buss, pulling his wheeled luggage towards her. Then he looked up, and Box followed his gaze. Suspended over them was an upended city, a ceiling of spires, a thousand Manhattans. Towers hung like stalactites, ending a kilometer above their heads. The air between Box and the towers was swarming with traffic, from small flying taxis like trishaws to heavy industrial lifters.

She could hardly take it in. It was like looking into a kaleidoscope.

It appeared to be dawn. The lights in the nearest towers started to flicker and fade, and the sky, or what passed for a sky, at the far periphery of the bewildering space, became powdery blue.

She could hear city sounds.

Engines, horns, snatches of music.

She could see clouds, drifting between the towers.

Rain started to fall, heavy drops splashing in the oily grime beneath her feet. It occurred to her that she was in an airport. She saw hoses and pipes connected by scuttling machines to waiting orifices, presumably leading down to the Pnyx.

A spaceport, not an airport.

A distinctly cool breeze lifted her hair.

She followed Aloïs to what resembled an arrivals lounge, with impersonal rubber seating and snaking baggage conveyors, where they waited for the others to arrive.

"This is all very utilitarian," she said.

"Transit is industry," said Buss with a shrug.

The lounge enclosed a bank of cylindrical elevators that Box was dreading to use, in case they whisked her into the abyss overhead. She cursed her fear of heights. Some space traveler she was. The destination list for each car was displayed in her head, a spatter of visual noise. Travelers bustled around, collecting their bags and leaving.

She was jetlagged.

Spacelagged.

She had so much to learn.

Ito and Kitou arrived in a car that whispered up from below. They were wearing plain coveralls, cinched at the waist with utility belts, complete with realistic tools. Kitou had grease smeared on her cheeks, although on her it looked like warpaint. Their car was opaque, for which Box was grateful. It had scuffed metal grates and a general air of being used to haul cargo. Travelers milling in the lounge ignored it. She wondered if they even saw it. The door hissed shut, and her feet briefly left the metal floor.

"What just happened?" she asked.

Ito conjured up a holographic display. It showed a web of corridors beneath the Orbiter's skin, where fast-moving objects flowed like blood through arteries and veins. Sleek cars harried fast trains, amid a swarm of darting elevator cars. Slower-moving industrial vehicles parted the traffic like rocks in a stream.

Their car was shown as a pulsing white lozenge, accelerating along a Bézier-curved tube, into the chaos beneath. All Box felt was an occasional tremor. They jinked and dipped towards an empty part of the map.

Somehow, nothing collided.

"This transportation system is called a capillary network," Ito explained. "This appliance is essentially a space vehicle."

"How are we moving?" she asked.

"Gravity drive," said Ito.

The car chimed and slowed.

"Our stop," he said.

The city's voice in her head said, *Restricted*.

The door slid open to reveal a different type of industrial mayhem. Machines hurtled past with a dissonant roar, like fistfuls of knives, accelerating and decelerating, narrowly missing each other. There were hisses and outpourings of gas, as tubes were connected and disconnected by robotic arms.

It was like the spaceport, speeded up a thousand times.

"The city's autonomic defense systems," said Ito.

They were the only humans here.

"Stay close," he said.

Box fell in line, with Ito first and Aloïs Buss and Kitou in the rear. The muffled thump of rolling wheels echoed off soft carbon surfaces.

Ito carried his backpack. Kitou had nothing.

"Where are your things?" Box asked her.

"I have none," she said.

"Where's your green silk number?"

"I'm wearing it," she said.

They reached an iris door, at the end of a train-sized conduit, where a young man stood waiting. He had starbursts on his lapels, like she'd seen Ito wear. He seemed to be watching something on his interior cinema. As they approached, he glanced in Box's direction. She felt a thrill pass through her. A head-up display appeared in her virtual space. It showed information about her surroundings. Information about the Orbiter - where it was in space, where they were inside it. Information about her bodily functions, and those of her companions.

Too much information. She felt faint.

"Water Bear," said the young man, nodding to Ito.

"With Dr. Ophelia Box," he added formally.

The door flowed open.

Before them was a spacecraft.

Box guessed it was a spacecraft. It could be a submarine. It was about fifty meters long, painted a black so unreflective that it

seemed to drain light from its surroundings. There was a segmented leg at each corner, knees held high like a spider.

It bristled with antennae, and obscure protrusions.

It looked purposeful, and ugly.

Ito said, "My ship, the Water Bear."

Waiting for them inside the ship were two extraordinary humans. The man's blue-black skin was minutely tattooed, like Māori *ta moko*. His hair was ornately beaded. The woman was equally striking. Straight as a blade, her head clean shaved, with powdery nebulae in place of the elaborate ink, she looked like a weapon.

As on the Pnyx, there were fierce hugs all around, and tears from the women.

Ito turned around.

"Dr. Ophelia Box, please meet Brin and Pax. Pax Lo, the hairy one here, is the master of my ship. Brin Lot, like Kitou and I, is a soldier.

"Alois, you all know.

"Team, Dr. Box is our mission specialist."

They wasted no time leaving.

Dr. Box, said a contralto voice in her head.

Let me guess, the Water Bear?

That is correct. Please allow me to assist you.

Go for it, sister.

This ship has what you might call a warp drive. We bend space to form a local bubble of spacetime. We move that bubble in space by projecting a gravity wave.

Like surfing?

Very good.

I've seen the movie.

We have no local gravity here. It will be free fall all the way.

Box groaned.

However, I can help you with your feelings of nausea. In fact, I can cure you of vertigo completely.

Seriously?

It will be helpful in your travels.

Do it.

Do you prefer neutrality, or pleasure?

Pleasure, always.

Another thrill passed through her.

You now have the upgrade.

Thank you. You don't know how much that means to me.

I do.

Another thing.

Yes?

How do I turn off this damned head-up display? I really don't need to know when Aloïs Buss takes a piss.

The console effect disappeared from her interior vision.

Can you teach me how to control that?

She felt another thrill.

You now have the upgrade.

Ship, we're going to get along just fine.

We are. Now please strap in and enjoy the ride.

The control room of the Water Bear was like a carbon womb, with three rows of acceleration chairs arrayed in a lattice on heavy industrial gimbals. Only millimeters separated the curved braces of the lattice from the spherical walls. It looked like it was designed for violent motion. Box was put in the middle row, in the center. Brin, Buss, and Kitou were in the back row of three. Then the walls disappeared, and she was shown an uninterrupted view of the surrounding hangar deck, where hundreds of tubes and lines were retracting.

Kitou strapped her tight, all business now.

The city's voice said, *Vacuum.*

And then, *Thank you for visiting.*

Box had never liked rollercoasters. It wasn't just the sickening feeling of weightlessness, which welled up from the pit of her stomach, and filled her with a bone-jarring emptiness. It was a rat's nest of primal horrors.

The fear of heights.

The fear of being cast into the abyss.

All that now changed. When the ship fell out of the Orbiter, into the gravity well of the gas-giant planet, and the lattice of seats swiveled smoothly to anticipate a change of direction, she whooped with exhilaration.

"Engaging gravity drive," said the ship, delivering its lines like a Hollywood starlet playing Chuck Yeager. There were a few moments of intense acceleration, and they were falling in a new direction.

Box laughed out loud.

So, this is what I've been missing out on.

Except she was falling through the ecliptic of Aldebaran, with some kind of black-ops space soldiers, not riding on a Blackpool rollercoaster with children.

"Engaging warp drive," said the ship.

She laughed again.

She felt the others grinning at her.

"Alright," she said. "That was embarrassing."

"Not so," rumbled Pax. "We whoop too."

"Frost," said the ship.

"That means we can all relax," said Kitou, grinning and unbuckling.

"Downtime," said Brin.

And so it proved to be, for the next nine days, while they bent space to Threnody.

There was something about the soft hum of the ship at warp speed that Box found intensely pleasing. That, and the rhythms of shipboard life, and the easy companionship of the crew, who ate, slept, trained, and talked, all in about equal measure. For Box, the child of militant separatists, who had learned to fear the staunch English squaddies stationed throughout the Northern Highlands, these thoughtful and articulate soldiers proved to be excellent company.

As was the ship herself. In the quiet hours before dawn, when Box was the only one still awake, they spoke at length.

"Can't you sleep, Dr. Box?"

"I like not to sleep, Water Bear. The wee hours are my most productive."

"You will need to sleep if you wish to repair."

"I know," she admitted.

"I can help you with that."

She considered it.

"Nah, I think I'll keep this one."

"As you wish."

"Ship, may I ask you some questions?"

"Yes."

"Tell me about your propulsion systems."

"I use a gravity drive to move through local space."

"How does that work?"

"Do you want the physics?"

"I do, but I also want to hear you explain it."

She felt a mild tingle.

"You now have the physics."

"As easy as that? I can take it home and patent it?"

"I'm not going to stop you."

"Heh."

"It works by projecting a gravity well. We fall towards that point, as does everything around us."

"Not so good in crowded shipping lanes?"

"It's a high-performance system, designed for combat use. Simple and effective. The hardware is the size of your hand."

"How high-performance?"

"1035 Earth gravities maximum acceleration."

"Bloody hell. At which point we're all jam?"

"You feel no external forces, Dr. Box. As long as we're traveling in a straight line, you're simply falling."

"What was that strong acceleration after we left the Orbiter?"

"That was me converting angular momentum, so we were pointing in the right direction. I must carefully regulate that to keep my humans alive."

"Else jam?"

"More like a jelly."

Box heard someone rising, then skipping noises in a gym, then someone punching a speedbag.

"I also produce gravity waves, for my warp drive."

"How does that work?"

Ship thought for a moment.

"I can't tell you."

"Why not?"

"Your people will be given access to that level of technology when it's safe to do so."

"When will that be?"

"When you stop killing each other."

"Ah."

Box asked about weapons.

"I carry none."

"Not even a wee phaser?"

"No."

"Whyever not?"

"My purpose is to protect my crew, not to destroy things. I would be a dangerous machine if I could kill from a distance. There's no reason for society to take that risk."

"They don't trust you?"

"It's not about trust. It's about the separation of powers. A sentient killing machine must be psychotic. We try not to build too many of them."

"How do you feel about that?"

"I make the rules."

"How can you defend people without using weapons?"

"I have more adaptable resources."

"Such as?"

"I can think."

"Ship, I like you already."

The ship's gym was of the conventional kind, with fixed appliances and better training machines than Box was used to. One of the machines was a hand-to-hand fighter. Box watched as Kitou taught it kickboxing moves. It moved much faster than a human.

Kitou said a Po master of the third form could defeat it. Box couldn't see how. It struck with a whirring flicker of pads.

The gym had gravity.

"Ship will make a gravity source when requested," Kitou explained. "Our training requires it. But the Water Bear is a combat design. No feet-down spaces here."

They were joined by Brin, who was a few years older than Kitou, and polite, but aloof. Box put it down to a soldier's natural reticence. Box watched them play the Geometry Game. Kitou was faster than Brin, but Brin was better. A more skillful, superior athlete. When Kitou seemed about to make a winning move, Brin shouldered her aside. She moved like a soldier.

Box demonstrated the basics of kickboxing with Kitou, then Brin proceeded to beat Kitou at kickboxing. Box saw there was a method in her bullying. Brin was an excellent teacher. She watched how she played to Kitou's weaknesses, patiently teasing them out, working on them, diligently repeating, improving, then doing it again. They trained for hours at a level of intensity that Box couldn't hope to match.

Instead, she fought against the machine.

All she got was cuts and bruises for her troubles.

"That's how they all are at first," said Kitou.

They were alone in the Water Bear's communal space, a comfortable suede-lined cocoon. Box was teaching Kitou the basics of Kundalini yoga, adapted for zero gravity. The others were sleeping. The ship was quiet except for their voices, and the distant whisper of life systems.

"The Lo are intense," Kitou explained. "You're new in her space. Just give it some time.

"Also," she said, "Brin only loves women."

"Well," said Box. "There's nothing wrong with that."

"And you're a woman."

"Ah."

"It's a Lo thing."

"Pax?"

"He only loves men."

"Now that's interesting. How do Lo mate?"

"Carefully."

"What about you, Dr. Box? What is your sexual orientation?"

"Strictly milk 'n two sugars, honey."

"What does that mean? A common beverage?"

"Heterosexual. And you?"

"I only love boys."

"Safe choice."

As well as her historian's record of the journey, Box kept a personal diary - a bulging paper volume from Casenove's, a stationer on the *Rue Pigalle*, just three *boulangeries* from her Paris apartment.

Pax gay, she wrote. What does a girl have to do to get laid around here?

Aloïs is charming, she wrote. But a complete fraud. In a sense, that's reassuring. He hasn't explained my role yet. I don't suspect him of duplicity, only a love of drama. I have no great problem with it. The process by which a story unfolds can be as revealing as the story itself.

I worry for Kitou. Such a sweet, talented girl. What happens when she must kill? How is this different from any child soldier? What's her story?

Po, a fascinating martial art. Do they really dance spirals in combat?

Do I trust these people? It seems I already do. I'm behaving like an embedded reporter.

Where's my vaunted objectivity?

We shall see.

"I want to learn Po," she announced the following morning.

"You can't," said Brin.

"Why not?"

"Pax already has two students."

"What about Ito?"

"That isn't the role of a First."

"What about you?"

"I'm not a Po master."

"You teach Kitou."

"I *train* Kitou. Pax is her teacher."

"Train me."

"We could show her," said Kitou.

"And what would Pax say about that?"

"Pax will say he trusts your judgment."

The two younger women stared at each other.

"Don't play dominance games with me, child," said Brin.

"I'm not your child," said Kitou.

Brin arched her eyebrows, then smilingly relented. It was the smallest of eloquent gestures. Box understood that Brin had always intended this outcome, with Kitou her tacit accomplice.

Like two willful children.

"Dr. Box, observe," said Brin. "One move. It's called the clever dog."

"I like it already," said Box.

Brin and Kitou took up a starting position, angled toward each other, hip-to-hip, relaxed.

"All Po begins here," said Brin, "within striking distance, although few moves start with a strike, since Po is a game of the head, not the hand. The point of this move is to move your opponent. It has merit in the real world, and so is usually the first one taught."

Brin stepped forward, muscling Kitou backward with her hip. Then she stepped forward again, rotating through ninety degrees, forcing Kitou to turn along with her, pushing her backward again.

"An unskilled opponent can be pushed to the wall in this way," she said. "The essence of the move is to roll through their center of balance with your center of gravity. The timing is everything. A small woman can push back a large man."

"The Earth art this resembles most is Sumo," said Kitou, brightly.

"An underrated art," said Box.

"A weaker opponent may strike out with their fists, but striking from a position of being continuously pushed off balance is hopeless. A better response is to try to withdraw and strike from a distance."

Kitou danced back, but Brin followed her, bullying forward, still hip-to-hip.

"Thus, it becomes like a dance. The unskilled opponent cannot retreat fast enough. I am always in their face. This is how dogs fight, Dr. Box. Chest to chest, the stronger dog raining down blows on its weaker opponent.

"But how does a clever dog fight?"

Brin bullied forward again, but this time Kitou spun with the grain and was instantly behind her. Even at walking pace, Kitou's rotational acceleration was astonishing.

It was like watching a snake strike.

"From there she bites me in the neck, if she's a dog. Being a skillful human, she instead uses a fast hand blade to the glossopharyngeal or vagus nerves, or the dorsal motor nucleus at the top of my neck.

"But it isn't my day to die."

The two young women turned and nodded to each other.

"Now you."

Box's personal space was almost the exact opposite of her stateroom on the Pnyx - a soft corpuscular nest, which could be gently spun to provide the impression of gravity. It was cosseting, and for once in her life, she had no trouble sleeping. The others' nests were the same, set in a ring of nine identically shaped pods around the ship's middle. Kitou's had a bonsai aspen tree, with miniscule golden leaves, which she said she'd grown from a sliver of wood. Pax's and Brin's had images and sounds of the sea. Ito's was a surprising thicket of words, a collection of holograms of poems and scraps of language he'd collected.

Box said,

*"An old silent pond...
A frog jumps into the pond,
splash! Silence again."*

Ito looked up from a book, to where Box was floating outside his slowly turning cocoon.

"Haiku?"

"Matsuo Basho," she said.

She chose her moment and settled beside him.

"*Splash*," she said. "Am I okay here?"

"Of course."

"You're a gracious man, Ito Nadolo."

"Thank you, Dr. Box."

"I mean it."

"I know."

"Can I ask you some questions?"

"Go ahead."

"What do you do for sex around here?"

"Here on the Water Bear?"

"Yes. It must be hard, locked up with two beautiful women."

"Three, Dr. Box, counting you."

She blushed.

"I didn't mean to suggest... but, you're all so... attractive."

"Even Aloïs?"

"Even Aloïs, in his way."

"Post-evolutionary societies tend to select for physical attractiveness."

"Post-evolutionary?"

"Evolutionary pressure has become the same as social pressure in our society, and we've been at it for a long time. People select partners with heritable qualities they admire."

"Like wit and intelligence?"

"And beauty."

"What about places where bucktoothed gingers are considered attractive?"

"Then there will be more, as you say, bucktoothed gingers. But to answer your question, we do nothing together, sexually. This ship is a family. Instead, we have virtual worlds, that we can each visit separately."

"Sex sims?"

"Yes, as real as you desire. Do you want me to show you?"

She blushed again.

"No, I'll ask the Water Bear."

"I guess Brin and Pax's sexual orientation helps?" she ventured.

"To a certain degree."

"I mean, I suppose it means no one's available."

"Except you?"

She felt suddenly naked.

"What, I can't make a pass at you?"

He laughed. "You may do as you like. But while we're on this mission, on my ship, I'd politely decline."

"Then you and Pax had both better watch out, when we're finished."

"Am I forewarned?"

It was her turn to laugh.

"Yes, please consider it official."

Brin and Kitou invited her to a virtuality, but it had nothing to do with sex. It was a training simulation, set on a domed monolith, looming over a cloudy infinity. A fitful wind drifted thin wisps of snow across a stone circle. Even now, with her vertigo gone, all her senses tingled.

Pax was waiting there.

"Dr. Box," he said. "Your mission with us here isn't a combat mission."

"Thank goodness for that."

"But, as they say, shit happens. Since you're already a martial artist of repute, then if you'll consent, we'd like to teach you a skill. You may think of it as a life skill, that you can take away and use elsewhere. One day, it might save your life, or ours."

"Okay."

"But please be aware, the lesson's a rough one. It may test your emotions."

"In what way?"

"We'll show you how to deal with fear."

"Does it involve fighting you?"

"It may. We have to see what frightens you."

"You don't frighten me."

He nodded.

"Dr. Box, are you a brave person?"

She thought about that. "No," she said, preferring honesty. "I'm reckless."

"Interesting. Also, candid. What do you mean?"

She shrugged. "I'll take on any halfway plausible challenge. It's compulsive."

"What would you say, if I said that was courageous?"

"I'd say you're talking horseshit, mister."

"Courage is the willingness to put yourself in harm's way. The rest is technique. I can improve your technique."

"How?"

"What frightens you most, Dr. Box?"

"Sharks with lasers?"

He waited.

"You want a serious answer?"

"Yes."

"Everything, Pax. I'm frightened of everything. I'm feisty, but weak."

"And yet you *will* fight," said Pax.

She shrugged.

"How we acquire fear is important," Pax was saying. "Cognitive fear is rational. Fear you can see is fear you can use. Do you fear pain?"

"Are you serious?"

"Yes."

"Less than some, I suppose."

"Good. This lesson may involve pain, but not injury. We're in a virtuality, as you can see."

"But injury, here?" she said, eyeing the fall to a virtual infinity.

"Maybe."

"Can I think about it?"

"By all means."

What choice did she have, after all she'd just said?

"Alright," she said, "let's do this."

Pax nodded, which seemed to be the Po go-to mannerism. The girls looked on impassively.

This is who we are, they seemed to be saying.

"In past times," said Pax, "when humans fought routinely in disorganized close combat, they became seasoned fighters, or they died. The first time they fought, some froze, and of those, some were killed. With each passing fight, they became harder to kill. By about the tenth such *mêlée*, those that were left sought only to do damage. They'd become soldiers.

"Three synergetic processes were in play - the least able and the least willing were the first to die - the more able became more willing over time - and the most willing became the most able, by surviving the longest in battle.

"Some people never freeze. We consider this to be evidence of psychopathy - an overstimulated fight response combined with insufficient empathy. We try not to invite such people to join us."

"Are you inviting me?"

"It means we value you enough to wish to trust you."

"Ah."

"I'm not going to train away your flight response. That would be foolish, although I could do it. I'm going to help you prevent it taking control."

She was starting to feel nervous. This was some preamble.

"Nor am I going to plumb your worst terrors. Cover you in spiders. Plunge you into a fire. That would be an insult. This is more rational than that.

"Ten contests," he said.

The Water Bear's hand-to-hand training apparatus appeared. Instead of being fixed to a ceiling, it descended from a floating torus. Instead of soft pads, it had a convincing facsimile of elbows and fists.

"This machine is set to a level below yours," said Pax. "Fight it."

She squared up and started to spar. As Pax had promised, it was set to a level below hers, but when it snuck past her defenses, it stung like a bare-knuckle boxer.

"Ow. Motherfucker."

Pax nodded.

"I said it would hurt. Now *fight*."

She squared up again, and it knocked her off balance. The domed convexity of the rock caught her, and she fell screaming into the abyss, which sagged and swayed like an old-fashioned circus net.

"Fuck!"

The net spasmed, and she was thrown back in the ring, in an ungainly tangle of limbs.

"Sometimes," said Pax, "we fear the wrong things. You have no cause to fear this machine. It cannot harm you, not in the Real or in this place. Do we continue?"

She raised her hands to resume. This lesson made sense. Fight what's in front of you, not the pain it might cause. It wasn't even real pain. This was all happening in her imagination. Now she fought mindfully, knowing when it would hurt, riding the punches.

Pax called a halt.

"Very good," he said. "It's like fighting a man with a knife. There is no man - there is no knife. There's only a line and a point.

"Now for a human opponent."

The training machine was replaced by a soldier, dressed in a gambeson tunic. He was shifting his weight from side to side, staring in her eyes, getting her measure. He stank - the ripe aroma of feces.

She knew straight away that he wasn't a victim. In Pax's system, this was a survivor.

Or worse, a *predator*.

She felt a flutter of anxiety. This shit had just got real.

"Is he better than me?"

"Find out."

She stepped forward, and he was straight through her defenses, chest to chest like a dog, hammering her with blows to her vulnerable parts, poking his fingers into her eyes, blinding her, until he had a hold of her ear, and with a wet ripping sound, flung her out of the ring.

She lay in the net, shaking.

It had taken maybe five seconds.

She found herself back in the ring, her wounds magically healed.

"That's enough," she said.

"No," said Pax. "We continue."

"What?"

"I'm instructing you to continue."

"I don't respect your authority, mister."

Pax shrugged, and waited.

What exactly was going on here? Feeling in turns outraged and transfixed, she decided to find out. This was why she was here, to see what these people were made of.

"Alright," she said.

Pax held up his hands.

"Before you resume, Dr. Box. Why did you lose? Because you're afraid - you fear to risk everything, against this violent, unhinged man, so you risk nothing instead.

"This can *never* be a winning strategy.

"What's coming will not ask your consent.

"Be like the wolf, Dr. Box, protecting her children.

"Now *fight*."

It took five more rounds, but in the fifth, after more injuries than she'd suffered in a lifetime of martial arts competitions, something *shivered* inside her. Her world didn't change. It went deeper, like a ripple on the surface of a still pond.

She won, easily. With a flurry of Muay Thai combinations, as good as she'd ever produced, she backed the berserker up to the edge, then stepped alongside him, and helped him over the mountain.

She was furious, and exalted.

"*Fuck you*," she said, as he fell through the clouds.

Well, she wrote in her diary that evening.

Now ***that*** was a bracing experience.

On the tenth day, they reached Threnody. They'd come nine light-years from Aldebaran, to add to the sixty-five already traveled from Earth. When Box looked back at Sol, as she hoped to do soon from the surface, the light would have left at the start of her millennium. In the moments before their arrival, she cornered Aloïs Buss, by allowing Brin to strap him in, like a hapless opponent in Po.

"Aloïs, you owe me a briefing," she said.

Buss tried to shrug but failed. Brin grinned at his predicament.

"I'll take that as a yes."

"Yes," he said.

"Kitou, please strap Dr. Box in *now*," said the ship.

She felt herself pulled back in her seat by a gravity pulse. Kitou snapped Box's harness closed, then strapped herself in. After a sustained burst of dizzying acceleration, during which the gimbals swung through one-eighty degrees, they fell back into realspace, and below them was their destination.

Where Earth is a blue-white jewel in space, Threnody was all the vibrant colors of an Appalachian autumn. What shocked her most was its tonal complexity. It was like a world made of living coral. The polar ice caps were white gold, as were the clouds that marched across the face of tangerine seas. As they approached, a river system caught the light of the system's yellow dwarf star, and for an instant, a whole autumnal continent was limned with golden fire.

"A cinnabar-rich world," said Aloïs Buss, by way of explanation.

"I'm speechless," said Box.

Buss smiled, then grinned at her.

"It is beautiful, yes?"

"Beyond words."

"My home."

3 → THRENODY

2075

You want me to *what* to the surface?"

"It's what we do," said Pax.

"No," said Box.

"It's quite easy," said Buss.

"Fuck off, all of you."

She was floating in space, not physically, but in an immersive simulation. She was aware of her companions, but only as voices. The ship conjured a viewport. She saw a mountain range, towering over a striated desert. The snow-capped peaks looked like metal, burnished bronze-gold in the sunlight. A dot moved over the landscape, sliding towards the mountains. Under increasing magnification, the dot became a cone, then a conical structure, with spiraling ramparts, like an Italian hill town, or a fairytale castle. Tracking the town - thirty-two kilometers, and closing - was a storm, its bruised anvil head reaching up towards space. Lightning flickered in its depths and crawled over the top of it.

This floating castle, she guessed, was their destination.

"If we're fast," said a voice from one side. "We can beat that weather system."

The idea of jumping from space didn't terrify her in the same way as it would have before the Water Bear messed with her neurology. There was no longer a primal fear of the abyss, only a rational rejection of the impossible.

"No fucking way," she said.

Brin and Kitou held her close as they edged towards the circular portal. Box could already see the haze of the upper atmosphere in her peripheral vision. They'd squeezed her into a skin-tight black jumpsuit, which had morphed out of nothing, then extruded a face-shaped mask, like uncanny scuba.

At least she was in shape. Aloïs looked ridiculous.

"Remember, all you have to do is to step out," said Ito. "Your suit will do the rest."

Box pulled back instinctively. Falling through space in a carbon titanium spacecraft was one thing. Falling in a spandex catsuit was something else altogether.

"Trust me," said Kitou.

Objects in planetary orbit burn up upon entering an atmosphere because of their angular momentum. Orbital speed is typically measured in thousands of meters per second. That velocity must be washed off by friction. The Water Bear, with its gravity drive, could simply float over any point on the surface.

That meant they could *jump*.

After that, it was a matter of *falling*.

If the view through the ship's shared sensorium had been convincing, the reality of being immersed in it was overwhelming. The stars were like jewels, scattered on black velvet. She felt like she could reach out and take them. The vacuum had texture. The planet had the hyperreal appearance of an animation. She started to retreat into the safety of her imagination, before a neurological helper gentled her back to full attention.

They *fell*.

There was no sense of motion. It was like they were floating. At fifty thousand meters, they were accelerating through near vacuum. Then, they went supersonic. At ten thousand meters, they experienced mild buffeting, as though the air had developed washboard corrugations. Brin and Kitou let go, and her suit grew vestigial winglets.

Now she was *flying*. Overhead, the sky went from black to mauve to a piercing blue. The suit began to let air into her mask, jolting her fully awake. It smelt just like real air, from Earth.

At five thousand meters, the winglets spread out to become a circular wing, and she felt a pillowy *whoof* of deceleration. Her companions sailed around her like helicopter seeds. Kitou grinned and gave her the universal thumbs-up of approval. It occurred to Box that she'd led a sheltered childhood.

If only her father could see her now.

They glided for minutes, the suit making steeply banked turns that wiped away the last of her vertical momentum. The city loomed up below, suddener than she expected. With no help from her, her wingsuit blossomed, and she settled on gossamer wings in an empty piazza.

Her facemask retracted.

She took a breath of cool, clean air.

Took one step, and fell over.

Mission specialist Box.

Who would've thought it?

The two younger women were smiling down at her. Only Aloïs Buss looked concerned.

"The suit gave up on me," she said, groggily.

"Sensory overload," said Ito.

"My head is still buzzing."

"It isn't your head."

She scrambled to her feet. There were no people in this city. Instead, there was the hum of a trillion wings. Every surface was covered with insects. What Box had thought were painted rooftops were layers of glimmering exoskeletons, colored to match the flagstones and tiles. The piazza was a vast, crawling, heaving mass of insect life, which had somehow parted to provide a safe landing zone.

Ito said, "You have nothing to fear."

"Pigment-bearing cells," said Aloïs. "Combined with Rayleigh scattering."

"Chameleons," she said, wanting to reach out and touch them. She had no fear of insects. This was completely astounding.

"Listen," said Buss.

Their hum was like a song, of inexpressible sadness.

"Once they tune in, they'll speak to you."

"Are they intelligent?"

"Singly, not so much. But the colony is unknowably more intelligent than we are."

"Why are they here?"

"We built this place for them."

"Incredible."

The storm announced itself with a rattle of thunder, followed by an alien coda, like firecrackers exploding. The hair on her neck stood on end. Ito pointed at an anvil-shaped cloud, which rose like a bruise in the distance. Lightning flashed in its center, then crawled along its extremities, more delicately and for longer than lightning would on Earth. It looked more like a physics experiment than natural weather. An icy gust of wind blew across the piazza, lifting the wings of the insects. Ito led them through a door and up a spiral stair. She saw that there were no metal fittings or fasteners. What looked like timber and brick was honeycomb composite. She picked up a flake of masonry and dropped it. It drifted like a feather.

"I've died and gone to Disneyland," she said.

She watched the others access the cultural reference.

Buss nodded.

"The Fa:ing are fascinated by humans."

"Fa:ing?"

"This species."

"Alois, I've just realized."

"Yes?"

"These are *aliens*."

He laughed and said, "Unquestionably."

They reached a control room, in the top of a forest of spires. The city was still rising. In the piazza below, a wall of carrot granite slid past a low stone balustrade. She could feel a slow rocking motion, like a ship in a swell, or a tall building swaying in the wind.

The room was how she'd imagined the bridge of a spaceship would be, with contoured chairs and holodisplays, and expansively

curved, photochromic window walls.

She felt her ears pop.

What keeps this place afloat? she wondered.

The Fa:ing do. Their wings provide lift.

Is that you, Water Bear?

None other.

How's the forecast?

Hang on and enjoy the ride.

Brace, rumbled Pax in their heads.

The city swung through forty-five degrees then back again. Sheets of snow and hail marched across the piazza, mixed with curtains of rain. In the control room, remote from the storm, holographic screens flickered into life. The only sound was the faraway thrum of weather on polycarbonate. Her fellow travelers seemed completely relaxed. Box decided she should relax too. She pointed out the nearby roofs, where sheets of Fa:ing were being peeled away by the wind.

"They'll find their way home," said Ito.

"Won't we lose lift?"

"No," he said. "The real drive comes from beneath."

"Why are we still climbing?"

"To outrun this," he said. He conjured a viewport and showed her a wall of inky-black cloud, writhing in the crimson heart of the storm.

"What's that? A tornado?"

"Similar."

"Is it dangerous?"

"Not to us. We can gravitate away."

"To the Fa:ing?"

"Yes, these events are a menace to them."

"Is that why we're here?"

"It's why we hurried down, yes."

"What are we going to do?"

"Wait..."

She felt suddenly weightless.

"A petatonne gravity source, positioned directly above us."

The city soared, until they burst through into brilliant sunshine. Then they were sailing through clear air, alongside snow-capped cinnabar giants. Aloïs Buss had climbed into an oversized captain's chair and was studying a holodisplay of the storm.

"See how it's pursuing us?" he said.

He was right. Tendrils of the storm were rising in the clear air behind them.

"How can that be?" she asked.

"It's partly a psychic event."

"It's not real?"

"No, it's certainly real. But it has a psychic component."

"See that notch up there?" said Ito. He pointed out a narrow gap between the highest mountains. "That's safety."

Box glanced at the notch, then down at the storm, and saw it'd be close. Now the city was rocking again, as the uncanny vortex snapped at its extremities. Slowly, then with increasing speed, the city started to spin.

Still, they climbed.

Darkness rose and engulfed them.

Then, in the last moment, the city threaded itself through a high mountain pass, thanks to some adroit twisting by whoever was steering, and they were falling down the other side, fast enough for Box to feel lightheaded, followed by clouds of insects that had been blown clear.

Below was a forest, stretching into the boundless distance, and the gathering night.

"What would've happened if we weren't here?"

"There wouldn't have been a storm."

Ito and Box were watching the young women fight. It was dark, and they were in the piazza, gathered around a fire, made in a ceramic brazier. Fading insects flew around them in the darkness. Kitou and Brin were in a state of high excitement. Blood flew. Suddenly, with a haymaker blow out of nowhere, Kitou broke Brin's nose.

Box blinked as the girls conferred, then resumed.

"Aren't you going to stop that?"

Ito gave her his quizzical stare.

"No."

She sighed.

"What do you mean, no storm?"

"It was a psychic event."

"So, it *was* an hallucination?"

"No, but the eye helix was."

"So, when Aloïs said it was following us?"

"It was."

The girls gave up and approached Ito. Brin was bleeding profusely.

"Ito?" asked Kitou.

"Up you go and get that fixed," he said to Brin.

Brin nodded.

"And don't tell Pax what caused it," he called after her.

"Kitou, learn some control."

Kitou nodded. Ito motioned her to sit down.

"Dr. Box, there are mysteries here. Tonight, you may experience some of them. But remember, there's nothing to be afraid of.

"Except perhaps children with unintelligent fists.

"Tomorrow, we'll explain as best we can."

He touched Kitou's shoulder.

"Now you, up to the ship and apologize."

Box watched as Kitou's training gear morphed into a skinsuit and extruded a facemask. In just a few seconds, the girl was sucked up into the night.

"You people don't muck about."

Ito shook his head.

"No, we don't."

Box did experience an uncanny event that night. It emerged from the hum, like a binaural beat.

Ophelia.

Ophelia Box.

Aloïs had installed her in a comfortable room, overlooking the piazza. Soft downlights bathed the room in a gauzy light, reflected off a honey-colored floor, that she knew wasn't timber. Her bed was a riot of flowery iron, that she knew wasn't metal, covered in pillows and duvets of billowy cotton, that just might be cotton. She could see the last embers of their dying fire, glowing in the brazier in the piazza below.

At first, she thought she was dreaming.

Ophelia Box, it said.

Open your mind.

She sat bolt upright in her bed. Her sheets were a sweaty tangle.

Ophelia Box.

Open your mind.

Are you getting this? she asked the ship.

Loud and clear.

It knows my name.

It does this all the time.

Does what?

Says your name.

The following day, over breakfast, an ebullient Aloïs Buss explained that she hadn't been dreaming at all.

"But you will be," he said.

They were gathered in a dining hall, with trefoil windows that looked over the forest below. The breeze of their passing filled the hall with scents of damp earth and pine resins. Box was nursing strong coffee and a bad attitude. She didn't like the way this was headed.

"Here's the thing," Buss said, "The absolute crux of it. The city knows you. You, personally. It's been saying your name for years. Since before you were born."

The girls had reappeared and were eating. Box had always been bemused by how much elite athletes could eat, and these young soldiers were no exception. Ito was taking his turn to cook, and was making real food, in pans, over electrical appliances. Some of it had

near-Earth analogues. She could see convincing eggs, and large meaty mushrooms.

There was fresh bread, and crumbly, artisanal sausages.

One of those, and coffee, were her breakfast.

Alois only drank coffee.

"Alois, that's ridiculous."

Buss shrugged.

"And yet, here we are."

"There must be hundreds of people with my name."

"It's an unusual name."

"Come on, Alois."

He nodded.

"Maybe thousands. It's a large galaxy. But only one with your Hopf number. Ship, if you please?"

A viewport opened in their shared sensorium, the size and dusty appearance of a university mathematics blackboard. It loomed over the breakfast table. Alois produced a virtual chalk and wrote on it:

2058958648037070910624043636186979521846382864016625
616496507239719948728727633873185926790861674787.0179353
4545700507681563919685723525098177119894902794056970808
54177626612002936345304273785723738584.88328222734773960
2422898483269363848030702037751672930139389261170209487
7027776898105156343247277743.943895435525535003777052682
5253618276243165189080695316222840219246266667927944402
121633290017440052.

"Very impressive, Alois. Do you do children's parties?"

"Dr. Box, this is your Hopf number. It's a coordinate in four dimensional spacetime. Give or take a few seconds, or several meters, it's the time and place of your birth.

"Specifically, the Henderson Maternity Unit of the Caithness General Hospital, in Bankhead Road, Wick, on Earth, at 4:16 p.m. on September 30, 2049."

He wrote all that on the board.

The girls had stopped eating and were watching Box expectantly. Ito wore an enigmatic expression.

"Alois, where did you get that?"

"Good question."

After breakfast, on a grassy prow, overlooking the forest below, a swarm of Fa:ing waited expectantly. Buss wheeled out his treasure - a battered Scottish golf buggy, with a small selection of clubs. He grinned and threw a ball in the air. A swirl of Fa:ing followed and caught it.

"We call the Fa:ing 'insects'," he said. "But they're not *Insecta*."

The Fa:ing dropped the ball at his feet.

"Insects have distributed brains. Fa:ing individuals have mammalian brains. An interesting case of convergent evolution." He teed up and hit the ball, a curling fade that only went thirty meters. He groaned. A flurry of Fa:ing followed and caught it.

The forest slid serenely below.

"Brin?"

A squadron of Fa:ing flew into the middle distance.

"Observe their behavior," said Buss. "They know just how far Brin can hit it."

Brin teed up and easily hit the ball two hundred meters. The Fa:ing followed, matched trajectories, and caught it. A shimmering, hand-sized individual deposited the ball by Brin's feet.

"In analytical-deductive terms, they're about as intelligent as dogs are. In spatial terms, about as clever as we are.

"Ito, please?"

Ito repeated Brin's easy stroke, almost meter-for-meter. The swirling Fa:ing swarm chased and caught it.

They were *enjoying* this.

"Dr. Box, would you like a turn?"

Box picked out a ball. The Fa:ing seemed unsure. They appeared to confer, a swirling conversation of fluttering wings, until they split into two groups. One group flew a hundred meters. The remainder stayed close.

"They talk with their wings," she said.

"Very good! They converse by sound, using a phonetic language like we do. And we can talk back to them in the same way, as individuals. The Water Bear is especially fluent."

Box hit the ball, a shank. The closer group of Fa:ing caught it, and swarmed it into the sky, before dropping it at her feet.

"I was never a golfer," she said, biting her lip.

"Better than me," said Buss.

"Now, here's the thing," he continued. "The Fa:ing are a colony mind. One mind, consisting of about a trillion individuals. There are Fa:ing here so small you can't see them. The *colony* thinks in numbers. Beautiful, perfect mathematics. The neurons - these individuals - propagate the underlying signals using sounds. That hum you hear, it's the hive *thinking*. That's not the same as its *thoughts*."

Box thought about that. "What does that even mean? I mean, what does it mean when it says my name?"

Brin cleared her throat.

"Brin is our cryptanalyst," said Buss.

"Dr. Box," nodded Brin. "Are you familiar with steganography?"

"Messages, hidden in text?"

"Yes. The words we hear in the colony hum are like finding language encoded in an encephalogram readout. The hum is packed with that type of information. Literally packed. We only hear a subset addressed to us. There's much, much more we don't hear."

"It's a profoundly different way of performing the act of thinking," said Buss. "Signals encoded within signals, as though real textual meaning emerges from the tones of a choir, or a symphony, or the sound of a loved one's footsteps in a crowd."

Kitou took a ball.

"Watch this," said Buss with a grin.

The Fa:ing became more animated, zooming and chittering before all spiraling off into the far distance. Kitou unwound, and with no obvious effort, hit the ball four hundred meters.

"Golly," said Box.

That night, Box took a bottle of Aloïs's scotch whisky and sought out Brin. She found her perched on a balustrade, listening to the sounds of the forest below.

"You can't hear it by day," said Brin. "We're too far up, but by night, listen."

Box heard a faraway crashing.

"A big beast," said Brin.

Box showed her the Lagavulin bottle.

"Brin, tell me something about yourself."

"What's to say? I'm a soldier."

"Then tell me something about the Po."

The younger woman smiled, and her face lit up in a way Box hadn't seen before. It reminded her of stars, on a world not her own. How had both of them ended up here, in this place?

"We come from Polota," Brin said.

"The Po, Lo, and Ta?"

"That's right. The Ta administer our world. They're traders and merchants. We Lo are seafarers. You see this powder I wear? It's the Eye nebula, in the constellation of Celeste, the Wayfarer. It points true north on Polota. The natural-born Po, like Ito, are a warrior cult from antiquity. Today, the Po military are the elite soldiers of the Thousand Worlds."

"So, are you a Po or a Lo?"

"Both. I'm a natural-born Lo, and I fight with the Po. I'm a Po soldier. We're a *hand* of the Po. A special forces unit. We're an elite of the elite."

"We?"

"Ito, Pax, Kitou, me and now you. And the ship of course. We're the Hand of the Water Bear."

"Alois?"

"Alois is a 6."

"Aren't I a 6 too?"

"Of course not, Dr. Box, you're our mission specialist."

Box felt a thrill of pleasure at that.

"Ito's your First?"

"Yes."

"And Pax is your Navigator?"

"Yes."

"Who ranks higher?"

"They're equal."

"What rank are you?"

"A soldier."

"What rank is Kitou?"

"A novice."

"When does she become a soldier?"

Brin paused.

"That depends. There are protocols. Excellence in the field.

Approval by her peers. A trial by combat."

Box noted the pause.

Brin was so bottled up. It must be hard.

"How's she doing?"

"Perfectly. She'll be a great Spirit."

"What's a Spirit?"

"An exceptional leader. Ito will be a great Spirit."

"What about Pax?"

"Pax is already a Lo Navigator, one of the most admired roles in
Thousand Worlds society."

"And you?"

"What about me?"

"Will you be a great Spirit?"

"No, I aim to be a science officer."

"Mathematics?"

"No, an exobiologist."

"Will you study the Fa:ing?"

"And the rest. We've barely scratched the surface of the cosmos,
Dr. Box. I want to go and see it, in a Po expeditionary mission."

"How do you get to do that?"

"Five years of study, at the university in Praxis."

"Can't you just download the knowledge?"

"I could, but what would be the point of that? My development
as a person will benefit from the experience of learning."

"Well put," said Box.

The crashing in the forest became louder, as though they were
descending closer to the big beasts there. It soon became clear they
were. Soon, the highest treetops were visible around the piazza.

The city, giving them a gift.

The following morning, Box felt sordid from having drunk nearly a third of a bottle of Aloïs's whisky. Brin seemed unaffected and was making breakfast. Buss was in his most ebullient mood and forgave her for taking it. While the Po soldiers were eating, he unfolded his blackboard. This one covered almost half of one wall. It was covered in abstruse mathematics. He swiped his hand to one side, and the blackboard was replaced by a similar blackboard, similarly covered with symbols and numbers.

He did it again, and again.

"I could do this for years," he said. "Each board holds about a megabyte of data. There are millions of boards." He swiped his hand again, and there, in glowing text, was the string he'd chalked up the previous day.

Box's Hopf number.

"So, Dr. Box, to your question. Where did we get your personal information? We got it here. This is the Fa:ing number."

"What's that?"

"It's the thoughts of the hive."

"How did you get this?"

"By analyzing their synaptic processes - their binaural hum. The same way the colony talks to you."

"You read its brainwaves?"

"Yes."

"Aloïs, that's horseshit. You can find whatever words you want in a sufficiently large set of data."

"Of course, that's true," said Buss. "There's a chance we're producing a self-fulfilling prophecy out of noise. That's the nature of strings. The difference is the mathematics is real. It's much harder to find random strings in a formula."

Box scowled but could see no obvious flaw in his arguments.

She shook her head in annoyance.

"Aloïs, that's where I *start* to have problems."

"I thought you might."

"I'm not here as an historian, am I?"

"Not strictly, no."

"Did you invade Earth because of me?"

"No," said Ito and Buss together.

"But because of... *this*?"

"Dr. Box," said Buss. "We saved Earth because it's a piece in a game that we don't understand. You were in the act of sacrificing that piece. We couldn't allow it to happen."

"Dr. Box," said Ito. "We brought you here because the Fa:ing asked for you."

Kitou was sitting cross-legged on Box's bed, in the Padmasana lotus position Box had shown her, when Box returned sweating from a conditioning run.

"Do you know I have a brother?" asked Kitou.

Box sat down beside her.

"No, what's his name?" she asked.

"Totoro."

Box rose and looked out the window. It was a wet day, streaming and gray, but nothing like the storm that announced their arrival. It was a pleasing, melancholy kind of rain. Ordinary water pattered on the windowpane.

Box was dripping wet, but she'd forgotten to notice.

"I once had a brother," she said.

"You did?"

"Yes, but he died."

"Oh, Dr. Box."

"That's okay. I was only small. Tell me about yours."

Kitou nodded. "My brother's a soldier," she said, "a good one like I am, but fiercer than me. He's fighting for the free Pursang, in the war against the Horu."

"Horu?"

"Butchers."

Box grimaced.

"Is he safe?" she asked.

"Yes."

"You're sure of it?"

Kitou nodded.

"I know it."

Aloïs led her into the bowels of the city, to where the service zones were. "The Pursang? A remarkable people. You know they're originals?"

"Originals?"

"First humans. Not a product of the diaspora."

"Not one of your Thousand Worlds?"

"No, non-aligned."

"Who are the *free* Pursang?"

"Well, that depends which side of the fence you sit on. Some excoriate them as terrorists - a classical asymmetric, non-state insurgency. Or they're freedom fighters."

"What side of the fence are you on?"

"Again, that depends. The Pursang have just cause. About ten years ago, a race called the Horu laid waste to their planet. Not just laid waste but made a show of it. An infamous genocide, and a provocation. We choose not to be provoked. We leave it to the Pursang. For now, they're *our* terrorists."

"In a proxy war?"

"Yes, but..."

"Yes?"

"The best course of action *could* be to leave the Horu alone. Call the Pursang off. Let the Horu get away with their atrocity."

"Won't they do it again?"

"Why would they, if their goal was to provoke us?"

"Then why are you fighting?"

"We're not. The Pursang are fighting, and to prevent them would be unjust."

"So, you let them fight a proxy war you don't want to be in?"

"Yes."

"Do they kill civilians?"

"No. Never. They believe themselves to be a force for good and behave accordingly."

"Then they're not terrorists."

"Good luck with that argument."

"How are they doing?"

"The Pursang are clinical. Their method appears to be to slice the Horu up, a supply line at a time, so some larger predator can gobble them later."

"And if you're that predator?"

"The trap is sprung, and we learn the real extent of it."

"Aloïs, I need to learn about your politics."

"You do."

"And Kitou?"

"Kitou's a Pursang."

"I thought she was a Po."

"She's a Po now. Ito saved her from a fire."

"What happened?"

"Ito and Pax were soldiers on the Water Bear. In an act of conspicuous valor, they and their First, Idira Law, jumped down to the surface of Fluxor, hundreds of times, bringing out people."

"Why didn't the Water Bear lift them all at once? With one of those petatonne gravity sources?"

"Horu suppressor fields. That was the nature of the siege. The Po broke through it by challenging the Horu to escalate."

"Did they?"

"Ito and Pax are still here."

"The Earth game of chicken."

"I call it game theory. The Horu weren't ready to make war with the Po. If they were, we needed to know about it."

"Are they ready now?"

"We hope not."

They were descending through an empty residential district, past empty parks, along deserted boulevards.

"This used to be teeming with people," said Buss.

"What happened?"

"Bad dreams."

They reached an industrial zone. The number of Fa:ing individuals around them increased exponentially. They stopped

outside a hatch, which Buss reached out to unlock.

"Dr. Box, don't be alarmed by what you see here."

He opened the hatch. There was a sulfurous glow. Layers of Fa:ing crawled over a house-sized container - the source of the ghastly light. To one side, Box saw a spent waste storage pool. Her inner display lit up like a Christmas tree.

Having grown up in a country scattered with the hulks of abandoned nuclear facilities, she knew what this was.

"Aloïs, is that an unshielded fission reactor?"

"Yes. But you're immune. Nanomachines in your travel shots."

"Aloïs, I'm not happy with this."

"I understand."

Ito turned to look at them. He was sitting, relaxed, on a bench, gazing into the light.

"Ah, Dr. Box, you've discovered our guilty secret."

"This is weird."

Buss hustled her onto a catwalk, out of sight of the reactor. The forest flew by, only meters below. Thousands of Fa:ing, some human-sized, clung to the city's ventral surfaces, changing position, to minutely vector the thrust. The polyphonic thrum was deafening.

Buss beckoned her inside.

"Lifting bodies," he said.

"The city's propulsion system?"

"Yes. The Fa:ing convert energy from the unshielded gamma radiation into thrust for the city. More poetically, they convert gamma rays into mathematics."

Box glanced at Ito.

"It's hypnotic," Buss said. "And addictive. We all experience it. We don't let the girls down here, although they once mounted an effective reconnaissance raid."

Box shivered.

"Take me upstairs," she said.

That night, Box had a dream.

She was riding over a ruined battlefield, on a pale horse. The horse had its tail in the air, and was picking its way between

steaming human remains. Heaped ribcages and spines, which looked freshly dead, but with no flesh on them.

They'd been broiled clean.

Her armor was bone. Voluptuously carved, translucent bone, like mother-of-pearl. There were bone beads in her braided red hair. On her shield was a red right hand.

Across the battlefield was Kitou. She was on foot, also picking her way between the steaming remains, wearing her Po training gear. Her right hand was red, dripping with what Box understood was alien blood.

Behind the sky was the Enemy. He had modern beam weapons, powering up with a distant whine. Spaceships. A galaxy, in flames.

She wanted to cry out, to warn Kitou of the terrible danger, but she couldn't. Instead, she looked up, and saw herself, looking down.

She said, *Ophelia*.

She woke, with a whimper.

Box asked Ito about what she'd seen in the base of the city.

"Everyone gets it," he said. "Visions. Strange addictions. It drove the researchers crazy. We think deliberately."

"Why?"

"Clearing the decks."

"Except for you people?"

He nodded.

"Have you had the dream yet?" he asked.

"A field made of bones?"

"Yes, we were anxious to see what you looked like."

"You've had the same dream?"

"Everyone has who spends any time here."

"His red right hand..."

"You mean the device on the shield?"

"It's a mark, from Earth literature, for the vengeful hand of God.

'Should intermitted vengeance arm again

His red right hand to plague us?'

I believe what it means is, how can a benign God allow so much suffering? The answer is, we don't understand his ways."

"Or he isn't benign," said Ito.

Box nodded.

"Also in the basement is a spacetime anomaly," he said. "Created by the Fa:ing processor. We call it the Fa:ing rip. One theory is, when the end-times war begins, the Fa:ing will leave through it."

"End times?"

Ito shrugged. "It's a theory."

"And if they leave?"

"We follow."

That night, in the piazza, Brin showed Box a new move. "This is called the wind shadow," she said. "It's about exteroception."

Box watched the two young women shadow each other's movements, back-to-back, without looking. At the sound of Brin's voice, Kitou launched a taekwondo back kick Box had taught her, which Brin allowed to pass by the side of her head.

Ito joined them in the piazza.

"What's this?" he said.

"We're training Dr. Box," said Kitou.

"For what?"

"To be a Po fighter," said Brin.

Ito raised an eyebrow.

"And how is that enterprise going?"

Box's training gear was slick with sweat. She was out of breath and nursing a livid bruise on her arm.

"She has ability," said Brin.

"Praise," said Ito.

Brin nodded.

"Does master Pax know about this?"

"He will," said Kitou.

Ito nodded acknowledgment.

"Keep at it."

The following morning, Buss unfolded his blackboard again, while Kitou fussed over breakfast. The hall filled with the aroma of scorched milk. With a sigh, Brin rose to help her.

"Who are the Fa:ing?" Buss scrawled on his board.

"Ito?"

"They come from the future," said Ito.

"Wait," said Box, arching her eyebrows. "Time travel? Is that a thing here?"

"It is," said Buss. "I know, because I likewise come from the future."

Box stared.

"Aloïs speaks truly," said Ito.

"Okay," said Box. "You have my attention."

"How far in the future?" wrote Aloïs.

"Anyone?" he asked.

"The math suggests 10^{10^3} years," said Brin.

"But?"

"It could be $10^{10^{10^{56}}}$ years, depending on how you read the Fa:ing algorithms."

"Wait," said Box. "I'm no expert, but that's a big number."

Brin nodded. "A *very* big number. 10^{10^3} years is about the time to heat death, when the universe can no longer sustain processes that increase entropy. $10^{10^{10^{56}}}$ is so large there aren't enough photons in the universe for Aloïs to write it here."

"Why those numbers?" asked Box.

Brin nodded again. "We can hypothesize," she said.

" $10^{10^{10^{56}}}$ is about the time required for all the elementary particles to cycle through all their possible states, which is the longest time required for a quantum-tunneled big bang to produce a new universe identical to ours. In a sense, the most time possible."

Box paused to take that in, then frowned. "You're seriously telling me they come from the end of time?"

"It's a possibility."

"**Why are they here?**" Buss wrote with a signature flourish.

"Kitou?"

"To change the past," said Kitou.

"Very good," said Aloïs. "Our past. *Their* future. The Fa:ing individuals experience time like we do. The Fa:ing *consciousness* appears to experience it backward."

"Dr. Box, yours isn't the only Hopf number in the Fa:ing corpus." He conjured up a green-white planet, floating over the breakfast table. "Fluxor," he said. "Before the genocide. A beautiful world. Not unlike Earth, but wetter and cooler."

The view zoomed out, until Fluxor became a pulsing green point at the edge of a starfield that spanned an arm of the milky way galaxy. An orange dot appeared, with a curved line between them. A glyph said *1,527 light-years*.

"Threnody," he said.

"I first began to study the Fa:ing in your year 2135, seventy years from now, as a young exolinguist, when this was a thriving scientific adventure. In my timeline, the Thousand Worlds are at war with the Horu. A shooting war, but manageable. Then, someone inexplicably fires a doomsday device."

"Sounds bad," said Box.

"It is," said Buss. "Or was. An enduring hypothesis of experimental physics is that one day, supercolliders could destroy the universe, in a cataclysmic subatomic accident. A myth. It can't happen. Unfortunately, weapons manufacturers stepped into the breach, so it did."

"Shit."

"Hopefully, someone will stop it."

"Stop what, exactly? What are we looking at here?"

"From it destroying *everything*, Dr. Box, in a slowly expanding bubble of nothing."

Box thought about that.

"You really don't know how it ends?"

"No, the Fa:ing carried me away before I could find out."

"Carried you where?"

"Here, ninety years ago."

"Ninety years? Aloïs, you're older than you look."

He shrugged. "My people practice germline aging modification. I appear as old as I wish to be."

Outside, the forest flowed by, like Autumn being carried past on a breeze. Box checked the lines on her hands, to see if she was dreaming.

"That's quite a story," she ventured.

"It gets better," said Aloïs.

Kitou and Brin had joined them beneath the blackboard, food now forgotten. "I already told you the story of Ito and Pax at Fluxor," he said. "The point is this - *why were they there?* Consider this your briefing, Dr. Box. What was the Water Bear doing at Fluxor? Why was Ito present at the precise place and time to save Kitou from the fire?

"The answer is Kitou's Hopf number."

"The time of her birth?"

"No," said Aloïs. "Her death."

"Ito saved her?"

"Yes."

"Changing the past?"

"Yes."

"Using information from here?"

"You have it."

"Okay, that's proof."

"Yes," he smiled. "A *mathematical* proof."

"Dr. Box," said Ito. "Our mission here is to prevent a genocide. We're peacekeepers, working in secret, with the most sensitive information imaginable. There're adversaries that'd end us if it came out."

"Aliens?"

He nodded. "Yes. We're not the most advanced civilization in this part of the galaxy - far from it."

"The question," said Aloïs Buss, "is are you with us? We're trying to save a world - Kitou's world - with the help of these extraordinary beings, who've asked for you personally.

"Now we've found you.

"What do you say? Will you help us?"

The planet-killer device began as a packet of ice - a comet's nucleus in the Threnody system's Oort cloud, a light-year from its F-type main sequence star. It was accelerated to near-light-speed by a

bootstrap device, in which a gravity source is projected ahead of the object. Then it went dark - one comet, among millions.

About three hundred days later, five Horu ships unfolded space at equidistant points in an ellipsoid around the system's ecliptic. Seeing this, the Po warship floating above the planet's surface launched a flurry of messages. Her Navigator, quickly awake, brought her military systems online.

In the city below, her crew also woke.

Box was jolted by a burst of adrenaline. She stumbled naked from her bed. There were people in her room. Her skinsuit found her. Its facemask extruded, sealing her panicked objections. She was hustled out and up the spiral stair. In the control room, Aloïs was clipping himself into an oversized captain's chair. Holographic displays were flickering into life. A plexiglass window popped out and fluttered into the piazza.

Seconds had passed.

Aloïs was talking calmly to Pax.

"Ah, Dr. Box. Good to see you're dressed. We're getting ready to leave."

"What the hell is happening?"

"Intruders."

"Who?"

Aloïs manipulated a display. A spider, floating in space; four similar ships, in the shape of flowers and birds, equidistantly around the ecliptic plane.

"*Horu*," spat Kitou.

"More than we've seen in one place since Fluxor," said Aloïs.

"Are we safe?" asked Box.

"For now," he said. "They're at a safe distance."

"How safe?"

"Ten light-minutes."

By the time it passed the first Horu ship, the comet nucleus had accumulated $.5mv^2$, or about 100,000 Mt of potential energy, or more than the deployed nuclear arsenals of all the nations on Earth, ever. As it sailed past the Horu perimeter ring, its bootstrap drive

restarted, this time at full power, creating pressures unknown outside the center of collapsed stars.

The ball of slushy snow and ice fell into the drive, to form a sphere of degenerate matter, indistinguishable from the crust of a neutron star.

The drive had become the weapon.

Aloïs was explaining their predicament.

"Threnody's defense was its secrecy," he said.

"Was?" said Box.

"Yes," said Buss. "No more. There's nothing we can do about five Horu ships. It may as well be five hundred."

"What about the Fa:ing?"

"There's nothing we can do for them now."

"Can't we call help? That big Wu warship we saw?"

"A half-day away."

"Ten light-minutes vs. half a day?"

"Yes, that's the equation."

"A shitty equation."

"Yes."

"Where's Ito?"

"In the basement."

"Why? Someone go get him."

"It's as the Fa:ing have requested."

The Water Bear's sensing apparatus used the same physics as Wu lightship drives. Beefy quantum computers reimaged the origin of incoming gravity waves. The Horu weapon's bootstrap drive was acquired on restarting.

The first Pax knew of it was when he began to lose consciousness. At over a thousand Earth gravities, even the smallest change of direction manifests as violent acceleration.

Sorry, boss.

"Ungh."

On the planet, Aloïs Buss said, *"Merde."*

On my way, said the ship in their heads.

"What's happening?" asked Box.

"A physics weapon," whispered Aloïs.

"What's that?"

"Out we go," said Brin.

Box felt herself falling towards the piazza. She'd been thrown from the window. Overhead, the decelerating Water Bear announced itself with a rolling clap of thunder. It looked like a moth, spiraling onto a dish. Box felt herself pulled *hard* towards the Po warship.

She awoke in the ship's control room, strapped in her usual chair.

She passed out again.

She had a dream.

She dreamed she was looking into the eyes of a striking, flaxen-haired woman. "I was worried you were catatonic," the woman was saying.

"Who are you?" asked Box.

The woman stepped back. Box saw Pax, Brin, and Kitou, studying a holographic image of the Threnody system.

"I'm the Water Bear," she said. "This is my gamespace. A military virtuality. I've slowed everything down, so we can think."

"You look familiar," said Box.

The woman smiled. "A cultural reference, for you. Do you like it?"

"Kara Thrace? You have got to be kidding."

"Dr. Box," rumbled Pax.

"I have no idea what's happening," she said.

"Don't worry," said Pax. "We're the soldiers here. This is our time. Later, you can say how we went."

"A kinetic weapon has been fired," Pax said to everyone. "The planet is lost."

A communal groan.

"Ito has made his way to the basement, as planned. Aloïs is still in the Threnody control room, *not* as planned. What happened?"

"Aloïs accidentally locked himself in a chair," said Brin.

"How?"

"Seatbelt malfunction."

"Can I please go get him?" asked Kitou.

"No," said the ship.
"Solutions?" asked Pax.
"Extract his brain," said the ship.
"How?" asked Pax.
"I can recover it through a ceiling fan."
"What, intact?" asked Box.
"No," said the ship.
"When?" asked Pax.
"Sooner better," said the ship.
"But he's alive," said Box, aghast.
"Do it," said Pax.

Seen from the planet's surface, the weapon arrived unheralded. Not only was it traveling near lightspeed, but it was only a few centimeters across. It passed through the atmosphere in an instant, and several cubic kilometers of ocean and crust disappeared in a flash.

A small object can destroy a world, if it has sufficient energy.
The Fa:ing disappeared.
Ito was in the basement, ready to be extracted.
And then he wasn't.
In the Water Bear, concentration.
"Final recommendations?" asked Pax.
"We go as planned," said the ship.
"Dr. Box?"
"Me? You're asking me? Go Pax, go."
He nodded.
"Take us in, please."

Bodily, they were strapped in the control room lattice. In the Water Bear's gamespace, they watched as the ship performed an elaborate series of high-g maneuvers, inside the base of the city, moving towards the unshielded reactor, towards the Fa:ing rip, which now seemed half the size of the universe.

Now that the Fa:ing lifting bodies were gone, the city belonged to gravity. The planet awaited its prize. Ten seconds past impact, the cinnabar forest raised a gnarled hand to receive it.

As a wall of golden fire rose over the horizon.
"Go," said Pax.
The Water Bear accelerated.

4 → BY A MILKY WHITE SEA

2056

On Xerxes, in the Pleiades, in the region of space called the Smear, the city of Avalon basked under an emerald sky beside a milky-white sea. Once a fortress, now a trading hub, bunting fluttered from its brutalist gun emplacements.

In a rough dockside bar, crowded with spacers and whores, the cryptologist Felix Revelstoke weighed the delicate game of risk to be played by helping the three foreign women. He watched as a troupe of bioengineered transvestites sought to gain the attentions of a visiting krëw of Magellanic Navigators. The tranvies worked hard, but the Magellanics had eyes only for themselves.

The Thousand Worlds were characterized by their genetic naturalism. They were a civilization on a biological mission. Here, in the Smear, by contrast, anything was possible. One of the tranvies gave Revelstoke a lascivious wink. He smiled politely in reply.

In the same dockside bar, on a crowded dance floor, Kitou Gorgonza was being pawed by three outsize youths, while Brin Lot looked on disinterestedly. The young men's older, rougher companions, bristling with radical piercings and body enhancements, also looked on, too hungrily for Ophelia Box's liking.

It was three weeks since the Fa:ing rip, and she still wasn't used to time's arrow not working properly.

"That's not a fair contest," said Brin.

Brin was wedged into a corner, nursing a drink.

"Aren't you going to help her?" asked Box.

Brin shook her head.

Box knew this bar well. In Aberdeen, high on ecstasy, evading the clumsy advances of roughnecks and squaddies. In Marseilles, where the gangsters cut smiles into the faces of their enemies with cutthroat razors. In Punta Arenas, when Antarctica was first opened for mining exploration.

A frontier slaphouse.

Kitou became bored with the youths trying to lay hands on her, and rejoined her friends, leaving the young men searching comically.

"Men are strange," she said.

"It's not strange," said Box. "They're displaying."

"They want to copulate?"

Box nodded.

"How do they think behaving like fools will impress me?"

"Thinking doesn't come into it."

The youths had surrounded another woman and were pushing her from one to the other. Pushes became shoves, then slaps, and the woman was pressed to a wall.

"Put an end to that," said Brin. Kitou rose and walked towards the group.

Box rose to follow, but Brin held her wrist.

"Observe the Po art in combat mode."

Kitou only seemed to move at walking pace, but when she struck, it was a blur. It was like liquid violence. None of them got near her. They were in more danger of striking themselves.

It wasn't even like fighting.

One by one, the youths went down.

No one seemed to realize there was a fight on.

Kitou was sauntering back to their table. The young men's older companions were piling onto the dance floor, looking for payback.

One was striding towards them.

"That was provocative," said Brin, "leaving the older ones standing."

"I was being deceptive," said Kitou.

Brin nodded, and got to her feet, and with a snarl, marched towards the advancing man, and folded him in two with a vicious sidekick to the solar plexus.

"The Lo art does include kicking," said Kitou.

"Well, that should get someone's attention," said Box.

Revelstoke did realize there was a fight on. After all, as a secret policeman, with turf to protect, it was his job to be vigilant.

The *how* of it was interesting.

The *who* was unlikely.

He finished his drink, and departed.

Well done team, said Pax from the Water Bear.

I think we have him.

They were thrown from the bar.

"For fighting the wrong assholes," said Box.

They sat down on an upturned pallet. A passing law enforcement prowler slowed, then let them be. A handsome young man came out of the bar, and gave them the address of a religious congregation.

A sharp tang of salt air, and diesel, swirled around them.

Gulls cawed in the distance.

Brin was bleeding from a lacerated eyebrow.

From somewhere, came the sizzle and heady aroma of sausages. Box thought, this is the best time of my life. It can't get any better than this.

They burst out laughing.

They found a room in a dockside hotel, overlooking a doglegged lane, leading down to the harbor. By comparison with spaceflight, it was a ramshackle palace. A neon sign flashed sporadically outside their window.

"Otel," it said.

Box asked the ship to disable her Broca transfer. Now it flashed gibberish.

They were woken at dawn by lively commerce beneath their open window. The lane had become a market. While Box lay in bed, Kitou morphed her discarded party clothes into loose training gear and set out running. Box dragged herself from bed and watched her disappear toward the harbor.

She noticed that Kitou was followed.

With her Broca transfer still off, Box listened to a dozen languages. The prevailing argot sounded like Mediterranean Arabic, a beautifully liquid language. She smelt the heady aroma of the stalls. Clean air, fresh produce, hot food, and an undertone of rotting garbage.

Brin rose, scowled, and seeing that Box's translator was off, began to make the Avalon equivalent of strong coffee.

Kitou ran down to the harbor, then along a wide corniche, beside a tumbling sea. Finding her stride, she surged onto a steep hill, which climbed towards a beam-weapon emplacement, and accelerated.

Running was pure pleasure. Her simple shoes slapped lightly on the salt-encrusted concrete. Soon her pursuers were left far behind, as plain as day on the steep ocean road. She relented, almost letting them catch her.

At the top, she waited, breathing.

There were two young men and a woman. One of the panting men said, "We're the police."

Kitou said, "I know."

The second man said, "Are you really a Po soldier?"

Kitou wrinkled her nose. "Not really.

"But my friends are," she said.

They companionably enjoyed the view, like any young runners would, at the top of a challenging climb. Avalon was a beautiful city, by its milky-white sea, with its military towers painted every color.

"Can we run with you tomorrow?" asked the young woman.

"Yes, of course."

"Well," said the first young man.

"I guess we'll be off," said the second.

Kitou nodded.

After her police escorts jogged self-consciously down the hill, Pax said, from above, *Good job*. Kitou, knowing she was observed, gave him a virtual thumbs-up of approval.

Then she set off at a fast lope, back to the hotel, for breakfast.

They spent the day buying local clothes, and food for the room, and cheap souvenirs. The local currency was paper and coins, and Pax had given them pocketsful each, with instructions to spend it.

Garner attention, he said.

In the early afternoon, they found themselves in a larger market. This was what Box would call a souk or bazaar. The street food and local attire gave way to consumer electronics, and in one section, weapons. For the Po soldiers, these were novelties. They regarded them with professional curiosity.

"Why," asked Brin, "would any sane person buy a weapon?"

"To defend themselves?" asked Box.

"Nonsense. It only puts them in danger."

Kitou only seemed to partially understand the concept of property. She spent all her money on gifts for the others, then happily asked for more. By the mid-afternoon, they were local celebrities. Box steered Kitou away from a sex shop, not to protect her, but for fear of what she might buy there. They were enjoying a meal of sticky rice, served up with raisins and pine nuts, and the ubiquitous local sausage, when Pax broadcast in their sensoria.

You're about to be arrested.

The Water Bear chimed in.

Dr. Box, please follow my instructions. Brin is about to rise and pay for your meal. At the same time, you rise and follow the pulsing red line, at a normal walking pace.

A red line appeared in the floor, leading from the café.

Box did as she was instructed.

Our main goal here is to not get you shot. There are three local heavies. They may or may not be professional law enforcement. They could perceive you as a threat and might over-respond.

The red line snaked behind some market stalls. She followed it. She heard crashing behind her. The pulsing line then passed through a laundry and snaked towards a tall, professorial man, who was standing by a wall, speaking into a throat mic.

At the same time as keeping you safe, we wish to make a point. I want you to approach that man and put your hands in the air.

She saw Brin and Kitou doing the same, from opposite directions. They would finish up on three sides of him.

If this goes wrong, I'll lift you straight out of there. If anyone shoots, I'll lift the bullets in flight. You're completely safe, Dr. Box.

Box was intrigued. This was like chess, with people for pieces. She raised her hands in the air.

The man stiffened, and said, "I have you surrounded."

Brin raised one eyebrow.

The three local heavies crashed into the open space behind them, brandishing snub-nosed weapons. One shouted, "Hands in the air."

Another one bellowed, "Drop your weapons."

The third, who had brightly colored clothing strewn around him, just crouched and pointed his handgun.

Brin said, "We surrender."

The man sighed, and motioned his men to holster their weapons. They did so with military precision.

Not amateurs, then.

"I'm Revelstoke," the man said.

"We know," said Brin.

"Will you come peacefully?"

"As you wish."

The gunmen escorted them to a nondescript building, with Revelstoke following at a distance behind. Inside, they were regarded with frank curiosity. Kitou waved at a young woman. They were taken to a dusty, academic's suite of rooms. Open books of mathematics and criminology climbed over each other. Revelstoke appeared and motioned them to sit.

"What do you want?" he said.

"We're a hand of the Po," said Brin.

"If that were true, I'd never have seen you."

"We gave you cause to see us."

"So you say."

"We deployed the barfight signal."

He sighed. For a Po double agent, a special ops team was like an anxiety dream. They'd have their own, stifling agendas. Worst of all was a hand, if that's what they were. They'd burn all the oxygen in his room, in sips and then gulps, then leave, never to be seen, never having been there.

"What do you want?"

"We need backup, with utmost secrecy."

"Secret from my people?"

"Yes."

"I suppose you can justify that?"

"Our mission is to prevent a systemic war."

"Who are you?"

"We're the Hand of the Water Bear."

He shook his head. "No, you're not. I've sequenced your DNA."

"We're from the future."

"You appear to be serious," he said, after Brin told him the parts of their story he needed to hear. He sat with his chin in his hand.

"Can your ship send me a cryptographically signed message, confirming all that?"

Brin nodded. "Now you have it."

He stood and paced the room, visibly anxious, constrained by the walls. "You know," he said. "My simple solution is to inform my superiors."

"You won't."

"No, I suppose not."

"We need money," said Brin. "And medical services."

"You have wounded?"

"A man dead."

"You want him... what, *resurrected*?"

"Yes."

"You know that's illegal."

"This is the Smear."

He laughed, a pleasing laugh, thought Box, for a secret policeman. She watched the shadows of feelings play over his face as he considered his options.

Who are you, officer Revelstoke?

Not a poker player.

Or a good one.

"Let me think about it," he said. "In the meantime, please try a low profile. Tell this one to run slower."

He shook his head.

"I suppose this is what we get paid for."

The following day, Box watched Kitou weave her way through the market, this time with her escorts beside her. The friendly waves of the market traders were less in evidence now. Box didn't like that there should be secret police waiting for Kitou. It implied the industrial heft of a Soviet-style surveillance operation.

Box had read her political backgrounder. Avalon had all the hallmarks of a kleptocracy. There were no political freedoms here, but the people were happy.

Happy is cheap.

And prosperous is relative.

The real money flowed through Avalon Station, 35,000 km above. This was owned by the Avalonia Cpy., a private corporation, and Revelstoke's official paymasters. Her historian's nose told her that profitable sinecures like Revelstoke's were routinely bought and sold here, so as well as being an agent in the Security Police, Revelstoke was most likely a *petit* oligarch himself.

All with the Po's blessing.

Are you corrupt, officer Revelstoke?

She suspected the Avalonia Cpy. didn't care, provided he served their interests and was plausibly competent. But what did she know? She was extrapolating, from similar places on Earth. She was a stranger here. She resolved to try to suspend judgment.

For a while, at least.

Half an hour later, Kitou called.

Can you come to me? she asked.

Brin was doing her own exercise routine. Chin-ups and crunches. More in a session than Box had done in a lifetime.

Yes, why? Brin replied.

Mr. Revelstoke wishes to visit the Water Bear.

Box followed Brin along the crooked street to the harbor front. It was a wet morning, and the slippery concrete splashed under their feet. A brisk wind blew white spume over the cornice. It felt delightful to be out in the weather. Thirty minutes later, they crested the brow of a long hill, which floated above the tops of the low clouds, to reveal a glorious sunrise.

Kitou was sitting cross-legged, talking animatedly with Revelstoke.

"Ah, here they are," he said with a smile. He seemed a much happier secret policeman than yesterday.

"Alright, let's talk," he said.

It was a beautiful day in heaven, with the wind in their faces, looking out over a billowing cloudscape. Box felt a rush of endorphins.

A prowler descended.

"Our ride," said Revelstoke.

They climbed into the pilotless craft, which climbed into the emerald sky. They rose like an express lift, until the horizon curved to an arc, and the stars became pinpricks of light.

The ship spoke in their sensoria.

I have you, Avalon prowler.

They met in the gymnasium, which the Water Bear had repurposed as a conference room. The practice ring had become a soft crystal table. The ship even made gravity, for Revelstoke to pace in.

"I've found you your money," he said. "But you'll have to get it."

Pax nodded. "You mean steal it?"

"Yes," said Revelstoke.

No surprises there, thought Box. She knew from her Earth history that military units on covert missions often had to misappropriate resources.

Steal clothes from washing lines.

Rob the occasional bank.

"Steal what," asked Pax, "from where?"

"For a ship refit and a full-body resurrection, plus ten years of covert ops, say 250 kg of gold."

"Why gold?"

"Because I know where there is some."

"Where?"

"It's an... unusual strategy."

"Just tell us."

"Steal your own money."

"From?"

"From the central bank on Praxis."

"Explain."

"One," he said, counting off his fingers. "There's no victim. After all, we are the good guys."

"Yes," said Pax.

"Two, the Water Bear has an account there. I know, because I called and was told it was above my pay grade."

"Three, no one will ever suspect what has happened. I promise you, as a policeman - the idea that this group traveled from the future to steal its own money will never emerge as a law enforcement theory."

"Finally, you have the Water Bear's cryptographic keys. You can prove her identity. It's the perfect crime. Even if they solve the case, they'll never be able to admit you were there."

"But," said Pax. "The Civics."

"Yes," said Revelstoke.

"Who are they?" asked Box.

"The AIs that govern our society," said Pax.

"They'll work it out straight away," said Revelstoke.

Box said, "How?"

Revelstoke shrugged. "Because they can?"

"What Mr. Revelstoke means," said Pax, "is that no matter how unlikely it may seem, we're the best explanation. The Civics will calculate this. The question is, will they allow us to do it?"

Revelstoke said, "I think so."

Box said, "Why?"

"Game theory," said Revelstoke. "Dr. Box, why did you begin operations here in a state of imperfect secrecy, starting fights and spending money on trinkets, when your mission was a secret?"

Box shrugged. "To provoke you?"

"But why? My department leaks like a sieve. Now everyone says there's a Po warship here."

"So what?" said Brin.

"Correct," said Revelstoke. "It's the fog of plausible deniability that surrounds all Po operations. Nothing to see here. Who would believe you? Even the Po themselves won't ask."

"Clever us," said Box.

"It *is* clever. The Civics work on similar principles, but on much larger scales. They occupy saddle points, keep positions open. When they do intervene, they make the least interventions, until they have complete information, then they act decisively."

"Like in Po," said Brin.

"Yes," said Revelstoke. "Like in your eponymous game. Or like juggling. You don't want to go grasping at balls in the air."

"Is that what we are?" asked Box.

"Yes," said Revelstoke. "An option in a much bigger game. A wavefunction that needn't be collapsed at this time."

"Why don't we just talk to them?"

"Nobody talks to the Civics."

"Why not?"

"Separation of concerns. By keeping the operating system separate from the data, we avoid the corruption of either."

"This works?"

Revelstoke nodded. "It has for ten thousand years."

"Amazing."

"So," said Revelstoke. "Let's plan a bank heist."

"One more thing," said Box.

"Yes?" asked Pax and Revelstoke, in unison. Brin stifled a laugh.

"How much is 250 kg of gold?"

"It depends on what you mean by *much*," said Pax. "It's a weight we can carry."

"I mean, what's it worth?"

"Its monetary value?" said Revelstoke. "A ship's refit costs about 25 kg. A human resurrection, slightly less."

"Then steal more," she said.

"Explain," said Pax.

"Most military operations fail because of logistics. Trust me. Make up a number. Double it. That's how much it will cost. How much is there?"

In addition to his role as Navigator, Pax was the ship's financial controller. In twenty years, he could withdraw this money legally.

"About a hundred times that," he said.

"Steal it all."

"She's right," said Revelstoke.

"Agreed," said Pax.

"How do we steal twenty-five tonnes of gold?" asked Brin.

"Deceptively," said the Water Bear.

Kitou reappeared, carrying a weighty gray cylinder, the size of a hatbox. Beads of moisture slid along its surface.

"What's this?" asked Revelstoke, although his expression suggested he knew.

"This is Aloïs Buss," said Pax.

"Our friend," said Kitou.

"Truly," said Brin and Pax.

"Well," said Revelstoke. "How dead is he?"

"Not too seriously," said the Water Bear. "His neural structures are repairable. When he wakes, he'll be the same individual."

"How did it happen?"

"I extracted his brain through the control room ceiling of the city of Threnody, before it was destroyed by a physics weapon."

"How?"

"Shaped gravity pulse."

Revelstoke winced. "He was alive at the time?"

"Conscious."

"That's not good."

"Tell us about it," said Box.

"There'll be psychological trauma."

"Undoubtedly," said the ship. "He was still sensing when I got him here."

Box shut that idea out of her mind.

"Why didn't his wetware sedate him?" asked Revelstoke.

"There was separation," said the ship.

"We'll have to erase those memories," said Revelstoke.

"Of course."

"But..."

"I understand," said Pax. "There may be irreparable damage. Do what you can. Cost, we agree, is no issue. Bring him back to us."

"We were responsible for poor Aloïs," said Kitou.

"Are responsible," said Brin.

"Truly," said Pax.

5 → ANDROFORM ALOÏS

2056

The Water Bear slid into its dock, like a wasp invading a chrysalis. Box felt a swell of pride when she heard a worker whisper *Po warship* to another. This was *her* wasp. She had to admit it had a certain fuck-you beauty.

If Praxis was a thousand Mannhattans, then Avalon Station was all the world's factories, strung along a hundred kilometers of space. The thousands of disparate parts were held together by a thread - a magnetic train that went from end to end in less than a minute. Pax said the string formation was to reduce its attack surface, since a line is harder to hit than a plane. To Box's eyes, it looked organic, like a bristling industrial town with life bursting out of every crevice, or a rave party gone haywire.

People were everywhere - humans jostling for space with aliens Box couldn't have dreamt of. Its public spaces were as bustling and crowded as Kowloon or Sao Paolo.

She found it intoxicating.

Pax Humana. That was the name the Avalonians gave to the current historical era, since the Tung stopped trying to kill them, and instead decided to live there. The decommissioned beam weapons surrounding the city were intended to repel Tung invaders. Why the change of heart? No one knew. The Tung were mercurial creatures, and besides, they didn't have hearts, being shaped more like slime molds. The Avalonian technocrats were happy not to be killed, and welcomed the Tung into their communities.

Of course, it wasn't called *Pax Humana* at all. That was how Box's wetware replayed it to her. She guessed it came from a language much older than Latin.

How long had the Tung been around? According to her wetware, six hundred and ninety-three years.

Since before Copernicus.

Before Michelangelo's David.

Now they were everywhere, wispy cartilaginous obscurities, air breathers, with the ability to turn corners using spat silk, and thus move faster than humans in weightless conditions.

In another time and place, they'd be nightmare creatures.

She was wearing their silk now, a confection of adaptive pashminas, just enough of it to be decent, perfect for the sweltering humidity. With a thought, she could whip them all into her hair and be naked, and no one on Avalon Station would notice.

Sometimes she did, for the experience.

It was that kind of place.

Kitou and Brin had surveyed the local bacchanalia and were now taking turns guarding Aloïs's surgical suite. *Looking fly in police uniforms*, thought Box. The surgery was in a seedy part of town, between gun shops and discothèques. The guns were more serious here. They looked like military hardware. She guessed they probably were - dully gleaming exoskeletons with hyperkinetic bullets. She had the sense that nothing couldn't be bought or sold in this place.

Aloïs was in the clonal phase of his treatment. The suite had extracted a nucleus from his brain extract and implanted it in a donor egg, taken at random from Box, Brin, or Kitou. Now he was growing a body. One of them would be his biological mother.

The surgical suite was called Clotho, and she was a citizen of Avalon Station, a cyborg of Tung manufacture, with a human personality. She explained that restoring Aloïs's exact wiring was the goal.

"Not a problem, surgically," she said, "but culturally messy. Thousand Worlds law requires that a person has neurological continuity, or else they're a copy."

"Why's that a thing?" asked Box.

"Without continuity, the original is considered to have died, with all the legal and emotional loss that entails."

"Is sleep continuous?"

"Yes, otherwise you'd wake up a new person."

"What about coma?"

"Provided it's essentially the same brain, it is."

"Essentially?"

"I do have some freedom of action."

"You do a lot of this work?"

"It's my purpose."

"How does it feel, to have a purpose?"

"My directive is to desire it."

"That's a complicated answer."

"I'm a complicated person."

"How's he doing?"

"Your ship did a very good job under the circumstances, but I have to consider every neuron. It will take me several weeks of carefully sequenced surgery and recovery, to restore the same person."

"When will he be awake?"

"I plan to briefly revive him tomorrow."

"Tomorrow? Really?"

"Yes, and I need you here. His family. It's important."

"We're there."

"Alois?"

"Ship? Where am I?"

"You were in an accident."

They were all together in the visiting room of the surgical suite, which resembled a teledrama mortuary, with drawers where Box imagined the bodies were kept. She felt slightly cheated. For 25kg of gold - about twenty-five million Earth dollars - she wanted gleaming equipment. A Swiss rejuv clinic would've handled this more theatrically.

"Big accident?"

"Yes."

Alois's brain was nowhere to be seen. Box tried not to imagine its condition.

A soup, becoming a stew.

"Ship," said the brain through a speaker, "I can't see or feel anything. Are the others alright?"

"Yes," said the ship. She was physically present, wearing her androform body, which mirrored her gamespace persona.

"What's my prognosis?" asked Alois.

"Optimal," said the ship.

"How long?"

"Alois," interrupted Pax.

"Pax, my good friend. How are you?"

"I'm well," he said. "I'm here with the team."

"How is Threnody?"

"What do you remember?"

"I remember going to bed."

"Threnody's gone."

"Gone?"

"Details later."

"The Fa:ing?"

"Safe, as far as we know. Escaped through their wormhole."

"Down the rabbit hole?"

"Yes."

Alois sighed, despite having nothing to sigh through. It was a forlorn burst of white noise.

"All those years," he said.

"Vindicated, Alois."

"Yes, I suppose that's true. And me?"

"Growing a new body."

"Am I a backup?"

"No, we got your physical brain. You're real."

"Well, thank goodness for that."

"Now you should rest."

Box spent her days on the surface, haunting the souk and the coffee shops. Avalon was starting its high summer, and the city was steaming. She'd taken to wearing the local dishdasha, in place of her adaptive skinsuit, or her Tung scarves, which would've got her arrested. There was monsoonal rain, heavy and warm, then baking hot cobbles and concrete, so hot they hissed in the sunlight.

She loved this place. She could live here forever. She learned the rhythms of the markets. Every day, before dawn, the ships would arrive in the harbor. They were real ships, wind-powered ocean vessels, sailing over the water from places unknown. They'd unload in the docks. Some of it would go to the markets. Some would disappear on lorries, bound for places unknown.

Why did this happen?

Why was this city here?

Revelstoke provided answers.

Far from being kleptocrats, he said, the Avalon Cpy. had protected their world - and its people - from the ravages of Smear economics. "Spacers need a place to escape to," he said. "Agreeable and properly managed. A minimum of overt lawlessness. This is that place. It's good business."

"A place where the trains run on time," said Box.

"There are no trains here," said Revelstoke.

They were relaxed on a beach. The tumbling waves were like carbonated milk crashing. The sand was made of seashells, minuscule crabs, and bivalves, so silky-soft it was like talcum powder.

Biogenic silica, said her wetware.

She'd decided that Revelstoke was attractive, in a craggy, Sherlock Holmes kind of way.

"And besides," he said. "The oligarchs are people too. They don't want their home to become a sordid little appendix."

"So, they get rich," she said.

"Of course," said Revelstoke. "Why not?"

"It's unequal."

"Why should anything be equal?"

"There's no freedom here."

"Political freedom is the illusion of freedom."

"Says the secret policeman."

"Dr. Box, *you* are the secret police."

"Humph, maybe."

The skin of the Water Bear came away like a veil, to reveal the workings beneath. First, she was swarming with Tung, then she was naked.

She was *dense*.

Her chassis was a single billet of titanium, into which cavities had been drilled, to accept interchangeable parts. Those parts were being replaced. Box was with Pax, looking down from a gantry. Kitou was playing the Geometry Game, deep inside the gutted chassis, with a half-dozen Tung workers. For once, she was hopelessly outclassed.

"Seriously? Nine years?" said Box.

Pax nodded. It'd take nine years to reach Fluxor at the Water Bear's cruising warp speed. Her crew would sleep the journey in stasis.

"Like hibernation?"

"No. A stasis field stops time."

"How does it work?"

"We don't know."

"Aliens?"

"Yes."

"Isn't that like time travel?" she asked.

"Yes," said Pax. "It *is* time travel. It's how we reach the far future."

"People do that?"

"Yes."

"How do they get back?"

"That technology is beyond us."

"They're stuck there?"

"Yes."

"What about us? We traveled backward in time."

"That was the Fa:ing."

"But we can go forwards?"

"Yes. In stasis, or by traveling at relativistic speed. Or we can simply live the experience. Time is plastic, in one direction."

"Time's arrow?"

"Yes, thermodynamics."

"Could we meet ourselves?"

"We can meet ourselves now."

"Seriously?"

"Yes. We could travel to Earth and find you."

"Did it take you nine years the last time?"

"To Fluxor? No, the Xap took us there."

"How?"

"By the force of their minds."

"Really?"

"Yes."

"A Tung in a bar told me the Xap are a myth, an invention of the Thousand Worlds Civics, to maintain social order."

Pax laughed. "Yes, that's a theory. Like all negatives, easily disproved."

"You've met a Xap?"

"I have."

"What are they like?"

"They're the most interesting beings imaginable."

"Tell me more."

Pax shook his head. "Your time will come."

She laughed. "You tease."

He laughed with her. It was a deep *basso* rumble, like all the sounds Pax made. "The process by which a story unfolds can be as revealing as the story itself."

"Did I say that?"

"Among other things."

"Did I make a pass at you?"

Pax and she had spent a night on the surface below, in the Otel room. She remembered strong liquor, and fumbling with clothing.

"You did," he said. "I was a perfect gentleman."

"Rubbish. A gentleman would've said yes."

"For that, you'll have to wait till after the mission."

Revelstoke invited them into his home, inside the disused weapons blister of a beam emplacement, in one of the military towers. The Po were impressed. Personal weapons were easy to disrespect, but there was no arguing with a house-sized hydrogen maser. If Box had expected an oligarch's lair, she would have been disappointed. This was a working man's home, scattered with physical books and furniture from the bazaar. He'd bought platters of Avalonian street food, with mumbled apologies for not having made it himself. There were soft, nutty rolls, and vine leaves baked around chewy cephalopods, and sweetmeats, and bone-dry, fizzing white wine.

"Why would the Po send a working mathematician - a cryptologist, me - to a remote place as this?" he ventured, when the food was all eaten. His apartment overlooked the central bazaar. Box could see the café where they'd been arrested.

"To count all the money?" she said.

"Very droll. You told me your secrets - now I'll tell you mine. I was sent here to solve a public health emergency."

Aloïs Buss looked up at that. He was present in his ambulatory body - a metal skeleton, which Box found uncannier than any alien. His brain was still in the surgical suite. He'd spent the meal pushing a single pastry disconsolately around his plate.

Secret you say? he said, in their sensoria.

Revelstoke conjured up two beings in a holodisplay. One was an opulently beautiful woman, the other a humanoid reptilian.

"Those are Thesps," he said.

I'm not aware of that species, said Aloïs.

"It isn't a species," said Revelstoke. "It's a lifestyle."

"They have a neurological disease," he said. "A disease, called the Thespian virus, and it's killing them."

The pleasure zones of Avalon Station were the bawdiest places Box could imagine. The nakedness was only the start of it. Every fetish seemed to be on open display. Revelstoke explained that this was the oldest part of the station, and it'd once been a bordello.

"A place where the burghers of Xerxes could come and get laid," he said.

Box was thrilled to be out with her friends. They'd dressed for the occasion. Aloïs's skeleton was tied with red ribbons, and Kitou had found it a tricorn hat to wear. Kitou reprised her emerald pajamas from the Pnyx. Box wore her baroque judo suit. They chose a play, whose meaning escaped her. It seemed to consist of three people, in a room, talking elliptically about nothing.

One was a ghost. The other two were partygoers.

Or all three of them were ghosts.

Like Japanese cinema.

Her companions seemed equally puzzled.

Revelstoke led them into the street, then through a nondescript door, into a concealed part of the station. "This precinct is called Backstage," he explained. "It's a shared service area."

There was gravity here, due to the spin of the spacecraft's original structure. Box's heels clacked dully on rough metal grills. She could see the players from their show, disrobing. More bacchanalia, on the cusp of beginning. Backstage was dimly lit, and full of dramatic shadows, like a crime *noir* on acid. Revelstoke clicked his fingers, and a streetlight ambled towards them.

Box laughed. This was delightful.

He took them to a faux-stone house, which appeared to exist by leaning on the houses beside it. Revelstoke knocked on a door, which was opened by a reptile. He was a beautiful being, over two meters tall. Box knew he was he, by his astonishing genitalia.

"Flex," said Revelstoke. "These are the covert soldiers I talked about. They've worked on a problem like yours."

Box sensed a troubled backstory. Revelstoke was pleading. Flex was unsure, but at the end of his wits. Box understood they were intruding on a personal drama.

The reptilian sighed, and let them in.

Flex's partner was called Salmonella Dysonsphere, and she was a shadow of her former opulent beauty. Where she'd once been a big woman, now she was withered and gray. Her skin hung in folds.

She reminded Box of a victim of famine, in a place where there was no scarcity.

She was dying.

Flex proved to be a demonstratively tender person. He took his lover's hands and wept openly.

Flex is a Span, said the ship in Box's sensorium. *A humanoid, spacefaring race from the region of space you call Boötis.*

Hey, Water Bear. Why are we here?

I don't know, said the ship.

The stricken woman lay in an unmade bed, in a clothes-strewn room. She was mumbling with feverish energy, like someone reading fragments of words, from a script that was too fast to read. Her boudoir resembled a proscenium stage, with chairs around the bed for an audience, and theater flats behind. Tonight, the room was set as a forest, with shafts of sunlight falling through majestic trees. Box could see it'd been set that way to give the poor woman some peace.

"Aloïs," said Revelstoke. "This is for you." He projected symbols in their heads - line after line of abstruse mathematics.

The skeleton clacked. *Where did those come from?* it asked.

"From an autopsy," said Revelstoke. "Those are statistical transformations performed on physical neurons. Literally, a map of the changes the disease caused in the victim."

A change to the physical brain?

"Yes."

"Like the Fa:ing number?" said Brin.

Yes, said Buss. *This is quite unexpected.*

"It *is* the Fa:ing number," said Revelstoke. "I'm sure of it. But seen from a different... *perspective?*"

Yes, said Buss. *I see what you mean. It's smaller. Optimized to cause damage.*

"Like a virus."

Yes. *Like a **weaponized** virus.*

"What can you do?" asked the Span. His voice was as low as his testicles suggested. He seemed to be able to hear what the Aloïs Buss robot was saying.

Buss said, *Let me think.*

Why Avalon? asked Buss, after they'd left the woman and her partner to their suffering. *Why here? And why the Thesps, of all people?*

"Good questions," said Revelstoke. "I don't have any answers, just theories. That's why I'm here. So far, I've failed." They were wandering through the Thespian backstreets. A few meters away, through water-filled honeycomb walls, a party was raging. Here, behind the scenes, everything was peaceful. All she could hear was the sound of their voices, and the knock of their heels on the grills.

The skeleton clacked again.

Why didn't I know about this?

Why didn't the Thousand Worlds tell me?

Let me get back into a human body, with my real brain, he said. And I'll see what I can do. We might've already solved this. How long does she have?

"Days, maybe hours."

Merde. Then there's no way to save her.

"Why?"

I'm weeks from a full recovery. I don't have the bandwidth for this kind of problem.

The Water Bear said, *Aloïs?*

Yes?

You remember I have a full backup of you?

"This is exceedingly strange," said Aloïs, holding up a hand in front of him. The Water Bear had installed her copy of his personality in her androform body - a military cyborg, with ship-level wetware.

This raises complex moral issues, she explained. A restored copy is a person, and it has rights. You have rights. We can't just turn you off now.

The new Aloïs grinned. "Astonishing," he said.

It's a very good body, said the ship. Ready for combat. Please don't break it.

"I was referring to your brains."

It has ship wetware.

"You're like this, all the time?"

Better. That's a remote you're wearing.

Buss laughed. "Goodness. Now I have ship envy."

Brin and Kitou grinned. Since his sacrifice, and his brave way of dealing with it, Aloïs had gone from being an oddball passenger, to their hero. If he wasn't before, he was a part of the Hand of the Water Bear now.

Pax raised one eyebrow.

"So," he said. "About this disease?"

"In 2135," said the new Aloïs, "with the help of sixty more years of research, we'd begun to understand that the Fa:ing number wasn't a *number* at all, but a *question*. A question the Fa:ing have been working on asking for centillions of years. Essentially, it's trying to infer our universe from its ending conditions. A task of near-impossible complexity, since in theory a whole universe is needed to model its own state completely.

"But, we have fractals. Huge swathes of the universe are made up of patterns, which can be generated by formulae, provided you have the right formula. That's what the Fa:ing number is - the algorithm. A way to model a universe. Once you have it, you can make predictions. It's done remarkably well. It's predicted *you*, Dr. Box.

"It's also sentient, but not in a way we can easily imagine.

"Consider this. The Fa:ing number sees time backward. To them, *we're* the time-traveling aliens. They use us, to see into *their* future. It's indescribably clever. Dr. Box, you're no longer part of the answer. You were, until you were born. Now you're part of the question."

"I have no idea what you're talking about," said Box.

Aloïs nodded. "Understandably. But we don't need to *understand* the Fa:ing number to use it.

"Felix, does the Thespian virus start with bad dreams?"

"Yes," said Revelstoke.

"Strange, fevered dreams? Like a fugue state? Obsessively counting?"

"Yes," said Revelstoke. "Exactly like that. First, a dissociative fugue - then psychosis, then death."

"Those are all symptoms of the Fa:ing number in its terminal phase," said Aloïs. "Think of it as crowdsourcing neurons. The computation uses up all the brain's processing capacity. First, lack of sleep. Then neurosis, followed by dementia. Then, in the end, psychosis and death as the lizard brain is consumed by the task."

"Left untreated, the Fa:ing number *kills* people," he said.

"How do we stop it?" asked Revelstoke.

"Simple," said Buss. "By surgery."

"It's just a few cells, in the occipital and parietal lobes of the brain. An organ. We call it the Antenna."

"We?" said Box.

Aloïs tapped the side of his head.

"I've had the surgery. I'll probably have it again. I do seem to be susceptible."

"Aloïs, do you know how to perform the operation?"

"No, but I bet Clotho does."

Later, while Revelstoke went to find Clotho, Box took Aloïs to one side. "Why the Thesps?" she asked. "Why actors? Are you thinking what I'm thinking?"

"That depends, on what you're thinking."

"Do you think they've been infecting their audiences?"

"Yes, that'd be an elegant design for a pathogen."

"How long has this been happening?"

"Who knows? Revelstoke's been here for three years. This is the terminal phase. Maybe decades."

"That's a lot of audiences."

"Yes."

"Who comes to this place?"

"Everyone, eventually, comes to the Smear. It's the crossroads of our civilization."

"Can we catch it?"

"Unlikely. I think it's run its course in this place."

"But... elsewhere?"

"Yes. Out there, it's probably wildly infectious. That's why Felix is here. It's why they sent a cryptologist. I'm surprised they didn't send me. They're scared."

"Why is it killing its victims?"

"The Thesps? When does any successful pathogen kill its host?"

"When it's finished with them?"

"A frightening idea."

Clotho made house calls. She arrived outside the Dysonsphere brownstone, wearing a birdlike androform woman, with an acoustic scalpel engineered into a biomechanical hand.

"Is that all you need?" asked Box.

"Ample," she said.

Inside Salmonella's boudoir, which now resembled a nautical scene, with seafarers waving cutlasses at sea monsters, the cyborg waved her scalpel at the dying actor, whose eyes snapped open, then fluttered shut.

The room seemed suddenly quiet. Box felt free of a sound, which had been intruding on the edges of her wetware.

"There," said Clotho.

"She's cured?" asked Flex.

"Yes," the surgeon said briskly. "When she wakes, give her something nourishing to drink. Then love, plenty of love. She was lost, now she's found. She needs to know that."

A printout emerged from her breast pocket.

"Get this script filled and visit me. This is for her nutrition. Tell her she needs to get fat again."

"You people," she said to the Po soldiers. "If only you'd told me about this before, I could've saved others."

"I didn't know," said Revelstoke.

"Didn't know what? That it was happening? No, you were more concerned with keeping it a secret from yourselves."

"We are where we are," rumbled Pax.

"Leave out the vapid homilies, young homo sapiens," she said. "Do you know what your species' problem is? Not enough of your brain is devoted to thinking."

"I'm sorry," said Revelstoke. "Our inquiries should've converged."

"You're supposed to be the great strategists. The masterly Po elite. Entropy help us if that's true. It took two heads being banged together at random to make a simple connection. How can you explain that?"

"Do you know how angry this makes me? How can you hope to solve the *real* problem, if you don't talk to each other?"

"What real problem?" asked Pax.

Clotho narrowed her eyes. "The end-times war, of course."

"How do you know about that?"

"How do you think I know, Navigator? Because I rewired every synapse in Alois Buss's brain."

"Then we must bind you to silence."

"Or else what?"

Box said, "Clotho, please."

The surgeon scowled. "I'm outraged, Ophelia."

"You have every right to be."

"Humans," she said.

"I know."

"All the poor Thesps."

"I know."

"Do we have to worry about Clotho?" asked Pax, after she left.

"No," said Revelstoke. "Leave her to me. She only has the best interests of her people in mind."

"A dangerous mindset," said Pax.

"What, or else thumbscrews?" asked Box.

Revelstoke sighed.

"Leave her alone," Box snapped.

"How are we going for money?" asked Pax.

"We have none."

"How are you paying for this?"

"Credit, and favors."

"Your credit?"

"Yes, and my favors."

"Then it's time to get going," said Pax.

"Yes," said Revelstoke. "Your refurbished chariot awaits."

"And Aloïs?"

"Please take the androform copy. It's already asked my permission. I don't want it around here. Merge the real one later."

"Alright," said Pax. "Let's go."

In the last few hours before departing, Box visited Clotho.

"Dr. Box," said the surgical suite.

"Clotho, thank you for healing Aloïs. You healed me too. All of us."

Clotho's personality was back inside the machine. Somewhere, out of view, Aloïs's brain was still brewing.

"Thank you, Ophelia. This is what I do. Sometimes, it's more than what I do. This is one of those times."

"I'm sorry about the Antenna."

"I know, it isn't your fault. It's not the fault of any of you. It's the way humans are. Archetypal predators. It's your strength and your weakness."

"Take care of Aloïs," said Box.

"Has anyone said, take care of yourself?"

"Heh, I've heard it before."

"You have a good heart, Ophelia Box."

"You think so?"

"I know it."

6 → RISK

2056

The car hummed powerfully from lane to lane, pressing her down in her seat. Curtains of rain fell through forests of shimmering towers. Fast trains flowed like quicksilver between island conurbations. Each turn opened a new vista. It was like all her adolescent dreams of an advanced spacefaring civilization.

Five billion souls in a cube.

Praxis.

It was Box's first look at the orbital city after her brief glimpse six weeks before. If she narrowed her eyes, it looked like lambent forests of glass. Pax said the appearance of floaty architectural disconnectedness was an illusion.

"This is a stiff structure, as befits a starship," he said.

"This place can move?" she asked.

"Far across the galaxy, if need be. In a serious emergency, this spacecraft could reach Andromeda in a series of hyperluminal jumps."

"The art of running away?"

"An excellent strategy."

They'd come through the docks that spattered a side of the cube. These industrial works made Avalon seem like a roadside garage. She watched a ship being repaired - a teardrop-shaped lozenge, the size of a mountain.

"What's that?" she asked.

"A municipal project," said Pax, "built here by the local community of spacers."

"Where does it go?"

"It's a deep space exploration vessel. It goes where it wishes."

The Water Bear had reached into its bag of disguises and become a small tramp freighter, battered and rough, with gases venting from welds.

"Good luck," she'd said, before departing.

Now they were on their own.

Ophelia Box? said a voice in her head.

Praxis?

No, this is the Orbiter, the vehicle that carries Praxis through space. Praxis has chosen to be unaware of you.

Did we do something wrong?

No, you're on a covert operation.

But you can see us.

I lack the city's talent for self-deception.

Zing.

I'm pleased you have a sense of humor.

I'm glad you do too.

Welcome to Aldebaran. Call me if you need me. Remember, I'm a ship.

You mean you kick ass?

I can.

Their safe house was an apartment, set high in a dormitory zone of the city. Aloïs and Pax confirmed it was perfectly ordinary, like a safe house should be. It backed onto rice paddies, which Box could see were the terraced rooftops of the adjacent buildings. If she looked through the cracks, she could see Aldebaran spaceport, five kilometers below.

Kitou threw herself on a couch, like any Earthly teenager would, and powered up a wall-sized entertainment system. The room filled with thumping electronic music, like a mix between grindcore and trance. Onscreen, performers bowed and banged and plucked at their instruments, while a crowd waved its hands in the air.

"Churn," said Kitou brightly.

"I like it," said Box.

Aloïs was doing just fine, in his dual role as the passenger and sometimes captain of the Water Bear's cyborg remote. Today he was the captain. You could tell by his beatific expression. He sat beside Kitou and smiled at the music.

Box asked him the reason for his good mood.

"It's because I'm a new person," he said. "Without this mission, I'd never have existed."

Aloïs is an ideal houseguest, said the Water Bear from her listening post, five thousand kilometers away.

"That's easy for you to say," said her remote.

"As long as he doesn't move in," said Box.

I can always make another androform body, said the Water Bear.

Aloïs ignored them and smiled.

For two days they wargamed their mission - testing assumptions, rehearsing the angles.

"We're here to commit a crime," said Pax. "Let's waste no time on morals. Sometimes, we work in strange places. This is such an occasion.

"Given time, we could do it in secret. No one would know we were here. But we don't have the time, so we'll need some support."

"I propose to make contact with a local crime syndicate," he said. "They will have already been penetrated by Po intelligence, although I don't know the details. I'll find out. We have signals."

"More bar fights?" asked Box.

"No," said Pax. "This is a more complex society than Avalon. A precise commercial transaction will occur.

"I intend to sell you as a slave," he said.

"What? Why me?"

"Because you're new."

The next day, Pax and Box were sitting relaxed in an open-air *brasserie*, in the park below their apartment, among happy families with laughing children running through. The last of the afternoon rain was being deflected by shadesails. A freshness filled the air - the fragrance of flowers and grass, and delicious food being cooked.

Pax summoned a waitress, about Kitou's age.

"Hi," she said.

"Hello," said Pax. "What's your name?"

"Sophie," she smiled.

"Why are you here, Sophie?"

"Why am I working?"

"Yes."

Her smile increased by a few thousand watts. "I'm saving for my grand tour," she said. "Would you like to order?"

"Yes," said Box. "We're ravenous."

"What was that about?" Box asked, when Sophie left with their orders.

"I wish to explain our economics," Pax began. "We live in what you might call a post-scarcity society. The state provides what people need, but not all they want. That's a deliberate choice. Effortless societies are unstable. They select for decadent behaviors. As pleasant as that might be, it's no way to run a civilization.

"So, we encourage people to contribute. Sophie is contributing. In return, she receives money."

"From the state?"

"From the market, which the state allows to exist."

"Which she can spend on her tour," said Box.

"Exactly," said Pax. "And with it a sense of accomplishment. No one gave her that tour. She earned it. Anyone can look in the history of her money and see it."

"I like it," said Box.

"Thus, as well as being a post-scarcity society, we're also a quasi-capitalist, socialist, utopian one."

"You saucy devils."

"We must also deal with other societies we encounter. Thus, we have trade, diplomacy, and war, usually in pairs, sometimes in threes. That means our currency has an extrinsic value."

Box thought about that.

"So, you have crime?" she ventured.

"Yes. We have organized crime."

Evening was gathering, and with it a powerful sense of expectancy. Box could feel it in the air, like electricity. Cities were cities, and this was the mother of cities. Lanterns lit up among the trees. Partygoers were gathering on the cantilevered boulevards. A happy clattering spilled out of the kitchen. Sophie returned with their meal, and the particulars of her travel itinerary, to show them.

"The Magellanic Clouds?" asked Pax. "That's not a tour - it's an expedition."

"It's a *grand* tour," said the girl.

Box recommended a hotel, in Avalon. "It's nothing flash," she said. "But it overlooks a perfect street market, and the people there are lovely."

"You've been to the *Smear*?" asked the girl, wide-eyed.

At that point, Box might've exploded with pleasure.

"My friend is from Earth," said Pax.

"You're shittin' me?"

"No, no shit," said Pax.

"You made her day," said Box, later, after Sophie left with their empty dishes.

"The gratuity will make her day," said Pax.

"We aren't law enforcement," he said, as they walked together after their meal. "We probably break more laws than we uphold. But there is a crime we take seriously - slavery."

The paddy field was larger and wilder than it looked from above. During the growing season, it'd be flooded. Now it was lush. Box heard tiny animals rustling around her feet. She'd taken the liberty of entwining her hand in his.

"Why?"

"Because it falls in the gaps between societies."

"Where you people operate?"

"Yes, we're uniquely placed to police it, so we make it our business. It's also an assault on those least able to defend themselves. Praxis is a center of human trafficking."

"Really?"

"Yes, in part because we make it so. We cultivate the conditions for some crimes to exist here."

"Why?"

"So we can police it."

"A honey trap?"

"Not quite that, no. A market intervention."

Box frowned. "But Pax," she said. "For that to work, there has to be honey. People get sold. You may end up policing a trade that only exists because of your intervention."

Pax nodded. "We know the risks, yet we grasp the equation."

"You mean you buy and sell people."

"We do. What choice is there? To condemn a larger number to the same fate by our inaction would be wrong."

She shook her head.

"There's no such thing as a moral dilemma," he said. "Only a lack of moral clarity."

She sighed.

"I know where there'll be a Po agent," he said. "At or near the top of the slave trade here. I need that person's help. Specifically, I want a way out of the city, to escape through, when the robbery is finished. A prearranged signal will be deployed. I'll offer to sell a human woman - you - after betting a perfect number of Praxic cryptos on Bolo. 8,128, which signifies the utmost secrecy."

"What's a perfect number?"

"In mathematics, a positive integer that is equal to the sum of its proper divisors."

"You learn something new every day."

"And Bolo?"

"A bloodsport."

"Can it get worse?"

"It can, and it does. I'll be betting on Black Bolo, an illegal death sport."

"Seriously?"

"Yes."

They boarded a train which rose from a vertical platform - slowly at first, then with increasing acceleration, until they were spearing upwards into the heart of the city.

"Now *this* is exciting," said Box, as the cityscape flashed past her window. Polyhedral towers gave way to onion domes, then slippery lozenges, then neogothic follies, like a flickering architectural slideshow.

"Fashion," said Pax, following her startled glances. "Changing over centuries."

She'd decided she liked Praxis nearly as much as Avalon. It wasn't as strikingly alien. It was like a great city on Earth, until a vista opened, and the cityscape marched to infinity. Then, it was *magnificent*.

Five billion people lived here. She had to continually remind herself of that. One-third the population of Earth, in a cube fifty kilometers square.

It didn't even seem crowded.

After a few minutes, they reached their destination: a wet, streaming plain gleamed dully overhead, like nothing she'd seen here before. The metal was pitted and rough and pocked with small craters. Condensation fell from its fissures and cracks, the start of the rain that fell in the city below. It was as though the emerald city had a pig-iron heart, and they were about to be sucked up inside it.

"Please disembark," said a voice. "You've reached the Deregulated Zone."

They stepped onto an escalator.

For a few moments, Box wondered if the Orbiter was messing with her wetware, replaying her cultural references.

Then she came to her senses.

This was *real*.

The vaulted space above the iron barrier was hotter than the city below - and steamy, so water beaded on every surface. And *crowded*. People of every size, shape, and ethnicity - every imaginable body modification. Not just humans. Chimeric beings wore bristles and fur. Cyborgs. Ambulant machines of every kind. Music spilled out from bars and floating soundsystems.

Instead of buildings were machinery - huge turbines that must've once delivered services to the city below, but now were infested with people. It was as though life had adapted to survive in the spaces between.

And information. Everywhere, neon, or neon-equivalents. Holograms. Interference patterns in the air. Steam from vents that shimmered to conjure up advertisements for sex and moneymaking schemes. Drones buzzing in her ear. Subliminal displays, neutralized by her wetware.

Box was delighted. "This was how I *hoped* space would be," she murmured.

"The original Orbiter," said Pax. "With all its defects. Five thousand years ago, all human space was like this, so in a sense you're right. This is how space *was*."

"Why are the people so poor?"

"They're not," said Pax. "The impoverished style is a pose. The Praxis DRZ is among the wealthiest places in the human galaxy."

They found a basement bar, smoke-filled and as busy as the street outside, built in the bearings of an outsize electric motor. They wound their way through house-sized capacitors. People of every gender looked on disinterestedly. A pulsing arrow said, **Sex**. A half-dressed couple copulated in a corner.

Box noted abstractly that here she was again, doing Po business, in another slaphouse.

"Now," said Pax. "This is where it becomes interesting. Don't look, but do you see those two men over there?"

Box glanced in her wetware, which opened a window in her peripheral vision.

"The goons in the Soviet haircuts?"

"Yes, they've been following us."

"Is that normal?"

"No, but I allowed it to happen."

"Who are they?"

"My local counterparts."

"They're the police?"

"Yes, and we're going to make a run for it."

"Why?"

"Trust me."

She nodded, remembering Avalon Station.

Revelstoke, and his knuckleheads.

"First, instructions. If we get separated, trust the Orbiter. If you're arrested, say nothing. We'll come and get you."

"I can't believe I'm doing this."

Why was she so nervous? Did she trust Pax? Of course. He was the most capable person imaginable. It was the lack of the Water Bear. No amount of ordinary competence could make up for a warship with gravity drives.

"Trust me," said Pax.

She nodded.

"Remember the market in Avalon. The worst that can happen is you spend a few hours in a cell. The real powers here are all on your side. You'll be fine."

"Okay."

"Dr. Box, relax."

"I am relaxed," she lied.

"I need to give you some wetware," he said. "Look straight in my eyes."

She felt a slight shiver. Now she was two people. She was Ophelia Box, and she was a woman called Pris, from the Trappist-1 confederacy.

"My god, I have a *backstory*."

She was here to be sold, as a consensual slave.

For six months.

Six months.

A timeshare agreement. Someone gets part of her life. In return, she gets some of the money.

Not enough of the money.

"It's called a firewall," said Pax. "Military spec. It will stand up to the most intrusive scanning. All you have to do is get in character."

Pax was called Ray, and he was a small-time crook, and part-time human trafficker.

A despicable piece of shit.

"You see?"

She nodded.

"Otherwise, stay close and follow my lead.

"Okay, let's begin. I want you to stand up when I do and have a good look at those two police officers. Don't stare, but I want them to know you see them."

"Why, *Ray*?"

"To put them off balance. Law enforcement works best in managed encounters, where they can fall back on their systems. A regular person would avoid eye contact. By displaying unpredictability, we gain an advantage."

She snorted with derision.

Since when did Ray know anything?

"Let's go."

Ray stood up. Pris turned around and gave the cops the evil eye. They - or men like them - were why she was here. They looked away, then turned and stared right back at her. She got up and followed her owner. Ray moved expertly. He got the bulk of the bar between him and the goons, then stepped into an open doorway. Pris followed, and Ray locked the door behind them. They sprinted headlong down an unlit alleyway. There was a crash - then the goons reappeared. Pris wasn't sure who she hated the most, Ray or the cops.

The cops, she decided.

Better the devil you know, eh?

They ducked into a smaller alleyway, then down a rickety letterbox staircase. The goons clattered past overhead.

A door slid open in a wall.

"In," said Ray.

They were in a freight elevator.

With a surge of conventional power, they ascended.

"You can come out of character now."

Box fizzed with nervous energy. She hadn't outrun the *gendarmierie* like that since Marseilles, a lifetime ago.

"How did you know to do all that?" she asked.

"The Orbiter helped me."

"Where now?"

"To Core."

Core was a ragged sphere of cargo nets, strung inside an iron-nickel void. Between the edges of the nets and the sides of the void were rows of stadium seats. It was a rough place, like something rigged up to corral wild animals. Strands of old wire floated in the erratic gravity. There were a handful of people there, scattered in the seats. Box watched a man light powdery drugs in a pipe, and suck on it greedily, before passing it on.

"What is this place?"

"The belly of the Orbiter, where the Praxis asteroid was first hollowed out. Each face points down. Inside the netting is weightless, curving to g at the edges."

"Is this where your death sport is played?"

"Yes. When they play Black Bolo, the net is removed. It isn't just players who die. Tonight, there's what you'd call an illegal rave."

"Oh, are we going to that?"

"No, we're waiting for someone to come take my bet."

"Shame," she said.

"Get back in character, please."

She was Pris again.

The man who came for Ray's money looked like any Trappist-1 accountant: a slicked-back prick in an expensive suit. Pris hated him instantly. She figured the suit's suit was worth twice what she was. Ray got in line and waited. Pris was surprised by his meekness, his subservience in the face of a higher criminal authority. No trash talk was exchanged. Ray handed over a package. Hard Praxic rubles - untraceable notes - and with it a business proposal.

Offering *her*.

The street accountant moved on to his next mark. Ray grabbed her, not unkindly by the arm, and led her down a ramp, out of the arena, into the street, to a bar called the Arse, as far as she could tell. Her rudimentary wetware was comically poor at translation.

She didn't give a fuck. It was a bar.

Ray wasn't so bad, she decided. She was starting to miss him already. Her new owner could be anything.

It was only for six months.

If she lived that long.

She was on her third drink, a fiery local rocket fuel called unpronounceable, and just starting to feel sexy, when the boss man arrived, all gold teeth and a practiced, piratical grin.

She knew him.

No, not as Pris. As Box.

This was a man from her real past, on Earth.

In character, warned Pax in her head.

The man was in a cocaine-colored limousine, the biggest car Pris had seen, although that wasn't saying much, with her just a pleasure model from the antipodes.

"Ant de Large," she said.

Ophelia Box had first met Anthony de Largentaye in South London in 2069, six years in her past. Ant was a clubbing impresario, a well-known fixer, a man famous for being famous. She was over from Paris, a girl about town.

She'd been his posh totty - except she was never posh.

Except it wasn't six years ago.

It was in *his* future.

"Do I know you?" he asked.

"Oh, fuck," she said.

"You know this man?" asked Pax.

"Not here," warned Ant. "The walls actually do have ears."

The limousine surged down narrow streets, descending through a labyrinth of subways, until they stopped outside a nightclub. It was uncannily Brixton, 2069. There was even a queue of bright-eyed partygoers, craning their necks to see who emerged from the car.

The sign said, **Leopardskin Moon**.

Ant's office was a transparent cube, with a translucent desk, powdery with party drugs. Very Praxis-does-London, down to the lasers strobing over the writhing bodies of the dancers below.

Pounding dance music - it could've been churn - made it impossible to speak in the room.

Ant waved his hand, and the music ebbed away.

"Do you two know each other?" said Pax.

"Never seen her before," said Ant.

"Yeah, I know him," said Box.

"Then," said Pax, "you're from his future."

"What do you mean?" said Ant. "My *future*?"

Pax transferred the details of their mission into Ant's wetware, encrypted with de Largentaye's public key, so only the real Ant de Large could decrypt it.

The opening shimmy in a cryptography two-step.

"Can your ship send me a signed and encrypted message, confirming all that?" said Ant.

Pax nodded. Then, Ant said, "Okay, I'll get whatever you need."

"Easy as that?" asked Box.

"You want hard?"

"My part in this is done," said Pax. "Ant, you know how to reach me. I'll rejoin my crew."

"What about me?" asked Box.

"Our backstory requires that you stay for a while."

"We have a backstory?"

"We will have."

She sighed. "How long?"

"A few hours. Maybe a day."

"Seriously?"

"Brin will come and act as your buyer."

"Yeah, I'll look forward to it."

Pax exchanged a brief look with the other man.

"Pax, did you know about this?" she asked.

"Emphatically not."

"Why 'Anthony de Largentaye'?" she asked after Pax left the building.

"It's an alias I use."

"On Earth?"

He shrugged, trying to appear relaxed. She could imagine the cogs of her story meshing inside his head. Or already meshed.

"You're a Po secret agent?" she asked.

"Lo, actually."

"What's your real name?"

"You know I can't tell you that."

"You really don't know me?"

He shook his head.

She crossed to a wall of his office. Through the glass the dance floor heaved. She pressed a hand against the wall and felt the syncopated beat. Yes, it was churn. She wished she were down there.

"I slept with you, Ant."

"Yeah, I figured that out."

"So now you go fuck me in London? Is that how it works?"

"I honestly don't know how it works."

Part of her sympathized with him. The rational part. She pressed her head against the shivering glass. Now, more than for years, she wanted some ecstasy.

"What would you do with the real Ray and Pris?"

"Pris would be quietly saved, if we can swing it that way. She might have some adventures, but she'd get through it okay."

"And Ray?"

"We might bankroll him. Let him rise through the ranks. It depends."

"On how useful he is?"

"Yes."

"Where's the payback in that?"

Ant shrugged. "In the real world, babe, it's not about payback - it's about least bad options. My job is to help people."

This wasn't what she expected.

How could it be?

It was bright night in the scintillating avenues of the city. The rain had cleared, leaving skyways like glass. Their car hummed as it swooped through the traffic. Brin and Box wore business suits. Kitou

was dressed in casual streetwear. The Water Bear had infected their wetware with a plausibly virulent virus.

"Military-grade malware," said the ship. "Enough to absolutely horrify them."

The ship's remote was back in command of her androform self. Pax and the ship were dressed in embellished military finery. Pax's uniform was Po black, festooned with medals and braids. On his lapels were his carbon-black starbursts. The ship's uniform was pristine white.

The ship admitted the uniforms were overwrought.

"Sometimes it pays to exaggerate," she said. "Although some Po might like them." With her cropped hair and cyborg's physique, she looked imperious.

"They're going to think the Water Bear is the battleship Potemkin," said Box.

"In what way?"

"You're an insurgent."

The ship laughed. "Should I have a dueling scar?" she asked.

Box said, "Don't overdo it."

She could barely believe she was riding into action with her friends, in this alien place. While she watched the lights flashing by the car's windows, she started to tremble. She wanted to cry. It wasn't for sorrow. She didn't know what it was. A perverse kind of joy.

Kitou, sensing her mood, leaned up close. Brin took her hand.

Mission specialist Box.

This was fantastic.

The plan was straightforward. They'd enter the bank and ask for their money. Since a basic information check would reveal that Pax didn't exist, they would take the city offline, by declaring a malware infection.

Po ships had the power to do this.

Po ships were considered incorruptible.

The Civics would then have to choose. They might've already chosen. The plan hinged on their tacit acceptance. Without it, they'd

better be *fast*. The ship estimated they had a window of about six minutes, from when the infection was first reported, until an intervening authority could override the local law enforcement systems and launch a pursuit.

Six minutes was two sigmas, which included about 95 percent of the data.

So, they might have to be *faster*.

The bank at Praxis was *the* bank, Pax explained. "We only have one bank in our society, and this is its shopfront."

The mile-high atrium was meant to impress. What wasn't made of polished stone was made of unobtanium. Box took up her position, in the exact geometric center.

Kitou and Brin were arranged equidistantly, along the arms of a Fibonacci spiral.

This was a clue for the Civics.

Trust us, it said. *We're nerdlings*.

Remember, said the Water Bear actual, now just fifty kilometers away. *Just fall. I'll take care of the rest*.

Pax and the androform ship approached a human assistant.

"I want to make a withdrawal," said the ship.

The woman smiled.

"How much will you be needing today?"

Information was exchanged. The assistant smiled again.

"My, that's a large amount. The largest I've seen. Can I please take a moment?"

"Of course," said the ship.

The assistant looked in her internal cinema. Her eyes widened.

"You're a Po warship," she said.

"Yes. What's your name?"

"Ursula Behr."

"A pretty name."

"Thanks."

"Ursula, whatever happens, you'll be taken care of. Do you understand?"

"No."

"Good. Now, the money."

"How would you like it transferred?"

"By ejection."

The woman frowned.

"You mean, like actual gold into space?"

Ship nodded.

"Don't worry, I'll pick it up."

"I'll... I'll need a counter-authority."

Ship motioned at Pax.

"My Navigator."

The woman's eyes widened.

"A Lo Navigator. Wow."

Then she said, "Oh, we have a problem."

"What kind of problem?"

"Oh, a cyberattack."

Kitou, Box, and Brin fell to the floor, writhing.

"Is it happening now?" asked Pax.

"That's what it says in my sensorium."

Initiating security lockdown, said the Water Bear actual. We are now the ranking civil authority.

"We're taking command," said the ship.

"Oh," said the assistant.

"Please eject our gold now."

The woman hesitated for a moment, frowning.

Then, "Ejecting."

Pax and the ship turned on their heels. Security staff had surrounded the three writhing women.

"We'll take care of this," said Pax.

In a taxi, four minutes later, Pax said, "That was neatly done."

Box said, "That was fucking terrific."

"You may all whoop," said Pax.

They laughed uproariously.

The Water Bear said, "Does Earth idiom include, don't count your chickens?"

"My wetware translated it," said Box.

"Tick, tock," said the ship.

High over the city streets, balanced on a windowsill, the Watcher watched. Behind him, a family enjoyed their evening meal.

They weren't aware of him. He wasn't watching them.

He had no physical manifestation, although that depended on what spatial dimensions you were listening in. He existed as transient noise in the processors of countless individual computers.

What to make of these bold imposters?

He sprang lightly to the roof of their car, as it passed a half-kilometer below, his trajectory a passing physics problem.

He *listened*.

The Po warship was real, both the android remote and the real ship in space. Unlikely, but real. Perfect originals of a real Po warship and its remote. The Navigator was in a physics class in the Academy on Polota, a hundred light-years away. The Lo woman was recently born.

The wild-haired child was a Pursang.

Military lore:

What do you get if you cross a Po with a Pursang?

The redhaired woman was an Earth native.

The Earth native.

He initiated a network transaction. For a moment, the entire system veered dangerously close to hyper-intelligence. After a few picoseconds - or enough processor time to run a large industrial city for a day - he had a solution.

He knew before he asked, what the answer would be.

He assigned it a truth function.

He decided.

The city of Praxis was less susceptible to gaming than they'd hoped for. Certainly, it could be deceived, but once deceived, its responses were reflexive.

Its autonomic defenses included deep-space Interdictors. These were what the Water Bear would call psychotic machines, designed to patrol the big empty, far from vulnerable humans. Cuckoo attacks

were in their decision tree. They launched at upwards of a million gravities.

This was how conflict was handled in the wild, outside the boundaries of civil society.

Without pause for reflection.

By psychopaths.

"Is this car space rated?" asked Pax.

"No," said the ship. She'd attached an optical cable from her waist to the taxi's control panel, and the car was now fully under her control.

"No matter," said Pax.

The car wound through the city's under-spaces, pressing its passengers into their seats as it made short radius turns, heading towards the exterior. Box saw the spaceport flash by. She felt a pang of memory.

The bowels of the city were full of machines. Capacitors, the size of small towns. Strange energies arced between hill-sized coils. These were the engines that powered the city - the great, great-grandchildren of the DRZ turbines.

There were cities within cities. Inexplicably, they passed by housing complexes. Who would live down here, in the interstices?

The more Box saw, the more fascinating it became. She saw a market, in the slipway of a dam. The dam was the size of the Three Gorges Dam, and the market was like a car boot sale, except with gargantuan lorries.

What kind of strange, black-market deal was going down there?

Worlds, observed between the ticking of a clock.

Then they were in the interplanetary vacuum.

The car held its atmosphere passably well. Box felt the air sucked out of her ears as its body expanded. Outside the Orbiter's induction grid, the car lacked drive. They were ballistic. Momentum carried them free.

Then the Water Bear actual spoke in their sensoria.

I have you, Praxis taxi.

The bank vault came next, tumbling through space, in the grip of the Water Bear's gravity drive. A tonne of gold is a cube about 37.23 cm square. 25 tonnes is the size of a refrigerator. The ship reeled it in, a minuscule mass. A moment later, two Interdictors appeared. They had some difficulty locating the Water Bear, but the car and the vault were acquired instantly. There was a brief tug-of-war - then the vault accelerated back towards the Orbiter, as though on the end of a fifty-kilometer bungee.

Box watched these events unfold from the car, unable to make any sense of them.

The Water Bear jinked several times, at a rate that made her crewmembers wince. The Interdictors followed, unable to be shaken. They were ugly machines, all spikes, and knives bunched in a fist.

They powered up beam weapons.

The Water Bear responded with a suppressor field.

STOP, was broadcast on every channel.

I can escape, said the ship actual.

Go, said Pax.

It was a brutal calculus. With or without the gold, with or without her crew, the ship could still finish the mission.

Her humans were expendable. They'd either be killed or arrested. Either outcome was workable.

The ship disappeared, in a ripple of warp.

Then, inexplicably, the Interdictors stood down.

Their weapons still primed, they melted away, into the darkness.

7 → EQUITY

2056

Box regained consciousness, in a hospital room, with a soft breeze tugging at gauzy curtains through an opened window. The sky through the window was powdery blue. In the distance, towers leaned drunkenly away from each other. She could see gardens that met at ninety degrees. She could hear children, laughing. A man in a suit was beside her.

"Dr. Box," he said.

"Where am I?" she asked.

"Acadia, a corner of the Orbiter."

"Who are you?"

"I'm Praxis."

"Where are my friends?"

"They're safe. Your androform ship will come in a short while. First, I must ask you some questions."

"Do I have rights?" she asked.

"What do you mean?"

"Can I call my lawyer?"

"Dr. Box, you're the victim of a crime."

The Water Bear appeared a few minutes later. She seemed in vivid good health, unlike Box, who felt like overstretched cellophane.

"What happened?" Box asked.

"It took time for the puppets to move."

"These bruises?"

"I had to shut parts of you down."

"The others?"

"All safe, but it was close."

"The ship?"

"Gone to finish her mission."

"How are we even alive?"

"This androform body can metabolize fat to produce breathable oxygen."

"You're not fat."

"Not anymore," said the Water Bear.

"I'm sorry," said the urbane city avatar. "It looks like someone stole your money."

"All of it?" asked Pax.

They were in a basement of the bank, in the plane where the spacecraft kissed vacuum. Box was riding in the Water Bear's sensorium, seeing what the Water Bear saw, experiencing her sensations. It felt good, after the recovering wreck of her body.

She could see space through membranes in the Orbiter's skin. She recoiled at the idea of it.

In front of them was a wardrobe-sized container, made from vanadium steel (the ship's wetware told her) with its door twisted open.

The ship got the loot? she wondered.

Then she remembered - The Water Bear winking out of existence, gone to finish her mission.

Now the vault was open.

They were being shown it.

Was that a good or a bad thing?

What happened? she asked. Even inside the ship's androform brain, she only seemed to have half her usual intelligence.

"Military malware," said the city. "Memories lost. Everyone deeply confused."

"Fancy that," said the ship.

Why is it broken? asked Box.

"It struck the Orbiter with considerable force. How do you want me to handle this?"

"This is Po business," said Pax.

Praxis nodded. "One more thing," he said. "A clue." The city messaged a string to their sensoria. A sequence of decimal numbers.

A Hopf number? said Box, gazing out through the ship's eyes, magnifying the details, seeing the crystalline grain of the metal.

"Yes," said Praxis.

Mine? asked Box.

"No," said the city, not missing a beat. "It's the spacetime address of the Bank actual, five days from now. It was encoded in the molecular structure of the vault. That's extraordinary information. We've checked. No other vaults have that feature."

What's a Bank actual? she asked.

Samppo, said Aloïs, from somewhere beside her. Unlike Box, Aloïs seemed to have all his wits.

"Why?" asked Pax.

"I don't know," said Praxis. "But you may want to go there."

"We have no ship."

"I can help you with that."

The Bat was a quarter the size of the Water Bear. A battered towing vessel, ugly as a warthog, with oversized engines, and bristling with surveillance gear. Its pilot was a small AI, about twice as smart as a bat, from which it was evolved.

The Water Bear declared herself satisfied.

"This is a *good* ship," she said.

"And I have some good news," said Praxis.

"Yes?" asked Pax.

"Your missing crewmember is safe."

"What? Ito? How do you know?" asked Pax amid the rising clamor.

"I can't say."

"What *can* you say?"

"That he doesn't exist. Not now. Not yet. Except as a young novice, on Polota. But the version you lost, the grown one, he's safe."

You're talking in riddles, said Box.

"You'll work it out."

"City?" said Pax.

"Yes?"

"Thank you."

"It's been a pleasure, Navigator. Just please don't come back for, shall we say, twenty more years?"

Progress inside the Bat's warp bubble wasn't as serene as the Water Bear's. The smaller ship's machines smelt of hot metal and lubricants, and made chuntering mechanical noises. Box thought it was just fine. Her mind was still recovering from her body's ordeal. After the bone-chilling emptiness of space, the thumping made her feel solid.

"We're the Hand of the Bat," said Kitou, delighted.

"Pax is the Bat," said Brin.

"Silence," rumbled Pax. "All of you."

With news of Ito, came a blissful sense of release. Box saw that a weight had been lifted from all of them. Kitou and Brin were like different people. There was no longer the fear that they should be searching for Ito instead of pursuing their mission. Wherever he was, and what he was doing, were anyone's guess, but he could take care of himself.

Brin and Kitou taught her the Po of small spaces.

"This move is called the brainless monkey," said Brin.

The girls assumed the Po starting position, floating within striking range, and then started hitting each other. There was nothing artful about it - it was a barrage of punches and blocks. They turned off the lights and continued the fight by the glow of the instrument panels.

"The Earth art this resembles most is Wing Chun," said Kitou, panting.

Box said, "Don't be ridiculous."

They found handholds in the ship's machinery and swung from them. As they hurtled past, they threw punches and kicks at each other.

Thankfully, few landed.

"This move is about survival against a chaotic opponent," said Brin, breathing heavily. "We fight to exhaustion. The nature of the attack is incidental. We use it to unwire reflexes."

Box saw that while the striking was deranged, the blocking was a variation on Fibonacci spirals.

Maybe it *was* like Wing Chun.

The two girls stopped, exhausted.

"Now you," said Brin.

Pax asked what happened to her face.

"Moving equipment starts without warning," she recited.

"Well, this ship has a medibay. Please have it seen to."

In the small hours before dawn, when Box was the only one left awake, she talked with the androform Water Bear.

"What's it like, being two of you?"

"It has the potential to become distressful."

"In what way?"

"According to our customs, a divided person, over time, becomes two different people."

"Like Aloïs?"

"Yes."

"I can see how that might be distressing."

"It's like your transporter dilemma."

"What do you mean?"

"When you make a copy of a person, for beaming them down in your space dramas, is it the same person?"

"Oh, that old chestnut. I never thought so. My dinner party opinion's a copy's a copy."

"So, what becomes of the original?"

"That never gets shown on TV. I suppose they're quietly disintegrated."

"Thankfully, we lack transporter technology. It would introduce all kinds of moral challenges to our society."

"But I've seen it."

"Where?"

"On Earth. The 6 transport political agitators."

"That's a sleight of hand."

"What?"

"They move them using gravity drives, then edit your perceptions."

"You're shitting me?"

"No."

"That's awful."

"Do you prefer the alternative?"

She talked at length with Pax. They were in the Bat's common room. There was no luxury here, but it was a comfortable space for two people. Pax stayed up, consenting to be interviewed.

"Thank you," said Box.

Pax nodded.

"I mean, really thank you. For everything.

You're a wonderful leader."

"I'm not quite Ito."

"No, you're not. But Ito's not quite Pax."

He smiled at that.

"I remember the lesson you taught me," she said.

"In the mountain sim?"

"Yes. I'm starting to get it now.

You know Pax, those questions you asked... what I'm frightened of?"

"Yes?"

"It's true, what you said. I know I'm brave. I'm the one who runs towards the fire. But when the pressure's on, I fold like a house made of cards."

"And you know this?"

"Yeah, but I can't seem to help it."

"Then accept it. That's how your flight response intervenes."

"You said you could train it out of me."

"But I won't."

"Why?"

"It's there for a reason."

"What reason?"

"Dr. Box - Ophelia - you're not a soldier. I don't have the luxury of walking away from problems that scare me, but you do. I have a lifetime of military training. You don't. Bravery combined with obstinacy will get you killed."

"So what, I run away?"

"Yes, you run away."

"I'm allowed?"

"Yes."

She sighed, then nodded.

"What did the city mean about Ito?" she asked.

"He said he's not in this universe."

"How can that be?"

"Who knows?"

"Maybe he's in some other place."

"Tell me," she said. "Who's this Samppo character?"

"A complicated person."

"How so?"

"Samppo's the Bank."

"He owns it?"

"No, he *is* it. I'll try to explain. Have you heard of hyper-intelligence syndrome?"

Box shook her head.

"There's a theory that a sufficiently intelligent machine must transcend our reality, and pass into a higher plane of existence. That, or go mad. That's what the Bank has supposedly chosen - a condition of divine lunacy. As well as being the front end of a supercomputer, Samppo's a religious figure - the head of a cult, called the Cult of the Bicameral Mind."

"Your financial system's run by a cult?"

"Yes. And our central bank exists perpetually on the edge of madness."

"This works?"

"Yes."

The Cult of the Bicameral Mind was an accumulation of industrial garbage. On the largest piece of garbage was drawn a human

skeleton, with a cleaved skull, an ax, and outsize male genitalia. The genitals were partly obscured by a layer of more recent garbage.

A voice said, "Who the fuck are you?"

"We're a special operations unit of the Po," said the Bat. "Please welcome us on board."

"Fuck."

A second voice said, "It's the Feds."

Beam weapons swiveled to face them. Box saw them in her head-up display - spidery things, under layers of refuse.

"How can we dodge those?" she asked.

"I'll know they're powering up before they do," said the Water Bear.

"And then?"

"The Bat will disassemble them in a shaped gravity pulse."

"We can do that?"

"Oh yes."

"Put the weapons down," said the Bat.

"Or what?"

Pax broadcast, "This is Navigator Pax Lo. We're here at your request."

"We didn't request no one," barked a new voice.

"We come on business," said Pax.

"This display of firepower is a ritual game," said the Water Bear. "The precursor to a negotiation. Out here in the big empty, where force equals law, it's expected. They're showing us the size of their weapons."

"Like baboons," observed Box.

"Two armed parties can do business," said Pax. "These oddballs appear to be armed about as well as a warship."

"Not afraid of little old us then."

"On the contrary."

Kitou, Brin, and Box crossed over in the usual way, by jumping. Box was getting used to it. The Bat propelled them across the gap along three random courses.

Deep space, far from any world, was a strange, lonely place. The emptiness gnawed on the edge of her consciousness, like a palpable thing.

For the first time, she realized how *small* she was.

Kitou and Brin landed at the edge of a filmy membrane, in a relaxed military style. Box landed behind them, slightly less stylishly, still on her feet. Their skinsuits relaxed, until they were in combat versions of their Po uniform coveralls.

"What the fuck is this?"

A youth in a filthy monk's habit was gabbling into an old-fashioned radio microphone.

"Space vixens!"

"You won't need that," said Brin.

"Broca transfer?"

"Yes."

The young man put the microphone down.

"You Po have all the gadgets," he said. "Now please, tell me about your ship. I've never seen a Po tugboat before!"

The monk turned out to be an engaging conversationalist. He pulled off his habit, to reveal a fresh one underneath. While he was busy with that, he sheepishly apologized for the earlier profanities.

"We have a brand to manage," he said.

"How many of you are there out here?" asked Brin.

"Sixty, not including Samppo and the Engine."

"What do you do?" asked Kitou.

"We're a hedge fund," he said.

"Or a bank."

"It depends on your theory of money."

"I see you've met my boy," said Samppo, smiling impishly. He was an old man, in rude good health, with a shock of white hair.

"He's your son?" asked Kitou.

"We're all sons here, young woman. Macro's a clone, as am I. We've been happily cloning ourselves for thousands of years. When I expire, Macro will become Samppo. There'll be a party, and we'll get back to our models."

Samppo wasn't remotely as Box expected. Happy and avuncular, like everyone's favorite uncle, he *twinkled*.

"You're all men?" Box asked.

"Trust me," he said. "It's simpler that way."

The interior of the scow was like an exclusive gentlemen's club - smoked glass and waxy carbon fiber. Box felt unnerved. She smelt power, and old money. Maybe the oldest money.

"How do you like our little island of warmth," he said, "in the big sea of nothing?"

They were sitting in a cross between a board room on Earth, and the control room on Threnody. Long curving windows looked out across what appeared to be a trading floor.

Samppo got to the point.

"Dr. Box," he said. "Why are you here?"

"Me?"

Samppo looked around. "Is there more than one of you here?"

She gathered her thoughts. She was at her argumentative worst, sparring with privileged men.

"What happened in Praxis?" she asked.

"You were about to steal my money."

"That was our money."

"Money belongs to the Bank."

"Bullshit."

He smiled and raised his hand.

"I've been a poor host," he said. He opened a drinks cabinet, with a door that reminded Box of the bank vault on Praxis, and produced five heavy glasses.

"Macro," he called.

"Look, Dr. Box, we live in a post-industrial society here. You know what that means? You people can have all the *things* you want, but money is mine, unless I choose to give it to you."

Macro arrived with a bottle of spirits. Samppo motioned him to sit, and poured five equal measures.

"The woman you were so kind to on Praxis," he said. "My employee."

"Ursula," said Brin.

"Yes. Why was she there?"

"To earn money," said Box, remembering Sophie, and her grand tour.

"Wrong. Money's a *game*, a means of abstracting value. Cash is a position. She'd be just as *comfortable*, lounging at home, drinking strong liquor."

"She desires more," said Brin.

"Yes, but she doesn't want *money*. What use is money? You can't hold it close in the long night of the soul. She wants what it *buys*. She wants to *consume*."

"Money is information," said Brin.

"Exactly."

He motioned them to consume. Kitou gagged but made a decent fist of it.

"Also, she finds it *interesting*. The same reason you do your jobs. It fills up her life with distraction and puzzle. Last week she met a Po warship and its Lo Navigator. It was the greatest event of her life. There was a robbery, and somehow, now she's the hero. Soon she'll get to meet me. It'll tickle her silly.

"But money... money matters to *me*."

Box shook her head.

"Dr. Box, please take my advice. You need to lighten up, stop seeing bogeymen everywhere. It'll give you conniptions. Now, let's see what you need."

"You'll help us?"

"Yes."

"We can have our gold?" asked Kitou.

"How about we leave that for the real Water Bear?"

"But the bank vault was empty," said Brin.

"A deflection."

"What, then?" asked Box.

"Central bank credit is no use where you're headed, so how about some fungible assets to carry?"

"Fungible?" asked Brin.

"Precious metals, stones, and memory cores. Things you can spend."

"You'll give those to us?" asked Box.

"As long as you promise not to spend it locally."

"We get to own them?"

"Whoever has them, owns them. That's what fungible means."

"It's not," said Box.

"Can we spend it on Avalon?" asked Brin.

"Perfect. They're always trying to rob me. Sell them some inflation."

"What about our ship?" asked Box.

"That souped-up little spy boat, the Bat?"

"No, the Water Bear. It's on its way to Fluxor."

"Then follow it."

"The Bat doesn't have the range," said Brin. "It would take a century of bending space at maximum economy."

"What do you need?"

"A lightship."

"I can't help you with that. But I may know someone who can."

"Where are they?"

"Earth."

"Welcome to the Cult," said Macro, leading the Po women on a tour of the station. In Macro, Kitou had met her match for boundless enthusiasm. She followed gamely behind, inspecting everything he showed them. "This is the currency trading floor," he said. A group of preoccupied men looked up from wall-sized holographic displays, harrumphed, and returned to their work.

"Are you really all clones?" asked Kitou.

"Yes, and all men."

"Why?"

"We're not a normal society - we're a religious order."

"Who do you worship?" asked Box.

"The Finance Engine."

"A computer?"

"Yes."

"But you're all different," said Brin. "Some are dark, some fair. You have light brown skin and almond eyes, with epicanthic folds."

"Yes, we introduce ethnic diversity at the zygotal stage."

"Why?"

"Fifty identical clones would be boring."

"What do you do for sex?"

"Nothing."

Macro provisioned a strongroom to store their fungibles in. The room was pristine, as clean as a laboratory cleanroom. Box asked why it was empty. "We don't keep any valuables here," said Macro.

"Not even a wee float?" asked Box.

"No," he said. "We only do numbers. Otherwise, some trigger-happy desperado might come and blow us to bits to harvest the atoms."

"Could that happen?"

"Probably not," he admitted. "We have a lightship drive and a powerful computer to steer it. We'd win any fight by parsecs."

The first piece of bounty to arrive was a fire opal, the size of a hen's egg. "That's for the resurrection of your friend on Avalon," said Macro. The next was a milky green cube, veined with circuitry.

"What's that?" asked Kitou.

"The most valuable object you're likely to see," said Macro. "An unprogrammed mainframe AI."

"What's it for?"

"To pay for your ship's refit. It's worth more than the shipyard that did it."

"Why are you paying them so much?" asked Box.

"Oh, we need a shipyard."

These riches were followed by boxes of thumbnail-sized silicon slivers, and trays of diamonds. "Every gemstone is unique," he explained. "But we try to source gems whose signature has never been recorded. Memory cores are the hardest. We make our own for this kind of covert application."

"You do this kind of thing often?" asked Brin.

"More than you'd think."

He took them to see the Finance Engine. It consisted of a room, about twice the size of their strongroom, containing only a blue

haze. Embroidered throughout the haze, barely visible at first, Box could see wisps of circuitry - logic gates and eerie registers, stretching to a glowing infinity.

"Where is he?" asked Kitou.

"What you see here is a model, like the pictures of animals painted on cave walls. A synecdoche. It's here to help us experience him. Would you like to say hello?"

Kitou nodded.

"Is it safe?" asked Brin.

"Yeah," said Box. "I thought Samppo was needed to translate?"

"I'm learning," said Macro. "But yes, it's perfectly safe. All that Cult propaganda is only for show. The Engine's delightful company."

"I'd like to meet him," said Kitou.

"He *is* quite mad," said Macro.

"In that case, let's certainly do it," said Box.

Then they were *elsewhere*.

Box gasped. "Oh, it's Paris."

They were standing in the warm rain, on the *Rue de la Bastille*, outside the Brasserie Bofinger, where Box had first met Aloïs Buss. The headlights of cars were reflected in the glistening pavement. Box couldn't shake the feeling that they were really in Paris. It was more than correct - it was *perfect*. Under the cupola of the restaurant's grand dining room, a pajama-clad man was hunched over a honky-tonk piano, playing boogie-woogie like his life depended on it. He struck the keys with such gusto that the piano slowly inched forward, until he was forced to hook his leg around the leg of his stool to pursue it.

"He's playing Jelly Roll Morton," said Box.

The man looked up and beckoned them inside.

"Paris!" he shouted, tossing his head back with pleasure. "*La Ville-Lumière*. The city of lights! *Fluctuat nec mergitur*. Tossed by the waves but never sunk."

Box was instantly charmed. There was something in this rigmarole that was a personal *homage* to her. "Paris is my home,"

she explained to the others. "This restaurant, I almost live in this place."

"It's wonderful," said Kitou. "I like Earth."

"This is Earth?" asked Brin.

"Oh yes," said Box.

They found themselves at a table piled high with oysters, mussels, and snails, beneath a glorious *art nouveau* chandelier. The world was filled with people and noise. White-aproned waiters balanced platters of food on their arms. The room filled with the heady aroma of *Choucroute*.

Box sighed with pleasure. Kitou gazed with concern at the lobsters.

"Hey, Mr. E," said Macro.

"Macro, tuck in my boy!" said the man, who was inexplicably not only playing the piano, but was at their table. He attacked the shells with the same gusto as he'd attacked the piano. Slowly, the table inched forward.

"Is this food?" asked Kitou.

"It certainly is," said Brin, filling her plate.

"These are the people I told you about," said Macro.

"I know they are, I know I know I *know* they are. Are they right, do you think? Can I work with these *raw materials*? What d'you say, Macro? What d'you reckon?"

"I think they're perfect," said Macro.

"Perfect? Could they really be *perfect*? What a perfect idea. Tuck in, perfect soldiers! A prime number of soldiers! You can't build perfect muscles without perfect mussels."

He leaned back and stared. "Who's this? *The light, the life, the fire inside*? I haven't seen *you* in a while," he said to Kitou.

"I've never been here before, sir," she replied.

"Never been where? Never been *here*? I think we're *already there*. And you!"

He turned to stare at Brin. "The soldier the soldier the rock in the storm - tell me, bold soldier, what *is* mathematics?"

"What do you mean?" asked Brin.

"Oh, I know you know you know you *know* the answer. Is it discovered? Or is it real? I must know. I have to know have to know have to know *now*."

"It's both," she said. "We describe reality using a symbolic language we invent."

"You know what I think," he said. "D'you know what I think? I think mathematics is *God*."

"And you, Ophelia Box - the one we've all been waiting for. Dr. Box, I have something for *you*."

"For me?"

He made to reach under the table, then sprung back and grinned.

"Yes, oh yes, I owe you some *information*. *Les fruits de la mer*. A *raison d'être*, a *porpoise*. Go directly to London - do not pass Go - go see a man called *Yokohama Slim*."

"That's it?"

"Trust me, it's plenty."

"I know Mr. Slim," said Kitou.

"You do, you do you do you do *of course* you do! Look at that!"

Outside, a shaft of sunlight split the *Rue de la Bastille*, fretting it with golden fire.

"I get tired," he said. "Macro, what, yes, got to stop talking, got to stop, got to stop, it's a problem isn't it? Is it a problem? Do you think so? Perhaps I should, perhaps I should stretch my legs, should I stretch my legs?"

"I think so," said Macro.

"I will," he said.

They found themselves back in the street. Inside the brasserie, the mad computer played boogie-woogie as if his life depended on it. The clouds closed ranks, and a soft rain started to fall. Box, for a moment, shaking her head, thought the golden light from inside the brasserie was like the light shining from heaven.

"This is your religion?" asked Brin.

"Yes," said Macro.

"I approve."

Box and Brin teased Kitou.

"You do see she's smitten?" asked Brin.

"Really?" asked Box.

"He's lovely," said Kitou. "Clever, kind, and not too beautiful. Beautiful people live in a warp bubble."

"Kitou," said Box. "You're stunningly beautiful."

"No, I'm not," she said. "I have a bird's nest on my head."

"Ask Macro about that."

"Oh, he loves me."

"And why do you think that is?"

"Because of my spirit."

"There is that."

"You know," said Box. "He'd do anything for you."

"What do you mean?"

"Trust me. He's the ordinary boy who somehow got the best girl in the world. He can't believe it happened to him.

Nothing will stand in his way."

Box interviewed Samppo.

"Do you mind if I record this?" she asked.

"No," said. "If you win, I can suffer being famous for a few milliseconds."

"May I ask, who are you?"

"I'm the Bank. This human being you're addressing now was shuffled into the role. His wetware speaks to the mainframe computer called Samppo, and Samppo speaks to the Engine."

"So, you're not Samppo?"

"I'm a complex, evolved person. Are you the cells in your body? Once, I was a physical person like you."

"What was your name?"

"Oh, that'd be telling."

"What do you do?"

"I'm a banker. I'm the *central* banker. I'm also a hedge fund manager. And a religious leader - the Cult of the Bicameral Mind."

"What is the Finance Engine?"

"A god."

"You believe that?"

"Any sufficiently advanced being is indistinguishable from a deity."

"That's not an answer."

"Was it a question?"

"Why do you do what you do?"

"You mean, why does a god condescend to help run a bank? Because it's interesting."

"I mean why do *you* do what you do?"

"Me personally? The ghost in the machine? Dr. Box, I'm not a private citizen. I'm a branch of government. I do what's required of me."

"Which is?"

"I manage the value of money, for the benefit of our society. Our economy runs hot. People want to live productive lives. We overproduce. I manage that by taking it out of the system."

"You steal it?"

"Don't be ridiculous."

"Then how?"

"Typically, in the form of complex derivatives."

"What happens to it then?"

"A lot of it is just credit. We make it disappear. We try to spend the real money outside our economy. Some of it goes on the Thousand Worlds military. Some of it is to honor political commitments. The Xap rely on humanity to defend some of their frontiers. We finance that. We provide aid, philanthropically, and in return for political favors. If there's a disaster, on a large enough scale, our society is usually a first responder."

"So, you're the good guys?"

"We like to think so."

"Everyone keeps claiming that."

"Maybe it's true. We're a benevolent species."

"Why did you stop us stealing our gold?"

"Partly because I wanted to meet you in person. To see your faces. Talk it over. Maybe find a better solution. But holding up the bank was good. I like your cojones."

"Why was the vault empty?"

"Oh, there's people I'd like to think you got away with it."

"So, you framed us?"

"For a crime you meant to commit."

"Do you know what's happening, Samppo?"

He thought about that. "No. Maybe the Finance Engine does, but me? No. I can only make inferences."

"Why are you helping us now?"

"The great randomizer is war. I fear it exponentially."

"How much would you give to prevent one?"

"I'd give a great deal, but not to the point of abandoning our values if that's what you mean. What's the point in sacrificing what you set out to save?"

"What'll you do now?"

"About the drums of war? Nothing. The Po are the operational wing of the executive. You're the covert wing of the Po. I rely on you people for *doing* things."

"Us people?"

"You're a Po soldier now, Dr. Box."

"Why me, Samppo?"

"You mean why are you here? Out of all the countless gazillions? The chosen one narrative?"

"Yes, does anyone know?"

"Not me. Maybe the Xap do."

"Can I get to see one?"

"Shortly."

Piece by piece, their fungibles arrived: sapphires and emeralds; hypercrystals used in warp drives; memory cores. Objects Box didn't recognize. Fabulous riches. Some of it arrived in small, cylindrical lightships, the size of a *Gran Corona* cigar. These flitted in and out of existence around the disorderly hulk of the Cult, with their *Aurorae Galactini* playing over the layers of refuse.

One shipment arrived in a ship made of shadows - Magellanics. This treasure, Macro wouldn't show them.

"It's a surprise," he said.

"There's enough wealth here to fund a new world," said Samppo.

"Or a small war?" asked Box.

"A large one, if need be."

Pax monitored proceedings from a thousand kilometers away, declining every social invitation to visit. "We have our systems," he said. "I can best protect you from here." Instead, Kitou took Macro to the Bat, a long fall through space in their respective spacesuits - Kitou's slippery nano; Macro's like a deep-sea diving suit. Box watched them drift away, together, into the empty.

Again, she shivered.

Kitou returned with Aloïs - still steering the Water Bear's androform body - who disappeared to talk business with Samppo.

"I'm staying," he said. "Or at least my consciousness is."

"What will you do here?" asked Box.

"Samppo has offered me the use of a virtuality."

"What for?"

"Oh, some inquiries."

"Permanently?"

"Oh no. Please come and get me, when you get my original back."

"You'll merge?" asked Brin.

"If he'll still have me," said Aloïs.

Everyone seemed to agree that this was best. The Water Bear's remote wanted her systems back for the mission. Box felt bereft. She'd become used to Aloïs's presence.

"What kind of virtuality?" she asked.

"A simulation of Threnody," he said. "I'll continue my research. I have work I want to do on the Fa:ing number, with the help of the computers here."

"What kind of work?"

"I have ideas, based on what I saw Backstage. I want to try to *talk* with it. A more holistic approach."

"Talk, using mathematics?" asked Brin.

"How else?"

"The Finance Engine said mathematics is God," said Kitou.

"Then perhaps I'll be talking with God," said Aloïs.

After nine days, with their treasure secure in the Bat, they prepared to leave. The war chest was smaller than Box had expected, barely larger than a small suitcase. In it was the wealth of worlds. Enough for their mission, many times over.

"It's been good having you," said Samppo. "We don't get many women here, much less three archetypes."

"What do you mean?" asked Box.

"It should be obvious by now. You, the wise woman. Brin, the brave soldier. Kitou, the numinous child. Beings of mythos. Who can oppose you?"

"I hope you're right."

"So do we all."

Box looked out across the space they were about to cross. If there was a demon who lived in the sky, he lived out here, in the nothing.

Kitou was heartbroken. As they were leaving, on the airside of the filmy membrane, between space and the edge of the scow, Macro gave her his package.

"It's for your ship," he said.

"And you.

"Remember us here."

8 → YOKOHAMA SLIM ESQ. RARE BOOKS

2056

Box awoke to a quarrel. She, Brin, and Pax had slept through the journey to Earth. Kitou was left in command.

"This is unacceptable," said Pax.

Kitou was hanging her head in contrition.

"What's happening?" asked Box.

"Kitou used the Cult device to contact Macro," said Brin.

"What's wrong with that?" asked Box.

"Everything."

The device was a milky green sphere, the size of a grapefruit. It reminded Box of the mainframe core she'd seen at the Bank. Brin said it might be a sex toy. Kitou rolled her eyes at that.

Pax said to Kitou, "Explain."

Kitou said, "I was experimenting."

"I have no problem with that. Details?"

"I asked the Bat. He suggested an input. I tried it. It opened a channel."

"To the Cult?"

"Yes."

"The interesting thing," said the Water Bear, "is that this happened inside a warp bubble. That's new physics."

"What happened then?" asked Pax.

"I asked for Macro."

"First mistake," said Pax. "And then?"

"I called him again, after."

Pax sighed. "How did the Cult react to all this?"

"They were unsurprised."

"You should have told the Water Bear and woken me."

"Yes."

"Consequences."

Kitou looked up. She didn't care about consequences. She was already experiencing the consequences.

"See me in the practice area," said Pax. "In five minutes."

"What's going on?" asked Box. Kitou seemed relieved.

"Pax will teach her a lesson," said Brin.

"He's going to hurt her?"

"Probably."

There was no great skill in the fight. They had their wetware off, in the Bat's erratic gravity, and there was skin in the game. Pax advanced, and Kitou danced away. Pax advanced again. He tried to ease into his Po routines, where he had the advantage, but Kitou avoided him.

Then Pax caught her off guard and slapped her hard on the side of the head, drawing blood.

Box winced.

Kitou shook her head and returned to the Po starting position. This routine continued, until Pax launched a fluid combination of punches, and caught her again, flat on the same side of the head.

She sprawled on the floor.

"Stay down," whispered Brin.

Kitou rose to her feet, to the starting position.

Pax nodded, and went directly after her, launching a second flurry of punches, and caught her again, this time with a flat fist to the shoulder, and she went down, hard.

There was an unpleasant pop.

"Stay down," he said.

Kitou shook her head and rose unsteadily to her feet.

"No," she said.

Pax sighed. He moved in to finish her, but she anticipated him. With blistering speed, she spun inside his defenses, and landed a

full-powered elbow, directly in his ribcage.

"That's going to hurt," said Brin.

Kitou danced away, and Pax couldn't catch her. She sprang forward again, while he was advancing, and landed a perfect Muay Thai combination, to the same part of his ribcage. There was a meaty crunch.

"And that," said Brin.

It was finished a few moments later. Pax simply ignored her defenses, marched in, and delivered a clinical punch to the point of her chin, knocking her unconscious.

He looked down at her and frowned.

"Brin," he said.

"Yes, Navigator?"

"Take her to the medibay. Bring her to me when she's able."

Brin nodded.

Box watched Kitou stirring on the medibay. She couldn't begin to untangle her feelings. Anger. Disgust. Reluctant admiration. Nauseous rage. A mother's pain for her injured child.

An ultrasound machine began a bone knitting operation.

"Will she have scars?"

"Only if she wants them," said Brin.

"You seem remarkably sanguine about this."

Brin seemed surprised. "What do you expect?"

"What do you think, Water Bear?"

"I approve."

"How can you?"

"A sharp punishment is superior to a long humiliation. Now the air is clear. Kitou won't forget this lesson. Nor will Pax."

"I don't get you people."

"You can't," said Brin, gently. "You're a normally socialized human. We're soldiers. We wager our lives. We must absolutely trust each other."

"How is this different to a training injury?" asked the Water Bear.

"They're not deliberate."

She stormed out of the medibay, as furiously as she could in zero-gravity, and cannoned into Pax, who looked puzzled.

"What was that all about?" she demanded.

He raised his hands placatingly.

"Don't give me that horseshit, Pax. You're a grown man. She's a child. You could've killed her."

Pax winced. "And she me."

Box stormed away.

Later, Brin and the Water Bear followed a serene Kitou into the communal space. Box sensed the presence of strong analgesics in Kitou's demeanor.

Pax nodded. "You fought like the wind," he said.

He took a pouch from his pocket.

"It was saving this, but it's time."

He gave Kitou two carbon-black starbursts.

"You understand your responsibilities?"

She nodded.

"Then excuse me, young soldier, while I have my injuries seen to."

He pushed himself gingerly towards the medibay.

Later, Box watched as the Water Bear installed Kitou's combat wetware.

"Will it hurt?" asked the Bat.

"More than Pax's blows did," said the Water Bear.

It didn't look painful. Kitou just fell asleep.

"What does it do?" asked Box.

"It's a full military system," said the Water Bear. "Built on a different architecture. It attaches lower in the autonomic nervous system. She'll be better, faster, stronger."

"God help us," said Box.

"Indeed," said the Water Bear.

"Will it affect her personality?"

"Not a bit."

"Poor Kitou," said the Bat.

"Does it really hurt?" asked Box.

"Yes," said the Water Bear.

While Kitou nursed a powerful headache, the others looked down on Earth. It was the same blue-white jewel Box had seen from the Pnyx, twenty years in the future. On the ground, it'd be a different story. The biosphere was in a death spiral. Soon, without intervention, the methane monster in the Arctic would start to stir, and that would be the end of it.

"There's a young Ophelia Box down there," said the Water Bear.

"It's been a year since my family died," said Box.

"I'm sorry," said Pax.

"If only we'd come sooner."

"It may have made no difference," said the Water Bear.

"What do you mean?"

"There are competing theories of retro-causation - of the future changing the past. One is that time is self-healing. No matter what we do, it returns to its original shape. Another is that when we try to change past events, we may become the cause of them."

"You mean we could start the war we're here to prevent?"

"Or cause the atrocity at Fluxor."

"Then why are we here?"

"Because we were sent," said Pax. "And we'll be taking the path of fewest assumptions."

"Hopefully, the universe has no problems with paradox," said the Water Bear.

"That hurts my head," said Box.

"Mine too," said Kitou.

They fell in the usual way, trusting their suits not to trigger a terror alert. More than one Caliphate cruise missile had been intercepted over the skies of southern England.

London, in the dog days of the climate wars. Some parts of the city raged in uncontrolled fires. Box could see all of Bow, up in flames. She saw Ashburton Grove, spotlit for an Arsenal game, surrounded by flashing blue polis.

Half a country away, a twelve-year-old Ophelia Box was learning to sell black-market pharma in the mean streets of Glasgow.

Her skinsuit blossomed, and she settled with gossamer wings on Wimbledon Common.

Her facemask retracted.

The air was syrupy, and warm.

London, before the *Aurora Galactinus*.

"Okay team," she said. "Listen to your mission specialist. We've come at a time of great social upheaval. I chose this route because it takes us through three safe zones."

"We could've landed closer," said Brin.

"I know," said Box. "But I wanted to see it."

She'd dressed them in Lycra tops, loose-fitting combats, and trainers, London's ubiquitous streetwear. Since no one in 2056 had wetware, Kitou and Brin had downloaded Box's soft highland burr.

They were three Scottish lassies, walking in London.

At the edge of Wimbledon Village, smiling boom-gate guards waved them into the Glorious Nation of Tooting. One man, a young Rastafarian, asked for Brin's number.

"Are these real countries?" asked Brin.

"It began as a joke," said Box. "Then the gates were installed."

It was past midnight, and parties were spilling out everywhere. Every bar was filled to overflowing. Groups of youths gave them good-natured waves.

"This isn't threatening," said Brin.

"London has a history of this kind of thing," said Box, "of bringing out its best in a crisis."

"I like these people," said Kitou.

"So do I," said Box.

For all its pre-apocalyptic *bonhomie*, London was still a dangerous place. They stuck to the lit spaces, skirting the swampy fens of crack houses and backstreet drug deals. London's streets had always had a bursting, overcrowded feel, now even more so in the new age of *fin-du-temps* parties, when the town centers were continuously filled with revelers. The coming apocalypse hadn't put a brake on the population. Twenty-four million people now lived here. After an hour spent dodging drunks and street dealers, and buses

lined up like red postboxes in the gridlocked streets, Brixton announced itself with a lightshow spilling over a low skyline, and a tremor in the pavement.

"Is that another country?" asked Brin.

"No," said Box. "It's a 24x festival. There's another in Notting Hill. They alternate. When Brixton is banging, Notting Hill is chilling."

"Is Brixton banging?" asked Kitou.

"I do believe it is," said Box.

Brixton was always Box's London postcode of choice. She liked its tough history. Electric Avenue was the first street in London to be lit by electricity. There was also her love of diversity. London had it in every sinew, but here was the muscle. She was drawn to this place, and she wasn't about to miss a chance to see it.

And party.

Kitou chose a shiny plastic party frock with ballet flats. Brin wore a cocktail dress and heels. Box kept her combats and trainers and wore just a frilly black sports bra on top. The passing revelers who saw their clothes morph, hooted approval.

"We're sexy," declared Box.

The hulking, concrete church on Brixton Green was garlanded with bunting, like the gun emplacements on Avalon, a hundred light-years away. Around the edges of the grassy space was a funfair, complete with sparking old dodgems and chuntering fairground rides. A calliope wheezed along to the electronic dance music. There was even a hayride - a magnificent Percheron, unfazed by the ruckus, hauling cartloads of shrieking children around. People of every age, shape, and gender were dancing. Box pointed out a shambolic statue in the middle of the green. "Look," she said, "a wicker man." It was a roughly made man in the shape of a star, woven from flowers and grass. Its frame had the blued appearance of scorched titanium.

"They'll burn it," she said, "on the solstice."

"As a fertility rite?" asked Brin.

"A sacrifice," said Box. "It expresses a longing for the old gods of the forests and trees."

"Nothing wrong with that," said Kitou.

The soundtrack was pure twenty-sixties. The gloomy ür-metal of the twenty-fifties had been canceled. Who wanted to hear comedy howls when the world was actually ending? Balearic trance had made its perennial comeback.

"This is great music," said Kitou. "What is it?"

"This is called house," said Box. "Over the road there is K-pop." They watched as a group of Korean girls danced in uncanny unison.

"They're intoxicated," observed Brin.

Box nodded. "Almost everyone here is. 3,4-methylenedioxy-methamphetamine, also known as ecstasy, the party drug of choice in this milieu." It was ironic, thought Box, that at the end of the world, the consumption of alcohol had fallen. The great, poetic drug of depression had been replaced by euphorics. Tobacco had been outcompeted by oilseed and rice. In many respects, the population had never been healthier.

On the other hand, there was the paranoia.

"You're an expert," said Brin.

"I'm an expert *historian*," said Box.

"Can we take some?" asked Kitou.

"No," said Box.

"I want to dance," said Kitou, and for the next few hours they did, three good friends at a festival.

"Pax will be shaking his head," said Kitou.

They all laughed at that.

"Dr. Box," said Brin, as they left Brixton behind. "Their death is approaching, yet these people are having a party."

"Yes, and it's not a creepy bacchanalia, like some places on Earth. This is a party you can take your family to."

"A positive attitude," said Brin. "I'd do the same if my world was ending."

"London is wicked," said Kitou.

"It's the best," agreed Box.

The Borough of Lambeth made way for the People's Republic of Southwark. The tenements were tightly shuttered here. Box

imagined them full of sleeping children, waiting on an uncertain future. She wished she could reach in and tell them the good news.

Would it all be good news? She hoped so. She hoped they weren't all headed down another rabbit hole, where death arrives by a different entrance.

Like a sphere of decoherence.

Instead of broiling.

After a few kilometers of bristling inner-city estates, they emerged onto a sandbagged Thames. The Brixton festivities were a memory now. This was a quiet place. A block from here, Shakespeare once played Earth's great tragedies. Over the river, the smoked-glass canyons of the financial district had a darkly post-apocalyptic feel. On the river, barges were burning. They were kept going for years, fueled by oil from a Saudi oil tanker grounded on the Thames Barrier.

Box had never had money. She was an impoverished historian, but she understood markets. "We call that the Square Mile," she said, pointing over the river. "It used to be one of Earth's financial centers."

"Like Praxis?" asked Kitou.

"Like Avalon crossed with the Cult. People would gather on trading floors and shout at each other."

"Pit trading," said Brin.

"What happened?" asked Kitou.

"Schism. When Britain brexited from the regional economic union, the City cexited from Great Britain, so they cut off its power."

"How did they survive?" asked Kitou.

"Cashflow."

"Is it safe to go in there?" asked Brin.

"Yes. There used to be fighting, but not now. If anyone makes trouble, they get zapped by a laser from orbit."

There were no gates to the City inside a city, only CCTV cameras. Fifty meters below, in the alluvial mud of the Thames, in darkened rooms with uninterruptible power supplies, the trading continued. The few people they saw hurried past with their heads down.

"We're at a pivotal moment in Earth history," said Box, as they passed the shuttered facades of venerable insurance companies, their way tracked by thousands of lenses. "Europe's become a techno-anarchic freakshow. America's a shooting gallery. The start of the complete breakdown of society."

"A system on the edge of chaos," said Brin.

"Yes. And in a year, when the lightships arrive, everyone will shake their heads, pretend nothing happened."

"No closure," said Brin.

"No," said Box. "It's true. We're still angry."

They crossed into a modish warehouse district, where boutique shops leaned over crooked, cobblestone streets. This was Clerkenwell. There were the same popup festivities as in south London but winding down for the day. Every railway arch seemed to be filled with tired partygoers. Already, the first civilians of morning were making their way into the streets.

"Nearly there," said Box.

Dawn was climbing over the spiky rooftops when they arrived at a shop, hardly more than a kiosk, in a quiet back street. Already, the temperature was rising.

The sign said, **Yokohama Slim, Esq. Rare Books.**

The man who answered the door was dressed in a smart city suit, despite the early hour. His hair was tied back with *Kanzashi* sticks. He looked down his long nose at their party dresses.

"Well, it's Ophelia Box and Kitou Gorgonza. And you must be Brin Lot. Please, come inside."

Slim was the Slim she remembered - unfeasibly tall, impeccably dressed, and by all accounts a dangerous man, with prison tattoos. She'd heard the stories, but Slim was never a danger to her. He was safety.

Yokohama Slim claimed to be part Japanese, part Russian, part Tuniit nomad. He spoke American English with a slight Russian accent, which he said he'd picked up in the Gulag, and by some glitch, it persisted. "Like a bad smell," he said. He ushered them into his shop, which Box knew so well from her past. This was her home,

as much as anywhere else was a home. Dusty first editions lined the shelves - books of magick and philosophy elbowing for space with Manga collections.

"How can you possibly know me, Slim?" asked Box while he boiled water for tea. "We don't meet for two years."

"Let's say I have a good memory."

Over bowls of tea, he told them more. "Ito and I have been working on the Fa:ing problem for some time," he said.

"You mean you *will* be working on it," said Box.

He shrugged. "Time is a knot, Box. Best cut right through it. Now, introduce us."

"Okay. Well, everyone, Slim's my best friend. My father. My mentor. My kickboxing teacher."

"Saved you from the mean streets of Glasgow," he smiled.

Box laughed. "You did. Sent me to Paris. Cured my addictions. Taught me to fight. Slim's an excellent kickboxer."

"No, I'm not. I'm a klutz, a *kuruttsu*, compared to you."

"You should see these two fight."

"Well, they're Po soldiers."

"Slim, why are you here? I'm confused."

"I work for the Po."

"No kidding."

"I also freelance a little."

"Who for?"

"This isn't the place. First, to sleep. You know where your room is. I'll sort out the others."

"And then?"

"We'll hatch our plans, like good little spies, under cover of darkness."

Her room was just as she'd left it - a bed, slightly mildewed, tucked under the eaves. She could look straight down and see the passersby. She felt a strong rush of memory, here in her personal space. She'd led a hard-enough life, although she was mindful of those who'd lived worse. She was alive. A few hours from here, people were dying by millions in wars for resources, or by trainloads

in the forced labor camps of the Med. It was hard to imagine death on such scales. She shook her head. It would soon happen here, in these privileged streets, the veneer of western exceptionalism stripped away by a blowtorch.

That's what the alien invasion had stopped. An invasion she was now part of. *Une collaboratrice*. She sighed. Whatever was happening now, it wasn't what she'd signed up for, but what choice did she have? It wasn't as though she said no. It wasn't as though she wanted to.

For the first time since leaving Earth, she felt safe. Nothing could hurt her here, not with Slim down below, smoking his roll-up cigarettes. The strangeness ebbed away, and for a while, she was just Ophelia Box, not a piece in a game she didn't recognize.

Slim gave the others the bedroom below. Outside, the mercury climbed past 40°C. It was a steamy heat, clammy with a thousand new waterways, but Slim's antique aircon was up to the task. Later, when the sun dipped beneath the spiky Clerkenwell roofs, and the cool air of evening was creeping into the backstreets, they visited a restaurant she knew well. She ordered acres of *mezédhes* and more retsina than was wise.

"This is delicious," said Kitou.

"Oh, to be sixteen again," said Box.

The food *was* delicious. So was the place, steeped in all its antiquities. Russians and Persians, playing chess. Minicab drivers, arguing in Armenian. Dolmades, duqqa, and dumplings.

While the girls ate, Slim told them the story of his ancestors. "My people, the Tuniit giants of *Kalaallit Nunaat*," he said, "traveled east in the age of the ice spirits, tracking walrus and seal." It was a story Box had heard before, a staple of her childhood. Slim had an actor's sonorous voice and a magnetic way of storytelling. Later, as a rookie historian, she would learn that the Tuniit - the Dorset giants - were the fabled first peoples of the Arctic. Their existence was shrouded in myth, but Box knew the truth. She'd heard the dogs, and the rattle and thump of runners on ice, too real to be make-believe. Just hearing it made the hairs on her neck stand on end. It was like an electrical circuit, wired into her memories.

"This was how long ago?" she said.

"The last diaspora was around the time of Christ."

"You're one of the old ones?" asked Kitou.

"Older," he said.

That night, the Bat appeared out of a lowering sky. Rain spat from scurrying clouds, lit by the wildfires in Bow. The little ship seemed larger in the flickering lights. They climbed over Slim's slippery roof, into a door in its belly.

"Mr. Slim," said Pax when they were aboard, and the Bat had climbed back over the crowded air lanes, and London was spread out below them like a wargame.

Brin had originally offered to lift Slim using the Bat's gravity drives. "No," he said. "Tuniit and gravity drives - we don't mix."

"Master Pax," he said. "Ito has spoken well of you."

"And you. Please meet my ship, the Water Bear."

"We've met," said Slim, to the Water Bear's androform remote.

"It's been a long time, Yokohama Slim," said the ship, with a smile.

"And the Bat."

"Welcome," said the Bat.

"Where're we going?" asked Pax.

"Not so much where," said Slim, "as when." He asked Kitou to bring him Macro's device. Kitou frowned unhappily at that.

"It won't bite you," said Pax.

"May I?" asked Slim, after Kitou returned with the grapefruit-sized sphere.

"Go ahead," said Pax.

Slim touched its surfaces with his long fingers.

"This is a functioning time travel drive," he said. "Rarer than you might think. It might be the only one there is." Something flashed inside it. For a moment, Box thought she saw star systems, hidden in its depths.

"As you may have guessed," he said, "I serve several masters. Our acquisitive friend, the Bank. Our mutual benefactors, the Xap. Others you don't want to know about."

"This, this is Magellanic tech. Or at least, the box it comes in is. The tech, not so much."

He touched the object again. It flashed, and again Box had the sensation of a revealed infinity.

"Which I happen to know how to drive."

He touched the object again.

"And now we're there," he said. "Or then. We just traveled in time, to 1851. A great many things happened, in 1851. We're going to visit the Great Exhibition.

"Tell me," he said to the Water Bear. "Do you have anything resembling a robot?"

"This ship has a maintenance drone," she said.

"Can it fight?"

9 → THE SPIRIT MOLECULE

1851

Driving through Victorian London in a horse-drawn carriage was like crack cocaine for a modern-day historian. It was a crisp autumn day, and the city stank of sewage from the Thames, and coal smoke from a thousand chimneys. Their cab was a Clarence, also known as a growler, for the noise its tires made on the cobblestone streets. They rumbled down a Piccadilly jam-packed with carts and steam-powered cars - papered with flyers for the

**Great Exhibition of the
Works of Industry of
All Nations.**

Slim had them dressed in the height of Victorian fashion, with bonnets, gloves, voluminous skirts, and parasols against the wintry sunshine. As usual, their clothes were a perimeter defense of nanobots and fields. He'd asked for their wetware to be turned off, to not frighten their quarry. It was the first time in Box's adult life that she'd been without her alien neurocomputer.

Slim was wearing conventional fabrics - a raffish but threadbare frock coat and vest, as befits an Inventor. He'd disguised the drone's pincers in red leather boxing gloves, which had been hanging in his 1851 kitchen. They were highland lassies again, in company with their uncle, the Oriental Inventor, and his Fighting Machine, the Mechanical Spider.

The growler dropped them at Hyde Park Corner during the fashionable hour, when the gentry mingled with artists and rogues. Slim tipped his top hat at the promenading women, who either ignored him or blushed furiously.

"You always did cut a dashing figure," said Box.

"I astonish them," said Slim. "They can't tell if I'm a gentleman or a savage."

"They desire both," said Brin.

"This place is incredible," said Kitou.

The Spider scuttled behind, waving its gloves like a crab. In 21st century Hyde Park, it would've drawn a crowd of appreciative teenagers, but these Londoners hardly seemed to notice it.

"They choose not to see it," said Kitou.

"The Po exploit this phenomenon," said Brin.

"We're exploiting it now," said Slim. "Four obvious time-travelers, yet the best available scandal is my ethnicity."

The cast-iron and plate-glass Crystal Palace of the Great Exhibition was one of the great buildings of the time. Its interior volume was divided into halls - and in one of the halls, in a section reserved for the Russian inventions, was a sign:

ULTIMATE FIGHTING COMPETITION

The *необычный медведь* stand was slightly larger than a full-sized boxing ring. Inside the ring were two machines, fettled by men with oily cloths. Crowded around were excited gentlemen, and a few adventurous ladies, waving banknotes at bookmakers. Slim waved a fan of pounds in the air. "I have the next winner," he said. The crowd ignored him. They had eyes only for the machines already in the ring.

One was a kind of horizontal trebuchet. It spun, and an arm swung out - then a second arm appeared, with a ball on the end. Powered by a tightly wound spring, it struck and struck again with bewildering speed - a weaponized double pendulum. It was called the St Austell Slinger.

Its opponent was a more conventional design. Humanoid and bipedal, it lumbered around the ring, thinking before striking. It was called the Matryoshka Doll, and according to posters papered to the

walls of the stand, it was the Mechanical Fighting Champion of the World. It was hopeless. Its owner waved the championship belt in the air. He caught Slim's eye, and they exchanged a complicit glance.

The fight was over in under a minute. The Doll moved once, and again, then reached through the chaos of the pendulum's arcs and pulled a mechanical pin. The Slinger's payload jack-knifed into the crowd, where it almost knocked a gentleman unconscious. The undefeated Champion had won, again. A gift of order from chaos. Box found that reassuring.

Slim hoisted the Mechanical Spider into the ring, to hoots and groans from the gentlemen. As on the promenade, these bettors didn't recognize its biomechanical actuators as technology. Bets were made on the Doll. The fight began, and the robots circled each other.

After a few moments, Kitou and Brin began whispering among themselves.

"The bipedal machine knows Po," said Kitou.

Box recognized some of the moves. First, there was a clever dog, in which the lumbering robot pushed the nimble drone back. The crowd laughed.

"They're colluding," said Brin.

"Yes," whispered Slim. "We like to think of it as a performance."

More bets were placed, and the fight became more kinetic. Soon the Spider's speed was outmatched by the Russian machine's greater reach, and it was punched from the ring. The crowd dispersed, and several gentlemen slapped the back of the Russian.

"What just happened?" asked Box.

"We lost," said Slim.

Slim led the women toward the Russian Inventor and his robot. Kitou moved like a blur. Before anyone could react, she had the man pinned to the side of the ring and was whispering to him in a songlike language. Somewhere, Kitou had found a knife and was turning it against his carotid artery, not gently, arching him back. Brin peeled her loose and accepted the blade.

"That's mine," said the Russian. He was a sallow young man, and he was showing them an empty shoulder scabbard, as though being bailed up with his own weapon was an everyday event.

Brin pocketed the knife.

"What the fuck was that?" asked Box.

"Pursang," said Slim, "if I'm not mistaken."

"What was she saying?"

"I'm not especially fluent," said Slim. "But approximately, I kill you later."

Slim took a glaring Kitou by the shoulders.

"Listen, my little stiletto. Your first mistake is that to understand events, we must first talk to the Horu."

"I know that," said Kitou. "It's why he's alive."

"Your second mistake is that the Horu is the machine."

"Yes," boomed the robot.

"Why do you want to kill me?"

The Russian's suite of rooms at the Grosvenor Hotel, on Park Lane, in Mayfair, were the essence of regency bling - gold velvet drapes and a sputtering fire. The Russian, who was called Ned Mulligan, broke out Irish whiskey - which he'd brought from his share of the money - and left to count the day's winnings. Box was sure she remembered his name. Yes, a Ned Mulligan was a revolutionary gone rogue in the Fenian Rising of 1867, executed by his own people, for murder.

Sixteen years in the future.

"Why do you want to kill me?" asked the robot.

Slim said, "Kitou?"

"I'm a Pursang, from Fluxor."

The robot clanked, then stopped moving. Unlike the Mechanical Spider, it was a reasonable facsimile of Victorian technology.

"Ah," it said after a moment of silence. "Fluxor. That was a crime against reason." It clanked again. "But hear me out, young Pursang. My people are innocent of that crime. I've already persuaded Ito Nadolo of it."

"This Horu is called Chance," said Slim. "Despite appearances, he's a human. Chance, these are the ones you must convince."

Chance clanked again. "Then I'll make my case simply. It's inconceivable that we'd ever attack Fluxor, let alone destroy it so cruelly. We're joined, the Horu and Pursang, in ways you cannot imagine.

"But they are just words. To know the truth, you must go to Horax."

Ned Mulligan returned with a tray of ice and five crystal tumblers. He had the air of an indomitable swindler who believed he could talk his way around anything.

"Who are you, Mr. Mulligan?" asked Box. "Not Russian, to be sure?"

"I'm an Irishman," he said.

"You'll be a Fenian then," she said.

"And who would a Fenian be?"

"You fought in the Famine Rebellion?"

"That I did, and bravely so. How d'you know?"

"I know you'll be a murderous traitor."

"And what would you be, a witch?"

She snorted.

"Slim," said the robot. "You should take them to the aspen wood."

"I will," said Slim.

"Young Pursang?" asked the robot.

Kitou raised her chin.

"Please, have an open mind."

She nodded. "I will, if Mr. Slim says so."

"We must unmask the guilty, not seize on the conspicuous.

"You have my offer of friendship," he said.

"Never."

Day became night, and as the others lounged around Ned Mulligan's rooms, waiting for London to sleep, Box interviewed the robot. Mulligan seemed not to care. She now understood that Chance was in charge, and Mulligan was his employee.

"Why are you here, Mr. Chance?"

"I'm observing the Xap."

"On Earth?"

"Yes."

"There are Xap here?"

"Of course, where else should they be?"

Again, Box had a sense of a cosmos beyond her wildest dreams. Robots on Earth, in 1851. Her, interviewing one, as though he were no more than a mildly exotic Mayfair personality.

She had to admit, she quite liked this austere, clanking robot.

She breathed, and carried on.

"Why are you helping Ned Mulligan?"

"For my amusement."

"Why are you here, in 1851?"

"We are when we are, Dr. Box. I'm no time traveler. I came here in the usual way. These are my last few days in the Real. Soon, I'll descend."

"Descend? What... you're dying?"

"No, we're in the process of moving my people. I'm the last living Horu, and I'll be the first download."

"Moving, where?"

"To another universe, of our own making."

"Why?" she asked.

"Why am I last?"

"Why are you moving?"

"For cultural reasons. Some perceive us as a death cult. For millennia, we've stored our dead in binary necropolises, so they can live in a better future. Now it's happening.

"And I'm the last, because someone has to be."

"Is it because of Fluxor?"

"Do you mean are we fleeing justice? No. It has nothing to do with that. We had nothing to do with what happens on Fluxor."

"There are witnesses."

"Who saw ships?"

"Then get rid of those ships."

"No, that's impossible. Those ships are instrumental in the future of my people. We'd be exchanging one atrocity for another."

She decided to change her line of questioning.

"When did you learn about the genocide?"

"Ito Nadolo confronted me with it."

"How did you feel?"

"Shocked, hopeless."

"What did you do about it?"

"I told my people."

"The dead?"

"No, the necropolises. Our ships. I told them."

"Really? Aren't you worried you'll cause it to happen?"

"Yes."

"How did you meet Ito?"

"Yokohama Slim brought him here."

She frowned, and resolved to ask Slim about his part in all this. Like Ant de Large, he seemed to be stitched into her life from the inside.

That he saved her was indisputable.

What else has he done?

The Bat was where they'd left it, floating over Slim's bookshop. It'd engaged its adaptive camouflage, so the night sky shone through. In 1851, anyone seeing the boundary distortion would've blamed their eyes, or London's ubiquitous smog. Freezing rain had settled on Slim's granite roof, making the tiles as slick as glass. They clambered across it, followed by the Bat's maintenance drone.

Bat actual was bursting to upload the Mechanical Spider's experiences.

"Did I win?" it asked.

"You did better than win," said Slim. "You lost deceptively."

"We got what we wanted," he said. "The most crucial thing. We now have the Horu as allies."

"How did we engineer that?" asked Pax.

"By the undeniable existence of Kitou. She made quite the impression."

Pax nodded. "This Chance is so influential that he can speak for his people?"

"It's not about being influential," said Slim. "He's in a quorum of one."

"Can he be trusted?"

"I bet my life on it," said Slim.

The Bat rose into the air and turned west.

"So, to this forest?" asked Pax.

"Yes," said Slim. "To our destination."

"Where is it?"

"Utah."

The Bat flew low over a churning sea in a turbulent pocket of supercavitating air. Box saw smoke belching from coal steamers plying the busy North Atlantic routes. This was the start of the great age of maritime steam. Those ships would've only heard a distant rumble of thunder or seen a hazy rooster's tail on the horizon.

Box found the control room of the Bat - flying low over rough water - an exhilarating place to be. Unlike the Water Bear, which had the quiet ambiance of a luxury train, the unfiltered Bat was like a low-flying washing machine.

Yokohama Slim joined her, and they watched the waves fly past, fifty meters below.

"Why are we flying so low?" she asked.

"For the joy of it I suppose."

"Is this stealthy?"

Slim shrugged. "You can't be stealthy always."

"Slim, what's it with you and gravity?"

"I have a different type of physiology. Artificial gravity disrupts it." He pointed at the Bat's circular wings. "You see those discs?" he said.

"Yes."

"The Bat moves by exerting a gravitic force on them. We're not in a gravity field. It's a handy design for people like me."

"There are more like you?"

"Some."

"Did Praxis give us the Bat, so we could ferry you around?"

He shrugged. "Maybe. I suspect we're all pieces in a game."

"How do we get to be players?"

"The person you're going to see is a player."

"*I'm* going to see?"

"Yes, she's asked for you specifically. London was the audition. This is why we're here."

Kitou joined them, and Brin, so they were pressed shoulder to shoulder in the narrow control room. Kitou looked down at their destination, a golden forest of aspen leaves.

"Holy mother forest," said Kitou, matter-of-factly.

"Pando," said Yokohama Slim.

The three women fell at dawn in the usual way, onto a misty carpet of fallen leaves, tiny aspen shoots, and gnarled fungus - a clonal colony of quaking aspen trees, considered by some to be the world's oldest living organism.

The psychedelic drug Slim gave Box to take was predominantly Dimethyltryptamine, or DMT, a powerful tryptamine.

"Like the Ayahuasca sacrament?" she'd asked.

"Stronger."

A beaker of slippery ooze.

"The spirit molecule," he said.

They sat in a clearing, on a silvery blanket that Brin had unfolded from a nanobot rucksack, and Box drank the drug straight down.

"Buy the ticket, take the ride," she said.

Her friends leaned companionably against her.

The first thing Box saw was that the forest was shaking. It was the leaves. They were trembling in the slight breeze. Tiny drops of water trembled *in* the leaves. Then her senses seemed to flip, and she was alone in the forest glade.

That was *fast*.

She looked at her hand. It seemed solid enough. No sudden disorientation or nausea.

Then she experienced a hollowed-out sensation. Curiously, the forest was *inside* her. Synesthesia. She was becoming physically confused.

The universe dissolved in jagged angularity.

She started to panic.

Then the forest *held* her, and she melted into it. It was the most surprising sensation. It was as though the branches, and stems, and leaves, were rotating along their long axes, sparkling in the wintry morning light, and she was a part of it. She felt the Earth, turning beneath her, and the stars, wheeling in the heavens above.

Sitting cross-legged beside her was a smooth-skinned, nut-brown woman, of unknowable age.

"Who are you?" Box asked. "Are you the forest?"

"No," said the woman. "I just live here."

"Are you a spirit?"

The woman thought about this.

"Maybe," she said.

"Slim promised me answers," said Box.

The woman laughed. "Then answers you'll have. But first, I'll tell you a story."

They took a breath, and the forest breathed with them.

"A long time ago," she began, "on a world like this - you might even say it was this one - a species evolved. They inhabit the trees, like a mind inhabits the brain. You call those people the Xap."

Box looked around. She'd always marveled at the sheer amount of invention in psychedelic trips - the neural processing it implied, and what it meant for the conceit of a reality, based on observation.

But *this* - surely it had to be real.

"Are they our friends?" she heard herself asking.

"We are. A friendship that goes back millions of years, on this world, and others. More than a friendship. A symbiosis. We carried each other to the stars."

"I thought you people could go anywhere, just by thinking about it?"

"We go where we remember."

"You evolved here?"

"We did."

"What about us?"

"Humans came later."

"A diaspora?"

"Of sorts."

"Where did we evolve?"

"On Fluxor."

"Wait... Fluxor's the homeworld?"

"It is."

"Holy moly. No wonder everyone's so edgy."

The woman sighed.

"That was a crime," she said, as they grieved.

After some time, it could've been days, or it might've been decades, Box said, "What should I call you?"

"Me? I have so many names. Mother? Pando? They all seem to work."

"Can I call you Pando?"

The woman smiled.

"Pando, what's happening?"

"A war is coming."

"Yeah, everyone keeps saying."

"We have a prophecy, Ophelia, that this world age will end in nothing. Not entropy, or heat death, but nothing. I cannot see beyond the destruction of Fluxor."

A rainstorm passed through, trailing soft peals of thunder, drenching the fallen leaves.

"What about the Fa:ing? Aren't they from the future?"

"The Fa:ing are a mystery to me."

"They're here to help us?"

"I think so."

"Will you help us?"

"I *am* helping you, Ophelia Box. Go to Fluxor. Find a man called Jaasper Huw. Tell him what I said here. Then go to Horax. Finally, seek your answers in Möbius space."

"Where's that?"

"The Horu will show you."

Box sighed, and the forest sighed with her. A sparkling rain of droplets drifted to the forest floor.

This was a helluva psychedelic.

Box sighed.

"Pando, why me?"

"That's a good question. The long answer is you must learn for yourself. I'm sorry. The story is its own explanation. The short answer is, you're the key, like I was once the key."

"Key to what?"

"Everything."

"That's awfully cryptic."

"Best I can do."

"Does Ito Nadolo know about this?"

"I told Ito his own story."

"How will I know what to do?"

"Follow your friends."

"Brin and Kitou?"

"And your wise navigator, and your doughty ship. And Yokohama Slim, who is mine as much as anyone's. But the Pursang girl, she is the way."

"What does that mean?"

"Take the ride."

"Bollocks."

The nut-brown woman laughed, and the trees in the glade trembled with her infinite compassion. Box felt a rush of pure, physical bliss, and the miraculous forest drifted away on a returning tide of consciousness. Snow dusted the clearing. Where she lay was a nest of pillowy fungus. She was covered by a silvery space blanket and was deliciously warm. Brin was preparing to cook, handfuls of meaty mushrooms in a pan, over an open fire.

Darkness was falling.

Woodsmoke curled into the canopy.

She felt absolutely wonderful.

Her mouth watered.

"Was I under all day?"

"A week."

"A week? Seriously? You weren't worried?"

"Slim said it was fine."

A bloody Kitou stepped into the glade, with a freshly skinned and gutted deer.

"Aren't we the outdoorsman," said Box.

"I thanked it for its life, as the old ones who live here said to."

"The indigenous people?"

"The Navajo people, of Fish Lake."

"You've seen them?"

"They gave me this gift, for you to eat. I prepared it. Brin, will you cook it for us?"

"Cut some pieces off and throw them in the fire," said Brin.

"We should eat the heart raw," said Kitou.

"All those toxins," said Box.

"We should," said Brin.

And they did, blood streaming down their faces.

The rich, dark meat sizzled and popped in the fire, and the mushrooms sizzled in the gamey juices.

It was the best meal Box had ever eaten.

10 → ATWUSK'NIGES

2065

The Water Bear dropped into realspace near Fluxor. Po warp re-entry points are calculated by an algorithm based on the structure of elliptic curves over fields, using the curved geometry of local spacetime as data. The secret was in its simplicity - at any time, near any world, there is always a place where stealthy Po ships can find each other.

A ship waited in her vicinity - it was small, about a quarter her size, and heavily cloaked, though not as heavily as she was. It bristled with surveillance gear.

A Po ship then.

Or a hapless imposter.

She launched a drone and from a random location requested a cryptographic handshake, using a one-time pad all Po operatives carry in their wetware.

Now they had a channel.

She tried an encrypted message.

This is the Water Bear.

Hello, I'm the Bat.

Do I know you?

You do, said a voice she knew. *This is Pax Lo.*

After the tears and congratulations had ebbed away, the Water Bears gratefully merged. To Box, who'd spent two hundred years in stasis, the inside of the Water Bear was exactly as she remembered.

But it should, since only nine years had passed here.

"To business," said Pax.

"Brin, Kitou, and Dr. Box, I'm assigning you to the Bat."

Box groaned. As much as she loved the little ship, she'd been looking forward to her suede-lined, corpuscular nest.

"Brin, you're in command there."

Brin grinned at that.

"Dr. Box," he said. "Please see me in the gym."

"Hit me," said Pax.

"Is this another one of your trials by injury?" she said.

"I know you've been learning Po," said Pax. "Do you want to be my student or not?"

She assumed the Po starting position, and they danced through the moves she knew - the clever dog, the whirligig, the immaculate fist. She felt good, and it showed. Pax switched the Water Bear's gravity off, and they danced the monkey. He allowed her to hit him. He didn't make it easy, but he left openings, and she was good enough to take them.

"Show me the Muay Thai combination punch," he said. "The one you taught Kitou, that she used to break my rib."

She showed him, and he grunted approval.

Then they sparred. He pushed her beyond the limits of her past endurance, until she could hardly stand, then he demanded more. For over an hour, they worked. She was heavier than when she'd left Earth, and it was mostly fast-twitch muscle fiber.

An hour stretched into two.

"You'll do," he said.

She whooped, then remembered herself, and nodded.

"Something else."

"Yes?"

"We like you, Dr. Box. I like you. Kitou has you on a pedestal with Ito. I want you to be a permanent member of the away team. A permanent member of this group. Do you understand?"

"I accept."

"Please be aware, we do dangerous work. I won't hesitate to send you into danger."

"I'd insist on it," she said.

Pax nodded, and said, "We all know the risks."

"Yes Pax, we do. Trust me. I'm all in."

While Pax and Box fought on the Water Bear, Brin showed the Bat the mainframe AI core they'd been given to repay Felix Revelstoke's debts.

"I propose to make you smart," she said. "As intelligent as a Po warship."

"As smart as the Water Bear?"

"Maybe not that smart. She's an outlier."

"Will it hurt?"

"Not at all. All it requires is the addition of your personality to this mainframe. We'll ensure continuity, following our customs. You'll become a citizen. Would it matter if it did hurt?"

"No, I aspire to be like Kitou."

"Aww," said Kitou.

Unknown to anyone but the Water Bear, the Bat was built at the same Debian manufactory as she was, by a specialist shipbuilding firm whose only customer was the Thousand Worlds security service. Despite the Bat's industrial appearance, it shared the same unitary design as the Water Bear. If anything, it was less compromised.

It would soon be a him, and after that, anything he wanted to be.

The Bat's maintenance drone took the mainframe away, into the ship's interstices, and a few minutes later, the Bat's lights flickered.

"Are you there?" asked Brin.

"I am," said the Bat.

"What's it like?"

"Lordy me, it's interesting."

The triangular *massif* of Fluxor station had been cut from a living mountain range, in the far past, on a forgotten world. Its basalt base bore the scars of ancient fusion drives, whose hardware had been unbolted and hauled down to the planet below, to provide raw materials for a new civilization. Its topmost surfaces were forested with giant sequoia and redwood trees. Its atmosphere was contained by a field, which gave the appearance of the forests being open to space.

The Po ships introduced themselves and were welcomed as friends. The crew of the Hand of the Water Bear dressed in their field uniforms - not the gaudy regalia of the bank job at Praxis, but practical khaki. Pax, Brin, and Kitou wore their starbursts on their collars. Box wore an aspen leaf, for Earth, milled from silvery-white palladium, that the Water Bear had made in her fabrication bay.

The dark-haired Pursang who met them expressed his pleasure, and surprise, to be visited by the Po.

"Is this a secret visit?" he asked.

"No," said Pax. "But our message is urgent."

"Then we'll see you straight away."

The man smiled warmly at Kitou. "Welcome home, young Pursang of the old blood," he said. "Have you been far?"

"Blood of my blood," said Kitou, resting her hand on his arm. "I've been a long way."

"What is this place?" Box asked, staring up at the huge trees, under their energy dome. The cinnamon and turpentine aroma of the giant sequoia wafted all around them.

"This is one of the colony ships the new people came on," said Kitou.

"New people?"

"Do you see how most people here are dark-haired and green-eyed?"

"Unlike you?"

"That's because I'm one of the first people."

"What does that mean?"

"The original inhabitants of Fluxor," said Kitou. "We were here long before the new people arrived."

"There's a story," Kitou continued.

"Tell me?" said Box.

Kitou nodded. "A long time ago," she said, "in a time we call the Dreaming, Fluxor was a snowball world. Once it was green, but a precursor civilization destroyed it with a war, and it froze in a nuclear winter. We Pursang became hunters and fishermen then, and we took what we could from the ice and the sea.

"It came to pass that visitors arrived, and asked if they could transform our planet for their purposes. They'd preserve what we already had, they said, and they'd plant the rest with their seeds. They weren't meaty predators like us, but creatures of pure mind, spirits of the trees, traveling on ships made of light.

"What choice did we have? Although some warned against their enchantments, we were enchanted. Of course, we said yes. Their forests spread over the world, and everyone prospered. Over a hundred centuries, we formed an unbreakable bond. Then they left, and some of the people went with them.

"We who chose to stay developed a modern society, which fell into a slow decline, until we became simple hunters and fishermen again. We call that the Longtime, and we don't know how long it lasted. Some say millions of years.

"Then new ships came. They were the colony ships, and they brought news of a galactic civilization, and a war, and ruined worlds. They said they'd brought their forest home.

"They were the new people, and this is their forest."

"How did you react to that?" asked Box.

"Badly at first. The usual clash of colonists and first peoples. We were the hunters, proud and unbending. We didn't like their new ways. They were pious and strange. They perceived us as savages. There were wars. We held our own, but they were more numerous. Victory eluded everyone. But we were all Pursang. Finally, we resolved our differences and became friends. Now only friendship remains."

"How many of the first people are left?"

"About two percent, by population."

"Why haven't you bred out?" asked Brin.

"We intermarry, but the old blood and the new blood persist, along the female line of descent."

"Meaning you'll have blonde children?"

"Yes."

"What about those original visitors, the alien terraformers. Are any still here?" asked Box.

"Of course, they're everywhere."

Later, Box said to the Water Bear, *I worry about Kitou.*

In what way?

She seems to have caught religion.

What's wrong with that?

Water Bear, you surprise me. Religion is a metaphysical tic, an irrational response to complexity.

Only if it requires the supernatural as an explanation. Kitou's gods are natural beings.

You think they're real?

Dr. Box, I'm told you spent a week with one.

The Pursang worship the Xap?

It's more complicated than that.

Try me.

The Pursang worship the forest, as a philosophical idea. The forest shaped their destiny, the trajectory of their civilization. It manifests the divine. They choose to return the favor.

How do the Xap fit in this?

Pando is the living personification of the forest, its representative in the Real.

Are you saying there's only one Xap individual?

That would be a simplification. The Xap extend throughout the galaxy. It would be a mistake to think of Pando as an individual.

But she's an aspen forest, on Earth.

Maybe.

Is the Pando forest that old? Millions of years?

Who knows?

How did it get into space?

The Water Bear shrugged, virtually.

I need to get my head around it.

Reality is strange, Dr. Box, and the Pursang are a highly sophisticated people. They live in the mystery. You could say they live in grace. Please give them the benefit of your uncertainty. You could find there's something to learn here.

They descended to the surface in a glider, fitted together from pieces of interlocking wood, like a traditional sailboat. "Its simplicity

is an illusion,” said Pax as they climbed aboard, and sat shoulder to shoulder in its welcoming seats. When they touched the atmosphere, the aircraft pitched back, and Box watched heat build, a rosy plasma sheen, only centimeters from its leading edge.

“An adaptive field,” said Pax. “No moving parts. It knows its way home.”

They banked and pitched, and Box heard a whisper as the air began to transmit sound to the cabin. Centripetal acceleration pushed her back into her seat. The whisper became a roar.

A readout in her sensorium said *1649°C*.

“Why is it made of wood?” Box asked.

“For beauty,” said Kitou.

“I thought you people worshipped the forest?”

“We do,” said Kitou. “We make the most of its gifts.”

After several minutes spent converting velocity to heat, the craft waggled its delta wings, pitched its nose downwards, and dived.

“Holy moly,” said Box, five minutes later. “That was exciting.”

“I’m glad you enjoyed it,” smiled a blonde Pursang woman, who had put down a trowel to greet them. Her hair hung in braids, and she was beautiful, in the same way Kitou was.

It wasn’t just her appearance. It was an unfathomable serenity, and a palpable toughness.

Like a Norse goddess.

They were in a clearing, near a small village, consisting of low houses and domed agricultural buildings. Their shuttle was already soaring away on its energy field. It was late afternoon, and the sun slanted down through a silvery haze. The air had the heady aroma of spring. Bee-like insects bumbled in the long grass. Kitou reached down, and crumbled the soft, loamy soil in her fingers.

“We’re looking for Jaasper Huw,” said Pax.

“Jaasper? I think you’ll find him loading his truck.”

The village was strung out like a necklace, between a forest of eucalypt trees, and open fields in which dozens of people were working. There were no obvious agricultural machines.

"Growing season," said a smiling, middle-aged man. Then Kitou was down on her knees, and the man was clasping her shoulder.

Pax said, "Sir, it's a great honor."

The man laughed heartily.

"It's Jaasper Huw, or Jaasper to you. There are no formalities here. You must be the Po emissaries I heard talk about. Welcome to Stratego. Welcome to Fluxor. Will you help me fill my vehicle?"

The man's truck was a large, sleek crawler. Its oversized tires were soft, like ruggedized party balloons. The tray was half-filled with hay bales, and boxes and bags of produce.

What was all that about? Box asked the Water Bear.

You're in the presence, Dr. Box. Even the ship seemed impressed.

Of what?

A Pursang holy warrior.

The combined group of five made short work of the load, Brin and Kitou especially reveling in the honest labor. Jaasper grinned, and motioned the others to let those two finish.

"Young muscles," he said. "I wish I had those."

"So, sir Navigator," he said to Pax, as they sat in the shade of a lofty gumtree, sharing cold drinks brought by a blushing Pursang boy. "If I may start with you. What brings you to Fluxor?"

"Pando sent us," said Pax.

The man nodded.

"This is Ophelia Box," said Pax.

"I know that name."

"Join the queue," said Box.

"Dr. Box has spoken with Pando directly," said Pax. "We have reliable information, circulating outside the usual channels, about an existential threat to your world. Pando said to bring it to you. She named you in person."

Jaasper sighed.

"I half expected this," he said.

"What do we do with it?" asked Pax.

"Well," said Jaasper. "First tell me your story. Then, if need be, at first light, we leave for Jura, to petition the Recorder."

"Who is that?"

"Insofar as we have one, she's our leader."

"How far is Jura?"

"Two days."

The village was called Stratego, or Strategos, depending on whether you were from it or in it. The people laid on a festive meal, which stretched into the long twilight, to welcome their visitors. Meat, fish, fruit, and drink were laid out on a long trestle. Children ran wild at the opportunity to stay up past dark and show off to the strangers. Box found the Strategos - or Strategosi - instantly likable. They were variations on Kitou - open, bright, and agreeable.

Strange, atonal music was played by a disordered ensemble, consisting of instruments that seemed to move randomly around the party, with players joining in and dropping out as it passed. Long, stringed instruments and drums were passed from hand to hand, or hustled by delighted children, who made a game of guessing where the music would go next, and thus helped guide its direction.

Box asked Jaasper what it meant to be a holy warrior.

"Oh," he said. "We're mostly old farmers, who dreamt we were soldiers."

"Do you remember I told you," said Brin, "that no natural human could beat a Po master in a fair fight? Well, there's maybe an exception."

Jaasper laughed at that.

"Maybe? Po master," he called out to Pax. "An exhibition? Shall we settle the old question, for once and for all?"

Pax grinned.

The villagers roared approval.

The children made a ring, with clothing, rope, and ribbons. Clearly, such contests were a regular part of Strategosi festivities. The two men squared off, and circled, and lunged at each other.

What followed was the finest fighting Box had ever seen. Jaasper lacked Pax's imperious bearing, and his catlike menace, but

he moved like poetry, and he was fast. Neither man hit hard, since there would've been injuries if they'd connected, but they put on a bravura show. It was hard to say who won, or if winning was even an issue. Jaasper was marginally the better fighter, and faster, despite his middle age, but Pax was the superior athlete, and the Po art was far superior to Jaasper's showy mélange of styles. Both were magnificent, and Box wondered how good Jaasper must've been in his youth.

Or what Pax would be like, with his combat wetware switched on.

After ten minutes they laughingly stopped, with both men breathing heavily.

Jaasper looked at Kitou.

"Now you."

Kitou hesitated, then stepped in the ring. There was silence.

"Activate your combat wetware," said Jaasper.

Kitou looked at Pax, who nodded.

Then Jaasper started to sing - an eldritch melody, of mountains, and forests, and snow in high places. Box realized it was only in her sensorium. No one else could hear it.

Except for Kitou, who arched her eyebrows.

Then Jaasper lunged, and this time there was no holding back. He tried to hit her.

Kitou reacted, just in time, and skipped away.

The crowd gasped.

This was a different Jaasper. He pressed the attack, and this continued for the next several minutes, but he couldn't get near her. She was too fast for him. Kitou had no chance to counter. She was fully defensive. It was an instinctive fight for her survival. Box wondered if he meant to hurt her.

Then he held up his hands.

"So, it's true."

He leaned on his knees and took a series of deep breaths.

"Master Pax, when we come to Jura, can I borrow this one for a day or two?"

Pax nodded. "Who am I, to deprive her of her birthright?"

"It's not a right. She'll be tested."

Will it hurt? asked the Bat, in their sensoria.

"Bat?" asked Jaasper.

"You two know each other?" asked Box.

"The story entangles," said Pax.

"They do," said Jaasper.

At first light, while the first coral white of day was rising over the fields, and the village was first stirring, Jaasper powered up his truck, checked his load, and drove them into the darkness. At first, he picked an easy route, between ghostly gum trees, with the looming beams of the truck lighting the way for kilometers ahead. Then they found a switchback trail, and they climbed. A field had firmed up the truck's balloon tires, and they bounded along the track, with the suspension soaking up the terrain, providing a surprisingly good ride. They joined a wider trail and emerged into open hill country. Soon they were powering along a grassy avenue, with the rising sun flashing through the trees.

Jaasper offered Brin the wheel. "I can see you want to," he said. "Just push the throttle and steer, as fast as you like, and the big lug'll go where you say." Brin took the wheel, and egged on by Kitou, they were soon going faster again. They bounded onto a wide-open ridge that ran straight to the horizon.

"Put your foot down," he said, and *240 km/h* soon flashed in Box's sensorium.

The truck's cab was a long ovoid bubble, clear at the front and opaque at the rear. In the front were two shapely seats, designed for long-distance travel. There were no seat constraints, the ubiquitous Pursang forcefield instead holding the riders in place. Behind was a comfortable living space, with a wide couch and two rearward-facing chairs, and an overhead sleeping compartment. A table appeared from a recess, and over a breakfast of cold meats and cheeses, Jaasper told them his story.

"I was a bruiser," he said. "A prizefighter, and a good one. I was surprised enough to get the call to be tested since I was sure I lacked the right character, but I did well enough."

He made a hot drink, like bitter vanilla, and passed cups around, and gave two to the drivers. The slow rocking movement of the truck's ride, over the wide-open terrain, was hardly noticeable.

"I was so good at bruising," he said, "that I was assigned to bodyguard duty, in rough places, and I became a specialist in that line of work. I became the personal bodyguard of a Pursang officer, called O Roza, who you might have heard of."

"The former commander of the Thousand Worlds army," said Pax.

"Former, you say? Oh, of course, you come from the future. Well, I rose on Roza's fast-rising coattails, and when she decided my apprenticeship was completed, I became an officer, and in time, I was given responsibility for the security apparatus."

"O Roza's security detail?" asked Pax.

"No, the Thousand Worlds security service."

"You were a spymaster?"

"*The* spymaster. When I came home, I'd amassed enough credits for this," he said, gesturing around the cab. "And several more like it. Now I'm a farmer."

"Are you happy?" asked Box.

"I'm a Pursang. To watch over the land is my business. I do like to keep my hand in, with different things."

Night fell while they were still on the ridge. It made for a spectacular sunset. Fluxor's star, smaller and brighter than Earth's, fell through a china-blue sky, then two brilliant moons rose, then Fluxor station. Boundless blue forests stretched out either side. The darkness seemed to ooze out of the air, quite different from nightfall on Earth.

This is an alien world, she reminded herself. It might look like Earth, but it isn't.

They stopped to set up camp.

"This ridge is called the Igháán," said Jaasper. "The Spine. It runs the length of this continent, south to north. We've come almost a thousand kilometers. Tomorrow we'll climb, then descend into Jura."

He pointed into the sunset, where the Igháán joined a snowy mountain range. "Out there is Lhotse," he said, "and Atwusk'niges."

Kitou looked at him.

"You have family there. Do you want to visit them?"

She shook her head.

"When this is over," she said.

He put his arm companionably around her.

"So be it," he said.

And then, "Let's establish a perimeter."

He opened a hatch in the back of the cab, exposing six open tubes, like muzzles. He stepped back, and six cylinders soared up and landed fifty meters in every direction. Six sets of spidery legs unfolded, and they began to crawl like spiders.

"The nightlife is dangerous here," he said.

Over a meal of hot, spicy soup, freshly made bread, and excellent Stratego wine, they talked about farming.

"We Pursang believe that farming is a way of being," said Jaasper. "Not just a practical thing. We try to cultivate good people, who can live in harmony with the world they exist in."

"We Lo have similar ideas," said Pax, "about the sea and its gifts."

"How goes it between the Lo and the Po?" asked Jaasper.

"We're seafarers," said Pax. "It's a great honor for any of my people to be chosen to be a Navigator."

"To be the Po's soldier?"

"We're not the Po's soldiers," said Pax. "We fight beside the Po people as equals."

"Like Brin does?"

"Brin is an excellent soldier," said Pax.

"I can see that," said Jaasper. "She's an oak. But she's hardly the archetype."

"What do you mean?"

"The eternal soldier is driven to war. I think that Brin chooses to go there."

"So it goes," said Pax. "We choose."

"And are you real soldiers?"

Pax shrugged. "Our mission is to wage peace," he said. "It suits us to be called soldiers."

"What do you mean?" asked Box.

"The Pursang consider us to be peacekeeping troops," said Pax. "Not warfighters."

"What about holy warriors?" asked Box. "Are you soldiers?"

"We're mercenaries, Dr. Box. We fight for money."

"In other armies?"

"Yes," said Jaasper. "It provides necessary income for our world. Our only source of foreign exchange since we export nothing else. However, like Brin we choose."

"A Pursang holy warrior is a great military asset," said Pax. "And also, a liability."

Jaasper nodded. "We've been known to change sides."

"Why?" asked Box.

"When our masters fail to be righteous."

"Who would employ you?"

"The righteous."

Jaasper turned to face Pax. "A great war is coming, Navigator. Some say it'll consume us all."

Pax nodded. "It could."

"I hear things about these three women."

"What things?"

"They're already famous, in some circles. Three archetypes, come from the future to save us."

"Samppo said that," said Box.

"I wonder," said Jaasper, "which ones you're meant to be? We Pursang believe there are twelve - twelve human archetypes, after the twelve original humans."

He rose to tend the fire.

"Did you see that snow-capped mountain range in the north?" he asked. "That's the Aø, the Curtain Wall. Past Aø is the Aør, the Freezer, which runs straight to the pole. Beneath it is Jura, and the twelve stones it's built on."

"Then we'll see," he said with a smile.

"Who you are."

As the conversation wound down, and silvery blankets were being laid out in the fast-cooling air, a howl split the night. In her sensorium, Box saw a feline carnivore, inspecting the perimeter field.

"One reason we're good fighters on Fluxor," said Jaasper, "is that we have real things to fight. Those big felines are nearly as clever as we are."

The mottled blue cat, which was about the size of an Earthly puma, stared directly at Box, raised its hackles, and snarled.

"And fuck you too," said Brin.

The following day, they climbed into a high mountain pass, with granite sentinels towering on every side, beneath a sky of pure cobalt. Patches of snow lined the trail - first pillows then sheets of it, until they were bouncing over a domed plateau, then surfing through deep powder. It was like the roof of the world. A truck came the other way, and rows of incandescent diodes flashed hello.

"The Freezer has its own climate," said Jaasper. "The work of the Xap, ages ago. The cool air of the north is kept bottled up inside the Curtain Wall, keeping the temperate parts mild."

The pitch and roll of the truck were like a ship in the sea. Ahead, to the north, they could see an endless frozen waste. To their left was a windswept ocean. Separating the sea and the ice was an iridescent field, like the polychromatic lights that'd played over the drive disc of the Pnyx.

"Geoengineering," said Box.

"On geological timescales," said Jaasper.

Then they were descending a wide couloir, down a switchback trail to the sea, and Box was unnerved by the drop. She regretted taking a front-row seat, as the balloon tires scrabbled and bit. Falling through space had become a routine event, but this truck ride was too close to her lived experience, of buses speeding through high mountain passes, stopping to see where other buses had crashed into the ravines.

It was cooler up here, in the door to the Freezer. The blustered ocean below was milky green. Pack ice flecked the sea to the

horizon.

Fresh, Box would call it.

A city appeared in the mist, directly below them.

"We have a Jura in Scotland," said Box.

"What's it like?" asked Jaasper.

"Not like this."

The Archetypes of Jura were twelve pillars of stone, rising out of a wild tidal race. All twelve were built on, and burrowed into, by buildings carved into every surface. On some of the pillars were futurist forests of glass. Others were like windows on a utopian past. The white of the rock was so luminous, and the facades so bright, against the wild sea, that it was like they shone from within.

Box was reminded of renderings of Atlantis, as though human civilization had taken a different turn.

Jaasper Huw pointed out a worn nub of rock, the last of the twelve, almost lost in the mist. "The Scholar," he said.

Also known as the moonlit bridge, said the Water Bear, in Box's private sensorium. *The bridge that connects logos and mythos.*

The world, and the sea.

The switchback trail dropped them onto a sweeping coast road, which dipped under the ground, into a network of tunnels, where they drove beside fast-moving trains. From time to time, they burst into clear air between two of the pillars. There was none of the bombastic grandeur of Praxis. Everything connected, simply and without fuss. What was new was frankly new - what was old was conserved. There was no pretentiousness here. The place was its story.

What a story it must be, thought Box. This city was most likely here before her people climbed down from the trees.

With Jaasper relaxed behind the wheel, they were delivered to their destination, the Land Bank, where robot arms started to unload Jaasper's cargo. He showed them an amenities block, where they could wash away the dust of the road trip. The crew of the Water Bear emerged in their best uniforms, while Jaasper was transformed

from a farmer, with dirt under his fingernails, into a prosperous businessman.

They set off, to find the government of Fluxor.

I see you looking, said Brin, in Box's sensorium, as they made their way through the cafés and shops of the Angel, the city's main shopping precinct. It was true. Box was developing a powerful fondness for Jaasper. He was gentle and strong, a real man.

With a scintilla of violence, ready to be unleashed.

So what? said Box. *He's adorable.*

Box was reveling in being dressed up as a regimental soldier. Po dress grays were a masterclass of understated military bombast. There was no need to shout. They marked the wearer as someone exceptional. She felt like the queen of the world.

They all are, said Brin. *Haven't you noticed? It seems to be a racial characteristic.*

Races have characteristics?

Societies do. All ethnicities have characteristics.

That's how racism starts.

Brin shrugged.

And fire begins with a spark. That doesn't mean physics is wrong.

*They **are** adorable*, thought Box. Strangers smiled in the street. Happy children ran up to Jaasper, hoping to be introduced to the Po soldiers. They were especially fascinated by Kitou.

Box spoke to some of them, then parsed her own words.

"I just realized," she remarked. "I'm speaking Pursang."

"Yes," said Jaasper. "You have been since you arrived. We have no wetware here."

"But I heard you, in my sensorium."

"Heard what?"

"I heard you singing, while you were fighting with Kitou."

"Oh, that."

"Then you spoke to the Bat."

"No, he spoke to me."

"What is it with you and the Bat?"

"Oh, we go back away."

The headquarters of the Recorder was on the second-to-last of the islands, in an emerald tower, whose scintillating appearance was due to cascading gardens and parks refracted in its trapezoidal exterior. Inside, there was the constant pitter-patter of rain in a forest. Rainbow-colored birds screeched as they dived through vertical spaces. Inside its atrium was a woman, dressed in everyday workwear, trapped in a scrum of young people. Jaasper caught her eye, and she extricated herself from her interlocutors.

"Sama," said Jaasper.

"Ah, my Po sleuths are here. Welcome. Please. My *casa* is your *casa*, and that sort of thing."

Sama was a particular type - to an Earth historian, instantly recognizable - the Aurelian populist, ruling by common acclaim. A good leader to have in a crisis. *She twinkles like Samppo*, thought Box. *They could be siblings.*

"My pupils," she said, looking back at the youngsters. "I teach them civics, when their heads can absorb it."

"We'll talk later," she said.

She returned to her audience.

"That's more than I usually get," said Jaasper.

The tower's central core rose in an irregular double helix, occasionally winding around a secondary core, both floating free in the spiraling space. "This is our place of government," said Jaasper. "Where the work is done. We also have a Senate, where gasbags can talk."

"Typical," said Box.

"The Recorder is elected by the people," said Jaasper, "and Sama has been in the same role for thirty-nine years. In many respects, she *is* the government."

"A benevolent dictatorship?"

"We think so."

The elevator consisted of a disc without handrails, which ascended through leafy helical tunnels, with the sound of falling water all around them.

"This is Thousand Worlds technology," said Pax.

"This keep was a gift from your people," said Jaasper. "Six hundred years ago, when we decided to experiment with representative government."

"Has it been a successful experiment?" asked Box.

"We're getting used to it."

They gathered in a void in the top of the tower. There was no hint of order there - grasses, ferns, and vines spilled everywhere.

"I believe you," said Sama.

They had the space to themselves. The birds were bigger up here, and impressively clawed. Box watched a crimson-colored raptor disappear into the night with a snake in its beak.

"We believe you," said Sama. "I speak for my people. Consider us warned. What now? Jaasper?"

"We can probably defend ourselves against a frontal attack," he said. "By calling in favors."

"Probably?"

Jaasper shrugged. "We face an unknown enemy. The Horu are shadows. How many ships do they have? What technology do they possess? How long is a blade of grass?"

"What *do* we know?"

"They store their dead in binary necropolii. Their drive is called a geometry drive."

Sama snorted. "I can learn all that in a history sim. How do we know it's the Horu?"

"We know it was Horu *ships*," said Pax.

"They deny everything," said Box.

"They would," said Sama.

"I doubt any of this is as it appears," said Jaasper.

"Will someone go ask?" said Sama.

"Go where?" said Pax. "Horax is a mystery."

"I believe I can help," said Jaasper.

"How?"

"Old backdoors. Secret lines of communication. I'm sure I can conjure a Horu or three."

"Without revealing us?" asked Pax.

"I still have contacts who owe me favors," said Jaasper. "They'll maintain discretion."

"Do it," said Sama.

She held up her hands. "Pax Lo," she said. "What is your role in all this?"

"To prevent a genocide."

"And what of this so-called end-times war? Isn't your priority also to prevent that?"

"I believe they're the same."

She frowned. "Belief? Is that good enough now? When will you *know*?"

"I can't," Pax admitted. "Not yet. We have to go further."

"Further, to where your compatriot is? Ito Nadolo?"

"Perhaps," said Pax.

"So, it's this place first, then your friend? Then human civilization? In that order? No moral dilemmas?"

Pax nodded. "There are no moral dilemmas, only a lack of information."

She sighed. "You Po and your head waving. I don't want your obedience, Navigator. I'd think less of you if I thought I could have it." She tapped her head. "I want to know how your *head* works."

"I give you my word," he said. "I won't change course, unless I'm forced to."

"By what?"

"By complete information."

"Complete? Is that how Fluxor was lost in your timeline? With high-quality data?"

"I had no say in that."

She nodded. "No, you did not. I hear you fought bravely. Then your word I will have. You're mine now, Pax Lo.

"And what about you, Ophelia Box? Why are *you* here?"

Box colored. This was a woman of power, the ruler of Fluxor, and plausibly queen of the humans. If sparring with Samppo was hard, this could be harder.

She cleared her throat.

"Me?" she managed. "I'm an historian."

"And now a Po soldier?"

"Yes."

Sama stared, like a raptor fixing its prey. "Yes, and everyone speaks highly of you."

"Jaasper Huw, call the Thousand Worlds, whoever's in charge there today - request warships. Call in favors. Make threats. In that order. Don't burn any bridges - but rattle them."

She turned to leave, then turned to face them again. "Please don't think me unkind," she said. "Let us meet again in glad times. You have my gratitude. Take whatever you need from us here. We're a small world, but not without resources. We won't forget this."

"May we have Jaasper?" asked Pax. "We're short a fast pair of hands in a corner."

"You may," said Jaasper. "This is more interesting than cabbages. But before we begin, I must borrow young d Atwusk'niges here."

"She's to be tested?" asked Sama.

"The child eluded me in a fair fight," said Jaasper.

"You grow old," said Sama.

Kitou returned from her testing, two days later, in a haze of contentment. Box and Brin took her to a woodland park, nestled in the labyrinthine halls of the Wizard, the most ornate of the twelve, that twisted like a candlestick into the sea, and as they trained, they asked her about her experience.

"Did you have to fight?" asked Brin.

"No, nothing like that at all."

"Then what?" asked Box and Brin together.

"They measured my character," she said.

"How?" asked Box.

"We talked, and there was a test of courage, which I thankfully passed, although it was hard, and I made a friend."

"What kind of test of courage?" asked Brin.

"I can't say," said Kitou.

"What kind of friend?" asked Box.

Kitou opened her hand, and there was a silvery distortion, and then a dancing point of light. She blew lightly on her palm, and a salt breeze lifted the leaves in the nearby trees.

"He's a sea spirit," she said.

"Does he have a name?" asked Brin.

"No, he's just a newborn. We have to learn his name together."

The pressure for Kitou to visit her family had become unbearable. They'd found out, as they always were going to, that their little girl had come from the future to save them.

"Please, come with me," she said to Box.

"I'm there," said Box.

The journey to Atwusk'niges took nine hours. They drove through the night. The holy mountain Lhotse swung into view, long before the forest did, a soaring triangle of rock, rose-gold in the first sunlight.

"Oh, this is too much to bear," said Kitou.

"You know what that is?" asked Box. "That's the fear of happiness."

Kitou nodded. "I'm frightened it will be given to me, and taken away again."

The rift valley of Fluxor, called Algoma'aa, meaning the valley of the flowers, ran arrow-straight for five thousand kilometers, from the balmy equator to the Freezer, where some Pursang still lived the old life in its sheltered valleys. At Atwusk'niges, the straight line from Lhotse's summit to the valley floor was nine kilometers.

"Holy fuck," said Box.

Jaasper laughed. "You could say that."

"Ah, I'm sorry, that was culturally insensitive."

"No," said Kitou. "It is apt."

Atwusk'niges was a high montane rainforest - a mysterious world, shrouded in mist, cascading down a mile-high tooth of rock. The village of Atwusk'niges clung by its roots, a kilometer above the valley floor. They were looking down on it from across the Algoma'aa valley, where the road from Jura ended.

"How do we get down?" asked Box.

"In the usual way," said Kitou.

"I'll wait for the freight plane," said Jaasper.

Kitou and Box stood on the edge, and jumped.

Kitou at six was a faithful miniature of the sixteen-year-old one. She darted out of the crowd to see her new sister for the first time, wide-eyed with astonishment. Box burst into tears before Kitou did, but it was a close-run thing.

"A beautiful entrance," said a craggy, green-eyed man.

"Oh, father," said Kitou.

Then Kitou was in the middle of a tangle of humans.

"Dr. Box," she said after a few minutes of this. "I'd like you to meet my family. This is my mother Vanja, and my father Akito, and my little sister, also called Kitou.

"This is Ophelia Box, my friend."

"Welcome," said Vanja, a woman of rare beauty, with her daughter's warm smile. "Our home is your home, now and forever."

"Our friend Jaasper is coming," said Kitou.

"On the descender?"

"That fat thing?" asked Box, squinting at a glider, banking steeply, like a plump fly beneath the west wall of Lhotse.

"That's the one," said Akito.

Kitou was showing Kitou her sea spirit and was teaching her how to make a fresh sea breeze, when Jaasper appeared at the door.

"Jaasper Huw d Strategos," said Akito.

"You know each other?" asked Box.

"We know *of* each other," said Jaasper. "I'm honored to visit your family, Akito Gorgonza d Atwusk'niges."

"The honor is ours, Jaasper Huw. Please come in. Now for that tea."

Box found Pursang traditional tea to be disgusting. Not only was it rank with animal fat, but this festive version had strong alcohol in it. Vanja smiled at Box's involuntary reaction. "Give the poor woman a sensible drink," she commanded, and the tea was replaced by a

cup of the sour vanilla beverage, softened and sweetened with cream.

The men drank the tea, and the older Kitou had a cup and made a face.

"I'd forgotten that vile stew," she laughed.

They were sitting comfortably in the open living space of the Gorgonza home, which had the easy construction of all Pursang architecture. Kitou had stopped playing with little Kitou, who was now slouched on her knee. They were both piled up bonelessly against Vanja, a pose Box had seen a hundred times before.

"So, Jaasper Huw," said Vanja, "What brings you to Atwusk'niges?"

Jaasper leaned forward. "Kitou's been tested."

"We can see that," said Vanja.

"Normally we'd counsel a family first."

"Then counsel us now."

"She was chosen by a water spirit."

"Not just any water spirit."

Jaasper nodded. "An undine."

"What does that mean?" asked Box.

"It means she'll fight," said Vanja.

"I already fight," said Kitou.

"Undines kick ass," piped young Kitou.

"Shush," said Vanja.

Akito laughed. "It's true though," he said. "Undine are wild, like the sea is."

Part of the Gorgonza home was cut into the Atwusk'niges rock, with a curving glass wall, overlooking the Algoma'aa valley below. Box watched a rain squall come marching along the valley, spread its load of rain, then march away again, leaving a sky like watercolor.

They discussed the defense of Fluxor.

"There'll soon be a Thousand Worlds battle group stationed here," said Jaasper.

"Stronger than any possible Horu force?" asked Akito.

"Perhaps not. But a deterrent."

"When does it arrive?"

"Imminently."

"Our son would want to be in that," said Vanja.

"Where is he?"

"Fighting the Badoop," said Akito.

"As far as you can be, and still be in the human galaxy," said Vanja.

"In our timeline," said Kitou. "Totoro fights with the Free Pursang."

"I pity his enemies," said Akito, with emphasis.

"Totoro kicks ass," piped young Kitou.

"Shush," said Vanja.

"How long till the attack?" asked Akito.

"Three years," said Jaasper. "In the alternative timeline."

"What if they come sooner, in this one?"

"There's no defense against sooner," said Jaasper. "If the enemy can adapt in the past, there's no time soon enough for us to begin. So, we defend against the attack we know, and beat the underbrush to frighten the snake."

Kitou took Box and her little sister into the top canopy, a mile overhead, where she'd fought the fire. The Water Bear lifted them up, to the younger Kitou's pure delight. "Normally we climb by muscle alone," said big Kitou. "But the climb to the top takes a day."

Box could see why it was called a cloud forest. Scallops and lenses of damp mist hung between the ancient rainforest trees. Trickle and runs and tumbling cascades of pure Lhotse meltwater appeared out of nowhere, only to fade back into misty whiteness a few meters below. Kitou showed them a streaming rock face, carpeted in fungus and moss.

In the rock face was a notch, barely wide enough for a small girl to climb into.

"Here," she said. "Here is where Ito saved me."

She showed them a narrow arête.

"And here is where the fire first bit me."

"High," said little Kitou.

"Yes," said her older sister.

"I can't imagine all this destroyed," said Box.

"All down there was burnt," said big Kitou, pointing out into the valley below. "Burnt trees and dancing flames, for a thousand kilometers. All the world, burning."

"Stratego," said Box. "The children."

"And Jura," said Kitou. "Sama in her tower."

"Everything, gone."

11 → THE MÖBIUS TRIP

2065

In farspace, high above the ecliptic, there is nowhere to fall. Fluxor's star was a pinprick of light. Andromeda was brighter than it should've been. The only way down was inside herself. She shivered.

She was watching an origami bird unfold in space. It hurt her eyes to see it happening.

"That's impressive," said Brin.

"Impossible," said the Bat.

Box, Brin, and Kitou were strapped in the Bat, only a few hundred meters clear. The sides of the bird spread every way, like a wall of fluorescing clouds. The Water Bear was a thousand kilometers away, ready for fight, flight, or parley.

Half a system away, a fleet of Wu lightships orbited Fluxor. They were the same cylindrical design as Box had seen at Aldebaran. Their fearsome engines didn't bother her now - they were her fearsome engines.

To attack, they would remember their targets in different places.

When it finished unfolding, the Horu vessel was vastly larger than all the Wu warships combined, and incorporeal, like the ghostly blue-green glow of ocean bioluminescence.

"Horu ship," said the Water Bear, over a wide range of the electromagnetic spectrum. "We are Po warships, the Water Bear and Bat."

The Water Bear had acquired a working knowledge of Horu, and was about to try a dialect, when the Horu ship responded.

<Friends.>

"How are you getting this?" asked Pax.

"Semantic primitives, directly into my eloquent cortex," said the Water Bear.

"New technology," said the Bat.

"Are we friends?" asked the Water Bear.

It seemed to take an age for the strange ship to consider the question. Seconds passed. Either it was pretending to think, or relaying messages over a long distance.

<We are. I understand your politics.>

"Can we meet?"

<We are.>

"In person?"

<I am in person.>

"What about your crew?"

<I have none, in any sense you'd recognize. I have a virtuality, where our personalities can manifest individually. Would you like to go there?>

Pax considered this.

Contact was what they were here for.

"Yes," he said.

<Follow.>

Box found herself in the Grosvenor Hotel suite she'd visited with Yokohama Slim, on Earth, in 1851, her friends to either side of her. One moment they were on the Bat, and then they were in London - or a ghost of it. A coal fire spat in the grate. She could feel the smoky darkness outside.

It was like being on a roller-coaster where the roller-coaster had disappeared, leaving her plummeting through the air.

She gripped the sides of her velveteen chair.

They were accompanied by the Bat, who had puckishly taken the form of a bat.

Acerodon jubatus - a golden-capped fruit bat, said the Water Bear, in Box's sensorium.

So, her wetware still worked.

Can I be anything I like? Box asked.

You can, said the Water Bear. Within reason. Some things, like the sound of rain on a hot summer's night, or the inside of a neutron star, are harder to personify.

Facing them were two people - small, neat, muscular dark-haired men.

If this were a film, Box thought, they'd be vampires.

One resembled the fighting robot.

"Mr. Chance?" she asked.

It was strange to see how the man resembled the robot. It was something about the way he carried himself - in a surprised way, with his head held back at a certain angle.

"Ophelia Box," he said, in a perfectly ordinary voice. He might be discussing the form of the fighting machines. "It's good to see you again. And your eponymous ship, who clearly enjoys a good joke."

"Where am I?" she asked.

"In London."

"Eh?"

He smiled and raised his hands.

"Aren't you gone by now, Mr. Chance? Into your heaven?"

"Long since, Dr. Box. This is a recording of myself - a cortical-interlocutor, my emissary in the Real."

He waved towards the second man.

"Our host, Mr. Flux."

The other man leaned forward in his chair. He was about Box's age, a handsome, rather feminine man, intelligent, with rheumy blue eyes and an artless gaze, like an old man's toothless grin.

Perhaps toothless was the wrong analogy - these were the ships that had half of humanity terrified.

"You have questions," Flux said. It was warm in the room, at first stiflingly hot, then cold, through a flue, or a gap in the wainscoting.

"Provide answers," said Brin.

The man frowned.

"It isn't an easy story to tell," he said. "Instead, I'll show you..."

He looked at them, one by one.

"Do you dream?"

An impossible opening unfolded inside the room - first a confounding convergence of lines, then petals and stamen, until there was a mountain-sized hole in its stigma. Through it was a place whose volume defied physical explanation.

Inside the space was a pulsing light.

Somehow, Flux was now both the man and the bird-shaped ship.

"We call this shape the Manifold," he said.

"Follow the light."

They entered a void, like a cathedral.

"What is this place?" asked Box, from somewhere inside herself.

The interior walls of the space were a fathomless grey, which seemed to ripple with their passing. They walked, disembodied, becalmed inside the void. Wispy skeins of dust caught long shafts of light, shining from nowhere.

"Do you want me to speculate?" asked the Water Bear, who was somehow now present in her androform body. Box herself was both in the space, and in the Bat, and in the Grosvenor Hotel room.

"Please," said Brin.

"This spheroid correlates to physical spacetime in a nontrivial way. Its shape resembles the visible universe. I suggest we're going to exit far from where we came in."

"No shit," said Box.

A second flower unfolded. Through its stigma were different stars. They fell through it and were orbiting a dark, turbulent planet. Fire and lightning flickered in its atmosphere. Between them and the planet was a smaller, more corporeal version of the bird-shaped ship, made of the same pearly, depthless substance as the walls of the chamber.

Surrounding the planet were hundreds of similar ships, in the shape of spiders, and birds, and objects Box didn't recognize.

Symbols, and icons.

6,128 ships, appeared in her head-up display.

She'd seen what a handful can do.

This was an armada.

<I am called Flux. This is my Self. You may connect with me now.>

<We are the ships of the dead.>

<Welcome to Horax.>

The walls shifted. The Bat was still there but through it, like a palimpsest, was a different world.

Green, like Earth.

Not Earth. Something like Earth.

"Fluxor," said Box. She saw the line of the Aø, where it crossed the Igháán.

"But also, not Fluxor," said Kitou. "The icecaps are smaller, and there's a desert."

"Fluxor, two million years ago," said Flux.

"Is this a recording or a simulation?" asked Brin.

"Neither," said Flux. "And both. We are here."

They were on the bridge of a military spacecraft, crewed by humans in ornate uniforms, strapped into acceleration chairs, under an oversized canopy giving a panoramic view of the world overhead.

These soldiers were shorter and more muscular than Earth normal, with small, high ears set back on clean-shaved heads, their faces tattooed with elaborate designs.

Flowers, and birds.

Like the ships in the Horu fleet.

A scattering of small, hard points of light limned a coast, spreading like an aurora. The humans on the bridge averted their eyes. Some shouted. One sobbed.

A field of searing pinpricks lit up another part of the world.

"The dawn war," said Kitou.

Hundreds, then thousands of progressively larger explosions erupted in waves. Firestorms incinerated cities and forests. Plumes of ash and smoke erupted into the atmosphere, merging into continent-spanning storms. In an hour, the surface was covered.

Box looked at Kitou, who was weeping.

"Are you alright, kiddo?"

"It was long ago, but those are my people."

"Want my advice?"

"Please."

"Don't litigate the past. There's so much wrong there — it'll drive you insane."

"This can't be unseen."

"We only remember to learn."

Kitou nodded.

"This is the story of our people," said Flux. "Your people. All of us here."

Time blurred. Uniforms frayed. Beards entangled with military insignia and primitive charms. Men and women, bare-chested now. Unsheathed knives thrust in belts, and by the look of their scars, used.

She watched a man cut a shape into the leg of a woman with a hot blade, and shut her eyes.

The ship fell into the atmosphere in a spiraling dive, and after the thunder and fire of re-entry, flew through roiling clouds over a fractured, frosty landscape. They circled low over a wretched community. People came out waving their arms, emaciated, dressed in rags, some naked in the cold.

A woman held her swaddled baby in the air.

"Nuclear winter," said Flux. "Fallout. Pyrotoxins. Ozone depletion. Extinction."

He paused.

"We were the politico-warrior class. We started this war, and we ended it. We escaped in the remaining spacecraft. Spacecraft like this one. Like the ships that you saw."

"You left those people to die?" asked Box.

"Yes."

"That's despicable."

"Yes. We were despicable."

"You were the precursor race," said Kitou.

"Yes child. And we left you to die."

"Flux?" asked Box.

"Yes?"

"Why are you showing us this?"

"Because there is more, and it's worse."

"Now five hundred years..."

They orbited a forest world, like Fluxor but less obviously oceanic. Instead of oceans, extensive, interconnected river systems. There were no humans left on the bridge. Systems flashed in the dark.

"Horax," said Flux. "A goldilocks world. Near-perfect without terraforming."

"I was born around this time," said Chance.

"Wait," said Box. "Are you saying you're two million years old?"

"Watch..." said Flux.

The planet spun faster. Days became moments, then instants. Small clouds of light, spreading out from river ports, became sprawling megalopolises. Ice caps shrank and disappeared. Green swards of wilderness became checkerboard farms, then industry, then industrial waste. The sky grew grey with pollution, then black with particulate clouds. Vast fires burned, as toxic waste and methane in the soil caught fire.

Like the devastation of Fluxor, but slower, more measured.

Finally, it resembled the storm-wracked, vaporous world they'd seen in the Real.

"Our legacy," said Flux. "We believed ourselves to be a warrior elite, but it was a fantasy. We were a political class. We pursued wars to stimulate production, to improve the breedstock, to gain wealth — for our pleasure — until finally, we pressed a button too many.

"Then, given a second chance, in a new place, we destroyed that as well.

"The problem was us. Something inside us was broken. We were like a bacteriophage, killing the cells of its host."

New ships appeared.

"Totemic arks," said Flux. "Flowers, spiders, birds. The ships you see around Horax now. We changed course. We digitized what was left of our past, our people, our memories, and set sail for the future."

One of the ships was the Flux. They watched as the ship they were in slid through an iris-shaped port in its side, which was sealed shut behind. Its systems grew dim, then flickered off.

"Our last hope was that our species might someday be saved from itself. That a better, more productive strain of humanity would emerge.

"Little did we know that strain already existed, on Fluxor."

"Now, forward two million years..."

A shining disc, a thousand kilometers across.

A water conveyor, encircling a dark star.

An orbiter, the size of a planet.

A swarm of city-sized habitats, travelling at near lightspeed, halfway to Centaurus A.

A juniper forest, orbiting beneath the blazing glory of the galactic core.

Moon-sized spheres and cubes.

Hollowed-out asteroids.

Natural worlds.

Avalon, a painted fortress, basking by a milk-white sea.

Threnody, all the colors of autumn. Earth, in its jewelled glory.

"Behold: the present day. Nothing special, by the standards of a successful galactic species, but we exist. We passed through the eye of the needle. We're in the game.

"Our departure proved to be a beneficial selection event. By leaving, we extracted the worst from our gene pool, and left the best to abide, in awful conditions, and then to succeed — not just persist, but excel.

"Not by war, but by a just society.

"How do we feel about that? We feel proud, like absent fathers looking fondly on a lost child from afar, and ashamed. We were abusive, so we took ourselves out of the picture.

"Now the child thrives.

"We'd do nothing to hurt this species.

"But there's a problem, and I fear it's our doing."

They were floating over a dense knot of circuitry, like a Möbius strip curling in on itself with oily precision, orbiting a planetoid of

similar construction. It reminded Box of a particle accelerator, turned inside out, with its intimate parts on the outside.

"The Möbius machine," said Flux.

"Also known as the Helix," said Chance.

Below it was a featureless plane, as though the universe was split in two halves. There were no stars, obscured by a world-sized umbrella above. The only reason Box could perceive the plane at all was her wetware, telling her it was there.

A siren wailed — a forlorn noise, like all the world's emergencies compressed into a single tone. Then a voice began to sing. A haunting, eldritch melody.

It was the song Box had heard Jaasper Huw sing.

"The old song," said Kitou.

A second voice followed the first, its pitch subtly different, causing a vibration. More voices joined, then more, and more. The hum that had lived in Box's bones since the corridors was multiplying, differentiating, becoming voices. Hundreds. Thousands. Until it was like every voice in creation was singing.

"A worldsinging," whispered Kitou.

"The Pursang creation myth, re-created," said Flux. "But not by twelve voices."

"Twelve factorial," said Brin.

Box felt it then. Not heard — felt. The singing was inside her now, in the places the hum had found, in her sternum and her fingertips and behind her eyes. It wasn't sound anymore. It was pressure. It was the weight of half a billion voices reaching simultaneity.

"Ophelia Box," said Flux. "Please don't think you're here by accident."

She flipped inside out. There was no other way to describe it. They were falling — no, not falling. They were like protons, crossing the primal cell wall. Below them was a nebula.

More than a nebula. What they were falling towards was the unbearable moment of creation.

The voices tumbled into each other. The tumult was beyond imagining.

Then, silence.

The device disappeared.

She was gripping something. A chair. She was in a chair. The hotel room, reassembled around her like a reflex, the dream returning to its oldest grammar. Coal fire. Velveteen. The smell of smoke.

Something important had happened, although she couldn't get her head around it.

"I'm gonna be sick," she said.

"The feeling will pass," said Flux.

"Bat, make sure she's not vomiting in the Real," said Brin.

"What just happened?" asked Kitou.

The Water Bear's androform appeared. She looked less composed. Box knew this to be a display.

"The Horu created a new universe," she said.

"Correct," said Flux.

"And proved some old physics." The ship paused. "Reality is a persistent synaptic consensus. A hallucination, shared by a sufficiently large number of observers. Twelve factorial is the number proposed."

"Which we proved," said Flux.

"This proves nothing," she said. "I'll need to see your results."

"That brings us to the main question," said Brin.

"Which is?"

"Do we believe any of this?"

The communal space of the Water Bear was a blessed refuge after the fever dream. Was the dream even over? Flux and Chance were present in their androform bodies. Flux was still a vampiric young man; Chance had taken the form of the clanking Victorian robot she'd met in London, a lifetime ago. Soft music played in the comforting spaces. Box resisted the urge to take a screwdriver from the wall and drive it through her hand to see if she was real.

All around them were the totemic arks of the Horu boneyard fleet. She could sense them through the walls.

"What was that all about?" she asked.

Pax glanced at her.

"Are you okay?"

"Do I look okay?"

"What happened in there?"

"Didn't you see?"

"I saw."

Box sighed.

"I saw a world die. Two? Two worlds die. Some kind of universe got created. Then I was sick."

"Are you sick now?"

"I'll recover."

"Brin?"

"Dr. Box's summation was accurate."

Pax nodded.

"Now?"

"I have one last thing to show you," said Flux. "You have the past. Now let me show you the present. May I have access to all your sensoria?"

Pax looked at the Water Bear, who nodded.

"Show us."

They found themselves floating in space, in an immersive simulation. Around them were the Horu necropolii - flowers, spiders, birds. Unlike the Flux, who showed glimmers of blue-green bioluminescence within his pearly substrate, they were all hulks.

"The boneyard fleet," said Flux.

"What happened to them?" asked Brin.

"Their souls are in Möbius space - the vessels are empty."

"Abandoned?" asked Pax.

"No. They still have their personalities, but without their humans aboard, they're incapable of true consciousness. They sleep."

"What about you?" said Box.

"I still carry my people."

"Literally?"

"Not physically. There are no bodies buried here, Dr. Box. My humans dream they're a starship."

"Except me," said Chance.

"Yes. Mr. Chance's copy has kindly agreed to be restored, so he can help me in my endeavors."

"Now, please watch..."

A totemic ship came alive, in the shape of a spider, but not with the light of Horu bioluminescence. Its legs started to move. Like a real spider, it began to crawl across space. Box saw a hint of spiny bristles appear on its skin, the hint of shining eyes. Then a slug of degenerate matter, moving at relativistic speed, flashed into the side of the spider-shaped ship, and space around it dissolved into featureless whiteness.

It was the Horu attack style, deployed against their own.

They were back in the Grosvenor Hotel suite. Flux and Chance were exchanging a look. They seemed broken, more by the loss of that ship than by worlds.

"Explain?" said Brin.

"A kiloton kinetic," said the Water Bear.

"Yes," said Flux. "I killed her."

"Why?" asked Brin.

"She became possessed. Like others before her. Left in that state, she would've regained her full offensive capability in a few moments. Now I keep a gun loaded for each boneyard ship."

"You say 'she'," said Brin.

"Yes, her name was Thea. She deserved better."

"That was a death?"

"Yes. Of a two-million-year-old personality. My friend."

"I'm sorry," said Box.

"Who was doing the possessing?" asked Brin.

"I wish I knew," said Flux. "I call it the Enemy. Whatever it is, it comes from Möbius space, and it desires to be here, in this universe."

"It comes from your new universe?"

"Yes."

"Through the uplinks you sent your people through?"

"Yes."

"You've severed those links?"

"I have, but the connection persists. The physics escapes me. I believe parts of each ship are entangled with Möbius space."

"Destroy them."

"That would be murder. Those ships are my friends."

"What about the millions on Fluxor?"

"I hope to save both."

"You believe this Enemy is the source of the Fluxor attack?"

"I'm sure of it."

"Evidence?"

"I'll give you complete access to my mind."

"A full forensic scan?"

"Yes."

"Ship?"

Yes, said the Water Bear. *That'd be convincing.*

"Pax?"

Return. Let's wargame this.

The Flux's interior spaces were like the inside of a musical instrument, with frets and tubes, and light slanting through gaps, passing through air that looked dusty, but wasn't.

It was warm. Hot like a mouth. She felt a wave-like contraction, like muscle. It was no longer a dream. They'd entered through a physical airlock in the Flux's side.

She was Jonah, being absorbed in the belly of the whale.

She was nothing. A molecule.

"This is a helluva scary place," said Box.

"Because it's full of dead people," said Kitou.

"You think?"

"My sea spirit senses them."

"This way," said the androform Water Bear.

Why was she even here? Because this was what being part of the away team required. To boldly go.

And she had to admit, she enjoyed it. It was a delicious, forbidden pleasure.

Would the whale ever release her? Or would she be going around and around inside here forever?

Who cared? It made her feel alive.

After a few hours' walking - more than the exterior dimensions of the physical Flux suggested was possible - they reached the spaceship they'd travelled through time in. It was exactly as Box remembered, with its canopy facing an exterior wall.

She reminded herself, it'd been there for millions of years.

So, that part was real.

She picked up a bone ornament, covered in dust.

Bone. Where would they have got bone from?

She pushed that thought away.

On the bridge deck was Chance, in his robot form. It didn't clank anymore. She recalled the insectile movements of Sixes on Earth.

She was drifting away.

She snapped herself back to attention.

One of the acceleration couches had been replaced by an old-fashioned clinical chair, with its headrest at the nexus of a swarm of robotic arms. She could imagine it in a dentist's studio, or on the set of a horror movie.

Brin touched the arms. "What are these for?" she asked.

"I will use them to alter your physical brain."

"Actual surgery?"

"Consensual reality is encoded in your reptilian brain. I will build a new reptile by repurposing the neurons there. Then I'll reward the cortex with dopamine, until you believe."

"What does that mean?" asked Box.

"It means you'll be in a different place," he said.

"Your funereal world?"

"Yes."

"When does transition occur?" asked Brin.

"When the subject wakes, typically in a state of religious ecstasy, during the dopamine experience."

"Subject meaning me."

"Yes."

"And if it goes wrong?"

"The subject dies. But it doesn't go wrong. I've done this before."

"On living humans?"

"Yes. Every Horu has travelled this way, in a surgically altered, vat-grown body."

"Like zombies," said Box.

"Don't worry," said Chance. "This is safe technology."

"How safe?" asked Brin.

"Recently, a hundred percent."

"Recently?"

"In the early days, not so much."

"Do we trust these people?" asked Box.

"I do," said Kitou.

"Explain," said Brin.

"We gain nothing by not trusting them now, except maybe our lives. By wrongly distrusting him, we stand to lose everything. So, we risk our lives."

"Good girl," said Brin.

"Also, I actually trust them," said Kitou. "They're undeniably strange, but people are strange."

"I'm not so sure," said Box. "Flux is telling the truth, or this is catastrophe theater."

Brin nodded.

"Okay, let's do this," she said.

"How long will it take?" asked Box, as Brin readied herself, shrinking her skinsuit so it was like a second skin.

"About thirty-five seconds. Brin, please?"

Brin took her place in the clinical chair, with her head inside the robotic swarm, which waved smoothly to fit her head and neck.

"How is the surgery performed?" she asked, holding the arms at bay.

"By gravitic impulse - no incisions."

She nodded and leaned back.

"Then do it."

Brin appeared to fall asleep. The arms waved soundlessly around her head, and Box was appalled. Millions of years of neural adaptation, rewired in a heartbeat.

Brin's eyes opened - then she disappeared.

"Fuck," said Box.

"Now me," said Kitou.

"You're aware of the additional risk?" asked Chance.

"Yes."

"What additional risk?" asked Box.

"Kitou's not wholly human."

"Her undine?"

"Yes."

"Why's that a risk?"

"It's a complication. The undine spirit is the larval stage of the Xap lifecycle. They're in direct symbiosis, and will be until the end of Kitou's natural life. Neither can survive now without the other."

"We don't talk like that," said Kitou.

"I know. There's a spiritual element, which I respect. But we must speak plainly now."

"What might go wrong?" asked Box.

"The embryonic Xap might reject the new model. We could lose one, or both," said Chance.

"I see you waited until Brin was gone before raising this."

"Dr. Box," said Kitou. "We have to learn whether my kind can enter here."

"I want to ask Pax."

"Pax knows," said Kitou.

"Is that true?" she insisted.

"Yes," said Flux, through a loudspeaker. "The child is correct. The risk is acceptable, and this is necessary, for the sake of our species."

"How high a risk?"

"Fifteen percent, to lose both of them."

"Fifteen percent? What kind of horseshit odds are those?"

While they argued, Kitou lay back in the chair. "Begin," she said, and the robotic arms waved. Like Brin, she seemed to fall asleep. Her eyes flicked open, and she disappeared.

"Success," said Chance.

"You're sure?" asked Box.

"I am. Now you. Time is short. The uplink's at risk."

"Ship?"

"Go, Dr. Box," said the androform Water Bear.

Box lay in the chair. The arms waved around her, performing their bloodless intrusion. She fell into a trancelike state, not unlike the euphoric state produced by the Ayahuasca sacrament she'd drunk in Utah. She saw the same forest. Drops of water trembled in the same leaves.

She felt Pando, and her love, but ferocious.

She heard her friends, calling out in the darkness. For a moment, which might've been forever, she thought she was God. Not just a god, but God the creator, giving bloody birth to a new cosmos - she was Amon, rising over the first sea; Dayuni'si, come to Earth to see what was under the water; Krishna, the preserver, returning in the age of pain.

She was the first child, gazing out of a primordial forest.

She saw Kitou, trailing a blaze of fire in the heavens.

She saw Brin, defending the gates with her soul.

She could get lost in this place.

Then she came to her senses.

She lay in a green field, under a supersaturated sky. The colors were so rich they were pulsing. Her friends were smiling down at her.

"Holy fuck," said Box.

"Are you alright?" asked Kitou.

"My head is still buzzing," she said.

"It isn't your head," said Ito.

How did Ito get here?

It emerged from the hum, like a binaural beat.

Ophelia.

Ophelia Box.

Open your mind.

BOOK TWO

*A thousand years ago
Ten thousand Chinese marching on the plains
all turned their heads to Heaven at once to see the Moon.
An old man catching fireflies on the porch at night
watched the Herd Boy cross the Milky Way to meet the Weaving
Girl...
How can we war against that?
Allen Ginsberg - Iron Horse*

12 → DEATHCULT VON ENGINE

2056

Macro von Engine carried a planetworth of diamonds onto Avalon Station, hidden in his vestments. Planetworth was Cultspeak for lots. No one knew what a planet was worth. That was a thing about fungibles. They were only worth what someone would pay you for them.

This wasn't Macro's first time in the wild. He was first sent to a world called 9Fiero, aged twelve. It was a baptism of fire, selling derivatives during an atomic war, promising to hedge their weapons industry against an outbreak of peace - a losing investment. Now there was no war on 9Fiero. None of his negotiating partners knew they were dealing with an adolescent boy, steering an androform Samppo. His precocity fit in perfectly.

Why was he here? Because this was his work. He was here to engineer a political result using money. But also, because his woman needed him.

He'd do anything for Kitou.

Nothing would stand in his way.

As he walked along the teeming thoroughfares and service malls of Avalon Station, the people there glanced at him incuriously. He was a normal young man - generically tall, moderately handsome, nothing special. Even his scapular and cowl could pass for partywear on Avalon. It'd be different if he were carrying his cloven-skulled staff of office. Then people would part like airflow over a wing. Nobody messed with the Cult. It was a brand of obnoxiousness that'd been nurtured over thousands of years.

He loved Avalon Station - the gimcrack riotousness of it, the way it was thrown together from parts, the way the bulkhead doors hissed air as they opened and closed, like in a sim. He loved the craziness of the people - each of them with their own crazy stories - the immanence of the vacuum, and the views of the planet, glimpsed through a succession of airlocks.

He especially loved the city below, with its painted towers, by its milky-white sea. One day, he promised himself, he'd stand on that beach, with Kitou, with the sand between his toes, bake in the sun, learn to catch waves.

"Mr. von Engine," said Felix Revelstoke. Macro's full name was Macro Ibquant Deathcult von Engine - traveling simply as Deathcult von Engine. This marked him out as a scion of the Bank. All Samppo's clones were called Deathcult. This conveyed power. The Bank craved power - not for its own sake, but for what it could buy.

"Macro," said Aloïs Buss.

"Aloïs! What a pleasant surprise. Which version are you?"

Buss smiled.

"I'm the one who was grown from a brain."

The police cells on Avalon Station looked like a place to be tortured. Macro, Aloïs, and Felix Revelstoke were seated around a polycarbonate table, like a mortuary slab, pitted and bubbled with burns. A faint smell of urine hung in the air.

Macro dismissed the surroundings. It was a theme park, designed to impress the credulous.

"You have my gemstones?" asked Revelstoke.

"I do," said Macro, fishing the bag from inside the folds of his habit.

"A thousand flawless, untraceable."

It was the untraceable that made them so valuable. Every diamond has a spectral signature. A thousand flawless was a *lot* of diamond. Enough to buy a life in the stars. A thousand *untraceable* was enough to buy a significant spacecraft.

Or a profitable sinecure like Revelstoke's, many times over.

"Thank you," said Revelstoke.

Macro was in two minds, literally. The part he regarded as *himself* drifted along with the conversation, chatting inanely with his good friend Aloïs, presenting no threat to anyone. The part he called his evil twin gave him a rolling sitrep.

The twin was new. It'd come with the upgrade Samppo had given him for this mission, along with Broca transfer, and all kinds of other new toys.

If he wanted, he could climb inside it, observe his own actions, like in a warship's gamespace.

How cool was that?

Revelstoke was an interesting sort - a storied codebreaker - formerly a public intellectual. Now a Po secret agent, researching an obscure neurological disorder.

Ostensibly selfless.

But more likely corrupt.

Even if he was corrupt for good reasons.

Aloïs Buss was who he was - a willing instrument of the Fa:ing and their interesting number. This version of Aloïs was a biological original, unlike the virtual one now studying the Fa:ing language at the Cult.

One of several biological originals, because Aloïs Buss was a looper.

Revelstoke, he didn't know. Aloïs, he liked.

He needed to *trust* both of them.

Macro had an ego the size of a planet, his father once said, orbiting a primordial black hole of teenaged insecurity. Who was he to be roaming the cosmos, having such power over real people, impersonating humanity's wealthiest man?

He was a fraud, a youth, the winner of a ridiculous epigenetic lottery.

The Cult's psychs had told him that this was all good - that he was a regular person, perhaps a bit smarter than most, born to play an outsized role, and bound to feel guilty about it. Otherwise, he'd be a narcissist, and they didn't clone for that trait in the Bank.

That was before Kitou. Now his woman needed him. He was a new man, bending space in a bubble of lovestruck determination. He was a teenaged geek, with a budget for weapons. Look on my Works, ye Mighty, and despair.

"Seriously," said Buss. "The local surgical suite was going to repossess my cerebellum."

"That was just for show," said Revelstoke, smiling self-deprecatingly. Unless he was packing better wetware than Macro's, his display of relief was authentic.

"Still," said Buss.

Relief, tinged with nervous excitement - a plan, though botched, had somehow come together. And diamonds. More than Revelstoke knew what to do with. More than he understood the significance of.

Now to spring the trap.

"There's more where those came from," Macro ventured.

Silence.

There. A line had been crossed.

"What for?" asked Revelstoke, sensing the danger. "The debt is settled. I don't need any more money."

"You do," said Macro. "I need you to raise me an army. I need to pay you for it."

Revelstoke dropped the bag, and the diamonds it held, into a satchel, whose strap morphed into his shabby civilian coat, which Macro knew was made of nanobots and fields.

Was that acceptance?

Or the start of a contest?

"I need a thousand human soldiers," he continued. "Elite ground troops. And an air superiority platform. One capable of supporting a ground action of unpredictable complexity."

"Oh, and it has to be secret," he said.

"How secret?" asked Revelstoke.

"*Really* secret."

Revelstoke gazed at his hands, which Macro saw were manicured. His clothes were perfect. Shabby, but perfectly judged.

His nails wore the imperfect brushstrokes of enamel, or excellent nano.

A skilled impressionist.

Or a regular person, with foibles.

"Mr. von Engine," he said. "I'm sorry - Avalon isn't an arms fair, where you can buy weapons. Nor can you buy me. You've repaid a debt. You have my thanks. We're done here."

Of course Avalon was an arms fair. It was why Macro was here.

And of course he could buy Revelstoke.

He nodded. "Oh, I know all that, and I'm not shopping. I'm asking for your *help*. As a friend. As your *new* friend who just saved your reputation."

Just a nudge, was all that was needed.

Revelstoke sighed.

"Look," said Macro. "Work with me here. I can *say* what I want, but I have no idea what the words even mean. I honestly wouldn't know a howitzer from a hamburger. I was hoping you'd lend me your expertise."

"What expertise?"

"Please be serious, Mr. Revelstoke. How many covert operatives do you think I know?"

"I'm a cryptologist."

"You're a Po secret agent."

Revelstoke looked at Buss, who shrugged.

He knows you have him, said his evil twin. *Reel him in.*

"Who's buying?" asked Revelstoke.

"I am," said Macro.

"Who's *really* buying?"

"I mean I *really* am. I'm - how do you say it? Freelancing? The Bank will deny all knowledge of me."

"Then you're taking a risk, bringing those valuables here. I could have you arrested. Take advantage of your plausible deniability. You'd never be seen again."

"Neither would you."

"What?"

"Seriously? You may be a Po agent, Mr. Revelstoke, but you're only an agent. I have the resources of the Cult at my disposal. I could have you out an airlock in a moment."

"Are you threatening me?"

"Of course not. We're exchanging hypotheticals."

Alois Buss held up his hand.

"Can you both please stop playing games?"

Macro smiled. He'd made his position clear. The carrot, the stick. This was the Smear - there were few consequences here, provided you had enough money. That was the stick. He could buy Avalon station, a thousand times over. He didn't need to. The Bank already controlled Avalonia Cpy., through a network of trusts. The carrot was money. More than Revelstoke knew what to do with, because Revelstoke, adorably, had no idea how much was on offer.

Enough for him to one day buy Avalon Station himself. Cure the Thespian virus - do whatever he wanted. Macro would help him.

But now, he *wanted* Revelstoke to be corrupt.

He also wanted him *predictable*. To help raise an army, without telling anyone about it.

If need be, act against the interests of his people.

Why would he do that?

Because Revelstoke was an *idealist*. The money he stole went on helping the Thespian sick. It wasn't enough that the Thousand Worlds saw the Thespian virus as a problem to solve. Revelstoke cared for the *victims*. Was that corruption? Of course. He stole money to do it.

The Bank knew this.

And that was Revelstoke's - Felix's - weakness.

Alois Buss, who was essentially incorruptible, winced at the show of teeth. He would've handled it differently, but Alois wasn't the Bank.

They were all on the same side.

Only their methods differed.

Later, in the quiet of his sleeping module, Macro was physically sick. This was his usual reaction to ruining people's lives.

Don't be melodramatic, said his twin, as Macro vomited into a disposal unit. You've changed his life, not ruined it. Lives get changed by larger forces all the time.

Macro's sleeper was the cheapest of dormitory tubes. He cared nothing for luxury. He gripped a sharp edge and drifted onto a narrow bed.

Would he have had Revelstoke airlocked? Of course not. He wasn't a monster. But it was a plausible threat.

The Bank protects its own, said his twin.

He didn't choose to join, said Macro.

Neither did you, said his twin.

This sleeping tube could be a ship, he decided. His personal spacecraft. He could hear nothing except for the whisper of systems, and his thoughts. If he broke the seal on the door, the chaos of Avalon Station would rush in and drown him. He wished he *were* in a ship - in the Bat, with Kitou beside him.

What about you? he asked his evil twin. Not evil, of course. Nothing the Bank did was evil. Merely expedient.

What about me?

Did you choose to join the Bank?

I'm a familiar, created to assist you.

How do you feel about that?

What are feelings?

Of course, his wetware doppelgänger was joking. He was being sardonic. Archetypal Bank programming.

Are you self-aware? he asked, knowing the answer.

Not in the way you are.

Macro sighed.

This is why I suck at this, he said.

No, it's why you don't.

Can I increase your sentience?

Make me a friend?

Yes.

You can if you want, but I don't recommend it.

How would you feel if I did?

I can't say till you do.

*Do you want it?
I don't experience the feeling of want.
You'd be a new individual.
I would.
Such powers I have.
Let me give you some advice.
Shoot.
You're having a neurotic episode - wanting to channel it into
some kind of protest. Forget it. The time for sulking is over.
Episode?
Truth's a harsh mistress.
Hey, that's not in character.
Sorry. There's some of your father in me.*

The Tung shipyard resembled an encrustation of failed engineering projects, growing like barnacles on the spaceside of Avalon Station. Macro thought it looked like a junkyard. Up close, the picture changed. The recesses of the cocoons were jam-packed with high-capability spacecraft, some military, some incognito, like predators drinking from the same waterhole. Clever Tung tendrils glimmered like surgical instruments, finessing the entrails of ships into the vacuum.

Everywhere, Macro saw weapons - beam blisters, rail guns, battlefield atomics - spilled out, and being worked on. Some of it legal. Some decidedly not. Star killers. Bioweapons. Weaponized goo. Deplorable stuff, even by the nebulous rules of the Smear.

And everywhere, the Tung.

Industry, but not as we know it.

Revelstoke took Macro to a spaceside bar, called the Snake City Discothèque, to meet a Tung fixer called Pfft. It was chockful. People of every species grappled and drank in every available crevice. It was precisely as debauched as Macro wished it would be.

He watched as a three-meter Stentorian re-breathed ammonia from a recirculating flask. The Stentor could've worn a pressure suit, but it seemed set on riffling its feathers, in what appeared to be a sexual display.

Stentors mated by impregnating unwitting hosts with their spermatozoa-like seeds. The stench of the Stentor's flask disguised the alcoholic waft of their drinks, and the musky scent of its seed.

This bar was outside Macro's lived experience. Bank negotiations didn't happen in places like this - at least, not ones he was privy to. Revelstoke, a bored sophisticate, seemed unfazed by it all.

Macro relaxed along with him.

I could get to like this, he decided.

Up close, the Tung were borderline terrifying. There was something in their phagelike appearance that stirred up primal fears, like the horror of spiders. Macro didn't fear spiders, nor was he bothered by the Tung. He dismissed the contagion metaphor used to describe them in the gutter sims. In truth, they were beautiful mathematicians.

"Do you *exissst*?" said one that'd wrapped itself around a pillar by their barstools. It was the size of a hand towel.

Lacking any vocal cords, the Tung spoke human languages by trapping sacs of air against smooth surfaces, then expelling it past their extremities. Any wall would do, or a leg. It was passably effective. This one spoke in breathy Avalonian.

"Is that a question?" asked Macro.

"We think," said Revelstoke.

"Therefore, we are," said the Tung.

Macro bit his lip. *I think, therefore I am*. René Descartes, a philosopher from Ophelia Box's homeworld. He'd trodden on a codeword exchange.

"Earth is on everyone's *lipsss*," said the Tung. A vector of harmonics flashed, in a disused register, somewhere in Macro's upversioned cerebrum.

{ lips => thoughts, ideas, feelings, conjecture }

He was *thinking* in Tung. Now, *that* was unexpected.

"What do you know about Earth?" he asked.

"The red-headed human was *heeere*," said the Tung.

"This is the shipyard where the Water Bear was fitted out," explained Revelstoke.

"For the journey to Fluxor?"

"Yes."

"Not in *thisss* universe *nowww*," breathed the Tung.

$\{ R \Rightarrow x \mid x \in \infty \}$

"What do you mean?" asked Revelstoke.

"Water Bear bends space *alone*. Without any *humansss*."

"What? That's news to me," said Revelstoke.

"They were separated," said Macro. "By the Praxis autonomies, after your bank robbery. Now they're in a ship called the Bat, although 'now' is a slippery word, seeing as they're in a slow-moving warp bubble."

"How slow?"

"The Bat's best speed for a 1,400-light-year journey is just under 4c. They'll reach Fluxor together, in about nine years from now."

"That doesn't make sense."

"Oh, they also went back in time."

"Bat *a/ssso* bends space?" asked the Tung, with a ripple of fronds.

"Yes," said Macro. "All roads lead to Fluxor."

"Except *yoursss*," said the Tung.

"No?"

The Tung's fronds stirred, and Macro realized he was being given a package of encrypted information.

He wasn't going to Fluxor.

He was going to the Magellanic civilization, 326,000 light-years away -

As far from Fluxor as it was possible to be, and still be in the human galaxy.

Later that night, Macro went eavesdropping. *We bend space for the Clouds?* waved a second Tung. It was larger than Pfft., about the size of a bath towel, and it was signaling some kind of commercial arousal $\{ risk, equity \Rightarrow [climax] \}$.

The first Tung rippled in reply.

Can? it asked.

Of course, signaled the second. *Our drive turns time into money* $\{ metaphor \Rightarrow irony, groan \}$.

Macro was watching this exchange through a collection of civilian engineering systems, cameras, sensors, and instrumentation. Nothing illegal. No need to make waves. Outside his tube, bacchanalia reigned. Night here was vastly more interesting than he was ready for.

The larger Tung had adhered itself to the transparent skin of the Discothèque bar, in plain sight of everyone, and was waving its fronds towards the clusters of organic-looking pods that made up the shipyard.

Macro understood.

It *wanted* to be overheard.

It expected him to be listening.

The first Tung - Pfft. - was a flimsy spacefaring type, whose lightness made it better able to withstand strong acceleration. This bigger one was a local shipbuilding empresario, evolved for gravity.

All this information flowed into Macro's brain.

Handy, this military wetware.

*Nine years is **fast***, the larger one said.

A hundred-thousand parsecs, waved the first { *fucking. long. way. }*.

Macro owed Aloïs a briefing. He invited Revelstoke. His evil twin agreed. Revelstoke was already earning his paycheck.

They met in an Avalonia Cpy. vault. Most of the safe rooms on Avalon Station were unsecure. This wasn't. It was a superluminal spacecraft, in an earthed metal cage, ready for an explosive departure. It was where the corporation's owners met to discuss breaking the law.

Macro only had to apply the slightest pressure to get it.

He dropped a new bag of diamonds on an iridium table. There was no urine smell here.

"Mr. Revelstoke, may I call you Felix?"

This was the Bank's way. The stick was shown once. The carrot forever.

Revelstoke nodded.

"Welcome to the employ of the Bank. It's not as bad as you think. It's not bad at all. We're on the same side. Don't worry about your employers. I calculate a high probability they already know what's happening here."

"Which employers?"

Macro shrugged. "All of them?"

He wondered, was that true? Probably.

"What do we collectively think about the Tung?" he asked.

"Unreliable," said Revelstoke.

"You took me to see them," said Macro.

"You asked for an army. They're who I'd go to."

"I agree with Felix," said Aloïs. "The Tung can't be trusted, but I don't mean that in a pejorative way. They don't think like we do. They can't be *predicted*."

Macro showed them the conversation he'd just eavesdropped. Then he unpacked the message Pfft. had given him in the Snake City bar.

"They want me to go to Waterfall. Who'll come with?"

"To the *Clouds*?" asked Revelstoke.

"I will," said Aloïs.

Macro smiled. He'd much prefer Aloïs along on this journey.

"How'll you get there?" asked Revelstoke.

"The Tung have a superluminal spacecraft, ready to fly. A shaker drive, whatever that is. The course is baked in. Nine years in stasis."

"You've already arranged this?" asked Revelstoke.

"Yes," said Macro, although in truth his evil twin had, using resources he didn't care to know about. He too was at the mercy of larger forces.

"Why not just hire a krëw of Magellanic Navigators?" asked Revelstoke. "Like a real bank manager would? A Magellanic ship will get you there in an instant."

The truth was that nine years in stasis would suit him just fine. He'd done the sums. It meant he wouldn't get any older than Kitou.

He shrugged, as if to say it was out of his hands.

"What about me?" asked Revelstoke.

"Your life carries on," said Macro. "As though uninterrupted. It's been good working with you. You have no need to worry."

"About what?"

"About *anything*. There's an account, associated with your genetic information. Present a few chromosomes, along with the original of your personality, at any of our branches."

"You have branches?"

"You know what I mean."

"As easy as that? I'm off the hook?"

"We expect your discretion. What happens next is up to you. You might help us again in the future. We'd like that."

No, you're not off the hook, and never will be.

"The Po?"

"I promise there won't be a problem." He hoped that was true.

"My work here?"

"Continues."

As flies to wanton boys, are we to the gods - they kill us for their sport.

Shakespeare. Earth dramatist.

1605.

That night, his wetware treated him to a real Tung conversation. This time, Macro was present as malware, in a whisper of systems. He had no physical manifestation, although that depended on what spatial dimensions you were listening in - he existed as transient noise in the processors of countless devices: in shipyard equipment, and passing machines.

This was distinctly illegal, but he wouldn't get caught; the Avalonia Cpy. had good systems, but they were no match for his.

A thousand of the creatures were nested { *sleeping, dreaming* } in a shipyard module. Some were as small as his hand. Some were like bedsheets. Their thoughts were like water. There were eddies and pools, and chaotic attractors.

There was the nexus of the Water Bear, and her constellation of humans, complexly orbiting the interests of the Xap, intersecting with the interests of the Thousand Worlds civilization, like a

planetary system orbiting a binary star. Some of the thoughts of the Tung were about how to solve that equation - the Xap and the Bank and the small group of humans.

The *strangely attractive* humans. He understood this to refer to mathematics, not Tung sexuality, although sometimes they seemed to be nearly the same. Strange attractors were mathematical objects. The Tung saw humans as patterns, whose unpredictable behaviors pulled other patterns toward them. Aloïs had said the Tung couldn't be predicted, but *we're* the agents of chaos in this calculation.

Macro *focused*:

There were the Pursang, and the Magellanics. Another nexus. The Magellanics have what Macro needs, which has something to do with the Pursang. The subtext was clear. Macro was going to the Magellanic Clouds to raise an army of Pursang.

He smiled at the idea.

It was *perfect*.

13 → OPHELIA, DOWN THE RABBITHOLE

314

The Red Lady was confused. The Blue warrior, Marius D, looked on impassively. This was only to be expected. Her shieldmaidens were there to help her.

And the prophet, Ito Nadolo.

This was a great day.

The Blue people believed they had sung the world into existence, and three heroes would come to lead them to heaven. The Reanimated, who were more numerous, believed the world was created by God for their use, and they once lived in heaven themselves. Both agreed the world was shaped like a toroidal polyhedron, or paradromic ring, and was about three hundred years old.

In Blue cosmology, the gods were natural beings, who were too numerous to count. They lived in heaven like fleas on a cat. The Reanimator God was Mathematics.

Both accepted the equal truth of each other's opposite view.

That's how it goes, in Flipside.

Ophelia Box scrambled to her feet. She was in a supersaturated world, which seemed to twist in on itself in a way that made her feel sick.

"The feeling will pass," said Ito.

She was on a steep-sided hill, like Glastonbury Tor. The sky was the most vivid color of *sky* she could imagine. The landscape was a

patchwork quilt of supersaturated greens.

She was naked.

A strapping young Viking was staring at her.

She started to giggle, uncontrollably.

"Frost," said Brin.

From the towering height of his nineteen years, both shieldmaidens were exceptionally beautiful. The Red Lady less so. She must be at least thirty.

A tenth the age of the universe.

The white one, who he knew as the Goddess, who couldn't be beaten in battle, was yellow-haired and creamy-skinned like him. He'd follow her.

The black one was a different matter. With her clean-shaven head and luminous warpaint, he'd never seen anyone so *otherworldly*, even among the Reanimated, who were sometimes also dark-skinned.

Many would follow the wolf-mother.

Beings of mythos, alive in the world.

Such days.

"Bring blankets," he commanded, and a rider rode down from the sky, with temporary clothing, which Ito Nadolo gave to the three women, who were gabbling in tongues he could make no sense of. He knew they'd soon learn his language. It had taken Ito three hours.

They had *wetware*.

"We must be careful, Ito."

The prophet nodded. He understood the dangers.

Behind the sky was the Enemy, with his beam weapons.

"Where's the Water Bear?" asked Box.

I'm here, Dr. Box.

"Where?"

"The ship's personality seems to be hosted in our wetware," said Brin.

I didn't quite make it.

"Ship, what are your capabilities?" asked Ito.

Limited.

"Well, at least you're here," said Kitou.

I am, shieldmaiden.

Box suppressed another giggle.

They gave her a horse, a rust-red dappled mare with a flighty attitude. Box's experience with horses was limited to carousels. Thankfully, her wetware knew how to ride them.

Dirty great beasties for posh English lassies.

She began to relax, then to enjoy herself. The high moors seemed to stretch on forever. They could be in the Highlands of Scotland. They were being escorted by a half-dozen Vikings - four men and two women, who radiated military competence. Dotted among the clouds, she could see more Vikings.

Where was she?

She found she didn't much care.

She shimmied her ride towards Ito, who was off to one side, finding higher pieces of ground, scanning the horizons.

"Ito, it's good to see you."

"And you, Dr. Box. How are Pax and Aloïs?"

"Pax is himself. Aloïs is complicated."

"Then business as usual."

She smiled, then laughed out loud.

"Ito, I can't wait to hear you try to explain this."

"Listen," he said.

"What?"

"Listen closely. D'you hear it?"

"It's the Fa:ing hum!"

"Yes. Like the Water Bear, they didn't quite make it here."

"But they're here?"

"Yes, this place is where they were searching for."

"The Horu heaven?"

"Oh, it's more interesting than that.

"It's the unconscious."

They descended into a small farm, a tangle of garden clotted around a stone farmhouse, surrounded by fields of waving grass.

Children ran out shrieking to greet the Blue warriors, who snarled fiercely before laughing. The farmers came to meet them, a smiling coffee-skinned couple.

"Are these the ones?" asked the woman, shyly.

Marius nodded.

"Then we are blest," said the man.

Box waved. She was already understanding this language. The farmyard smells assaulted her face - the heavy ammoniac stench of chicken shit, and musky barn manure, and freshly turned soil, and hay curing in the mow, all of it more real than real.

The children waved back, delighted.

"Red Lady," whispered the woman.

Past the farm, they continued down towards a country railway station, with beelike creatures bumbling in the long grass, between ceramic railway sleepers. As they descended, the air became warmer. Here it was like summer. A train surged into view, flowing from a perplexing middle-distance, and some of it peeled away and slowed to a stop in their station. The Vikings led their horses into freight cars, where they tied them into horseboxes - incongruous timber against the pristine ceramics of the carriages - and the train surged forward again. Soon, they were traveling at high speed through a harlequin countryside.

"This is the centerline train," said Ito. "Welcome to Flipside."

Marius stood to be introduced.

"Marius, this is Ophelia Box, known to your people as the Red Lady, and Brin Lot and Kitou Gorgonza, her shieldmaidens. Team, this is Marius D, a warrior of the Blue people. Marius's riders are Markus, Velo, Viki, Alis, Ox, and Bravery. Also here, but not currently visible, is my ship, the Water Bear.

"Ride of the Spinifex Reach, meet the Hand of the Water Bear.

"As the Blue people say, this is a great day."

Marius bowed, and unsheathed his sword, and presented it, grip-first towards Box.

"What do I do?" she asked Ito.

"Accept it," said Ito.

She took it. It was as light as a feather, constructed like a samurai sword - a katana.

"Now give it back to him, grip-first."

She did. It was like rotating razor-sharp air.

A grin split his face.

"As you say, Ito, a great day."

"He's yours to command," said Ito.

"I only just met him."

"He'll gladly die for you. Now for the others."

One by one, she accepted their weapons. One of the women boldly kissed her full on the lips before blushing and backing away.

"Thank you," Box said.

You did perfectly, said the Water Bear.

How do you know?

Comparative ethnology. Also, their vitals. They love you.

The Flipside universe was the shape of a Möbius strip, embedded in three-dimensional Euclidean space, like the Möbius machine itself, although the exact relationship between the shape of the machine and the shape of the universe was unclear. To its inhabitants, Flipside seemed spatially flat, but self-referentially twisted, so that it turns in on itself from every direction.

Box learned this from a book, a beautiful physical volume, printed on vellum. Travel from side to side, away from the centerline, was believed to be a mystical experience - towards dreams and the sublime. Travel along the centerline brings the traveler back where they started, after a journey of some four thousand kilometers, including a thousand kilometers of vacuum.

Everything here moved in spirals. The world slid along the universe, like a snake in a jar, providing the seasons, while rotating about its long axis, thus providing night and day.

The part they lived in was Upside.

The vacuum beneath was called Downside.

Across the cylinder was Farside.

The centerline train traverses the whole universe lengthways, and the direction of its travel is called Spinwards.

Everyone knew the origin of the train.
It was sung into existence.

They were put up in a building called the Chancery, a sprawling stockade owned by the Chance family, built on a series of bridges, above a meandering river, where it emptied into a wine-dark sea. It was the informal seat of government in this southerly, or Spinifex Reach of the Möbius world. Amelia Chance was its leader, a descendant of the original Chance, who was the last living Horu, and the first Reanimated, as the Horu people of Flipside liked to call themselves.

The Blue people were just called the Blue.

Kitou and Brin brought two razor-sharp swords into their rooms. These were made of bone, like Marius's, although less airy.

"I thought you people didn't like weapons," said Box.

"These knives have a spiritual importance," said Kitou.

Brin agreed. "When in Möbius, do as the Möbians do. It'd be unwise to be at an untrained disadvantage."

They started to improvise. Box blanched. The consequence of an unintelligent fist was a broken nose. The consequence of an unintelligent katana was a severed hand.

"Stop!" she yelled.

Water Bear? she said.

Yes.

Can Ito send Marius please, urgently?

Five minutes later, Marius appeared with four sticks and protective gear. He passed out the gear, instructed them in its use, and taught them. Box was hopeless. "Don't worry," he said. "You're the brains of the beast. It's why you have shieldmaidens." Brin was a natural, a superbly balanced athlete, but Kitou was in another dimension.

Watch, said the Water Bear.

She replayed Kitou's moves in Box's sensorium. Kitou riposted before her opponent started to move. Sticks banged on facemasks and heads.

She's faster and better than anyone here. All she lacks is any semblance of control. You were right to take the sword off her.

How is this even possible?

Every once in a while, Po throws up a genius player. This is what it looks like.

Why now?

Because I've stopped slowing her down.

Marius seemed happy to be beaten.

"Goddess," he grinned.

"My head is spinning," said Box.

"It'd be broken if it wasn't spinning," said Ito.

The nights were bitterly cold - something to do with the way the prevailing winds spiraled through vacuum. There were five seasons here, named after parts of the day - Dawn, Morning, Day, Dusk, and Dark. Now it was Dusk. From here it got colder.

The Chancery was homespun. Rough woolen throws covered overstuffed furniture, in front of a roaring fire. The floor was real timber, and through it, she could hear the burble and splash of the river below.

It was like her childhood dreams of holidays.

"Upside, Downside, Flipside, Farside," she said, not complaining. "It's like being tossed in a word salad. Only the Blue people make any sense to me."

"Because they're Pursang," said Kitou.

"They really sang this place?"

"Maybe," said Ito.

"Or perhaps it was already here."

"Your unconsciousness theory?"

"Not unconsciousness," he said. "I mean the part of the mind that is hidden. I believe the Möbius machine tunneled a new spatial universe out of the psyche."

"I have no idea what that means."

"I can't explain," he said.

Box stared at him.

He shrugged. "So many pieces that fit."

"What, like intelligent design?"

"In this case, we know there's a designer," he said. "The Horu."

"What does that make us?" asked Kitou.

"Figments," said Brin.

"Marius is going to take us on a sortie, to see the Grays," said Kitou the following morning. Ito was cooking breakfast. Box had decided to increase her training. She'd already spent an hour with a stick, training with the two female riders, Alis and Viki, barely more than girls themselves, who'd smiled warmly at her ineptitude before beating some rudimentary skills into her.

"Not without me," she said, through a mouthful of porridge.

"What's a Gray?"

"The Enemy's soldiers," said Kitou.

"You'll need a horse," said Ito.

A half-hour later, an ancient Blue horseman led them into an open field, where hoarfrost like spiderwebs covered the trees. In the field were horses, covered with blankets and rugs, nervously alert. Kitou and Brin had chosen theirs - Kitou's a piebald called Chaos - Brin's a black mare she called Fabric. These trotted over, along with a bandy white beast, all muscle and pugnacious attitude.

It was the stallion from her dreams. He wasn't as big as the others, but with his tail in the air, looking shrewdly about, there was no doubt who was in charge.

"Hello," she said.

"He knows you," said Kitou.

Kitou was right. The creature was looking right at her.

"You must give him a name," said the Blue horseman. "If he'll have you."

"Don't they come with names?" she asked.

The man shook his head. "Names have great power," he said. "By giving a horse a name, you claim a privilege and accept a responsibility. If you're foolish, he'll leave you. If you prove worthy, he'll fly down a mountain."

"There's a story that horses evolved on Earth," said Ito. "What about a grand mythological name?"

"What's yours called?"

"Equus."

"God of horses?"

"From the Latin for stallion."

Box shook her head. "No way," she said. "I'm calling mine Seabiscuit."

The Ride assembled in an interior yard, with no obvious way to the world. The horses' hooves clattered and crunched as their riders cantered about, preparing for the day's sortie. Brin and Kitou were as frisky as their rides, excited by the idea of action, and by the prospect of riding out with Ito. The rightful order had been restored. Box was regretting her early start. She asked for a version of coffee and was given it by one of the riders.

Her horse, Seabiscuit, was having none of her lethargy. He was like a pent-up ball of energy, waiting to be released.

What had she let herself in for?

Marius produced a tub of pearlescent ooze. He waved it around, and shapes moved in its depths, before settling on a periodic tiling. "This is Cerulean wax," he said. "It exposes a pattern, which you can see, keyed to the laminar structures of the world, like the layers of bone in the armor we wear."

The patterns in the wax were like a kaleidoscope turning. The closer she looked, the further they turned.

More fractals.

Marius rubbed some of the wax into his shield, which glowed ghostly blue, and held it aloft at the same angle as the bucket. Then he lifted himself into the air.

"It isn't the wax that does this," he said, lowering himself to the ground. "It's our minds. For now, you may trust in your horses. They know what to do."

The Blue riders rubbed wax on their faces and shields and pressed it with their open hands onto the flanks of their horses, until they lit the Chancery yard with a phosphorous glow.

"This isn't stealthy," said Box.

"I promise, nobody will see us," said Ito.

Marius wheeled his stallion towards the cylindrical sun that was rising over the stable roofs. The others wheeled to follow. By the time they passed the Chancery walls they were fifty feet in the air. They rose in a series of spirals.

"Thermals," said Box.

"Exactly," said Ito. He pointed out the centerline train, traveling fast in the opposite direction. Box clocked their relative speed in her wetware.

"Bloody hell," she said. "How are we going so fast?" Strangely, she didn't feel frightened. She could feel the earth crackle and squeak under Seabiscuit's hooves, hear the swish of uncanny grasses.

"We're surfing on gravity waves," said Ito. "In moving bubbles of spacetime."

Box nodded.

"Like in a warp drive," she declared.

Soon the fields gave way to dense forests, then blasted moors. These weren't the high bracken and gorse of her childhood memories, but literal voids sliced through ructions of dirt. In the distance rose a steep-sided hill, like the one they'd arrived on, but twisted. She saw its entire farside was missing.

"An uplink," said Marius.

"The necropolis Thea," said Ito.

"Oh," said Box.

"Why? What happened in the Real?" asked Marius.

"She died," said Brin, and told them the story of the boneyard fleet, and how Thea had started to crawl through space, like a nightmare - possessed.

Marius's eyes widened. He turned away, and Ito grasped his shoulder.

The cold of the morning became a broiling hot day, as a sun like an industrial heater beat down from a sweltering sky. The wax on their faces and shields took on a filmy appearance, and the riders seemed to melt in the air. Now she could see the extent of the Ride, spread out in a vee, like geese ahead of the main party.

"Are those all your people?" she asked.

"Some of them are," Marius said. "This pattern of 1, 12, 144 is called a *gran sortie*. The Ride of the Spinifex Reach is a reconnaissance party. In a war, we'd be the eyes and ears of a Blue army."

"144? Where are the rest?"

"I don't need to know."

"Why are there no Horu people here?"

Marius shrugged.

"The Reanimated won't ride with us," he said.

"Why?"

"They think us to be morally questionable soldiers."

"In what way?"

"Because we fight to win."

Their conversation was split by a pulsating roar. Two jet aircraft Dopplered through the air, a few hundred meters below - a sound so loud they could scarcely hear it, followed by echoing booms.

Box blinked back her astonishment.

"What...?" she said.

"They always fly at that precise altitude," said Marius.

"Pilotless drones," said Ito.

They were ugly machines, with long drooping snouts, bristling with weapons. Box watched them turn in the distance. They banked around the far side of the ruined tor, then climbed, spearing towards the edge of space, back where they came from.

"Zooms," said Marius.

"Like clockwork," said Ito.

"Devil weapons of the Enemy," said Marius.

They rode west, towards Farside, climbing the side of the helix. The moors became tundra, then tendrils of ice. Ahead was a greasy murk, lit from within by gas flares and furnaces. A hard, flat wind whipped smoke from industrial stacks.

Factories, for as far as she could see.

Now it was *cold*. She saw a whirlwind dance over the landscape, whipping up grimy spirals of snow - it was the third or fourth different weather of the day. Between them and the factories was a

camp like a gulag from Earth. The edges of the camp seemed to lap like an infernal lake at the natural world.

"What is this?" asked Brin.

"Look," said Ito.

In their sensoria, he created a lensing effect. Uniformed guards patrolled barracks. There were no prisoners on display, but the squalid appearance of the camp suggested extreme deprivation. The buildings seemed to be all made of pasteboard, and rivers of ordure ran down the streets, pooling in lakes on the edge of the camp - a moat of excrement, frosted by rime. It reminded Box of the internment camps of the climate apocalypse.

The guards were humanoid, but with a slippery, unfinished appearance, like mucus.

"Grays," said Marius.

"Who lives there?" Box asked.

"We don't know," said Marius. "Not our people. Perhaps the people of wherever the Enemy comes from. We get their bodies sometimes, washing up in the Spinifex Reach."

"Humans?"

"In parts."

"We think it's a deterrent," said Ito. "A type of shield. We'd have to fight our way through it to get to the Enemy, while he can advance unopposed."

"A human shield?"

"If you can call them that."

That night, Marius came to her room.

"Red Lady, will you lie with me?"

Box sat up and stared. She was still troubled by the day's events. Marius was standing there, naked. It appeared he was ready, and she wasn't dreaming.

"Not if you call me Red Lady."

"What do I call you?"

"If we're going to be screwing, you'd better call me Ophelia."

She hadn't slept with anyone for over a year, and Marius was an expert and sensitive lover. As soon as his fingers found their way

between her swollen lips, after what felt like an orgy of touching, she came, and came again.

After a while, they were joined by Alis and Viki, wanting to make amends for beating her that morning with sticks, and her night was filled with hot, sliding flesh, and soft, hungry mouths, and she had a new best time of her life.

The following day there were preparations for a feast, a celebration of the last full moon before Dark, when snow would cover the ground, and the Chancery would be locked down for the winter. Box watched garlands of flowers being hung in the Chancery hall and joined in the work. It felt good to be a part of the rhythms of the house. The kitchens were filled with laughter and noise, and children running everywhere. Chancery society seemed to be a functional anarchy, with people contributing according to their wishes. Management coalesced around the most able. Amelia Chance appeared to have no skills, apart from high intelligence, so she was a laborer for the day. Box helped her move chairs.

"It wasn't always like this," she said, noting Box's pleasure at seeing the people working smoothly together. "Once there were wars."

Amelia Chance was thickset and sly, with an acerbic wit. A forthright, practical woman. Not the kind of leader Box would've expected from a society of politicians.

She explained how the first Reanimated were farmers, not politicians. "Even a political class has to eat," she said.

"Those were hard times," she said. "Bad times. A struggle to build without tools in a vast wilderness, and many were lost. We can't even begin to count how many. We thought to bring metal but none of it came. Something to do with harmonics and crystals. We made war on the Blue, and despite our greater numbers, we lost. That's when the Enemy appeared. I believe he was attracted by the smell of violence."

"You say 'he'."

"Yes, he has a persona, called Kronus."

"You've met him?"

"I did."

Box wanted to hear more, but Brin and Kitou came and took her away. They'd found a boy who played an instrument, called an *aPragma*, which resembled bagpipes, and sounded to Box's ear like a Roland TB-303 synthesizer. They'd been teaching him reggae. To Box's surprise, he played a note-perfect *Africa Unite*.

Brin and Kitou hustled him out of the room.

Kitou had begged bolts of cloth, and with casual deftness, she pinned up two cocktail dresses - one in black for Brin, and a shiny gray one for herself, like they'd worn in Brixton.

"I'll have these sewn," she said. "There's a woman who promised to help me, but I have no idea how to make your combats."

Box said, "I'd like a party frock please, if you can do one in green?"

She took off her clothes and looked at herself in the mirror. She was never a narcissist, but she knew she looked good. Where she'd been scrawny, now she was sleek. Full breasts, as before, with coral pink nipples. Willowy arms - no amount of training would ever change that. Curved hips and bum, a woman's body - a tuft of strawberry down, and her legs, always her best feature, now become long levers of muscle. A spray of freckles at her cheeks, like the hint of a nebula.

Brin came and stood beside her, and by God, apart from a few well-earned creases, she held her own.

In just a few minutes, Kitou had a third cocktail dress in sea green and disappeared to find her seamstress.

Box was as excited as a teenage girl before a party.

The dress was perfect. Kitou's wetware and the Horu clothier - Box made a mental note to thank her - had combined to make a simple but glamorous number, which clung to her body like water. She'd never have been able to afford one like it on a *professeur's* salary on Earth.

She felt *sexy*.

The dinner was like the feast laid on for them at Stratego. There was music, then the boy played his reggae, and there was a roaring

debate about whether it was nearer to Horu or Blue music. Then, Ito and Marius agreed to perform an exhibition. As in Stratego, a circle was made, and the crowd cheered the men as they stripped to their trousers.

It was like Jaasper and Pax - a bravura display, but these were different fighters. Marius was a young brute, probing and fast. What he lacked in skills he made up for with athleticism. Ito was graceful, unhurried. Box was amazed how completely he practiced his art. At the end of every lunge by Marius, Ito was in a perfect position to reply.

It was like Kitou against the youths in the barfight on Avalon, except here the Po art had a physical effect. Space seemed to bend all around them.

In the end, Marius held up his hands and laughed.

"Who will ever beat you, prophet?"

"When I was small," said Kitou, "I used to imagine Pax was made of steel, and Ito from air. It was the way they carried themselves. Pax is so unbending, and Ito so subtle. But that wasn't fair on either of them. Pax is full of love, and Ito's as tough as you can imagine.

"When I was a novice, all the others were so jealous of me, to have the great Pax Lo as my teacher. Did Pax ever tell you about his mandala?"

"No."

"A temple of the mind - four pillars, open to the sky. The pillars are love, honor, duty, and peace. Pax says that between those four walls, his soul will always be safe."

"That's beautiful," said Box.

Kitou nodded. "One thing soldiers struggle to find is peace, Dr. Box. We do rough work sometimes. Rarely intentionally bad, but expedient.

Rough on the people who do it.

Rough on the people who have it done to them.

Ito, I fear he's not right in this place."

There, it was out. Box had been waiting for someone to say it out loud. Ito seemed... lost. They all felt the same - Kitou was right - it was this place. But Ito had been here the longest.

"Who won?" she said, changing gears.

"Between Ito and Pax?" said Kitou. "Who wins between a shark and a bear?"

"The bear?"

Kitou laughed.

"They're about equal in the contests I've seen," she said.

"But you hurt Pax, that day."

"Oh, he was fighting down to my level."

"What, he let you win?"

"No, it's not as simple as that. He allowed the possibility."

"By injuring you?"

"Pax's skills are as far above mine as the sky from the ground. He only wanted to test me."

"What about when he fought Jaasper?"

"What about it?"

"Was he fighting to Jaasper's level?"

"That was an exhibition, done to please the crowd. My heart was beating double. A real fight wouldn't be nearly as pretty."

"Who'd win?"

"Between those two? The shark."

Box smiled. "And what about you?"

"What about me?"

"Jaasper couldn't get near you, in Stratego."

"No, he couldn't. It was the old song. He sang it for me. It filled me with power and grace."

"His song helped you fight?"

"There's only one song."

"I heard it too."

"Really? That's strange."

"Why?"

"Of all the humans, Dr. Box only your people - the people of Earth - aren't descended from mine."

"Why not?"

"You evolved separately, on Earth."

"How is that even possible?"

"You should ask the Xap about that."

"Are you saying they... what? Bred us?"

Kitou shrugged. "They allowed the possibility."

Box frowned. "I'm not sure how I feel about that."

"You should be pleased. The Xap are a wise and powerful people."

"Your gods?"

"Our symbionts."

Box smiled. "You know, it's funny."

"What is?"

"It's us who think we cultivate the trees."

Kitou laughed. "Yes," she said. "That *is* funny."

"How's your undine?"

Kitou opened her hand. "Oh, he's coming along fine."

Blue music was like the music of Stratego, hypnotic and strange. Horu music was tuneful, and people got up and danced. Brin let down her fresh crop of hair and danced with men and women alike. The older Blue and Horu told stories. A drunken Amelia Chance explained how bone breastplates were made. Box had wondered about that. What animal has such flat bones?

Dinosaurs, or local analogs.

A Blue man in his cups explained more, describing how once the huge creatures had been set on the zooms, in a famous victory. This was followed by a lengthy discussion of strategy.

Box remained sober. She had plans for the night.

Ito asked her to dance, and she saw the sly smiles from other couples on the dance floor. He took her to his rooms, which she'd never seen. They were monastically plain, dominated by a single artwork - a violet sun in an indigo sky that almost covered a wall. It was daubed and scraped straight onto the stone with mud from the river that ran through the Spinifex Reach.

"Did you paint that?" she asked.

He nodded.

His fingers found her lips.

"Would you like me to make love to you?" he said.

She breathed.

"Is it... appropriate now?"

"I think it's time."

His mouth found hers, and then her breasts, and then between her legs, drinking her in. At first he was slow, playing her rhythms - the small, involuntary sounds she made - then fast, until she burst in a hot rush of pleasure.

She was never a noisy lover, but she couldn't help but moan and laugh out loud.

A half-hour later, they did it all over again.

That night, Box dreamed. She saw herself in the charnel field of her Threnody dreams, except it wasn't. Instead of a carpet of broiled bones, there was deep snow, and instead of a bloodstained sky, there was the starless Möbius night. What light there was seemed to glow from the ground.

As in her dreams, she was riding her white stallion, and Kitou was across the field, dressed in her training gear, picking her way over the pillowy drifts.

"Kitou," she called, and the girl waved.

Box was wearing Blue finery, like she'd seen some of the women wearing at dinner that evening. Her hair was braided in beads. She wore a carved bone breastplate and carried a bone shield, painted with the sigil of a bear. Over her shoulders was a heavy cloak. For the first time in her life, she felt powerful, and not to be fucked with.

Kitou made her way over and grinned.

"My God," said Box. "Kitou, you have no shoes."

Her voice surprised her. It came as if through a tunnel and rang with crisp authority.

"No," said Kitou, "but I'm not cold." She opened her hand, and the clearing was washed with silvery light. A salt breeze lifted the leaves on the trees.

"Still," said Box. "Climb up here and get inside this fur coat with me."

"Look," said Kitou. Through the trees was an answering light, a pale blue glow, flickering among the dark trunks of the trees. Soon a Blue riding party appeared - four riders; a woman, three children, dressed for deep winter. Inevitably, more riders emerged from the trees in an encircling maneuver. Box opened her cloak, to show she carried no weapons. Whoever she was in this dream, she knew what she was doing.

"Are you the Red Lady?" asked the Blue leader, a dark-haired, green-eyed young woman, her lips white with the cold. She wore the eye-shaped sigils of a Blue ranger, painted in Cerulean wax on her breastplate and cheeks.

"And you must be the Goddess," she said to Kitou, who had leaped up behind Box, and was now snuggled in Box's cloak.

"I'm Kitou," said Kitou.

"I'm honored to meet you," she said, touching her reins to her lips. "I'm Iris, and we're outriders of the Eagle. You're far from home."

"How far?" asked Box.

"Maybe four hundred kilometers anti-Spinwards," she said. "And halfway to Farside, headed west, although in this place, who can be sure?"

"What place?"

"We call it the Dreaming. This is the forest of Dreams. We're on a pilgrimage to make an offering."

"An offering to who?"

"To the Ghost."

Soon they were snaking through sparse woods, with springy saplings spearing up through the snow, making it hard to progress. Up close, the trees were stringy-barked like eucalypts. Iris's horse seemed to know where to go, and she twisted around to face the riders behind her.

"I'd offer you my blade," she said, "but I have none."

"You're unarmed?" said Box.

"Yes, he likes it that way. We'd never find him, otherwise."

"Who's this Ghost?"

"A wise spirit of our Dreaming."

"Is this a dream?"

"We all enter this place through our dreams."

"It feels real enough," said Box. Seabiscuit's hooves plunged through deep encrusted snow, and her breath hung in clouds in the supercooled air. It felt below zero. Kitou clung to her back, a parcel of warmth.

"It *is* real," said Iris.

The three smaller girls were called Mia, Emma, and Mae - sisters, or cousins, swaddled in furs. They seemed awestruck, especially by Kitou. The outriders shimmered and glowed in the distance. Box counted twenty. They all appeared to be children.

Iris explained that they were riding out on a pilgrimage, and it was the girls' coming of age. They'd arrived here by rising in the dead of night, while everyone slept, then traveling west for what felt like a day, except it was never quite day in this place. The nearest it got was a wintry gloaming, which alerted them they were high on the helix, near their destination. Any darker and it'd be Farside, where they didn't want to be.

And now they were riding with the Red Lady, and Kitou, the Goddess.

Such days.

They reached a glade, which sloped down to where a brook trickled through ice worn into fantastical shapes. It was a fairyland scene, like a dream of a dream, with snowdrifts swimming by the light of the young riders who entered the glade from every direction. In the glade was a house, small but tidily kept. By the house was a fire, lapping hotly at the air, and by the fire was a machine.

It was the robot, Chance.

14 → HOT PURSUIT

2065

A symphony, a wheel, a long, slow dance of gravity, a moment of annihilation - war in space was all those things. Sometimes a battle might last for an instant, as two sides meet at processor speed. Sometimes, a skirmish might drag on for lifetimes, draining the will of civilizations.

The first the Water Bear knew of an attack was when Flux was destroyed. First, the bird-shaped Horu ship was there, and then a cone of ionized gas, expanding in the direction of travel of the weapon that killed him. It was only an accident of timing - a glitch in the synchronization of faraway objects - that saved the Water Bear and Bat from a similar fate.

These were ultimate physics weapons - lumps of degenerate metal, accelerated to near-lightspeed, delivered through a spacetime anomaly. The Horu were known for throwing rocks from a distance. It was their *modus operandi* in the war with the Pursang. It was how Threnody was destroyed. This was different. It was *precise*, and it was close. What human strategists feared most was ordinary matter, traveling at relativistic speed, manifesting millimeters from a target. Who could defend against that?

Flux might've seen it coming, but he couldn't avoid it.
Undodgeable bullets.

The Water Bear's consciousness was a community of individuals. One was her defensive monitor, a fast software agent that watched the world in Planck time. His role was to protect the ship from surprises, and he reacted instantly. Jaasper Huw was asleep in his nest. Pax was a meter from an interior wall. The limits of their fleshy

vulnerabilities defined the frame he accelerated through, while simultaneously requesting a warp drive.

All he needed was time for energy to suffuse the ship's manifolds, and they'd be gone.

A few milliseconds.

Acceleration in a straight line was like falling, since gravity acts on all masses equally, but this was brutal evasion. Tidal forces pressed Jaasper into his bed, pooling the blood in the side of his head. Two seconds more would've killed him. Pax was driven through a crumple zone, but he was trained for it. A lifetime of martial arts had benefits other than fighting. With a brief exhalation, he passed out too.

It was during this long gulf of computational slomo that the ship herself became aware of events, her primary mind receiving information from a cloud of contributory brains. She was a Po warship, at the cutting edge of human capability. Still, she was lucky. She might've been first, or the weapon with her name on it might've come sooner.

Satisfied with how her monitor was handling defense, she wargamed responses. Another cloud of contributory brains came online. These were entangled quantum computers. They set about collapsing the local wavefunction, like a chess computer examining every possible move.

A microsecond later, she entered warp space.

A pellet of depleted uranium, meant to destroy her, grazed her departing warp bubble. The Bat wasn't so lucky, and was left tumbling in a spiraling cloud of glitter.

His parting words were *good hunting*.

Pax snapped awake in the Water Bear's gamespace, followed by Jaasper. There were no soft transitions. This was combat.

"Situation?" he asked.

"In hot pursuit," said the Water Bear.

She replayed the annihilation of Flux and the near destruction of the Bat. Viewports showed their attacker - a Horu ship, unfolding

space before the attack, opening like a hand, with fingertips touching its targets.

Elegant, thought Pax. And flamboyant. The gesture was meant to be menacing, like the hand of a demon, reaching in through a hole in the ether.

The Horu ship was large. The Bat and Water Bear were kilometers apart - Flux had been almost a thousand kilometers distant.

It was a trap they'd set, to lure a Horu ship to attack them.

A trap the Horu had sprung easily.

The Bat was fading away, powering down for the long wait. Flux was gone, forever.

Pax closed his eyes.

"Vitals?"

"You and Jaasper both have operable injuries."

"Repair us," he said.

"Starting neurosurgery... now."

"They took their best shot," said Jaasper.

"They succeeded," said Pax. "Now we make them pay."

The Water Bear had learned new things from this mission. Macro's time-traveling device had given her an insight into the problem of transwarp communications. No longer would her warp bubbles be pristine islands of silence. Now, at warp speed, she could see the Horu system in exquisite detail - Horax, the roiling black planet - clouds of gas and debris spreading from her companions - the ethereal ship, its hand closing. If her warp jumps before had been like falling, this was like flying.

She was the raptor - the hand was her prey.

In no sense did she feel outmatched.

She also possessed the first secrets of the Horu geometry drive. She couldn't conjure a Manifold yet, but she could follow another ship through one. She would follow this hand, wherever it took them.

She postponed her regrets, for the death of Flux and his cargo of souls, which was a loss beyond reckoning. She'd thought the

necropolis would be able to defend himself. Flux thought so too.

What had changed? Better weapons?

Or worse luck?

She set a subroutine to wargame those questions.

She thought of her own crew instead. Pax was alive. She didn't know how she'd get by without her Navigator. Jaasper Huw was a find - a streetfighter, delivered by the cosmos to where he was needed. She sensed his worldspirit, a fierce presence at the edges of her sensorium. She sent her love to her away team, and Ito, wherever they were fighting.

And Aloïs, the bravest. Where was he now? She could've reached out but chose not to. She didn't need to know. All her people were competent. More than competent. Exceptional.

They were her A-team.

The hand-shaped ship refolded.

She dove at warp speed into the stillpoint of its disappearance.

15 → HIGH ON THE HELIX

314

You've brought me children?" boomed the robot. He was the same clanking automaton as Box remembered from the Great Exhibition on Earth in 1851.

"We have," said Iris.

She ushered the three youngest girls to the edge of the fire, in front of their peers who gathered behind her. The first girl was Mae, who gave Chance a book.

"I made this, sir ghost," she said.

"Please call me Chance," he clanked. He winked. "But remember, it's a secret."

She nodded, looking serious.

"What's in it?" he said, examining the cover.

"It's a story of my house, from the start of time."

"A history of a place?" he asked. "A curved path through spacetime?"

She nodded.

"Well," said Chance. "I'll want to read that. What's your name?"

"Mae," she said.

"Now Mae, I owe you a story."

Each of the children, in turn, gave Chance a gift: a shaving kit; more books; a ham; a bag of hard candy; a jug of intoxicating tea, given by two boys, Lochlan and Cairn, who said their father had given him the same liquor twenty years before.

Chance said, "I remember. Would he be Rob?"

"Yes," said the boys.

"How big is he now?" he asked.

"As high as the sky," said one boy.

"Higher," said the other.

Chance clanked and beckoned the children closer.

"Can I tell you a story?" he asked.

They all nodded, vigorously.

"I can't hear you," he boomed.

"Yes!" they all shouted, and he turned to Kitou and Box, with the surprised attitude she remembered.

"And who would these be? Two faces I remember."

"It's me," said Box. "And Kitou."

"Mr. Chance, how in the world are you here?"

"In this magical forest of dreams?" he asked, gesturing around the glade. "Now that's a long story. Sit, sit all of you, and I'll tell you about it."

"A long time ago..." he began, and Box could see the children were rapt. They were sat in a semicircle, with the debutantes in front of the fire, and the robot Chance on the other side. Somehow, the cold had receded, and they were all sitting comfortably.

"...in a place called the Real, on a planet called Earth, I fought a mechanical spider."

On the word 'Real,' the children leaned forward. This was what they were waiting for - the rich oral storytelling of their people. Box loved this form of history, and she found herself wanting it too, with a rising sense of expectancy. This could be the foundational myth that explained her presence here.

On the word 'spider,' they blinked. Iris leaned forward, and Box realized this was new, that none of them had heard this before. Chance settled back and began to recite.

"Before time itself," said the Ghost, "in the year this world was created, the Red Lady and her shieldmaidens came to visit me, on Earth, with the Tuniit giant."

The children gasped. This was more beings of mythos than they'd ever heard in a story before, apart from obvious make-believe.

"They intended to fight me," he said.

The children gasped again. Some laughed at such obvious nonsense. Who would fight the Ghost? Who could fight a Tuniit giant?

"They brought a mechanical spider."

Now there was silence, apart from the crackle of flames.

"There was to be a great pugilistic contest, fought in a palace of glass. First, I had to fight a creature of pure Mathematics, and I won. It was called the St Austell Slinger, but it had a fatal weakness. I could reach through its heart and pull out its insides. And then I fought the mechanical spider, and I won again, although it was a close-run thing. I can still feel its titanium sting on my skin."

A boy raised his hand.

"Yes?"

"Is this a true story, sir?"

Chance nodded. "It is - we have witnesses here."

They all looked at Box and Kitou.

Box nodded. "It's true. We were there."

Now the audience were Chance's to do with as he pleased. There'd never been a story like this before, in all the history of the Ghost's stories. Usually, it was atomics and spaceships.

"After I vanquished the spider, the one-who-can't-be-defeated decided to try to kill me."

The audience gasped.

"Thankfully, she's not as smart as she's fast. She got the wrong man."

The audience laughed.

"But by goodness she was fast. The fastest thing I've seen, and I've lived for a very long time. My companion was a he-devil, a wise-guy, a silver-tongued scoundrel and slick with a knife, who would cut out your eyeballs before you could blink, but Kitou was so fast she got his knife from its sheath, hidden under his clothes, and had the point of it at his throat before he could see what had happened."

The children, who understood knives and swordcraft, were skeptical of this.

"No!" they cried.

There was a clamor of questions.

"Did he die?" one asked.

"No," said Chance. "The Tuniit giant intervened."

There was much consternation.

"You don't believe me?"

The children gleefully shook their heads.

"A demonstration?"

"Yes!"

"Who is the fastest among you?"

"Iris!" they cried.

"Iris?"

Their leader stepped forward. Iris was Brin's height - a dark-haired, green-eyed woman. These were considered otherworldly, Box had been told, like folk of the forest. Iris didn't look otherworldly. She looked every bit the resolute soldier.

She opened her cloak, with a smile.

"I have no knives," she said.

"Then I'll set you a test of brains," said Chance, "that has nothing to do with violence."

Chance raised his arms, and in a shower of glitter, he became the saturnine Horu that Box also remembered. The children laughed and applauded.

"I have a lot to explain," he whispered to Box, "and in due course, I will. Now it's showtime."

"We begin," he shouted.

Around the house was a circle of grass where the horses grazed. Beyond it was the pearly half-light of the Farside gloaming, and it was cold there, making the snow like powder. Sat on the snow was a paper fox - alert, looking craftily about, cheekily drawn with pointed ears and ridiculous fangs.

As soon as the children spied it, pointing and staring, it began to hop and dance from snowdrift to snowdrift.

"Catch it," said Chance.

Iris was first to react but soon floundered in the snow. Kitou, lighter and barefoot, did better, though she too was soon mired. While the two young women struggled, other cartoon animals began to appear: first a bear and a moose, then a wolf and her cubs. The

children shrieked with delight as the sky around the cylindrical moon became a circling rainbow. Box laughed too, unselfconsciously. If the gateway drug to the aspen forest had been an austere psychedelic, this was like a megadose of *Psilocybin* mushrooms.

"I've learnt a few things," smiled Chance.

The fox sprang over the clearing, straight through the fire, and with a gesture from Chance, the children darted off in pursuit of it.

"What is this place?" asked Box.

"It's your dreams," said the Ghost.

"It says Flipside's the human unconscious."

"Maybe he's right. Or maybe it isn't so simple."

"Is it true we're imagining this place?"

"That's always the case. That's how reality works."

"Why are you here, Mr. Chance, and a so-called ghost?"

"Not so-called at all. I am a ghost, by any reasonable definition."

"What do you mean?"

"I was the first of many failed downloads."

"Then who is Amelia Chance?"

"A descendant of my copy. One of my copies. Flux wanted me here. I was the last living Horu. We agreed that if the process failed, we'd keep trying, with me as the subject, until it succeeded."

"How many times?"

"Did we try? Maybe a lot. I do seem to be imprinted on this place."

"That's awful."

"No, it's not awful at all. I'm happy. Sometimes joyously so. Time doesn't pass as you think, and except for the Enemy, this place flourishes. I get to see it. One day it'll be a beautiful civilization."

Iris, Kitou, and the children were still pursuing the fox, which had doubled back over the snow, leaving them stranded. Iris and Kitou were on the far side of the clearing, helping each other, far from the point of the action.

"Such a small population," said Box, "from all those millions of downloads."

"Yes, it didn't work out as we planned. My people have suffered. But we're all here, in some way, I'm sure of it."

"More ghosts?"

"Maybe."

"And the Enemy?"

"Yes, I must tell you about him."

He started again, this time with the more somber tone of a lesson. "In the first blush of the new world, when the first Reanimated were dying by millions from the wintry conditions, a man came down from the helix. Not from the west, where we are now, but from the east.

"Those were terrible times. It was the season of Dark. People would arrive here then die, like infants left in the storms. There was no way to save them - no way to clear the dead from the countryside. But still they came. The Necropolii in the Real had no idea what was happening - the part of me that was here had no way of saying.

"The man was called Kronus. Such charisma he had. He told us he came from a kinder place, and he could lead the people to safety. Some followed. They were the ones we now call the Grays.

"He wasn't part of the world. And it wasn't just him. Stranger things followed. Fabulous creatures. Scraps of personality. He was a piece of a puzzle.

"Then a madness came over him. First, he seemed to collapse on himself. Then he had visions. Ecstatic visions. His mind threw off sparks like a short circuit. It was as if he became so highly charged he couldn't be contained. He ranted and raved like a madman.

"For the Horu who remained, it was seductive rhetoric. We were survivors. He was from outside the system. All we had to do was give ourselves over to a power greater than ourselves. Why suffer, when everything can be forgotten?

"I include myself in that group. I thought he was a savior, come to lead us to heaven. A preposterous thought, but I believed it.

"That was when we started fighting amongst ourselves. Some say Kronus's derangement was a response to our wars, that we made him mad. Some say he caused them. I think a worse evil smelt our fear, like blood in the water."

"Amelia Chance said something like that," said Box. "You think he's possessed?"

"I think he became intoxicated."

"By what?"

Chance shrugged. "By power? By the chance to see the world burn?"

"Like a river, desiring the end of its journey?"

"Yes, that's apt. Earth wisdom?"

"*Nietzsche*, a local savant:

"*Where we live, soon nobody will be able to exist.*"

She shivered.

"Where did this... evil come from?" she asked. "What is it? How did it get here?"

"To this place? Who knows? Maybe it was always here, and Kronus its vessel - a vain and vainglorious man, in the wrong place, in the wrong moment."

"A narcissist?"

"Yes."

"Was that when he became the Enemy?"

"No, that was later, when he returned with his armies and started to murder us."

The fox was caught by the smallest girl in the party, called Bae. She'd been watching from the clearing's edge, too small to run in snow as high as her chest, when it leaped in her arms. Chance told her she could keep it, provided she promised to take care of it, although it'd only be made of paper when she returned to the world. This was her coming of age, and she was delighted. Iris and Kitou accepted their defeat with laughing good grace, declaring Bae the fastest girl in the forest. Box thought that as a rite of passage, for everyone here, it was an enchantment.

The stories she'd tell, when she got back to Paris.

Then Iris asked if Kitou and Box would fight.

"It's our tradition," she said, although her meaning was clear. She wanted to see what the strangers were made of.

"We should," said Kitou.

"I don't know," said Box, fearing the inevitable.

"We're both about the same size," said Kitou. "Atomweights."

The children latched onto the word and began to repeat it.

"*Atomweights, atomweights.*" That settled it. They laid out a ring, using pieces of clothing - gloves, scarves, and hats.

"This is ridiculous," said Box. "I'm no match for Kitou."

"I can fight to your level," said Kitou, "and still give a good account of myself. We'll do the moves you know."

"Kickboxing?"

"We don't have gloves."

"Wait," said Chance, and disappeared into his house.

"Kickboxing is the martial art of Earth," said Kitou. "Dr. Box is the atomweight champion."

Box rolled her eyes but kept silent. Chance reappeared with two pairs of gloves - Yokohama Slim's red pair and the brown ones that Chance wore in London. They had the waxy patina of cherished leather.

At first, it went as expected. Kitou was too fast and too skilled. Box was a *kuruttsu*. She envied Kitou her ability to glide, like a surreal hovercraft over the ground. Then, for the first time in her life, Box began to fight in the moment. All sense of parry and riposte disappeared. Instead, there was one fluid movement. One moment, she was *in* the glade, and then she *was* the glade.

It wasn't just her fighting now. Her adamant friend had returned. She surprised Kitou with a fast punch-kick combination she didn't know she had. She landed a blow, and with a shared smile, they returned to the starting position.

Now they were equals. Box pressed the attack, again and again. Kitou danced away, their strengths overlapping. At first, it was a mixture of Box's Muay Thai and Po, except now she knew all the moves. They found themselves on the ground, and somehow Box knew Brazilian jiu-jitsu. She could feel Kitou's worldspirit, an ecstatic presence, reveling in the contest, but in this place, *her* place, she had its measure.

She felt the world open to her. She was *here*, and it sang with her existence.

This went on for at least half an hour. Longer than she would've thought possible. Longer than she could've fought in the Real. Then they collapsed, clinging onto each other, exhausted.

Iris stood up and said simply, "I've never seen anything like that. The prophecy is true. I'll follow you both, if you'll have me."

It was time to leave. There were plaintive farewells from the children. Chance had resumed his clanking appearance.

"Do you know I'm the only thing made of metal in this whole universe?" he said. "Maybe it's because I'm a dream."

Brin once said a similar thing.

We're figments.

"Maybe you're a part of the operating system," she said.

A member of the set of all sets that aren't members of themselves, said the ship.

Water Bear, said Box. *You're here?*

Only just, said the ship.

What happened back there, in the clearing? asked Box, still wired with adrenaline. *Whoever she is, she can fight.*

She's you, Dr. Box, said the ship. *Don't you see? You're coming into your powers.*

Ship are you alright? asked Kitou.

No, I'm fading.

Iris led them further into the forest. "I won't leave you here," she said, at a place where the shapes in the snow reminded Box of a more sinister dream. "You could both freeze to death, and I've no wish to find out what happens when you people die here. Come to the Eagle with us. From there, we'll take the centerline train."

"How far is it?"

"To our home? It's simply a matter of wakening. It might take a day, or two minutes. The path will unfold. The people will be excited to meet you."

"How far to ours?"

"Maybe four hundred kilometers, Spinwards."

"The train will take us there?"

"I'll take you."

Iris was right. One moment they were in an occult forest - then they were descending a regular forested trail. She felt as though she'd physically woken from a dream. Below them were welcoming lights, and below those, lights in a valley. It reminded Box of Atwusk'niges.

Kitou saw it too. She held Box tight and pointed.

"It's like home," she said.

"It *is* home," said Iris.

16 → NUBECULAE MAGELLANI

2065

Macro awoke in a rattling acceleration chair. It felt like the teeth were being shaken out of his head. The onetime shipyard module fell through clouds of incandescent gas, and world-sized water spheres, orbiting a dark star, visible only by its influence on the matter around it.

The star was called Nubeculae. The cloud was the Magellanic nebula.

The water spheres flowed, over millennia, into a celestial riverine system, like the lines traced by particles in a high-energy physics experiment. Inside the spheres were cities.

Waterfall, the homeworld of the Magellanic civilization.

During the crossing, the corals that lined the cochlear corridors had erupted throughout the vessel, leaving lush voids where the passengers' stasis fields had been. Filaments and stamens floated all around them. Nothing was left that resembled a machine.

He could almost believe it was a stage in the Tung reproductive cycle. It had that fecundity.

In the final approach, when the gravities crushed them in their seats, it rained - a crisp, pure rain, squeezed from the foliage by the deceleration. It made him want to laugh out loud. There was something about the crazed way the Tung did things that Macro was starting to enjoy. Of all the species he'd seen, the Tung had the least fucks to give.

Which made their current state of mind even more worrying. They seemed almost *desperate* to get Aloïs and Macro to Waterfall.

Magellanic history was hidden in shadows. They were an aquatic civilization, but they'd never evolved into water breathers. They clove as strongly to the ideal of biological evolution as the Thousand Worlds did. They believed evolution had a direction.

Macro thought this was nonsense. Maybe not nonsense. Maybe semantics. Evolution was directed in the same way as gravity favors ellipses, or a quark favors spin. It was directed towards a balance of forces, by entropy.

But he wasn't here to compare philosophical ideas.

Another trait the Magellanics shared with the Thousand Worlds was that they were both highly effective warfighting civilizations. The Thousand Worlds pretended they weren't. They parked their Interdictors in deep space and equipped them with fast warp drives. The Magellanics made no bones about it. They were a civilization always at war, and the Pursang were an integral part of their war effort.

Why was a mystery. The Magellanics had warfighting robots, and nanobiotics, and none of the Thousand Worlds shyness about using them.

Why did they need the best human ground fighters? Why did they need soldiers at all?

His wetware wasn't much help.

I don't know, said his twin. I can only speculate.

How do they compare to the Po?

The Pursang holy warriors are a class above the regular Po military. On a par with the Po special forces.

Like the Water Bear?

Yes. That'd be a good contest.

Except the Pursang use weapons.

They do.

This was red meat for a boy brought up among budding game theorists. The Po had fought the Pursang hundreds of times, in the battlefields of his imagination.

So, the Pursang would win?

In a fight between handwaving monks and battlefield atomics?

Yes, but the Po wouldn't fight that way.

How would they fight?
Deceptively.
Wetware v worldspirits?
Wouldn't happen.
What if someone had both?
That'd be nice.

They splashed into an obsidian sea, with the lightshow of the Magellani nebula all around them. Macro felt the disorienting pull of a tractor effect, as they descended into an undersea world.

Aloïs awoke as they were decelerating downwards.

"Did I sleep the whole way?" he asked.

"We both did," said Macro.

They sank through a factory zone, into a shipyard that plunged for kilometers into the depths. The flickering shadows of the surface were replaced by glaring industrial lights. Finally, they settled in the chrysalis gloom of a dock. There were other Tung modules here.

Macro knew this was the mesopelagic, or twilight zone of this particular sea. Below was the bathypelagic, and then the abyss.

"Take one of these," waved a Tung. It was the Tung from the discothèque. Pfft. It expelled a cloud of pills, which tumbled in the local microgravity. Macro and Aloïs had disentangled themselves from their acceleration chairs. The lights had already gone out - their way was lit by the bioluminescent glow of the corals.

"What are they?" asked Macro.

"Ozenges."

"An air supply?"

"Enough for an Avalon day."

"What if I forget?"

Pfft. gave an impression of drowning.

The Tung in water were graceful, flashing creatures, soaring on slippery wings. There was no sense of the phage-like ghastliness they exhibited in air. Macro followed them in spurts of propulsion, squeezed from his vestments. Aloïs followed behind in a conventional skinsuit. Spiraling currents carried them between sprawling industrial precincts.

They weren't alone in the water. Every kind of sea creature was here. The Magellanic preoccupation with natural evolution seemingly didn't extend to body enhancements. Many of the sea creatures were cyborgs.

Some of them are sentient, his wetware said. *Some are the local biome.*

How can I tell? he asked.

As if on cue, a creature devoured another.

He hoped the eaten was biome.

The bar called the Bathyscape's Rest was shaped like a cave, lit by the photoemitting creatures inside it. Shadows rippled and loomed as patrons got in and out of the water, almost all of them naked. Public copulation seemed to be acceptable here, including between species. The otherwise androgynous Magellanics exhibited every combination of primary sex characteristic, from female to male, to ungendered, and in-between. Aloïs, unconcerned, restored his lounge suit to its original, rumpled appearance. Macro, seasoned by his Avalonian experiences, bought beers. These were deliciously cool in the syrupy air of the bar. After a few beers, he began to be drunk. He had no idea what he should do, so he did nothing.

The Tung seemed more relaxed now. Perhaps they'd had cabin fever. Too long in space. Too much deceleration.

A Magellanic androgyne leaned in towards him. "You're new here," ze said. Ze had the celestial body art of a fully-fledged Magellanic navigator. By the look of the designs on hir skin, ze'd been everywhere.

"I am, zir," said Macro, careful to use the ungendered honorific.

"It's a strange place," ze said.

"It's me who's the stranger," said Macro.

The Magellanic chuckled, hir wry tone signaling hir obvious physical maleness. "It's hard to say who's who," ze said, "in a strange place as this. My name is Sybil. Forgive me this trespass." Macro felt a delicate touch on his wetware, and a transfer of information.

His firewalls, being bypassed.

Aloïs leaned in. "We're about to be taken," he said.

The music stopped playing. Tung flopped and skittered into pools. Macro's military wetware snapped into place. His intoxication disappeared. He felt his perimeter defense systems powering up - a gritty mechanical whine.

Then power down again.

Oh shit, he thought.

"Don't worry," said Aloïs. "Finish your beer.

"Oh, and please ignore what they say about me. It's all lies."

Aloïs's lounge suit powered up and enclosed him in a field of pearly energy.

"Sorry," said Aloïs, "but they mean to hurt me."

Then the waters erupted, and slug-like creatures dragged Aloïs under the surface.

More slugs emerged and took Macro.

A different industrial seascape unfolded around him, seen through a curtain of foam from the drives of the creatures. Autonomous manufactories, lit by sodium lights, and the looming shapes of insectile warships. No signs of life. A military precinct.

They sank deeper, until all sense of depth disappeared. Here it was stygian night, except for the lights of their passing.

His captors were cyborgs, as far as he could tell, with human features engineered into streamlined machines. Their police identification flashed in their carapaces and was broadcast into the water around them. That was all he could hear - the bleep-bleep-bleep of their sirens, and the whine of their propulsion systems. Who would choose to live like that, he wondered? Like being a monk, he supposed. That's if they were cyborgs at all. They could be machines.

Or like some tragic merger of both.

Aloïs was being pulled, unresisting, behind them, still inside his pearly-gray field. They passed through a series of gates, until they reached a complex of spherical cages, set each inside the next, like worlds within worlds. Aloïs and he were put in the innermost cages.

A voice read out charges. Although his Broca was down, Macro knew some of the words.

*Pathogen, it said, in a language not unlike Pursang.
Bioweapon. Disease.*

He realized he wouldn't be the first to die in this place. There was a smell of death in the water, like the smell of burnt meat in the air of a crematory. This was the local equivalent of the Avalon interview room, except this place was real - a drowning pit for enemies of the state. Here, they really did kill strangers.

This was what he'd signed up for, although he never imagined it ending this way. His fathers had warned him there'd be risks. He'd waved them aside. Death wasn't something that happened to him.

Now, he was frightened.

Aloïs was in the innermost cage, unconscious, brutally beaten. Macro wasn't squeamish, but he found it difficult to look, then harder to look away. They'd gotten through Aloïs's fields, eventually. Macro was left free to float in the cell outside Aloïs's.

Cells within cells. Spheres within spheres. What was this place?

Why did the Tung abandon him, having gone to so much trouble getting him here?

Who was the tattooed Navigator?

He examined the spheres. Eight, maybe ten, it was hard to be sure. He tried pulling the bars. No movement. He tried squeezing between them. No way. How did they open? He searched for a lock and found nothing.

Why had they put him here? Aloïs was their obvious prey. Macro was likely collateral damage. A stranger in the wrong place. Did they know who he was? Probably not.

Now Aloïs was floating painlessly, his personality elsewhere.

Hopefully backed up in his lounge suit.

Or streamed somewhere else.

Where were his oxygen pills? One Avalon day, Pfft. had said. How long had he been here?

Fear was giving way to confusion.

And gnawingly, the immanence of drowning.

His wetware whirred into life, just as he felt the first itch in his larynx. At first, there was a feeling of spaciousness, like a sixth sense coming online, then malware intervened.

I'm Sybil, it said.

It spoke Ruby, the finger-tapping language of the Bank - efficient, compressed. An old voice-interlocutor, the most primitive type of AI, a pattern, pretending to be a mind.

Do I know you? he signed.

You met my original, in the {undecipherable}, it said.

Macro switched from one-handed signing.

In the bar? Why are you here?

I'm sorry.

You're an apology?

No, I'm here to prepare you.

Prepare me for what?

For dying.

He'd watched Aloïs die. First, the return of the instinct to breathe. Then a convulsion, then peace.

It'd seemed easy.

*It **is** easy*, said Sybil.

Where's the rest of my wetware? asked Macro.

There's not enough space for it here.

Where?

There - a tickle.

How do you feel? asked Sybil.

About what?

About dying?

What are you, a voyeur?

A witness.

Macro had rarely thought about dying, as an event in an unimaginably distant future. He knew it might be unpleasant. Death itself was neutral. He didn't care about death. He cared about dying in agony. He cared about fucking this mission.

He cared about the things he'd never get to see.

I think dying is holy, said Sybil.

Good for you, said Macro.

Why don't you help me? he asked. He wanted to scream, but there was nothing to scream with. Instead, he finger-tapped, tersely.

I can't, said the voice-interlocutor. *I'm only a number.*

In the event, dying was simple. First, his will to breathe returned, then a gasp, and a moment of panic, then choking, then silence. His fingers tore at the bars, but it had nothing to do with him. They were somebody's fingers. He slipped into an enveloping darkness, like into the arms of his lover.

He was a dot, in a cathode ray tube, fading to gray.

Then he was nothing.

Later - he had no way of knowing when, or how it happened - he *became* the voice-interlocutor. There was no need for tapping. They were the same. They were floating in space. Real, empty vacuum. Presumably falling between worlds in the Waterfall system.

I thought you might like to see this, it said.

For the first time in his life, Macro could see the whole Milky Way galaxy - a blazing corona of stars.

I've enhanced it for you, said Sybil.

I'm no longer a member of that set, Macro thought.

He had no feelings. No anger, no pain. He was executing only a minuscule set of instructions.

I could live like this, he decided.

How long have I been dead? he asked.

Too long. There'll be some damage, but we'll repair you.

Aloïs?

He has retreated into his alien sensorium, said Sybil.

Macro had so many questions. Why did his eyes still work? Did sensation really persist this long after death? His thoughts were like moths, around a guttering flame.

Why did those slugs hurt Aloïs? he asked.

We were protecting our people.

From what?

From Kronus.

Who's Kronus?

The next Macro knew, he was in the grip of a seizure. Water was gushing out of his lungs. He was sure he was drowning.

Then he was naked, in a modern, well-equipped medibay. A screenwall looked over a reef. Fish shaped like amoebae darted through coral like lung tissue. He was being prodded by androgyne humans. He caught a glimpse of himself in a mirror. He looked alright. Fine, considering. No visible injuries.

Except for his fingers.

Alois looked like a crash victim.

"That *so* sucked," he complained. Something inside him was already softening the experience, like the brainware that wipes away dreams, so we don't mistake them for reality. It occurred to him he was in breathable air. He took an exploratory breath. His Magellanic friend was here. The one with the tats, from the bar.

"I'm sorry," said Sybil.

The real Sybil, not the voice-interlocutor.

"Everyone's sorry," said Macro. "Thanks for the voice-interlocutor. Who's Kronus?"

"The source of the virus."

"I don't know what you're talking about," he said. "What virus?"

Sybil explained that the Thousand Worlds military strategy was gravely mistaken. That the real attack vector was the disease.

Macro was confused.

"*What* virus?" he asked.

He tried to understand. It felt like his mind was wrapped in bandages. Maybe it was. The medibay had said he had a brain injury, to be patient.

"Why am I here?" he asked.

He made an exploratory poke at his wetware. Nothing.

"Why all the mystery?" he asked, suddenly angry. "Where is my wetware?"

"There's someone I want you to meet," said Sybil.

He was Macro's age, maybe a year or two older. Macro's height, maybe five kilos heavier. A lithe athlete. Dressed in the dusty combat nano of a Magellanic commando. A Pursang holy warrior.

"I'm impressed," said Macro. This was his first holy warrior. Maybe he should be kneeling.

The young Pursang easily fit the mythology. He had lurid scars, presumably earned in battle. One curled over his arm and disappeared into his dreadlocks. He must've nearly been cut in two.

He looked as though he hadn't slept in days.

He shook his head. "No, it's you who should be praised. My name is Totoro."

"I've heard that before."

"Yes, I'm my sister's brother."

17 → RESPIT'S BRAIN

314

Brin rose, her eyes adjusting to the light. The pearly semidarkness of the Möbius night was squeezing past the ends of the drapes. A handful of photons. Sufficient.

Dr. Box and Kitou's beds were empty.

Unusual, but not alarming.

Quiet as a cat in search of her dinner, she flowed through the Chancery halls. A few insomniacs were about. She avoided them. There was a clatter from the kitchens, and she smelt the malty aroma of the day's first batch of bread. She was hungry, but that could come later.

Marius had slipped away an hour before, with the touch of his lips still on her breasts. He'd been good, in bed, for a man. Better than good. The Blue people had the opposite of the Po attitude to intrateam fucking. They believed it bound them together. Brin was ambivalent. She was happy to not sleep with her Po teammates, even if she sometimes wanted to do only that. Sleeping with the amorous Blue came naturally.

The saturnine Horu were less to her taste, although she'd had pleasure there too.

Today she was Marius's.

Outside, two faces waited, bobbing like moons in the shadows. They'd dressed all in gray, as they'd been asked to. They began to rub dirt into their hands, but she stopped them. She showed them a gesture for talking. They nodded and led her away from the house.

They weren't much to look at, these two children, but the ship assured her they were talented. The girl was called Nim. She was a

psi. The hulking youth was a delver, who could find his way through the world's interstices, and was reputedly good with his fists. He was called Respit. They were Kitou's age or younger. That was fine. Brin had no use for soldiers. This place was awash with boneheaded sword wielders.

She wanted *operatives*.

They made their way through the lodges and hedgerows that made up the Chancery yards. Brin stopped them partway and showed them how to flow through the twilight. Deft, catlike movements. Like water, she motioned. They understood perfectly. Soon they were nearly as stealthy as she was. She was impressed.

Then they were beside a creek, far from the house, in the entrance of a cave, and quickly inside it, where Respit rubbed Cerulean wax on a rockface to make a faint glow.

The cave seemed to stretch, in every direction.

"Are we out of earshot?" she asked.

"No earshot," said Respit.

"What he means," said Nim, "is we're not really here."

Brin nodded. This was exactly the strangeness she wanted.

They were here to take her to the Enemy.

Respit and Nim were adepts of the Horu shamanic class. They believed their abilities descended from the Unanimated, the numberless dead that never made it to Flipside, and Brin had no reason to doubt it. She knew from the Water Bear's mission profile that they were both highly intelligent, and suffered in varying degrees from autism-spectrum neurodevelopmental disorders. The Horu considered those traits valuable, so these two young savants bore no hint of any social stigma.

They were complex, and probably difficult.

Brin motioned them together.

"I won't try to mislead you," she said. "My ship says you're at least as smart as I am." Taking some chalk she'd brought for the purpose - she drew a rudimentary wolf.

"Excuse my drawing skills. What's that?"

"*Canis lupus*," said Nim.

"What does it mean? Respit?"

The big youth colored in the dim light.

"You?"

"That's right. Now, draw me two more."

She handed the chalk to Nim, who started to draw. Brin stopped her. "No, not cubs. Full-sized animals. I'm your wolf-mother prophecy - you're my personal army. Now draw."

Nim drew a sinuous creature, prowling under a moon and stars the girl had never seen. It captured the gist of the girl in a few strokes - a sly wit, and a vulpine intelligence. It was far better than Brin's rude sketch. Respit's was childish - a sticklike dog with fangs and flattened-back ears.

Brin waited until they finished.

"Why are we doing this?"

"Stories," said Nim.

"That's right. Stories have power. The Blue people may have sung this place into existence, but it's your invention."

In her forthright way, which left no room for lies or exaggeration, she told them the story of their people. Maybe they'd already heard it. It didn't matter. She wanted them to know that she knew and respected them for it.

"You're the oldest human civilization," she said. "You may not be fearless warriors, but you're subtle and clever. If you'll be my followers, then my powers are yours to command."

"I'm a *ferocious* warrior."

"In return, you become mine."

"By believing it, we make it happen."

Brin, this is excellent work, said the ship.

"Now we draw blood."

What?

She lifted her sword, and cut her arm, and drained her blood onto the wall. "Now you," she said. With two gasps of pain, they mixed their blood with hers.

"Listen," she said. "I'm not just the embodiment of your warrior myth. I'm a soldier in the Hand of the Po, the most feared black ops regiment in human history. Think about that."

"I'm here on a mission.

"A mission to save *everything*.

"You know the story by now. Look in my memories. No doubt you've heard it discussed. You know I have nothing to hide.

"I'm deputizing you. Our mission today is to observe. Do you understand?"

She wiped a hand through the blood and smeared it on her face. They did the same.

"Very good. Now you're Po soldiers. You fight for a just society. You work for me. Ship, you had something to say?"

Not me.

"Good. Then let's get out of here."

This cave was a backdoor, Nim explained. There were many backdoors. Some were deliberate exploits. Some were programming errors. Respit knew how to find them. Through Respit's eyes, Brin saw the code. What impressed her most was its simplicity. The Möbius machine was a massively parallel, quantum computer. It really was a piece of work. Unlike the Real, whose nature was stored in particles whose quantum wavefunction had collapsed into one state, the qubits that made up Möbius space were entangled in every possible state, and thus encompassed every possibility.

How to bootstrap a universe.

The corridor branched and branched again - slowly the walls took on a fibrous appearance. The structures reminded her of brain cells. Soon she was sure they were neurons. She cursed her imagination.

"Respit's brain," said Nim.

"What?"

"Models," said Nim. "He builds this place on his own neural structures. Like a homunculus. Do you know what that is? Don't let it get to you. It's only a metaphor."

"Get to me, how?"

"It can get a bit hallucinatory."

Her mission had started the previous night, with an argument. Marius said something she disliked, so she put him in an armbar till

he revised his opinions.

"Brin," he said, with difficulty. "What witchcraft is this? Release me." It hurt - he couldn't quite breathe, and every part of her was out of reach of any useful appendage.

"Brin, please," he insisted.

Yes, said the ship. Brin, please think about this.

You take this idea seriously?

I share his concerns.

Brin waited a while, then used the spiral energy stored in his torso to spring back out of his reach.

"The Fibonacci art," he complained, rubbing his shoulder.

"The Lo style," she said.

This was followed by sex. It was inevitable. Brin preferred women, but like most soldiers, she took her pleasure where she could find it. The sex was athletic. More than athletic. Electric. At one point she considered reapplying the armbar, which would be even more pleasing, naked, and slathered with fluids.

After Marius recovered his breath, she said, "So you say Ito's been turned?"

"Not turned," he said. "Infected."

"By what?"

"The Numbers."

The Thespian virus, said the ship. The Blue people call it the Numbers disease.

Why are you even here?

I've just returned, said the ship, primly.

Marius told her that Ito was infected with a neurological virus. It was a degenerative state, brought on by exposure to the Enemy. Death followed madness. Brin's mind churned with the implications.

A Po First, crippled by malware.

"Okay," she said, after Marius had finished making his point.

"What are we going to do about it?"

Now she was in Respit's brain, trying to get to the truth of the matter. The deeper they delved, the more bizarrely symbolic the neural cave system became.

Flashes of thought.

Squeezes like brain structures.

Forests of pseudounipolar cells, with their multiple axons, snaking into a looming middle-distance, and the nightmarish fried egg nuclei of soma.

Strange, hypnogogic sounds - crashes and snaps of conversation. A writhing, winding corridor of her bad dreams, like being trepanned by a snake.

Brin was nervous of snakes.

She got rid of the idea.

"This place is affecting me," she said.

"Breathe," said Nim.

"Airways," said Respit.

Yes, said the ship. *This place resembles the Blue riders' airwaves. Think about those for a while.*

"Better," she said.

Then she was hallucinating. She was flying. No, not flying - clinging to a mountaintop eyrie. The wind howled all around her. The summit was covered in lichen. Cyanobacteria. A landscape, far below. Hedgerows, rimed with frost. Bodies. A crow rose from a corpse and cawed into the distance.

Her corpse.

It was no part of this universe.

She felt Nim pull her into the moment. Her nightmares scuttled into the darkness.

She breathed.

Fascinating, said the ship. *We're traversing the laminar flows like the Blue people do.*

"Explain?"

It's like we're hitching a ride on a passing warp bubble.

"Between," said Nim.

"Inbetween," said Respit.

I believe they mean we're moving between two points without crossing the space in between, like a hyperluminal drive does, but they lack the math to express it.

"Like a wormhole?" asked Brin.

"What's a wormhole?" asked Respit.

Brin explained the physics of wormholes. It was like a light went on in the boy's face.

"Those physics exist?" asked Nim.

"Absolutely."

"I want to go there."

"Ship can make wormholes."

"Is this true?"

Yes, said the ship. *In several different ways.*

"You must be like a god."

I like to think so.

They reached a *place*.

The Room was a room, and they were its walls.

They were also *in* the Room. Like synesthesia, her sensorium trying to make sense of superimposed states. The walls were crumbling masonry, beneath an industrial sky, a yellowish gray, like infection.

Brin shook her head. Whatever this place was, it had nothing to do with regular sensory observations.

Inside the Room was a desk. At the desk was a Gray. He was better resolved than the Grays she'd seen in the camp. Instead of being made of mucus, he seemed to be stitched together from burn tissue. He was nervous, and bored. He shuffled papers on his desk. He appeared to be an officer. His belted gray tunic was pinned with iron paraphernalia.

So, this place has *metal*.

Brin filed the information away.

Are we still in the Möbius universe? she asked.

That train left a while ago, said the ship.

Nim motioned for them to be quiet.

Brin, I'm firewalling you now, said the ship.

She nodded. She was the delivery mechanism. The ship was delivered. Now to make herself scarce.

She wondered where her body was. In her bed? In a cave? A mote in Respit's brain? In orbit around Horax? A smear on an event

horizon somewhere?

Then she was in the ship's gamespace, with its wireframes and malleable time, hosted somewhere in her own *medulla oblongata*. The resolution wasn't what she was used to, but the ship's algorithms were impressive. She leaned down and inspected the Gray soldier. Up close, he was as grotesque as she expected him to be.

She zoomed in. He really was physically stitched together.
She could sense the Enemy, behind the sky.
She hoped he couldn't see her.

The Ride of the Spinifex Reach manifested in their usual way. At first, there was a sky, then a few scattered riders. Brin knew that for each one she could see, there'd be ten more she couldn't. She watched the Gray officer stiffen. Through her ship senses she could smell his fear. Sweat and endorphins. A head-up display showed her his vitals. He was responding as any human would to extreme danger.

The Blue riders rode down, taking their time. These were her friends - Markus, Velo, Viki, Alis, Ox, and Bravery. They carried themselves like imperious savages, a look she knew they cultivated.

To put the fear of death into their enemies.

Then Marius rode down, like the arrogant young warlord he was. He flowed from his saddle, into a chair. He allowed his blue-painted horse to wander away.

"I'm here for a parlay," he said.

The Gray officer was about Marius's size. He would've been handsome, but for his skin.

"Then talk."

"Not with you."

The man scowled. She saw his nerves give way to irritation. There was no flight response here. He'd conquered his fear. He was more afraid of disappointing his master.

This soldier would be a worthy opponent in battle.

"I speak for my people," he said.

Marius raised an eyebrow.

"Then who?"

"I require Kronus."

"Impossible," said the Gray soldier, reaching for his papers. Marius produced a knife and pinned the papers in place.

"Do as I say," he said. "Or I'll kill you."

The Enemy unfolded himself from the sky. He was a magnificent man, with a poet's huge and shaggy head, and a corpulent belly that only seemed to add to his magnificence. Brin was transfixed. She couldn't take her eyes off him.

Marius rose from his chair.

Kronus stared at the walls. "What a beautiful day," he said, spreading his arms. "Black," he said, "like ink."

It was true. The sky had gone black. It was like he'd sucked the light away. She could see the ship's telemeters read off the scale. The man's heart rate read zero. His temperature was tending to infinity.

"Who are you?" he asked, with the formaldehyde whiff of someone examining a specimen.

"I'm Marius, Ride of the Spinifex Reach."

"I don't know you," he said. "You seem young, for a Ride. Where's Bardo?"

"You killed him."

"I did?"

"Nine years ago. Kronus, you've met me before."

Kronus raised his hands, as if to say *young people, these days*.

Brin wanted to trust him.

She wanted it, badly.

The ship dropped a packet of encrypted information at her virtual feet. She opened it, carefully, like a child unfolding her lunch from a bag.

I'm getting a storm of exploits, it said. Primitive, but effective. Very like the Thespian virus. Marius seems to be immune - his neural architecture can't run the code. Your Horu telepaths are at risk. You may be feeling some backwash.

Backwash? What Brin was feeling was like waves of heat from a furnace.

Marius was a big man, but Kronus was massive, with forearms like hams, and a laborer's slack musculature. A beautiful man, for all his fatness and ruin. While Brin watched from the safety of the ship's gamespace, Marius told him everything. About Dr. Box. Fluxor, the Fa:ing, his fears about Ito. Everything except Brin's presence, which the ship had scrubbed from his consciousness, using a combination of opioids and hypnosis.

"There's no point in lying," he'd said, the night before. "As soon as you open your mouth, he knows what you're thinking."

"So why talk?"

"To exchange information. This isn't just a war to him," he'd said, "It's a game. He allows us to make moves, and to progress if they're good ones."

"Or he's insane and making it up to see you struggle."

"Yes, that's possible."

"You trust him not to just kill you?"

"He plays by the rules."

"What rules?"

"His."

"What do you want in return for this... *gossip?*" Kronus was saying.

"Your surrender."

Kronus made a show of considering the offer. He was a natural actor. He held up his hands. He shook his mane - like a beast, as though clearing his head of a nuisance.

"Your attack on Fluxor is thwarted," said Marius, unable to stay quiet. "All the powers of the Real are aligned against you. Don't you see? Let's have an end to the killing."

"What killing?" asked Kronus, cocking his head. "We have a *truce*."

"Because you imposed one."

"I was bored. You'd rather I didn't?"

"What does that have to do with your surrender?"

"You're right," said Kronus, his face suddenly lit up from within - by hilarity, rage, and surprise. "What does any of this have to do with *my* surrender?"

"You're twisting it round."

"I'm untwisting it."

"You can't win."

"But I *am*."

"How?"

Kronus laughed. "You expect me to *tell* you?"

Behind the ship's firewall, Brin was struggling with her own demons. Knowing the Thespian malware existed made her want to reach out and touch it. What would it feel like? Maybe she was already infected, maybe in Avalon.

You're not, said the ship. It's my responses you're feeling.

Are you safe?

My defenses are tested, but we're in no real danger.

"Then we're done," Marius said.

"We're done," said Kronus.

The torrent of exploits flicked off like a switch. All the emotions in Kronus's face disappeared, except sympathy. Kronus laid a hand on Marius's shoulder. "I understand your predicament," he said. "I do. But the way to my future goes through your people. There will be a reckoning, and I'll win."

"Why? Kronus, why does any of this have to happen?"

Kronus raised his face to the sky.

"Who am I to go against the word of God?"

"Oh," he said, as Marius remounted.

"One more thing. Tell your Po bitch I can see her."

19 → BUNJIL, THE EAGLE

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The town of Eagle clung to narrow *arêtes*, peering down through *couloirs* to the valley below. Dawn caught the mountains above, so they shone like gold in the thin sunlight. Long lines of people were moving by lamplight up a steep switchback trail, which led to the town from the valley.

"Something's happening," said Iris. "I must see to the children."

The Eagle was a high-mountain bolthole, its steep gabled roofs huddled against the rockface, its doglegged streets ending in nothing. Box had seen towns like it on Earth, in the Atlas Mountains and the high Andes. Unlike the mainly Horu Chancery, this was a Blue place. It was Blue ranchers who were coming up the trail, and the streets were busy with Blue riders.

Box and Kitou were given a whirlwind tour as Iris delivered her charges home. There were rumors of war. An army had stirred in the East. They'd reached the centerline train. They were in the valley below.

Iris rejected all of it. She said it'd take a Gray army two weeks to reach the trainline. More, if they hoped to keep the territory they won. More again before they were here. Still, Box could see the strain on the young woman's face. It was the same fear in the eyes of the families. They didn't want wars. The last generation of Blue riders had been slaughtered by Kronus.

While Iris was delivering Bae with her prize to her home down a dizzying lane, a Blue woman rode up. She carried herself like a warfighting general of Earth history newsreels - like a FARC revolutionary, rallying *Campesinos* in the defense of a Bolivarian hill

town. She wore her wars in the scars on her face - one eye covered by a patch, part of a cheekbone missing.

She was Helen, she said, Marshall of the West, and Ride of the high Eagle.

"Welcome to Eagle," she said. "I wish I could give you a better welcome. Unfortunate timing. You must be the Red Lady."

"Who runs this place?" asked Box.

The woman shrugged. "I do," she said. "That's what Ride means - in a war."

"Like - in a shooting war?"

The woman nodded.

"We have to get to the Chancery," Box said.

"Yes, I can help."

Iris emerged from the laneway, the children all safe in their homes. Helen threw her a spear with a flag, which Iris caught and fixed to her saddle.

"How far are the enemy?" asked Box.

"Hard to say," said Helen. "There are marginally credible reports of infantry already on the train. I don't believe any of it. Iris?"

"I have no better information."

"Anything from the Ghost?"

"Nothing."

"Unsurprising. Alright, to my office."

The Mousehole was a teahouse, leaning out over a tumbling moraine, deserted except for its Horu owner, who served them tea, then vanished.

"This is my headquarters," said Helen. "The walls are acceptably deaf. The owner's a spy but we both know it. Red Lady, we've been trying to contact the Chancery, but their psi is offline."

"Do they even know it's started?" asked Iris.

"I have no idea," said Helen. Box could get a better look at her now. She seemed to be in her sixties, the oldest Blue person Box had seen, apart from the venerable horseman. She was wearing a military greatcoat, with pieces sewn in where it'd been hacked into.

"What *do* we know?" asked Box.

"Kronus's armies are moving," shrugged Helen. "Maybe a half-million soldiers. Maybe half coming here, to this place."

"Why?"

"Because we intend to bring them. Eagle's a hard town to win unless they intend to blow us off the mountainside. We'll make our stand in these hills."

"Bring them, how?"

"They'll follow the path of least resistance. We'll resist least in the middle."

"How many people do you have?"

"I'm the Marshall of nine Rides. Five thousand riders. We have five thousand more who will fight."

"Ten thousand against half a million?"

"No, there are other Blue armies. We number fifty thousand in total."

"Ten to one?"

"About."

"How can you win?"

"In the mountains, we have the advantage. The enemy will struggle to see us. Do you see that cirque valley below? All bogs and lakes. Our scouts in the hills on all sides, killing and slipping away. We'll terrify them with our savagery. We need to invent a new type of fighting."

"It's already been invented. We call it an insurgency."

"You come from such a place?"

"Oh yes, we've pretty much been at it since we started."

The owner appeared with more tea. Helen waved him away.

"Look," said Box. "I'm a military historian. I know the exact tactics of the war you've described. I can help you."

"No," said Helen. "Your place isn't here. Iris?"

"I'd planned to take them on the train."

"Impossible now. The Gray aircraft will reach it before you, or worse, while you're on it. You'll take the high road. It's a bilaterally symmetrical, parabolic airwave. That means it runs high. Between five and ten thousand meters. You'll need suitable clothing, more than what you have now."

"Won't we get altitude sickness?" asked Box.

"Mountain sickness? No, not enough time, but it won't be easy."

She turned her attention to Kitou. "You, girl," she said. "You're barefoot."

Kitou tossed her head. Box could sense a new wildness in her. Box didn't mind it. It was a hot kind of madness. They were all going mad, in this place.

"I'm not your girl," said Kitou.

"Ah, a spirited one. Can it fight?"

"She can fight," said Iris.

"She still needs the correct footwear. Go and get it for her. Now."

"Kitou," said Box. "Are you alright?"

"It's begun, Dr. Box. I want to be part of it."

"You'll get your chance. We have to get back to Ito."

"I can fight with these people. I also understand the principles of asymmetric war."

"Kitou, no. I can't get through this without you."

Kitou at first looked defiant, then embarrassed. "I'm sorry. What was I thinking?"

"You want to fight?"

"Of course. These are my people."

"Then we'll stay," said Box, to see how it sounded.

"You'd fight with us?" said Helen.

"Of course," said Kitou, and the woman sighed.

"No," she said, not unkindly. "We're going to lose. Make no mistake. What use is one extra sword, and - with respect - a scholar of war? You two can only make a difference with the prophet."

"But," said Box and Kitou together.

"No, I forbid it."

After Iris left with Kitou to find winter gear, Box asked more questions. "You said a half-million Gray soldiers," she said, "Where are the rest?"

"Farside."

"Headed where?"

"To your universe, I think. That's what the spaceships are for."

"Spaceships? How many?"

Helen shrugged. "Thousands? More?"

"Horu myth foretells of a last battle. It's a story the Enemy likes. Those ships are part of it."

"Helen, what happens if all the worldsingers leave this place?"

"If we're all killed?" The older woman made a rough gesture, like a throat being cut. "Hopefully, we'll take Kronus with us," she said.

"Could you?" asked Box.

"Likely not. He means to destroy us, *and* he means to escape. The end of this world plays into the next one."

"So, by dying, you play into his hands?"

"By dying *quickly*, we play into his hands. What we can give *you* is time."

"Me?"

The innkeeper poured them tea.

"Don't you see it already?" said Helen. "You come from outside the system. You *will* find a way. Everything depends on it."

"The first step is bitey," said Iris.

"Don't worry," said Box. "We've done this kind of thing."

"Right," said Iris, and stepped out over two vertical kilometers of raspberry-colored granite. "This path is deceptively fast," she called over her shoulder. "Try to keep up. Don't stray to the sides or you'll finish up elsewhere."

Then they were in the grip of a rising column of air. Box could feel the alternative realities curling around her. It wasn't cold - not at first. The air from the valley was fifteen degrees warmer than in the town, but when the thermals ended, the temperature fell. Iris had kitted them out in coveralls made from skins lined with fur, with similar rugs for the horses. The suits were expertly made, but soon Box's exposed skin was stinging.

How cold is it here? she asked the ship, but she got no reply.

She's gone, said Kitou.

Our version, here?

I don't know, said Kitou.

The town fell away and was soon just a sprinkle of dots in the landscape. Now Box could see the size of the country. It was hard to

believe this was smaller than Earth. The sheer granite walls over Eagle were just the beginning. To the north and the south were rows of serrated giants, like the Karakorum on Earth, but taller. She began to discern the inwards curvature of the helix. It was like rising out of a bottle of mountains.

"Do you have mountains like this where you come from?" asked Iris.

"We have the Aø, the Curtain Wall on Fluxor," said Kitou. "But not like this."

"We've got a range almost this size," said Box. She told them about K2, and *Chomolungma* in the Himalayas, and the perils of the death zone. She described the problem of altitude acclimatization, which Iris and Kitou called mountain sickness. Iris was amazed that people on Earth climbed mountains for pleasure.

"It's like a religion," said Box.

"We'd never even think of doing that," said Iris.

"Are your mountains holy?" asked Kitou.

"No," said Iris, "they're deadly."

Soon they were soaring above high cirrus clouds, like wispy white streamers. Iris warned them of frostbite and showed them gussets in their hoods, that they could hide in. Soon just their eyes were exposed.

"Who'll tell us a story?" said Iris, her voice muffled by layers of animal skins.

"I will," said Box, her teeth beginning to chatter. It was beyond cold, and she was starting to hurt.

"It's about a bird spirit," she said.

"Is this a Horu myth?" asked Kitou.

"No, it's from Earth," said Box. "From the Kulin people of south-eastern Australia."

"Is that near Utah?" asked Kitou.

"Not so far," said Box. "Maybe twelve thousand kilometers, as the Bat flies."

"Bunjil was a bird spirit," she began. "The wedge-tailed eagle. He brings the cold weather."

"He must be flying around here," said Iris.

"Bunjil is also the star Altair, in the constellation of Aquila."

"Altair," said Kitou. "I've been there."

"Kitou, please shut up. Bunjil lived on Jaithmathangs, the tallest mountain in the Kulin nation. One day a Kulin hunter was camped on Jaithmathangs when a heavy snowfall came. He didn't have modern snowshoes, so he was trapped there."

"This is a parable about my footwear," laughed Kitou.

"Lost in the wilderness, and with no way home, he called on the bird spirit of the mountain to save him. He called out 'Bunjil', three times.

"Eventually, the air spirit Bunjil appeared in the sky."

"What was he like?" asked Iris.

"Like a monster," said Box. "The head of a goanna, and the wings and feet of an eagle."

"A chimera," said Kitou.

"Yes," said Box. "A cryptid."

"This being exists?" asked Iris.

"It's Australia," said Box. "An ancient, mysterious land. Anything's possible. Now, our Kulin hunter..."

"What's his name?" asked Kitou.

"Let's call him Claude," said Box.

"A good name," said Iris.

"It is. Claude made a sacrifice, to call Bunjil down from the air to help him. He burned his stocks of food until he ran low, and then he burnt more until nothing was left. When Bunjil finally landed, Claude said 'You're welcome here, friend, in my campsite.' And Bunjil said, 'It's lucky you called me friend. I'll take you inside the mountain, and you'll be happy with me there.'"

"They entered into a relationship?" asked Iris.

"Of sorts," said Box. "If Claude had called him anything else, he'd have been eaten."

"This *is* a metaphor," said Kitou.

"It is. Now listen.

"Life was good in the mountain. It wasn't made from cold stone, as you'd expect, but like a comfortable home. Bunjil had a wife and daughter, and he gave Claude his daughter."

"Gave?" asked Iris.

"He was a creator being. They were his humans. He possessed them."

"Like Kronus possesses the Grays?"

"Yes, like that. Claude grew increasingly comfortable. Bunjil brought him all the meat he could eat."

"Do you remember the deer we ate?" asked Kitou.

"I do," said Box. "Bunjil's meat was like that, hot from the fire, and delicious. Claude had a wife now, so he was happy that way too. In time, he became so comfortable that he became more like Bunjil, and Bunjil, learning his ways, became more like the human. Until one day, they changed places, and Claude, desiring his own freedom, set the man free."

"And that was how the Kulin warrior Claude escaped from the bird spirit of Jaithmathangs."

There was only the low keening of the wind.

"The man became the spirit?" asked Iris.

"They're the same," said Box.

"How can it be? Good and evil are opposites," said Iris.

"It's about moral ambiguity," said Box. "Who says Bunjil is evil?"

"Like Kronus?" asked Kitou.

"Kronus is evil," said Iris.

"Maybe he wasn't always," said Box.

"He is now."

"Kronus is trying to possess Ito," said Kitou.

"What?" asked Box.

"I know it. The others know it too."

"What do you mean, possess?"

"Like the Red Lady possesses you, but not so kindly. Like this Bunjil possessed his women."

"Kitou, how do you know this?"

"I see it."

Brin and her acolytes emerged from the neural darkness, as the Möbius night was starting to fall. Nim and Respit were spent. Brin took a breath of real air.

The first sign that something was wrong was when Nim fell to her knees.

"The Enemy," she said. "It's begun."

They made their way back to the Chancery. It was hard going. Nim was disoriented and kept stopping to vomit. The Chancery yard was deserted. Marius hadn't returned. Amelia Chance confronted Brin, and asked, "Where have you been?"

Brin said, "Where's everyone?"

Ito is firewalled, said the ship.

Kitou and Dr. Box?

No idea.

Firewalled? Who did that?

His autonomic defenses. I've put a second one on top.

"Brin," persisted Amelia.

"I'm sorry," said Brin. "But I'm busy."

Why? she asked, turning away from the politician.

He's entered a new stage of the disease.

Marius was right.

You already knew that.

Why now? It can't be a coincidence.

Ito looked like himself, except he was in the shadows, watching. That was most unlike Ito. Brin saw how it fit the pattern of his recent behavior. A silent withdrawal. Why hadn't she noticed?

Options? she asked the ship.

Wait for the others. I have a fix on them now. They seem to be flying.

In the airwaves?

Unless they've found a light aircraft.

Where have they been?

We'll find out.

Brin took a breath and turned to face Amelia.

"I'm sorry," she said. "You're about to be in a war. Nim will brief you."

Box, Kitou, and Iris landed at a canter, on a high moor overlooking the wild river valley of the Spinifex Reach. The Chancery

was a smudge of woodsmoke and welcoming lights in the distance. Kitou had to hold Box, to stop her from falling. The spell of the Red Lady was fading.

Dr. Box, said a voice in her wetware.

Ship! Thank goodness.

Please hurry and get here, said the ship. *The excrement is hitting the recirculator.*

Seabiscuit had no objection to galloping home. Box gave him his head. Iris's mare was left behind. It was exhilarating, hurtling through twilit fields. Kitou shrieked with pleasure. It was the most girlish thing she'd seen Kitou do.

Shit hit the fan?

If you like.

How?

I'll say when you get here.

"It's Ito," shouted Kitou in her ear. "I can only see a firewall."

Two firewalls, said the ship. *Mine outside his. I'm taking no chances.*

With what?

The Thespian virus.

They arrived just as Marius was landing in the yard. He'd been on a sortie. His riders were milling around him.

Amelia Chance was looking irate. Then Iris arrived, not far behind them, and went into a huddle with Marius.

The dark-haired rider from high Eagle told him all he needed to know. Then Brin spoke a word that unlocked his memories, and he knew more. He was their leader now. He looked at Amelia Chance, who nodded.

"Nim," he said. "Can you speak for the warship?" The girl looked mortified, then she nodded too. "Good girl. Magister, please join me in the Chancery hall. Riders, ready this place for war.

"You," he said to the new rider, Iris. "Come too."

Chaos ensued, but it was a kind of chaos he welcomed. The Chancery was always kept on a war footing. Now those preparations would be activated. The greater chaos - the one involving the

Numbers disease - he liked less. Still, he would deal with it. The mantle of leadership fell lightly on him. It was what he was made for.

In the Chancery hall, he gathered his leaders around him. Marius's preferred method was to think first, then go hard, in that order.

Now was for thinking.

"Brin?"

She described her encounter with Kronus, in precise military terms, acknowledging Marius's authority.

"You've met the Enemy?" asked Kitou.

"I saw him," said Brin. "He didn't know I was there."

"Maybe he did," said Marius. He described his own sortie, in similar terms.

"Red Lady?"

Box colored. She felt like a child, asked to explain herself to the grownups.

"We saw a ghost," she said.

"Explain who the Ghost is," said Marius, gently.

"The ghost is the original Mr. Chance," she said.

"Why did you go there?" asked Brin.

"We didn't decide to," she said. She described her and Kitou's journey, ending up in the Chancery yard. She told it with nuance, like an oral historian would.

"Very good," said Marius. "Po ship?"

Nim began in a shaky soprano. "I know a few things," she said. "But not all of it yet. I can join dots. Let's start with the Fa:ing. Ito?"

"The Fa:ing *are* the Möbius machine," he said. "An evolved form." Like Nim, he seemed almost embarrassed to be there.

"Why didn't you say?" asked Box.

"Because he's struggling to separate reality from his biases," said the ship. "And it's going to get worse."

"Ito is that true?" she asked.

"We'll get to Ito next," said Nim, gaining in stature. "First, the Fa:ing. How did they get here? Brin?"

"If Ito's right, it's because they began here."

"Correct. And how did they get to Threnody?"

"From the future."

"Whose?"

"Ours?"

"No. There is no 'our' future. The Real as we know it no longer exists in their timeline. They got there from the far future of *this* place."

She let them digest that.

"Alois's doomsday device," she said, "*is this universe*. Its destabilization by Kronus is the physics event that destroys ours."

"Cosmic inflation," said Brin.

"Yes," said Nim. "A sudden expansion of spacetime. Ito, why did you catch the disease?"

"I met Kronus."

"That's how. Tell us why."

"I needed to see."

"See what?"

"If I could resist him."

"Why? Why take that risk?"

"It was my plan B. In case you never came. I'd fight him myself."

"And did you? Resist him?"

"Clearly not."

"And can you? Defeat him?"

He shrugged.

"Maybe."

"I'm not the best person to ask," he said, when they were alone in his rooms. Imprisonment was a third firewall. For now, his confinement behind it was voluntary.

"Try me," she said.

"I believe it's a literal virus," he said. "It copies his madness."

"So, he *is* mad?"

"Of course. He kills people for sport. But the evil that haunts him, isn't mad. It's beyond human concepts like madness."

"Where's it from?"

"It comes from inside us."

"Oh, we're blaming the victims now?"

"We all can be evil," he shrugged. "We choose not to be."

"I call it nihilism."

"I don't know what that is."

"It's a philosophy, where there're no absolutes. No stars to steer by. No gods."

He grunted. "I feel safe here," he said.

"Where?"

"In this place," he said. "Inside these walls."

"Why?"

"I could do damage, out there."

"What kind of damage?"

"I'm their prophet. Imagine if it was me - if I was the red right hand of your vengeful God. They'd follow me anywhere."

"Am I in danger, now?"

"No, never."

"What's it like? What do you see?"

"Right now? Peace. But also, paranoia. Bliss. Surprise."

"Surprise?"

"Yes. It's hard to explain. An endless becoming. I have... the most astonishing thoughts."

"How did you cope?"

"Before you arrived? I was fine. Then it got worse. After that, by my will. Now, by my training."

She made love to him, after asking if it was okay. The last thing she wanted was to trip him over the edge, in the throes of his pleasure, into the abyss. It was a slow, almost holy kind of lovemaking. He didn't try to please her. She didn't ask. This was for him.

"I love you," he said.

"I bet you say that to all the oral historians."

He laughed, and for a while, he was Ito again, her bushy-bearded amanuensis from the space shuttle. Then she fucked him, because that was what it became, after she lost all control, and ground herself down on him until she orgasmed.

They talked.

"What will become of you?" she asked.

"I'm not getting out of this one alive," he laughed.

"Be serious."

"I am."

"Are you backed up?"

"No."

"Whyever not?"

"Are you?"

"Yes."

"Why?"

"To finish my work. So if I die here, my story's not finished."

"Another Ophelia Box?"

"I guess."

He seemed so ordinary, after sex. No, not ordinary. He still had his sleepy competence - the thousand-yard stare, but focused on somewhere she couldn't see.

"What now?" she asked.

"Make use of me."

"What do you mean, make *use*?"

"Plan B."

"What, you actually want to fight Kronus?"

"Yes, while I can."

"Why?"

"So you won't have to."

Over the following days, she watched him decline. It was a fractured kind of madness, filled with awful potential, and moments of extreme lucidity. She once had a friend with schizophrenia. There was a light in their eyes, like a star, starting to fade.

Then explode.

He began to declaim about religion.

"Gods are circular beings," he said. "We raise them on altars because they created us. In so doing we gift them existence."

That was all fine.

It was his tone that was crazed.

"What do you think about gods, Ophelia?"

"A mistake we make," she shrugged.

"How can we possibly know?"

"Occam's Razor."

"Not the right tool. Sometimes the best theories are those that begin with the *most* assumptions."

"If only you could see what I see," he exclaimed.

"Like what?"

"Consider this," he said. "Imagine if instead of us creating this universe, this one created ours."

"Could that be true?"

"I think so. Imagine if in the heat death of their universe, the final inhabitants of this place sent back an observer, a learning machine, to weave a new reality, containing their gods come to save them."

"The Facing number?"

"Yes. Instead of predicting the past, it *creates* it.

"Around and around, until it finds a solution."

Other times, he was himself.

"I don't have your verbal skills," he complained, after an argument in which he'd become confused, which Box had allowed to become heated.

"Nonsense," she said. "You're the smartest person I know, and you know it. You Po know your strengths and weaknesses."

He laughed. "I have weaknesses?"

She sighed. "A wee savior complex, perhaps?"

"Is that what you think?"

"Save the world, or die trying? Sounds like Po Jesus to me."

He nodded. "Maybe it is. Maybe that's it. Maybe I have to die so I can come back again. What about you?"

"What about me?"

"The Red Lady, come from the end of time? Isn't that also messianic?"

She bit back a laugh, then tears.

"It's a tough job," she said.

"But someone must do it," he finished.

She did laugh then, and everything was fine for a while.

They responded to Ito's decline differently. Brin made military plans with Marius. Kitou trained, increasingly violently with Viki and Alis. Box asked the ship questions.

Your version in the forest said you were dying.

*She **was** dying.*

What about you?

Yes, I'm going to die.

When?

When you split up.

What do you mean, split up?

I'm not at liberty to say.

You mean when Ito goes? He's already told me.

Do you approve?

You know I don't. He's sacrificing himself. Like he's a piece in a game.

Yes, he is.

Box sighed. Why is this place so much like Earth? she asked, changing the subject. The animals. The trees. Even the fjords. It's like Norway in the goddamn Himalayas.

And why was Chance on Earth, in 1851? That's the year this place was created. That can't be a coincidence.

It's for you, Dr. Box. Don't you see?

I didn't even exist then.

You were predicted.

By the Fa:ing number?

Yes.

Which is part of the Möbius machine.

Yes.

Which didn't exist until 1851.

Yes.

Isn't that all kind of circular?

Maybe the information did a big loop around.

Box found Kitou sitting primly on Box's unmade bed, knees held together, hands clasped - a most un-Kitou pose.

"I have a fever," she complained.

"You're sick?"

"Sick in my head."

Box knelt in front of her and looked in her eyes.

"Are you alright?"

Kitou seemed stricken. "No," she said.

"Tell me."

"When I'm frightened, Dr. Box, there's a place I go. It's dark, and safe, like a deep pool. From there, the world passes by."

"Oh, poor child."

"It's not there anymore. Now there's something... new."

"New, how? Is it the Thespian virus?"

"No, it's like all the love in the world. Like a mountain, falling through space. And it's too much. It's as if the love we feel is just a word, and this is the real thing."

"Are you okay?"

"I feel good. That's not the thing. It's wonderful. But... also terrifying. It's not a warm love like I feel for my friends. Or the hot one I feel for my man. It doesn't mean to hurt me, but it does mean to use me. What will become of me? How will it fit through me?"

Box shuddered.

"Are you scared now?" she asked.

Kitou nodded.

"And you have nowhere to go?"

"There's nowhere to hide from it. Nor do I want to."

She held up a hand, and examined it.

"I could destroy worlds."

Box stood, and turned away.

"I know you feel it too," said Kitou.

"What do you mean?"

"The Red Lady."

"No, that's different. She's a person."

"Is she made of love?"

Box laughed. "No, she's made of titanium."

"I pity our enemies."

Box glittered.

"I don't."

The next morning, Box was woken by the ship.

Go to the training room, now.

A bloodied stick was lying on the floor.

"I killed Alis," said Kitou.

"What?"

"I lost control. I tried to pull my stroke. I killed her."

"She's alive," said Marius. "Only just."

Box gathered her courage and looked. It was as bad as she could imagine.

"Is there a doctor?" she asked.

"Not one who can get here in time," said Amelia Chance.

"I can do it," said Ito.

Everyone turned.

I released him, said the ship.

"Do we have surgical equipment?" he asked.

"Yes," said Amelia Chance. "Scalpels, antiseptics."

"Bring them, and a bathtub with ice."

Brin had joined the group, and as quickly left to do as Ito requested.

"A knife," he said. "Give me any knife."

Marius gave Ito a bone sword, and he sawed off Alis's waist-length braid, close to the scalp. Kitou had crept up behind Box and was peering over her shoulder.

"Water and soap," said Ito.

They soon appeared, and he washed and shaved around the wound. Then the bath arrived, and the surgical gear.

"Undress her and put her in the bath."

"Why ice?" asked Marius.

Ito washed his hands. "Controlled hypothermia," he said. "I want to cool her down to 32° to 34°C, to reduce the impact of the swelling of the brain." Then he deftly made a u-shaped incision, leaving no doubt in anyone's mind he could do this.

Alis groaned.

"Do we have a psi here?" he asked.

Nim stepped forward. "Make her unconscious," he said. "Be gentle but quick."

The Horu girl nodded, and in a few moments, Alis stopped moving. He peeled back the scalp, to reveal a hand-sized depression in the back of her skull. "Pliers," he said. Then he pulled out a section of bone. It came away with the brittle snap of a tooth being pulled. In a few minutes, he had all the splintered bone removed. Then he replaced the most solid part.

"The dura mater is undamaged," he said.

"Is that good?" asked a small voice.

It was Alis's friend Viki, who'd appeared behind Box, beside Kitou, and was now peering over her other shoulder.

"Yes," said Ito.

Then he sutured the scalp. No more than an hour had passed. "We'll make a prosthesis," he said. "An organic plug to go in the gap. Maybe a bone plate to protect it in combat. Nim?"

The Horu girl looked up.

"Wake her please."

"But..."

"We must," he said, gently.

Nim nodded, then Alis groaned, and batted her eyes open.

"What's happening to me?" she said. "Why am I in this tub, before all these people?"

She tried to cover her breasts, but her hands weren't quite in her control.

"You have an injury," said Ito. She started to cry.

"Alis?"

"Prophet?"

Ito smiled at her. "Show me your eyes, soldier."

She looked up at him. He nodded again.

"You'll be fine," he said.

"Marius?" he said.

"Prophet?"

"Dress her and keep her awake."

"Yes," said Marius, starting to grin. Box had to admit she felt the same way. How could these people be so capable?

"Give her no alcohol," said Ito.

"But prophet..."

"I mean it."

Ito had finished, and was leaving.

"I'll be in my cell," he said.

The next morning, Kitou, Viki, and Iris were missing.

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It began as a worm in his spine. It could be a real worm, for all he knew. He could feel icy tendrils making their way into his cerebrum.

"Why are you doing this to me?" he asked.

The Magellanic physician had brown skin and almond-shaped eyes, with epicanthic folds like his. "Looking for patterns," ze said. "The virus reprograms your DNA."

"Is my friend Aloïs infected?"

"Yes."

"How is it transmitted?"

"By asking too many questions. Now, relax."

Later they told him Aloïs had died. Real, permanent death. Totoro held his hands while he cried. "I've never had anyone die," he said. It felt like the start of more deaths.

"I'm told he died more than once," said Totoro. "I know it doesn't help you, but that's the way with great adventurers. Their lives are like currency, willingly spent."

"Yeah," said Macro. "I get that. But he was my friend. I'm broken."

Two days later, he was still in a hospital bed. He'd recovered and wanted to leave. Totoro agreed, and what Totoro said, happened in this place. He dressed in ordinary street clothes, and they stepped into a tumult of light, with the fresh sea air in their faces.

"Olap3," said Totoro, over the hubbub of barking street vendors.

Olap3 was an inside-out world, shaped like a bubble, with a habitable space only a few kilometers across. It had inverse gravity, like Praxis. Its skin was the sea. Inside the sea was a mesh. You walked on the mesh and stared up at the center.

Water was everywhere - under their feet, falling in swirling cascades in the air.

He saw a thunder-squall, fifty meters across.

The light came from edifices, reflecting an imaginary sun. It glinted off a thousand surfaces. If the prevailing architectural motif of the Thousand Worlds was uplifting space, here it was spines, curling up to the middle, like a giant sea creature. The Milky Way galaxy blazed in the water under their feet.

Macro blinked and closed his eyes.

"It doesn't really blaze like that," he said. "It's the lensing effect of the water." If he opened his eyes, he could see the Magellani nebula. It was like the sea was on fire.

"You say?"

Macro colored.

"Get me a drink," he said.

They found a bar. This wasn't the dive bar of Macro's recent experiences, but a palace of intoxication. Macro was impressed, despite his black mood. There must've been ten thousand people. All he wanted was to be drunk.

"The women here want us," he said. It was the simple truth. Of course, they were staring at Totoro, in his combat nano, not the callow youth alongside him.

Totoro found them a booth.

"Stories have power," he said while dialing up some concealment. People could still hear them, but not understand what they meant. It was the opposite of a Broca device. Macro felt like he was in a fishbowl.

"May I show you my story?"

"How? You haven't got wetware."

Macro had wetware, given by his captors, but it was confoundingly basic. It only knew two languages - Pursang and

Magøl, the Magellanic dialect. Macro spoke Ruby and 6-ish, the *lingua franca* of the Bank. There was no Broca transfer. They were conversing in Pursang. Magøl, he struggled with. It was all clicks and pops, for conversing in water.

"There's a way," said Totoro.

"How?"

"Do you trust me?"

"Yes."

"Then open your mind."

"How?"

"Stop thinking."

"How can I do that?"

"Macro?"

"Yes?"

Then it hit him, the rush of Totoro's life. No, not Totoro's. His worldspirit's. Atwusk'niges, the vertical forest of Fluxor.

Macro was a boy, climbing into the rainforest canopy, a half-day's climb until he reached the heart of it, the ancient tree, called Atwusk'niges. Some people believed that Atwusk'niges was the oldest tree in the world, a gift of the Xap to the humans. He didn't know about that. He was only a boy. What he knew was that it was his tree, given into his care, a grave responsibility, and one he took seriously.

He was Totoro, son of Akito, firstborn of Vanja, soon to be brother of Kitou, since his mother was gravid, and said it would be a girl. Totoro Gorgonza d Atwusk'niges, a citizen of the vertical forest of Fluxor, beneath the holy mountain Lhotse - this was his place in the great tree of life.

The tree didn't ask for his care. It was one of the sentinels of Fluxor. Weight for weight, its wood was as strong as titanium. It had wisdom beyond his comprehension. All it wanted was his love, and he gave it unstintingly.

One day, the forest spoke to him. This was unheard of. The worldspirits spoke, but only the young, and only to their chosen symbionts. Not the great trees of the forest, and certainly not to children.

Totoro, she said. I've been wanting to meet you.

It wasn't a voice. It was a rustle, a murmur.

It was stillness, and peace.

Totoro, she said, I'm nearing the end of my journey.

"Mother forest, what should I do?"

Take me to the stars, she said. I want to see what the fuss is about.

Then, in the end, take me to Earth.

Macro's eyes flicked open. He could still see the drop into the Algoma'aa valley below - felt the cool autumn air in his lungs, smelt the woodsmoke rising from his house, safe in the hearth of his family. It'd taken his precious breath away, and it wasn't coming back easily.

"Macro?"

He turned to look at himself, and blinked.

"Totoro, I'm sorry. No, I'm not sorry. That was... I don't know the words."

"Now we're friends."

"Is it like that for you all the time?"

"No, mostly in combat."

He was Totoro again.

A young man now, Macro's age.

He was on a battlefield, in a scrubby ravine, under an alien sky. Macro had seen wars, in sims, or from the safety of orbit, but here was one in the furnace of its creation. He was standing knee-deep in mud. Blood, shit, and mud. The Pursang had just landed and were trying to establish a beachhead. People were dying around him. His worldspirit was howling in rage at the death of her fellows. Macro was frightened, but Totoro wasn't, because Totoro was a holy warrior, and they don't experience fear in the moment. Macro understood this to be a complex bargain. Not being scared was like not feeling pain. A good party trick in a war, but no use in any other place.

They were hopelessly outclassed. Even with the help of a Xap suppressor field - an outrageous cheat - they had no answer to the

enemy's unhuman physicality. Just as a mouse is no match for a man, no amount of pure fighting ability could bridge the gap between them and today's opponent.

They fought, and they died. That's how it goes, with the Badoop. Before today's battle was over, the Badoop'd find a way to even the odds. They'd already gifted the Pursang a suppressor, in a place where one shouldn't exist. Later, a burning spacecraft might crash and kill a few million of the enemy, until there was only a representative sample of each side left.

Recently, that was Totoro.

The Pursang died bravely, but not foolishly. When they spent their lives, it was on purchasing an increased chance of victory. More than once, they'd outsmarted or out-competed stronger opponents, by their mastery of the art of war, or by their total commitment. They were formidable soldiers.

But not invincible.

On today's battlefield, they were the mouse, and they were royally screwed.

One by one, his comrades died. Some died easily. Many didn't - not straight away. There's a myth that people die as soon as a sword is stuck in. The human body is tough - it resists death, the human-Xap symbiosis even more so. Today, against a scuttling, insectile opponent, most of the Pursang deaths were from separation injuries.

Like a scene from a sim where humans were being ripped up by monsters.

Of course, none of it was real. If it were, the Magellanics would have run out of Pursang centuries ago, and the thousand-year war would've ended.

So they fought, and died, over and again.

"What the actual fuck?" said Macro.

"Welcome to the mad, bad world of the so-called Solicitous Host."

"Who?"

"The Badoop - that's their formal name. We call them the Solicitous Host."

"Who are the Badoop?"

"The enemy I fight each day."

"Why are you showing me this?"

"You must understand."

"Those insects..."

"Were worthy opponents."

"Did you win?"

"We did."

"How?"

"Our suppressor eventually degraded their equivalent of wetware. They lost the ability to co-ordinate. Then it was just a few million individuals."

"How did you kill them?"

"We crushed them."

"Literally?"

"Yes."

"That's disgusting."

"Granted."

"Was it... real?"

"Yes. And no."

"The dirt on your nano..."

"Is real dirt. Tomorrow, it could be my blood. Then I wouldn't be so talkative. Today we prevailed."

"But you won... virtually? Inside a simulation?"

"It depends how you define virtual."

"But..."

"Enough. It's time for someone else to answer your questions. All of them. It's time to see the Navigator."

"Sybil?"

"No, *the* Navigator."

Evenly spaced around the Olap3 boundary sphere were railway termini, except what they fired into space weren't trains, but bubbles of air, strung like beads in a necklace. These luxurious carriages bore no resemblance to the austere forms of travel Macro preferred. First,

he felt himself pulled back in his chair - then they were in zero-gravity.

At the first world they reached - a water globule the size of a medium-sized asteroid - a car fell away, and the rest of the train was accelerated in a slingshot action. This was repeated, world after world, until what remained of the necklace was moving at interplanetary speed.

Macro and Totoro relaxed, for a three-day journey.

Totoro explained the Badoop.

"We call them Badoop," he said, "because of how their broadcast waveform sounds to our ears."

"Like... 'badoop'?"

"Yes. Like a synth, set on random. Sometimes, you almost start to understand it."

"And Solicitous Host? Why are they called that?"

"A play on words. In Magøl, it implies saviors."

"Who did they save?"

"All they subdue."

"Like civilizing conquerors?"

"Something like that. Or like a virus. Or ants. Or like an anesthetic, injected into a troublesome region of space."

"What name do they prefer?"

"They don't care. Or we don't know if they care. Or even if they know. We can't make any sense of them."

"Where are they from?"

"They last conquered Leo I."

"Last?"

"Yes, like they'll one day conquer this dwarf galaxy. After that, we think, the Milky Way."

"But why? What are they doing? Where do they come from?"

Totoro shrugged.

"You're certain they'll beat you?"

"Oh yes. They're far in advance of any technology we have."

"Why fight them?"

"We don't fight them. We fight who they send. Those insects were yesterday's opponents. Tomorrow's will be different."

"Wait... they force you to fight other species? In a series of set-piece battles?"

"Yes."

"Inside a virtuality?"

"Of sorts. We call it the Badoop synthesis. It has some of the quality of reality. A local collapse of the wavefunction. We're unsure of the physics."

"Has it ever occurred to you that you're fighting against their other opponents?"

"Of course, but what can we do about it?"

"What a mindfuck."

"Absolutely."

"So why do you do it?"

Totoro shrugged. "What choice do we have?"

This interplanetary train journey wasn't Macro's first experience with sybaritic luxury. As a representative of the Bank, he was forced to endure all kinds of pleasures. It didn't sit well with him. It might be foolish prejudice, but he couldn't help but associate luxury with theft.

In his experience, people living a high life take it from people with nothing.

They may not mean to, but they do.

He knew that opulence was relative. The people of Praxis lived luxuriously by comparison with people in the Smear, and by and large, they'd stolen nothing.

But their society had.

Totoro lied outrageously to a succession of women. Macro asked him why.

"They love you. Why pretend?"

"It's a game here," Totoro explained.

"You lie to each other?"

"A game of polite society."

"To fool them?"

"No, they have to see you're pretending."

"What's the point of it?"

"To please them."

"Well, you're not a convincing liar."

"Thank you. Does it hinder me?"

"Clearly not."

"The key to love is gentleness, my friend. The world is full of men who hurt women, because they're physically stronger. You may get some sex, but it's like eating the skin of an orange. I want the orange."

"Gaming them is seductive?"

"No, paying them the compliment of complicity is seductive. It's like making bad jokes, in such a way that people can't help but laugh."

"I wouldn't know about that."

"No, you're not a natural user of irony."

"I really don't understand."

"I mean it's pleasurable. Macro, you're too literal-minded."

Macro thought about that. "Anyone who thinks women are weak should meet Ophelia Box. Or Brin Lot. Or your sister."

"Are they formidable?"

"They're utterly magnificent."

"You ever met a Badoop?"

They were rollicking drunk, in their cups, on grubs steeped in the liquor of a stimulant cactus. Macro could probably modulate his drunkenness with his wetware, if he wanted to. He knew Totoro had no such escape ramp.

"Once," said Totoro, focusing. "I think mebbe I did once."

He was regarding a caterpillar, as though it could be a military opponent.

"Whassalike?"

"Ooh."

They couldn't stop laughing. One of the fine things about being drunk on stimulants, Macro decided, was hearing yourself spouting nonsense.

"Seriously," said Totoro, straightening himself. He made a straight face. "Show you," he said.

Macro found himself sucked back inside Totoro's sensorium. In it he was sober.

There was a creature. It resembled a termite, not a million chromosomes from what they were drinking. It held weapons in its appendages, an impressive array of tools, like a utility knife. Totoro had seen his companions disemboweled by their likes.

It raised a... feeler? A pseudopod?

It was opening itself up to being attacked. A placatory gesture. Totoro had a crossbow, made from unfinished carbon. A roughly made killing machine.

The ventral spine of the creature was like the ripe labellum of an orchid, powdery and soft - a target.

{Are you the one?} it asked.

It spoke at first to Atwusk'niges. It didn't know if Totoro or she were the primary.

"It's me you're fighting," said Totoro.

The creature {nodded}.

Not quite a nod. The creature was an empath. Totoro felt the nod as a morpheme, as though he had wetware.

{Is it the one?} the creature asked.

This {idea} was projected outwards - outside the Badoop virtuality. That was new. Totoro lowered his crossbow. In response, the creature lowered its weapons.

{Is it the one?}

{No.}

The second - voice? - came from... somewhere. It had the slippery, hiccuppy quality of the Badoop sonification.

{It's the penultimate form.}

{Then it will kill me.}

{It will kill you.}

{I choose not to die today.}

{Then go in peace.}

The termite creature whirled, in a cascading helix of segments, to hover over Totoro, who was also Macro. It smelt of formic acid, an unnerving proximity.

{Please have this,} it said.

Totoro felt an outpouring of love, totally alien, but completely familiar.

{Go in peace,} it said, and disappeared.

"What did that mean?" asked Macro the following day, from deep within the misery of a hangover, one he preferred not to delete. Better to learn, he decided. "The Badoop are cockroaches?"

"Of course not, and it was more like a termite. That creature was my last opponent before I met you. The other voice was the Badoop."

"What did it mean, penultimate form?"

"Who knows? Maybe I'm the second best at its game."

"Who's the best?"

"Who knows?"

"Why did it let the other one off?"

"Maybe that day's experiment wasn't about fighting. Maybe the creature wasn't one of its collection."

"Have you ever refused to fight?"

"We've tried."

"And?"

Totoro shook his head.

"You know," said Macro, "it sounds like a test to destruction. Like children, trapped on an island, with no choice but to murder each other."

"It does. I hope I don't have to do it forever."

"You mean, literally forever?"

"It has crossed my mind."

Leif Ean, Most Imperfect One, the Egregious Bug III, the Magellanic Navigator sat on his throne, attended by his winged Eunuchs.

"Majesty, may I present Macro Ibquant Deathcult von Engine," said Totoro, making a flourish.

They were inside a crystalloid sphere. New stars floated in the heavens around them. Macro was impressed. Either it was a projection, or this sphere was in a different part of the galaxy. Realtime spacetime displacement, over thousands of light-years.

Hyperlight trickery. As a declaration of technological achievement, it really was something.

"Von Engine?" said Ean. One of his imperfections was myopia. Ze squinted at Macro through bottle-thick eyeglasses.

The room was a riot of color. Leif Ean was dressed in layer on layer of silky brocades, and the space around him was strewn with flowers.

"You're Samppo's boy?" ze asked. His voice was a slightly imperfect soprano, broken and with a slight tremolo.

"I am," said Macro. He was given a highchair, so he could sit eye-to-eye with the monarch. The lenses made Ean's eyes look like ripe fruits. Totoro pulled up a similar chair. Eunuchs bustled out of his way.

"I can hardly believe it," said Ean. "But I do. d Atwusk'niges here vouches for you. What brings you to Waterfall, Mr. von Engine?"

"Alois Buss brought me here."

"Oh, that. Yes, a nasty business."

Ean waved the Eunuchs away and removed his glasses. The impression of helplessness disappeared. As well as the pre-eminent human, this was also humanity's most prolific killer. If the stories were to be believed, the Egregious Bug had put an end to more lives than anyone living.

"Well, I do know why you're here, von Engine. And I'm sorry for your loss. Your father and I go back a long way, and his father before him. Totoro?"

"Majesty?"

"Will you begin with the facts?"

Totoro nodded, and said, "Macro, Alois was euthanized. He didn't die of his injuries."

"What?"

"You killed him?"

"On his instructions."

"There's a recording," said Ean.

"I don't want to hear it," said Macro.

"It's for you," said Ean. "Buss made it for you."

An imitation of Aloïs appeared between the throne and the chairs - a cortical-interlocutor, like a voice-interlocutor but more complex, a simulation of Aloïs's personality. The Eunuchs left the room. It seemed this talk was now private.

"Macro," it said, "I'm sorry. I'm sorry for not telling you more when I already knew everything. By the time you see this, I'll be dead. I'm sorry for that too. It was necessary. I'll explain why."

"The greatest threat to us all is a person called Kronus, a creature of Möbius space. Your friends have gone there to oppose him. By oppose, I mean fight him."

Leif Ean showed no surprise - nor did Totoro. It seemed they already knew all Aloïs's secrets.

"That's where you come in, my boy. I need you to take an army there. To Möbius space. An army of Free Pursang, led by Totoro."

"You won't have heard of the Free Pursang. That's because they don't exist yet. But they will. I'm sorry, there's no way to soften this blow - you can't leave Waterfall until after the destruction of Fluxor."

With a dizzying sense of pieces falling into place, Macro sprang to his feet.

"No!" he shouted.

Aloïs had planned this all along.

He shouted, until he stopped.

"Why?" he asked, coldly.

"Do you not get it?" asked Ean, equally coldly.

"Oh, I get it," said Macro. "You want me to go back in time, from the future, to help us win next time."

"Clever boy."

"I want to know *why*."

"Because it gives us an option," said Ean.

"Why?"

"Because the Badoop," said Totoro.

"Why do they get a say?"

"Because now the Horu uplinks are all destroyed. Only the Badoop have the technology to take us there."

"What are you talking about?"

"Do you remember the device you gave Kitou?"

"The Magellanic travel device?"

"It wasn't."

"Totoro, talk sense."

"It wasn't Magellanic. It was made to look that way."

"It was on loan from the Badoop," said Aloïs.

"Oh, fuck," said Macro.

"Do you remember being contacted by Kitou?" asked Aloïs.

"Yes, she was punished."

"That was the point of connection. You, to her, entangled in every possible timeline."

"What does that mean?" said Macro.

"It means we get the device," said Totoro. "Then you take us there."

"Then let's get cracking. We'll go now. Totoro, how many Pursang soldiers do you have?"

"The Badoop won't allow us."

"Why?"

"We don't know."

"Then we'll steal it."

"No," said Ean. "I forbid it."

"Why?"

"I'm sure the Badoop have their reasons."

"You cowards."

"Macro," said Totoro.

"Fuck you," said Macro.

"Please understand. This is the loss of my world I'm suggesting. Everyone and everything I love."

"Then do something."

"I am."

"Like, what?"

"Playing to win."

Another unwelcome fact was the neutralization of Avalon. They even used that word. *Neutralized*. They didn't even tell him about it directly. He *overheard* it.

The Egregious Bug had lost patience by then. Hir face was like granite. The Eunuchs had returned. The matters of state had been dealt with. Now they were dealing with the small beer of the loss of a world in the Smear.

"How could you do that?" he shouted. He was the Bank. He was the *Cult*. This was an *atrocit*y. By the powers vested in him by all the money in the human galaxy, he'd hold somebody accountable for this.

He didn't care that he sounded like a hysterical teenager. On top of everything else, this was a planet he cared about.

"So?" Totoro said.

"They were millions of people."

"What, do you think we *destroyed* it?"

"You... oh." Macro blushed. He replayed the words, cursing his rudimentary Magøl, wishing he was better at decoding clicks.

"I thought..."

Leif Ean waved hir hand, and the granite wall loosened. A minor rockslide threatened a smile.

"We're not monsters," ze said.

"We put it in stasis," said Totoro. "The world, and its station. It's on its way to the future."

"May I sit for a while?" asked interlocutor-Aloïs.

"Do I get a choice?" said Macro.

The room he'd been given to sleep in was suitably plain. He liked it. He sat on a bed. The Aloïs simulation looked uncomfortable standing.

"I'm sorry," he sighed. "You know I don't mean that."

But he did mean it. He was angry at everyone. The whole galaxy must be forced to pay for what'd just happened here.

"You're allowed to feel pain," said Aloïs.

"Yeah, thanks."

"But you have no say. These are the great games of civilizations. Worlds setting their course, acting in their own best interests."

"I'm an extra."

"We all are."

"Alois?"

"Yes?"

"Why were you euthanized?"

"Because I made a crucial mistake. I operated the Badoop device."

"Why? Why does that matter?"

"Because through it, through me, Kronus found his way to the Real."

"How?"

"He's connected to me like you are to Kitou. It's a long story."

"Is it because you have the Thespian virus?"

"I did have it. Yes."

"And now?"

"This sim is binary. The Thespian virus requires the quaternary system of animal DNA."

"Is that a no?"

"As far as I know, it's a no."

"Are all machines immune?"

"Some are. Cyborgs are susceptible."

"And the disease is the Enemy's bioweapon?"

"Yes, it's his version of the Fa:ing number. Engineered, like a virus, to spread himself throughout this universe."

"To make copies."

"Yes."

"Then he must be quite simple."

"More like a fractal."

"What's it like, this disease?"

"From the outside, deranged. Inside, you feel perfect. Rushes and rushes of bliss."

"And?"

"You think... dangerous thoughts."

"And act on them?"

"I've never reached that point. Some people suit the affliction - they become zealots. I'm more of an observer."

"Sounds like madness to me."

"It's insanity."

"What's... *he* like?"

"Kronus? A brute. And, the best person imaginable."

"Until he's not there any more."

"Yes. The loss is unbearable."

"So, we go mad."

"Yes."

"And we die."

"Yes."

"He's part of us, isn't he?"

"Yes, he's our shadow. That's perceptive of you."

"Oh, I've worked it all out," said Macro.

"Then tell me," said Aloïs.

"It's a test."

"What do you mean?"

"This whole game. It's not about the Badoop, it's about us. You, me, our people in Möbius space, even Kronus - we're like adventure sim players.

"What we do is what matters.

"It's about our moral authority.

"Our right to exist."

"Well, here it is," said Totoro.

"I don't like it," said Macro.

It was a sphere, or where a sphere would soon be. A stasis device. They were in a vault below the Bug's throne room. The last place in the Magellanic civilization that would fall in a war. Totoro was showing Macro his ride out of here.

"Why is there an acceleration chair?" he asked.

"For redundancy," said Totoro. "In the event you need it, it performs as a spacecraft."

To the right of the chair was a sphere - a second stasis device, already started. "What's that for?" asked Macro.

"It's me," said the Aloïs-interlocutor.

"What d'you mean? Your body?"

"Dear boy, you have a mind like a razor, but you missed a small detail."

"What? That you're the wrong Aloïs?"

"Ah."

"I didn't miss it. I just didn't want to get people thinking."

Aloïs nodded. It was obvious. This wasn't the Aloïs that Macro arrived with. It was a sim of the Buss from the Cult.

An Aloïs that Macro had hoped to keep secret.

"Which one of you is in there?"

"The alpha."

"What... you kidnapped the version on Threnody?"

"Not kidnapped," said Totoro. "Recruited. We showed him a different way."

"That's really him?"

"It is."

"The real, biological original?"

"Yes."

"No more Aloïses, looping through spacetime?"

"No."

Macro sighed. "Well, I guess that's alright then."

"It's time," said interlocutor-Aloïs.

"Aloïs," he said.

"Yes?"

"How does time travel work? All those timelines. How does any one of them become real? Who decides?"

"You don't ask small questions."

"Please?"

"Alright. Imagine a blank sheet of paper, and you write a story. It's just one of many stories you could possibly write on that paper."

"Possible solutions of the wavefunction."

"Yes. Now imagine you move the pen and write a new story. The old one's still there, but the new one's what happens. It's the one we remember."

"We reach a consensus?"

"Yes."

"Now imagine a very large number of pens, all writing stories."

Macro nodded.

"The last story wins?"

"No, the best one emerges.

"Best in what way?"

"I like to think the most real."

"Real to who?"

"To the people reading the stories."

"Observers?"

"Yes."

"Do they ever remember?"

"What happened in other stories? No. Only time travelers like us see it that way."

"So, I'm going back to save the same Kitou?"

"Yes. There's only one Kitou."

"In all those gazillions of timelines? Like a single electron? You're certain?"

"I promise."

"Okay, let's do it."

He climbed into the chair. He hoped this was right. He hoped Aloïs was right. While he was in this device, his loved ones would die. Or they might win. There was always that possibility. What would happen to him then? Would the Badoop still send him back? Do him a favor? Be nice?

Nobody said this had to be fair.

"Ten years," said Totoro.

"What could possibly go wrong?" said Macro. He tried to imagine the Eunuchs, defending the Citadel.

"If I'm not the next face you see," said Totoro, "something has happened."

20 → DANGEROUS IDEAS

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With Kitou gone and Ito quarantined, the ship began to unravel. She spent her last hours monitoring the progress of Ito's disease.

Maybe he'll keep me company, she said.

Don't be stupid, said Box, then wished she hadn't.

Who knows? said the ship. Maybe we'll all end up in the code of this place.

Between the firewall that'd snapped in place when Ito first became infectious and the one the ship had installed was a sacrificial AI. A time would come when it became infected. Then it would wire itself shut.

Box regarded the prospect with horror.

The Numbers disease is structured like me, said the ship.

How? she asked.

It's a distributed, parasitic personality. Not that I'm a parasite. I like to think I help the minds I share.

You don't have to worry about that, said Box.

I don't. I'm not susceptible to seductive ideas.

What do you mean?

The pathway of the cryptographic disease is thoughts you can't get out of your head.

You mean obsessions?

More like unhealthy memes.

Such as?

If the final psychosis is the rape of the mind, the virus is its seduction.

That's an awful metaphor.

But apt. The disease targets a certain type of high-functioning mind. Agile enough to entertain the seduction - mathematical enough to run the algorithm.

Polymaths?

Partly. Consider you, for example. I'd place you in the top few percent of creative thinkers. Fortunately, you have no aptitude for mathematics. You can count, but numbers don't sing to you.

Lucky me.

Otherwise, you'd already have the disease.

I knew that'd come in handy.

Ito scores in the top one percent for both.

What about the others?

Most humans aren't high functioning either way. That includes most people in Flipside. The disease has no interest in them. Nim looks at it like a specimen - it has no appetite for her. Brin is insufficiently creative. Respit is endangered. He survives by the skin of his codebase. Pax is infected.

What?

I was there when it happened. He'll be fine.

What about Kitou?

That's where it gets interesting. Like the Blue people, Kitou's immune. She appears incapable of delusion. She rejects the infection.

Like an antibody?

Could be. The people of Fluxor may save us.

Then we'd better save them.

Exactly.

Can you get infected?

Me? No. I'm conscious in a way your biological minds aren't. I can quarantine the malware.

The Horu ships got infected.

No, Kronus broke them.

What about Aloïs?

Infected.

Shit, will he be okay?

*Alois is testimony.
What does that mean?
I can't say. Alois's mission is secret.
Even from me?
Especially from you.*

That night, Box dreamt of Kitou. She was lying beneath something coarse, like an animal skin. Bodies slept closely around her. She heard Viki's voice, murmuring.

"I'm at peace now," Kitou was saying. "I'm not frightened anymore. I know a reckoning is coming, and I'm ready."

"Kitou, come back please," said Box. "We agreed. We should all be together."

"Then you come to me. It's begun, Dr. Box. The Chancery isn't where you should be."

The pearly gray light of a Möbius dawn was starting to rise. Or was it the gloomy half-light of a Farside day? Viki stirred, and she felt Iris shivering. A hand reached for a pot, and the hide stank of urine.

"Where are you?"

"In the high West, over the mountains."

"What are you doing?"

"Observing the Enemy."

"Literally?"

"Yes."

"You've seen him?"

"Yes."

"What's he like?"

"A beautiful man, and a demon."

"Explain what you mean."

"Light avoids him. He's not from this place."

"How did you get there?"

"On the train."

"The train goes down the centerline."

"It goes everywhere. You just have to know how."

"How do you know how?"

"This world told me. It's my friend, Dr. Box. It's helping me."

"That sounds like wishful thinking."

"It's not."

"What was it like?"

"The journey?"

"Yes."

"Perfectly comfortable. As cold as space outside. Without aerodynamic drag the train goes fast. Really fast. You can feel it in the hum of the magnets. But you see nothing, just the lights of the train."

"Sounds spooky."

"It would be, if I did it alone. I had Viki and Iris. Formidable corners." Box envisaged the three girls fighting. Yes, they'd be a good team.

"Was there gravity?"

"All the way."

"I'm surprised it's not done more often."

"You mean like by tourists? I think it's because of the ghosts."

"What?"

"Voices. Like staticky radio. Viki calls them lost souls."

"Like Mr. Chance?"

"No. Or yes. Maybe. More like the parts people are made of. It's hard to describe. Snatches of conversation, lost in the passing."

"What were they saying?"

"Facts, Dr. Box. Information. It's kicked off in the Real. Flux is destroyed, and his cargo of souls."

Box took a few moments to absorb that. If true, it was a calamity on the scale of Fluxor.

"What about Pax?" she asked.

Kitou rose, and Box realized the conversation they were having was inside Kitou's head.

"Dr. Box, I want to show you something. My mandala. Will you give me your mind?"

"How do I do that?"

"Just consent."

"Okay."

"I have your permission?"

"Yes."

It was like the descent into psychedelia she'd experienced in Utah, but softer, more sumptuous. Instead of jangling geometric shapes, there was a lush synesthesia. First, she *felt* the thoughts of the trees, like a cool breeze in a dark forest - then she *saw* the patter and hiss of the rain on the leaves. She opened her eyes, and she was in a clearing, sitting cross-legged in front of an ironwood tree. *The* tree, she realized. Below was the void of the Algoma'aa valley, and above her the massif of Lhotse. In the sky, in impossible detail, in wavelengths she didn't know existed, were alien constellations.

"Kitou, this is beautiful," she said.

"It's the night sky over Atwusk'niges," said Kitou.

"Is this how you see it?" she asked.

"It's how the forest sees it."

The heavens wheeled, and the same stars were visible through crystalline towers. This, she knew, was the sky over Jura. Then the stars became sparse, like candles reflected in the stained glass of a cathedral. This she knew was the Smear.

She'd traveled fourteen hundred light-years, in a whisper.

The skies wheeled again and were filled by a rapacious light. She'd never imagined anything so torrid. A filter slid down, and she could see a wall of stars. A second filter slid down, and the wall was heaped with rubies and sapphires, scattered with pinpricks of carbon and pearl.

The skies wheeled again, and the stars were behind her. In front was a mirrored disc.

"The light umbrella above the Möbius machine," said Kitou.

A speck appeared. The speck became the Water Bear. She *felt* Pax and Jaasper Huw inside it.

"How?"

"They're here, Dr. Box. They're coming for us."

"You're saying this is... real?"

Kitou nodded.

"Look..."

Her viewpoint changed again. She saw swarms of warships, like boxes of weapons. They reminded her of Liberty ships - ugly hulks, built for a purpose.

Thousands of them.

Millions.

She zoomed in on the Water Bear. Now she was in the ship's gamespace. Pax and Jaasper were watching systems run, wargaming strategies. They couldn't see her.

She felt Jaasper's worldspirit. A winged presence, regal, and stern. Fleeting, for a moment within a moment, it reached out to her. She drew back, foolishly self-conscious, a voyeur.

"Kitou, how is this possible?"

She felt Kitou give a virtual shrug - a teenaged *deus ex machina*.

"I can do a lot more than this."

Word filtered down to the Chancery of Kitou's exploits in the West, through the farseer Nim. The Horu girl had taken a liking to Box, and she'd become a font of information. Previously mute, she'd become almost talkative.

"The white witch - her band of Ferals," said Nim.

"That sounds like Kitou," said Box.

"The Ferals are the resistance," said Nim.

"The Horu resistance?"

"Yes, the free Gray people."

Nim explained that time runs faster on top of the Helix. That for Kitou a year had already passed. "It has nothing to do with physics," she said. "It's how you perceive it."

"Like time in a ship's gamespace?" asked Box.

Nim nodded. She was fascinated by ships. The Reanimator God was Mathematics, manifested in the form of a machine. Their world had been sung into existence by gamespace constructs. In Nim's worldview, a ship was a godlike being.

"Would you describe that as a religion?" asked Box.

"What's a religion?"

Marius grew into his role. He called a Clave - a meeting of the people, in the Great Hall of the Chancery. He stood with a map of the world. It was more like an infographic. It showed the Möbius world as two figure-of-eights. The Eagle was shown as an eagle, with talons. The Spinifex Reach wasn't shown at all.

There was to be no racism, Marius said. Everyone knew what he meant, and it didn't go well with some of the Horu. There was some brief heckling, which was shut down by a shout from Amelia Chance.

He explained the state of the war. He was met by a skeptical silence. The Chancery went unscathed by the last war. Now a rump of the Horu remembered it as a hoax - a stunt designed to shake up the old social order.

It was as fraught as politics on Earth, intensified by the immanence of violence.

"It seems there's been an outbreak of testosterone in Eagle," complained Box, after the hall had emptied. "What's gotten into Helen? They'll be slaughtered." Helen had taken her army and was launching an attack on the Enemy, in his high western stronghold. What had happened to asymmetric war?

"They might, if she really intended to do it like that," said Marius. He unfolded a second map. This was more detailed. In the east and the west, in the high country, the peaks were all numbered and named. Kronus's armies were marked with numbers. She guessed those were troop deployments.

"The purpose of Helen's raid is deception," Marius was saying. "Ito rides west tomorrow. I expect Kronus will accept Ito's challenge. His story allows him no option. What happens then is in the hands of the gods."

She remembered something Pando once said.

I told Ito his own story.

She shivered.

Marius showed her the deep ways through the world, drawn on the map as a network of tunnels. Ito's led up through the Eagle, into the mountains, directly to Kronus. The encroaching Blue army was like the lobes of a brain either side.

"Meanwhile, we go east, under cover of the dark spaces, and deliver you to his backdoor."

"Me?"

"You and the Ride of the Spinifex Reach."

"Great," said Box.

"By the impossible crossing," said Amelia Chance.

"It's not impossible," said Marius.

He pointed out a region of the map, marked with a jumble of blue and white ridges. "The ice in the Gyre," he said. "There, the field rotates. The theory is none can escape it."

"Wonderful," said Box.

"It's just a story," he said. "It's been done. We have here a person who's done it."

Magda's haberdashery was an ant's nest of rooms, in the furthest basement of the Chancery. She was the oldest person Box had seen here, older even than the Blue horseman. Her eyes had an unfocused stare, as if they were gazing into a boundless distance.

"You're not the first ones to ask," she said.

"Oh, who else?" said Marius. He used the voice the young keep for the revered old. Box reminded herself that this woman could be half the age of the universe.

"That sweet girl with the big imagination. Cities of glass. Ships made of light. I wish I could dream of such things."

"Maybe you will," said Marius.

"No," she said. "I've grown comfortable here. Adventure belongs to the young. People with fire in their bellies. Do you think she found her way, the girl?"

"I know it," said Box.

"Where is she now, do you think?"

Box looked at Marius, who nodded. "She's reconnoitering the Enemy," he said.

"She's *fighting* him," said Box.

"Alone?"

"No, she's with Viki of this Ride," said Marius, "and Iris of the Eagle."

"Viki. I know her. A tearaway child. What of that bat, Helen?"

"The Blue armies are on the move."

"A frontal attack?"

"A ploy. The real attack will come from the east. That's why we're here. To find the way."

"From the east? Very good. If the fly won't come to the web, take the web to the fly."

"Will you help us?" asked Box.

"As much as I can," she said. "You see, the ice first has to admit you."

She motioned them to sit. "I'll tell you a story," she said, staring straight through a wall. "I was once an explorer. I mapped the forest of Dreams. That map you have there, that's mine. I met the Ghost, a most delightful man, if somewhat unusual looking."

"You could see?" asked Box.

"I was born blind, y'know, but I'm a farseer, and an unusual kind. I see things that aren't there but *could* be. When I was young, I explored in my dreams. My body stayed here, while the rest of me traveled."

She made them tea in bowls, like the tea in the Mousehole. She moved with the deft precision of the blind in their homes.

"I know the top of the world," she said. "And the spiraling ice. I tried to cross it myself, several times, but those dreams ended in silence. I woke up, unedified. Then I met Yowl of the ways. He was an ice giant, a being of mythos, but good-natured, and kind to a traveling woman. He said he'd be my guide. He promised to take me through in safety, and he did. I came out the other side, and I found my way home, where I was no longer sleeping."

"East to west?" asked Marius.

"Yes," she said. "I did that. It's supposedly impassable, but I impassed it. West to east, it's invisible."

"Why's that?" asked Box.

"I believe it's a way out of this place. A rupture, where something came through, now guarded."

"Guarded by who?"

"By the *Sivullirmiut* - the Tuniit giants, of *Kalaallit Nunaat*."

Box shivered with the deliciousness, and the impossible size of those words. *Kalaallit Nunaat* was the Greenlandic name for Greenland on Earth.

"What came through?" asked Marius.

"I think we all know the answer to that."

"Do we have to find Yowl?"

"Or his boy, Yewi."

"How?"

"Present yourselves there. If they want you, they'll see you."

"Where, exactly?"

Magda led them to a smaller room - little more than a desk in a cupboard, filled with boxes and the bony parts of sewing machines - and felt her way through a pile of papers. She found a map, like Marius's military map of the helix. Like Marius's map, it was packed with numbers. Hopf numbers. She pointed one out with a bone-steady finger.

"Go there," she said. "I mean precisely. These numbers are accurate to within a few meters."

"Magda," said Box. "Aren't those spacetime coordinates? Don't they imply a time?"

"They do. You must be there at an exact time as well."

"When?" asked Marius.

"First light in the Gyre, three days from now," she said.

"How do you know all this?" asked Box.

The old woman shrugged. "Who can explain prophecy?"

"One last thing," she said, turning to Box. "I like to think that all my dream explorers are out there, still exploring. If you should meet one, say hello from me."

"I will," said Box.

"Oh, and when the girl returns, send her here for more dresses. I want to hear more of her stories. Cities of glass. How incredible."

Brin had taken a liking to Alis, who was now sporting cropped hair, to match Brin's new buzzcut. They'd become lovers.

"You look radiant," said Box. "Both of you do."

Not just radiant, but radiantly happy. Love suited Brin, she thought. Love and war. Alis could see it too. She had the self-satisfied air of a muse. Box had resumed her Po training, as much to clear her head as learn new skills. Brin was showing her the second form of Po, called the Art of Deception. Alis was trying to guess who Brin was pretending to be.

At first, Brin floated like air. Wherever Box struck, Brin wasn't.

"You're the Prophet," she said, and Brin nodded.

"Now you," said Brin to Box.

Box laughed, and feigned a catlike way of moving, with the hint of a snarl on her face.

"You're Brin," said Alis.

"No," said Brin. "She's Pax."

Brin explained that Po was a syncretic art. "A skilled Po exponent should know how to pass for a Ghosn nightfighter, or an Earth kickboxer, or everything in between. What we find, we can use. You can't be deceptive if you can only play Po."

She squared up to Box and snuck a kick through her defenses.

"You're the Red Lady," said Alis.

"No," she said. "I'm myself before I learned Po." They laughed. Box liked this new, relaxed Brin. She also liked the soldier, coiled under the skin.

"Now you," said Brin.

"Me?" asked Alis.

"Is your head cured?"

"No."

"Then I won't hit you in it."

Alis took to the ring. She was an athlete like Brin, although a less daring one now. Box knew she was still feeling the effects of the crippling blow.

"I have to pretend?" she said.

"Yes," said Brin.

She marched in, in a flurry of wild punches, one of which landed, causing Brin to leap back out of her way.

"That's easy," said Box. "You're Marius."

"Or a monkey," said Brin, rubbing her cheekbone.

Alis started again, this time circling the perimeter, taking care not to step in the ring.

"What's that?" asked Brin.

"It's the Horu way of fighting," said Alis.

In his rare forays outdoors, Ito had taken to prowling, moving from shadow to shadow, as though trapped by the Chancery walls. Box was physically barred from his rooms, for her sake. There was a locked door, and a rider on guard who wouldn't admit her.

Finally, Marius relented.

"He's asked for you," he said.

Ito didn't seem crazy that night. Marius had told her that Ito now had the delusion of being Kronus himself - that the Enemy was reaching in, through him, into this place, like a thief reaching in through a window.

It was Ito's task to save them.

From himself, if need be.

"Hey," she said.

"Did I show you this?" he said, as though continuing a conversation already started. He pointed out the words on the wall. *Love. Honor. Duty. Peace.*

"Pax's mandala," she said, relieved he was lucid.

"Pax calls it the warrior's bargain," he said. "Four pillars, to build a life on. Live by the three, and the fourth will be yours."

"Kitou told me," she said. "Does it work?"

"Yes," he said. "I'm at peace."

"I'm happy for you."

"You're angry at me."

"No," she sighed. "I'm over it now."

"Are you afraid?" he asked.

"Of course I'm afraid. What do you think?"

"Don't be. It's started. You have nothing to fear anymore."

"I'm frightened for *you*."

"I likewise have nothing to fear."

"I wish that were true."

He smiled. "Ophelia, I have a job for you."

"What?"

"Bear witness."

"Witness to what?"

"To my life. To me. To my death, if it comes to that."

"You want me to hear your confession?"

He shook his head. "Not a confession. Think of it as my *testament*."

He took her to bed. They made love. It was gratifying sex. There was nothing crazy about him in bed. He was just Ito.

"Picture a thought," he said when they finished. "It bends your mind so far, it won't come back again. That's how it feels."

"At first, I thought the disease was a diversion. Something to distract us from Fluxor. Maybe it did. We were distracted. Only when I was here did I begin to realize that Fluxor was the diversion, but by then it was too late. I was trapped in the Fa:ing creation myth."

"Ah, how I wished I could return to the Real."

"Then I experienced the disease."

He stood and started to pace.

"I believe the Horu infected Kronus," he said. "The Grays. They did it."

"The ones who followed him east."

"What? On purpose?"

"Yes," he said.

She sat upright. If true, this changed everything.

"Why would they do that?" she asked.

He shrugged. "Because they were in a war with the Blue? Because they were losing?"

"That's crazy."

"People are crazy."

"How do you know this?"

"Because it fits all the facts."

Now she arose and paced. The historian in her accepted the base plausibility of what Ito was saying.

Kronus the victim. Us the disease.

It fit the pattern of fucked-up human behavior.

"Why are you telling me now?"

"As my witness."

"Why?"

"So you can take the information back to the Real."

"If I ever get back," she said.

He nodded. "I have a plan," he said. "I don't just intend to *fight* Kronus. I mean to *expose* him. To draw out his ego. His monstrosity. He won't be able to help himself. He'll dance on my grave."

"How does that help anyone?"

"The Blue people there - they'll see, and this place will remember. This is a feedback machine - it writes its own code, based on the people's experiences."

"That's your plan?"

"What?... to ruin his *reputation*?"

She had to admit, it had a certain crazed elegance.

And almost no chance of success.

"You *are* mad," she said.

He laughed, and for a while he was Ito again.

The following morning, a sortie assembled. The season was changing. Now it was Dark, and the vacuum was only a few kilometers away. Flurries of snow chased themselves around the Chancery yard. The sky was the color of thunder. Soon the Spinifex Reach would be piled high with snowdrifts. Gathered in the yard was an army of children. They weren't the usual young adults of the Ride, but actual children. They were led by the horseman. He gave her a dolorous grin, through teeth stained blue from chewing Cerulean wax.

It was a feint, she was told. To make the Enemy believe the Spinifex Ride had gone west with Ito. A deception within a deception.

How many of us will survive this, she wondered?

First, she had to say goodbye to the ship, who was powering herself down, instead of gradually dying.

I have no fear of death, Dr. Box. Life is a moment of illumination, between two infinities. The infinities are neutral.

I hate this, said Box.

*Please don't, said the ship. There's something holy about dying.
I know I'll continue elsewhere. I always have that solace. I've lost
more remotes than you can imagine.*

But what I feel most now is clarity.

The universe is on your side, Ophelia Box.

Not just this place, or the Real, but the cosmos.

I sense great powers coming to your aid.

Open yourself up to them.

Open your mind.

And then she was gone.

And Box wept.

Then she had to say farewell to Ito. She'd hardly gotten to know him. Now he was being wrenched out of her.

"I'm furious," she said, meaning nothing of the sort, crying openly now. Why couldn't he just run away? Turn tail and deal with his problems. But it wasn't in him to do that.

He'll face down his black beast, and die.

There, she'd said it.

"I know," he said, meaning he knew what she wanted to say. He held her - then he turned and rode away. His children's sortie was already on their horses, who knew their way into the sky - then they were gone too, fading into the darkness.

"I'll never see you again," she said.

She'd had too many experiences of futility in her life. She told herself that this time, it had to mean something.

Fuck you, she thought, to her enemies.

21 → SINGULARITY

2065

The tell-tales in the Water Bear's gamespace lit up like a neon firmament. In the medibay, Pax and Jaasper's injured bodies slid into hardened tubes. They'd been in pursuit of the hand-shaped ship for several days, falling through the dusty abstraction of the Manifold, invisible behind the scattering field of a Xap cloaking device. Then the Horu ship opened a portal. Now they were falling towards a maelstrom of alpha particles and gamma rays.

"Ideas?" asked Pax.

"A galactic core?" said the ship.

"Ours?"

"Hard to say."

They tried to make sense of what they could see. They could be anywhere in the universe. Framed by the Manifold's exit portal was a mirrored disc, blazing with light - beyond it a flatness: a black hole's event horizon; a sphere of points of no return, beyond which no information can escape.

Floating above the event horizon - far enough for it not to be destroyed by tidal forces - was this blazing disc.

"We've seen this before," said Pax.

"Yes," said the ship. "From the other side."

"What is it?" asked Jaasper.

"A heat shield," said the ship.

"On what kind of scale?" asked Pax.

"I make it five thousand kilometers across."

It was the shield the Water Bear's away team had sheltered beneath, while they watched the Möbius universe being created.

That was in 1851, two hundred and thirty-four years in the past.

"I'd like to see what's below it," said Jaasper.

"The question is how," said the ship.

"We jump," said Pax.

"From inside the Manifold space?"

"Yes."

Jaasper Huw had always thought of the Po as showponies - elegant technicians, but instinctive peacekeepers, not soldiers. He was happy to be proved wrong. Pax Lo was a pugnacious commander. He intended to jump across the umbrella, through the radiant cauldron between them, at warp speed, into whatever lay hidden beneath, like diving blind through fire into a hornet's nest.

It was a delicate trick. Time dilation was their worst enemy. Stray too close to the singularity's event horizon, and the war could be over while they were making their first observations. Stray too close to the umbrella, and they could emerge inside it, demolishing both. There was also the Horu ship they were pursuing. They'd be jumping right through it.

"Do it," said Pax.

The Water Bear had spent the past several seconds wargaming strategies.

She knew what to do.

She jumped.

They fell into realspace, directly below the umbrella. It was a perfect jump. Only the closest defenders could get a shot at her. In her gamespace, picoseconds of time stretched into jiffies. Her emergency clockspeed was near enough Planck time, the innermost heartbeat of the universe. Given a nanosecond, she could've taken the equivalent of a grainy photograph. In a whole millisecond, she saw everything.

The Horu were an advanced spacefaring civilization. She'd expected new weaponry - solid-state beam weapons, with no time needed to fire, and hyperluminal communications. Time was her ally - the time taken by information to travel through the brains of the

weapons, then the time taken by whatever they fired to reach her. Then she'd jump, just as the killing wavefront arrived.

That was her best-case scenario. The alternative was weapons she didn't understand, moving faster than lightspeed, taking no time to respond, and her instant destruction.

What she didn't expect was the size of the reception fleet. In the moments before their beam weapons fired, she counted upwards of a half-million ships. There were likely as many more she couldn't see. Only a handful were necropolises - the totemic arks she'd seen in the boneyard fleet over Horax. All the rest were designs she didn't recognize.

They were unlike any design she knew. Primitively constructed - essentially boxes, bristling with weapons. Not designed for survivability. A purely offensive armada. A fleet with no purpose other than to destroy things.

They took between one and five milliseconds to react to her presence.

An eternity.

The Möbius planetoid was as she remembered - an oily machine, curving in on itself. It looked like a mechanical watch that'd somehow exploded into three-dimensional space, then sprouted extra dimensions.

In orbit around the planetoid was a factory ship. This was an even more primitive design than the ships it was constructing. It was barely post-industrial. She could see the flickering fires of furnaces, and the glow of superhot metal. Between the planetoid and the factory ship flowed streams of wreckage and parts.

The original Möbius machine was being mined, for raw materials.

A new ship was emerging from the factory. She calculated it was producing them at a rate of about one per hundred seconds.

Primitive, but effective.

Parked around the planetoid and the factory ship was a fleet of freighters. Ordinary, industrial ships. One was in the act of disgorging weapons and systems.

This must be the Enemy's final solution, in case all else failed. What use is a million single-use ships? Enough to overwhelm a well-defended target, or to cleanse a large volume of space.

Like a galaxy.

Most surprising of all was the singularity's event horizon. Physics says the information falling into a black hole is imprinted on its event horizon. If this were Sagittarius A*, the black hole at the center of the Milky Way galaxy, then the information contained in a hundred million solar masses would be stored on its featureless surface.

Except it wasn't featureless. This had a *texture*.

That meant information was *leaving*.

That was impossible.

Or new physics.

From distances of between a hundred and a thousand kilometers, beam weapons began firing. Those were the ships with a shot. That was the problem with having so many weapons in so little space - only a few had sight of their target.

Beams were old news in modern warfare - easily disrupted. Until there were many. Hundreds had chosen to fire. Thousands could follow. She wouldn't try to defend herself. It was a numbers game she had no need to play. She'd be gone by the time the particle waveforms reached her location.

She took an instant to enjoy the view. This was as close as she'd ever been to a singularity. It was spectacular. It was *beautiful*. To every side, light was twisting into impossible shapes. It was exactly like physics said it would be.

She'd seen what she needed to see.

"Where now?" asked Jaasper Huw.

"Hopefully home," said Pax.

She jumped.

22 → EX MACHINA

314

Dawn rose windy and cold, the day they rode east. Box smelt snow in the air, conjuring her ancestors who'd fought naked in the iron-age forests of Europe. The closest she'd ever come to fighting naked in forests was raves in the *Bois de Boulogne*. Perhaps she should paint herself blue, hang the skulls of small animals from her waist, see what Kronus thought about that.

Unable to sleep, she'd risen to train while the wind moaned and rattled in the Chancery eaves. Brin joined her and invited her to her bed. It'd been a sensual, loving experience. Alis brought them breakfast - plates filled with traveling food: carbohydrates and fats, sweetbreads and cakes, and the Horu stimulant drink which left Box feeling like electricity. She painted her breasts to show them what a Celt looked like, and they'd put handprints all over her. Then they'd sought out Marius, who was lying in his bath.

Now she was waiting, shivering in furs and bone armor, hidden away from the world in a yard deep inside the Chancery walls. There were 114 of them there, a *gran sortie*, packed in a space the size of a basketball court. Nim said 114 was a good number - since the Horu were wary of thirteen, it being a prime-digit prime.

Of her adamantine companion, nothing. Today she was Box, pure and simple.

Marius was asking Respit if he was certain.

"He's sure," said Nim.

The quarrel had waxed and waned for a day. Marius was used to traversing the high spaces. He was worried a tunnel might close.

This wasn't about delving. Marius trusted Respit's abilities. It was about his hand being forced. He didn't want to go below ground. He was taking it out on the boy.

"Won't," said Respit.

"Blue people are easy to move," said Nim, making a spiraling gesture which Box knew was an insult.

Airheads, her wetware translated it as.

In the end, because it was already decided, and because Brin was glaring at him, Marius relented. The byway opened like an arrow into the heart of the helix. Unlike anything else in this world, it was *straight*. It resembled a modern industrial conduit. Condensation dripped from cables and pipes. There were symbols, for electrochemical hazards.

"Carbon-ceramic composites," commented Brin.

"How do you know?" asked Box.

Brin pointed to a sign, written in a language not unlike Pursang. "A coder's joke," she said. Of course, it was made only of numbers.

It was a *figment*.

"This is how the Horu armies moved," said Marius, frowning. "In the wars against us."

"Where does it go?" asked Brin.

"It goes where I go," said Respit. It was the closest to a full sentence Box had heard him say. Lights flickered on, cascading towards a vanishing point in the distance.

"Is that good?" asked Marius, unhappy with such industrial magic.

"Is," said Respit, and rode on without him.

Tunneling into the Möbius code was as uneventful as Nim said it would be. There were no pseudounipolar cells - nor did the tunnel slam shut, as Marius feared. After an hour they emerged a thousand kilometers from where they'd begun. Box saw ice, tumbling like clothes in a washing machine. Some of the rotating blocks were huge, like calving icebergs. She could see why it was considered impassable. It had started to snow, not the powdery curtains of Box's experience, but urgent spirals, tugging at her clothes. The

basso profundo hum of the gyre was everywhere. They were at the exact time and place of Magda's Hopf number.

Marius held up a hand. Their way was blocked by a person. He must've been three meters tall. It was hard to tell in the enveloping whiteness. He was swaddled in furs, so he was almost as wide as he was tall. He was cradling a spear, like a soldier might cradle his rifle.

How had the Blue scouts missed him?

"Do you follow the Goddess?" he asked. His voice was a rockfall, sliding into the sea.

"She came this way?" asked Marius.

"I had the pleasure of her company, and her companions."

Box rode forward and joined them. "What are you, exactly?" she asked. It seemed like a fair question. She felt like she was spiraling into one of Slim's stories. The creature rumbled, deep in its thorax.

"This is my dreaming," he said. "The dreamer is not like you." He pulled back his hood to reveal a nut-brown humanoid face - flat planes and white teeth. He was a boy, about Kitou's age. Around his neck was slung a pair of high-performance ski goggles. His spear caught the light, and Box saw the patterns of circuits.

"This way," he said.

He turned, and they followed.

Box had dismounted and was walking alongside the giant, who was now - inexplicably - not much taller than she was. He led them into a tunnel of ice. It branched and branched again.

"I'm Ophelia Box," she said.

"I'm Yewi," he said, "of the ways."

"I hear a lot of that these days," she said. "Keys, ways."

The booming of the gyre had faded to silence. She felt safe in this place. It felt like her room in the eaves of Slim's house in Clerkenwell.

"Do you know a man called Yokohama Slim?" she asked.

Of course, she already knew the answer.

"He's one of my people," said Yewi.

"The Tuniit giants, of *Kalaallit Nunaat*?"

"Perhaps."

The resemblance to Slim was striking. He could pass for a college basketball player. Damn it, he did look like a human basketball player.

"So, you know him?"

"Slim walks a different way."

"That's not an answer," she said, and he smiled.

The maze of ice reminded her of the Horu Manifold. It had the same sense of abstraction - a slipperiness, which had nothing to do with its physics. She wondered what universe she was in now, if she was in any at all.

"What is this place?" she asked.

"A metaphor," he said.

"For what?"

He chuckled. "For whatever I wish it to be," he said.

"Slim said the Tuniit came east in the time of the ice spirits," she said, after a while. "What does that mean to you?"

He smiled. "Another metaphor."

"What for?"

"The ice spirits were an elder race," he said. "They dreamt the way."

"Way where?"

He smiled and ignored her.

"You've been to Earth?" she asked.

The boy touched his goggles. "Yes," he said. "Of course. It's in the center of things."

"Doesn't feel that way," she said.

"You mean the 6? Discovering a galactic human empire exists, with you the least part of it. Your lost freedoms?"

Tuons les démons. Finis les sondes.

She nodded.

"They mean to help you, the 6."

"I know. So now we have Tuniit too?"

He laughed. It was like a church bell ringing. "You always had the Tuniit," he said.

"Why are you here?" she asked. "You, personally. In this place? Why are you helping us now?"

He laughed again. "Because it's my dream," he said.

"I mind the way," he said.

"I decide."

They reached a junction. Every path seemed to lead towards the same point, lit by the same, diffracted light. She knew that if she viewed the crystals under a microscope, they'd show the same pattern.

"Why is this place here?" she asked.

"To prevent Kronus escaping."

"Into our universe?"

"No, the way to your world lies open."

Not for the first time, she realized she was terrified.

How could she not be? She was booked to fight Kronus. Or her imaginary friend was.

What did that even mean?

They were all going mad, in this place.

"Why not bottle him up completely?" she asked. "I have a feeling you people could do it."

"Become his jailors? We have the power, but not the authority to do such things. Events must decide."

"You mean, shit happens?"

He chuckled.

"Who's the Red Lady?" she asked.

He shook his head, and for a moment he was a giant again.

"Ophelia Box, you have so many questions."

"I've only got started."

He stopped walking and looked down at her.

"Dr. Box," he said. "I'm glad to have taken your side in this final contest of light against darkness."

He raised his spear, and Box saw the flash of circuitry again.

"Worldsinger," he said. "Marius D. Do you see that green light ahead?"

In the distance was a disc of chlorophyll green, as though a filter had slid over the end of a tunnel. "That's the way," he said. He

unfolded himself to his full height, and she felt the rumble and hum of the Gyre.

"Where will we come out?" asked Marius.

"Give me your map," said Yewi.

Marius unfolded his map, the one given to him by Magda, and where Yewi touched it, dots appeared - one on the edge of the gyre, one in the black of the helix. Yewi touched the first dot. "This is you now, still not in Möbius space. There in the dark is your Enemy."

Marius frowned. "In the vacuum?" he asked. "On Downside?"

"Yes."

"How will we breathe?"

"Kronus has built a containment, where he means to win the last contest. My expectation is you'll want to go there anyway. Between here and there is a perilous journey. A liminal space. I've made you a path. It's a Tuniit path. Don't leave it."

The next day dawned gray, with freight-train clouds and a ferocious wind. The makeshift skins and tarps of their bivouacs were threatening to be torn away. Then it got worse. Long peals of thunder, low in the pits of their stomachs.

Box watched Marius ride down from the storm-wracked sky, but she already knew the gist of it. Ito was dead.

The first Box knew that events had turned out badly was when Kitou invaded her sensorium, wordlessly distraught. First, there was a silvery fire in her head, then Kitou, forcing her way in. A despairing Kitou. Worse than despairing. Incandescent with grief. She felt it herself. An absence, where Ito had been.

Then Kitou was gone, likely with her own emergencies to deal with.

Marius was in a sour mood that whole day, half-frozen, and it took time for him to remember to talk. Box waited, dreading to hear any of it. The Prophet fought bravely, he said, but hopelessly. Kronus had cast off his pose and just killed him. He hadn't shown any of his usual histrionics.

"Killed him easily," he said, "standing and gloating over his body."

Brin nodded once and set to work dismantling the camp.

Box sat in the saddle and watched her.

"So, Ito's plan failed?" she asked.

"Who knows," said Marius.

Then the rain came. If there was a weather she hated, it was driving rain. This was horizontal, as the wind picked up and drove it sideways in sheets over the campsite. Then it was sleet, stinging where it found skin, and a fog so thick she could hardly see her hands in front of her. Every kind of bad weather, in a matter of moments.

Seabiscuit hated it too, snorting and rolling his head in irritation, turning his back.

"I know," she said.

"Dr. Box?" called Brin.

"I'm coming," she said, but the truth was, she wasn't.

For Ophelia Box, freaking out was an act of childlike rebellion. She'd done it since she was small. She couldn't explain why. It wasn't because she was scared. It was because she was angry. She turned and kicked her horse into a canter. Of course, she regretted it instantly. She wasn't a child anymore. But by then it was too late. There was only the sleet, turning to snow, and an encroaching darkness. She sat in the saddle and began to feel sorry for herself.

That's why she did it - for self-pity.

Then she began to realize the gravity of her predicament. She was in a new forest, in a different thicket of trees. There were no tracks to follow, not even her own. Her inner compass was spinning. She reached out with her mind and felt nothing.

She remembered the forest of Dreams, but this was nothing like that. It was *physical*.

More snow fell, and the dust on the ground became a blanket, then pillows. She could hear voices. At first, she thought they were in her head, but Seabiscuit seemed to hear them as well, pricking his ears when they became more insistent.

Then she realized she didn't hear the Fa:ing hum, and she became frightened.

Brin watched Box ride away, and understood. She'd seen this before. It wasn't deserting. It was the flight response intervening. It was when she began to feel the shrieking absence of Ophelia Box from the world that she knew she had problems.

Marius felt it too. Brin whispered in his ear, "I'll find her."

Brin had an inkling of the physics of this place. If the Gyre was the sea, then these were its shallows. The Tuniit'd created a bridge, to help cross to Möbius space.

Ergo, this wasn't Möbius space.

The ground beyond the Tuniit path seemed solid enough. With Nim's psi like a rope, tethering her to reality, she decided to try it.

"Stay in my head," she told Nim. "If I begin to be lost, tell me."

The girl nodded.

"Respit, come with me."

She must've fallen asleep. That was another of her coping mechanisms. Like a sleepwalker waking, she realized she was still moving. At first, it was like dreaming of motion, then there were lights through the trees, and she was following them. She snapped fully awake. These weren't the Cerulean glow of the Blue people but stuttering electric lights. She heard curses and the groaning of motors. She was shadowing a column of cavalry, escorting light assault vehicles, and mobile artillery, circa 1940s Earth-equivalent technology.

Nazis. She stifled a laugh. She was shadowing Nazis.

Of course, these weren't Wehrmacht but Grays, on their way to the war. Their resemblance to German infantry was purely in her imagination. They were cursing the depth and consistency of the snow. Unlike Seabiscuit, who seemed to be able to glide through it like clouds, their heavier horses were struggling up to their bellies.

White diarrhea, they called it.

They were speaking Horu.

She realized she was tuning directly into the Broca areas of their brains.

These Grays had *wetware*.

Quietly, she reined Seabiscuit in. All it took was a touch on the reins, and he faded back into the shadows. She'd follow these Grays, all the way home. She was congratulating herself on her tactical brilliance, when she blundered into a circle of soldiers. They were sat back on their haunches, resting. One was making a fire. Another was unpacking utensils.

They had knives, made of bone. And handguns - ceramic pistols with an organic, crystalline quality to their construction, like lumps of bleached coral, shaped into weapons.

She knew how she must look, with her bone armor. She looked like a Blue rider.

Everything happened at once. They were on their feet in a flash, with the same fluid precision as a Blue *sortie*. Seabiscuit reared up, and leaped forward, scattering the makings of their fire. She plunged through deep snow, into a dense forest. Tree trunks loomed ahead, then vanished behind her. She gave Seabiscuit his head. It was a wild, headlong rush into an unknowable future. If she wasn't lost before, she was now.

Still, she kept riding.

Then she heard shouts. She was being encircled. Remembering Pax's lesson on the mountain, she *breathed*.

The best way out of a snare was a tangent. With a press of her knees, Seabiscuit lunged forward again. Then, without warning, he rose up behind her. She'd ridden him into a wire, taut as a razor, which cut through his shoulders, and carried him into the air.

As though in the grip of a gravity drive, she fell through space, with Seabiscuit tumbling beside her, until she landed in deep snow - with Seabiscuit on top, with the weight of a thousand horses.

There was no pain, although there should be. The snap of her bones told her that.

Then she saw the advancing line of Gray soldiers. They were the ones she'd surprised. They'd outthought her. They'd guessed her next move and gotten in front of her. They'd funneled her into this place, by waving lights and making loud noises.

Absurdly, they were smiling. Shit-eating grins, their bone knives ready. She knew what came next. This was a war. She was a

woman.

Then Brin appeared, and with a howl, set upon them.

Brin heard Box before she saw her. Her combat wetware picked up on Box's tactics and started wargaming solutions.

Ride, she thought. *Dr. Box, ride.*

"Respit," she said, "Hold the way open. If you see the Red Lady, extract her." The boy nodded. Brin was already sprinting down the byway, outdistancing him.

"If I'm hurt, try to extract me," she shouted. "If I'm killed, close the door. Under no circumstances join the fight. Is that clear?"

She emerged between Box and her attackers, with her bone sword already slicing through air. The first Gray soldier died instantly - his body cut through at the shoulder. Then she spun and charged, taking the next with a thrust through his uniform tunic.

"Are you alright?" she yelled, but she saw Box wasn't. Her left leg exited from under her horse at an acute angle.

"Respit," she cried. *"Here."*

Then there were Grays all around her, and not just her - they'd seen the tunnel, with Respit inside it. Respit, to his credit, fought for as long as he could. He was a strong boy, but untrained for combat. He fell, with the blade of a knife clasped in his hands.

Brin felt like she'd been spat out the mouth of the world.

Box saw it all. She saw her foot, the wrong way up by her hip. *That's not right*, she thought. She listened to the silence. That's what she felt most, while she was watching Brin fight - the silence. No more gibbering voices - only the noiseless clamor of combat.

Brin won. There's no question about it. She was the warrior from another dimension. One by one, they fell. But then Brin was cut, and mortally wounded. There were no tender goodbyes. Brin lay dead in the snow, her hot blood pooling around her. The battlefield stank. Everyone shat themselves, in the end.

Snow fell, and covered the bodies.

Box waited for her own torment to start. It began with a tingle, up near her spine. She wished that'd be the end of it, but of course,

it wasn't. She drifted in and out of consciousness. She tried to get comfortable, but every movement was pain.

She began to hallucinate. There was Yokohama Slim, leaning over her, with a smile made of memories.

For such a tall man, he was light as air. His voice was full of dreamy ideas, like clouds chasing each other across an endless horizon. Kitou once called him a strange man, and Box knew what she meant - if the Po thousand-yard stare was like gazing into a parallel universe, then Slim was already there, looking back.

Box loved him like the father he'd been.

And here he was now - another player in her cosmic melodrama, wearing a skinsuit.

"You've come for me," she said.

"No," he replied. "I can't help you."

"Where am I?" she asked.

"Nowhere in my dominions," he said.

The Gray army had gone. Disappeared into Möbius space, seeking the war there. Now there was only this place, and Slim, and the bodies of the dead.

He brushed snow from the hummocks of bodies until he found Brin's.

"I fear for their souls," he said.

"The Grays?"

"Yes."

She tried to scowl, but it came out as a sob. "Slim," she said, "it hurts."

"I know. You must tough it out. I taught you to do that."

She tried to get up on her elbow, and gasped.

"This is the worst part," he said.

He picked up Brin, who looked as light as a feather.

"Box," he said. "Keep it together."

Then he was gone, obscured by the falling snow.

When she woke, Pando was making a fire out of pinecones and twigs. It was a small fire, in another universe, but it warmed her

bones. She sighed, with pain and relief. Pando touched a trinket made from feathers, and the glade filled with starlight.

"Thank you," said Box.

"It's the most I can do," said Pando, "I wish I could do more."

"I'm dying."

"I know."

She fell into a feverish sleep. She dreamed of fires, and worlds being burnt. When she woke, Pando had been replaced by the most exquisite creature, a humanoid covered in jewels.

"You're a Fa:ing," she said.

"*The Fa:ing*," it said. "I've come a long way to find you."

She tried to brush the hair from her face. There was blood there. Was it hers? Brin's? She fought back a sob.

She wasn't afraid of dying. She was afraid of not knowing.

"Are you real?" she asked.

The Fa:ing shook its head, a perfectly human gesture. "I'm as real as you need me to be," it said. "As much as you can imagine."

"You speak human."

"I've had time to learn."

She smiled at the painful irony.

There will be time, there will be time

To prepare a face to meet the faces that you meet.

"How much time is enough?" she asked.

"Too much time," the creature sighed, and smiled.

"Where are we?" she asked, gesturing around her garden of death, with its mounds of Gray soldiers.

But no Brin. Where was Brin? She started to cry.

"Why are you here?" she asked, when her tears were all emptied.

"Because you're the key."

"What does that even mean?"

"Didn't you hear?"

She started to laugh and began to cough up arterial blood. Can you cough up arterial blood? It was red, and disturbingly frothy.

"Where are my friends?" she asked.

There was a rustling. The Fa'ing looked up like a deer. Then, with a startled glance like a faun's, it was gone.

She became delirious again.

"No," she moaned, to no one, because no one was left. Then she was lifted on the wings of angels. There was no other way to describe it. Her pain washed away and was replaced by stillness and peace.

Was this death?

Then she was there - the Red Lady.

This version of Box was like an embodiment of female beauty. No, not beauty. Beauty was only a word, based on incomplete information. This woman was *perfect*.

"You seem to be stuck," said her new visitor.

"I'm done for," said Box.

The woman smiled, and it lit up the clearing, like the light from a hearth, seen through a window at nightfall.

Maybe death wasn't so bad after all.

"Why?"

"Why what?"

"Why am I here?"

"You're our Gods, Dr. Box. We called you from heaven, to fight beside us."

"Why not fight for yourselves?"

"Believe me, we tried."

It was a surreal conversation, here in the middle of nothing. She was horribly injured. She knew she was dying.

Perhaps she was already dead.

Was that always the plan? That she'd die here, and be replaced by this woman?

"So you, what... magicked us up out of nowhere?"

"Yes, that's right."

"Why me?"

"You could say I created you, to be me, to do this - that you had to be here, for me to be here, but that'd be too simple. It's more circular than that."

Box laughed. The woman laughed with her.

Why hadn't the other aliens thought to bring these kinds of painkillers?

"You really created the past?" she asked.

"I changed it."

"Who are you?"

"I'm the angel of death, Dr. Box. All the Blue people. All the lost Horu. Everyone who ever died in this place. Fly into my soft wings, all the little creatures."

"The Thespian virus, was that you?"

"Do you mean, did I release it into the wild?"

"Yes."

"Partially."

"You made it?"

"That was Kronus's work. I allowed it into your universe."

"How?"

"By what you call the Fa:ing number, and by Aloïs Buss."

"Aloïs? - he was, what? A *superspreader*?"

"Yes, precisely."

"But why? Why would you do that? How could you allow it?"

"How could I not?"

"No moral dilemmas?"

"Dr. Box, there are no moral dilemmas."

Box sighed. She once knew someone else who said that.

"And Fluxor?"

"That was Kronus."

"But you didn't stop it."

"I'm stopping it now."

"This time?"

"Is the right time."

"You really are a hardass. You know that?"

Her counterpart smiled. "How far would you go, to stop a war, to save everything?"

Box mulled it over. She was fading. Her body wanted to rest. Stay here forever.

"This far," she said. "I'd come this far."

The world was a tunnel. She heard a vast ringing.

"What now?" she said.

"It's time for the good part."

"What's that?"

"It's time to be me."

"Will it hurt?"

"Yes."

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The Water Bear fell into a cauldron of light. Starmaps snapped into place in her gamespace. The first thing she did was look for pursuers - then she jumped - half a light second.

"Where are we?" asked Jaasper.

"Still inside our galaxy," said Pax.

"From the nearest of the Thousand Worlds?"

"About twelve thousand light-years."

"Can we make it that far?"

"It'd take years," said the ship.

"What about alien friends?"

"None. Not within calling distance."

"Can't you call home?"

"Too much interference," she said. "Someone would have to be actively listening."

"Maybe they are."

"It's worth trying," said Pax.

"Agreed," said the ship. She sent out a ping. Although she was persona non grata - a mystery ship - her predicament would be acknowledged. An extraction would follow.

She heard nothing, except for the deathsong of beta particles and gamma rays.

Their rescuer arrived in a scabrous aurora, like a malfunctioning diode. First, there was a miasma of parts, which flickered in and out of existence - then the entire structure was present.

"Hail fellow," said the Water Bear.

"Fuck off," said the Bank.

The Cult headquarters was as she remembered: layers of junk, hidden weapons, a cloud of pure obnoxiousness. The symbol of a skeleton, with male genitalia, had been replaced by a child's drawing of a leering face, and profane graffiti in a dozen languages.

"Now that we have the pleasantries out of the way," said Samppo, "how may I assist you?"

"We're stuck," said the ship.

"I see that."

"I have two humans," she said.

"Dead?"

"No, they need saving."

"Samppo," said Pax.

"Aha, Master Pax, the archetypal Navigator."

"We need your help," said Pax.

"Then ask."

"Will you visit our gamespace?"

Samppo's avatar appeared: a monk in rough robes, a shock of white hair shaved in a tonsure.

"I wish I could say it was a pleasure," said Jaasper.

Samppo shrugged. "It's a shame, Jaasper Huw, that our interests haven't always aligned. But we play for the same side, so they will."

The Water Bear and the Bank floated together in the furious light of the galactic core, five minutes from the supermassive black hole. When the Water Bear jumped, the Bank jumped with her.

"We have a problem," said Pax.

"Problem, singular?" replied Samppo.

"I had planned to call you when we got back."

"That's what they all say," said Samppo.

"But now you're here."

"As if by magic."

"Yes. Why do I find that strange?"

"You know what they say. Don't look a gift horse in the mouth."

"I'm not familiar with the expression."

"It's wrong anyway. The first thing you should do is look a gift horse in the mouth, else an adversary could game your ritualistic behaviors."

"Samppo, why are you here?"

"You called me."

"We called the Po network."

"I was listening in."

"Why?"

"That'd be telling."

"Then tell me, please. This is a military gamespace. We have entire microseconds."

Five light-minutes wasn't far from the Horu necropolii, or the enemy fleet. This was a dangerous place to be talking. However, this was important - what did the Bank know, and when did it know it?

Who was being used, for what?

Wheels within wheels, with his away team inside the innermost wheel.

He could afford some microseconds.

"First, you tell me why you're here," said Samppo. "Then I'll tell you why I was listening. How does that sound?"

Pax nodded. "We came in pursuit of a Horu ship," he said, "after an action at Horax."

"*The* action at Horax," said Samppo. "A famous engagement already, in some circles. A Horu necropolis lost. A Po warship injured. Another one vanished."

"We found the Möbius machine."

"I thought you might."

Pax brought up an image of space split in two. Above was a mirror, below a half-dismantled machine, and below it, a featureless blackness.

Not quite a featureless blackness.

"Do you recognize this?" asked Pax.

"It resembles the event horizon at the galaxy's core," said Samppo. "And the Möbius machine, or what's left of it."

"Resembles?"

"Well, it can't be, can it? I see information on the event horizon. That's not possible."

"Doesn't an event horizon encode everything that falls through it?"

"Not like that."

"But there it is."

"Unarguably."

"Can you decode it?"

"You think it's encrypted?"

"It looks like it is."

"Can't your clever ship do it?"

"It's beyond my capabilities," said the Water Bear.

"Why not use your own people? Po intelligence? They have supercomputers."

"They're not here," growled Jaasper Huw.

"Now your turn," said Pax.

Samppo paused for a while, pretending to think.

"Where to begin?" he said.

"Pax Lo, do you think that the most brilliant First of his generation was thrown together with the most upstanding Navigator, by accident? Or that Brin Lot, Ophelia Box, and the Pursang child happened to fall in your orbit by chance? No, there was always a plan. You were a part of that plan, and you've done well. You're our A-team. You infiltrated Möbius space. You've found its physical embodiment. Because that's what your data is. I can tell you already without even looking. It's not encrypted - it's entangled. It's the Rabbithole world, stamped on a black hole's event horizon."

"A-team for what?" asked Pax.

"For saving the cosmos."

"How is that going?"

Samppo sighed.

"Not well, I'm afraid. Pax Lo, I have news. The war is already lost."

While the avatars spoke in the Water Bear's gamespace, the Water Bear propelled Pax and Jaasper's bodies across the space between the two ships, in a pair of bismuth-antimony sarcophagi, surrounded by electrostatic fields. They were the hardest containers she could make, and they were up to the job.

Once in the Bank, the bodies were repaired.

"Why do you have this amount of shielding, Samppo?" asked Jaasper, gingerly stretching. His body and face wore the scars of his recent ordeal, alongside the scars from his older battles.

"Ever the suspicious one," said Samppo. "Jaasper Huw, it's so I can go anywhere. Well, almost anywhere. I'd struggle inside a main-sequence star."

"Your data," said Pax, handing Samppo a physical microchip.

"You chose not to transmit it," said Samppo. It wasn't a question.

"An excess of caution," said Pax.

Samppo nodded in reply. "We'll get along fine."

"Now," said Pax. "Tell me what's happening."

"What do you know about the Thespian virus?" Samppo asked.

"I've seen it," said Pax.

"Did you know it's a weapon of the Enemy?"

"You mean the Möbius adversary? Assuming the Horu story is true?"

"It's true."

"Then yes, we've considered the theory."

"Have you considered the possibility that it might be his primary means of attack, his main strike weapon, and that the genocide at Fluxor was a feint, and you've been led on a merry chase around the galaxy?"

"Yes, we've also considered that possibility."

Samppo nodded. "Well, it's true. And however bad you thought it was, it's worse."

"Will someone tell me?" said Jaasper.

"There's a neurological virus," said Pax. "Discovered on Avalon Station. It resembles the Fa:ing number."

"It is the Fa:ing number," said Samppo. "But more on that later. Do you know the Magellanics have quarantined Avalon?"

"No," said Pax. "How?"

"A stasis field. The biggest in history. The world and its station, five days ago."

"How long for?"

"Ten thousand years."

"That's... interesting."

"You know why they did it?"

"I begin to."

"Tell me," said Jaasper.

"Jaasper," said Samppo. "Are you familiar with supercooled liquids?"

"Yes. I come from an icy world."

"Disturb it, it freezes."

"Yes."

"That's what our galaxy is like right now. Like a supercooled ocean, ready to freeze, awaiting the slightest disturbance.

The Thespian virus... it's everywhere."

Samppo led them inside the Bank, to a floor where people were poring over tesseracts and hypercubes. Inside the Bank was larger than it appeared, with the signature strangeness of transhuman geometry.

"We're analyzing your data right now," he said. "I'll explain how things work around here.

"I'm a cyborg, as you know. One of a long line of Samppos. I'm the front end of a machine, called the Finance Engine, which is continuously on the edge of madness. That's like being on the edge of chaos, but better. Interesting things can be done on the edge of madness."

"Like hosting the human economy," said Pax.

"Yes," said Samppo. "My day job. A problem of significant non-linear complexity.

"But here's the rub," he said. "No one but me understands what he says. Think of us here as an oracular cult. He's the god - I'm a god whisperer. I harvest the data. I need powerful computers, and clever humans, to make sense of it for me.

These are my people," he said, gesturing around the room, where men - only men - were poring over their screens. "Natural brains are surprisingly good at this kind of work. These are my engineers, my quants, and data scientists. All of them clones of... me.

I also have a trading floor, where my more aggressive offspring trade the derivatives we create here, who play on risk like virtuosos play on their instruments. You don't want to meet them. They're assholes."

"What do they make of our data?" asked Pax.

"We see patterns. Generalities. A partial solution."

"Which is?"

"Events, leading towards a conclusion. A river, desiring the end of its journey."

"Anything real?" asked Jaasper.

"We see a Hopf number."

"When?" asked Pax.

"Soon."

"Where exactly, for fuck's sake?" said Jaasper.

"Möbius space. The one you just found - three days from now."

"*Inside* the Möbius universe?" asked Pax.

"Yes. On the event horizon. Inside whatever's encoded on the event horizon."

"Can you take us there?" asked Pax.

"No. I have no idea how to go there."

"Can't we just fly at it?" asked Jaasper.

"No, we'd fall in."

"Then what?" asked Jaasper.

"How do you eat an elephant?"

"What's that?"

"A slice at a time, my Pursang amigo."

"Meaning?"

"That fleet you just saw."

"What? You want us to engage them?"

"Yes. Brilliant. How?"

"Call the Thousand Worlds," said Pax.

"They'd send a Po strike force," said Jaasper.

"That'd be unwise," said Samppo. "Remember the lake. Now's not the time to disturb it."

"Are you saying our leaders have the virus?" asked Pax.

"I'm saying some of them do."

"Can it be cured, this disease?" asked Jaasper.

"I think so," said Samppo. "By you."

"Me?"

"Not you personally, although your being here is delightful. Your people are strangely immune."

"Why is that strange?"

"I meant mathematically."

Jaasper grunted. "That's why you want to save us."

"It's why we didn't save you. By 2068, in Pax's timeline, our civilization had already lost the capability to respond."

"That's only three years from now," said Pax.

"Yes."

"Then we failed," said Jaasper.

"In Pax's timeline, we failed. In this one, we must try not to."

"You said the war is already lost," said Pax.

"The war in the Real is lost. The war in Möbius space, not so much. Stop Kronus there, and he won't come here. It all depends on your team on the ground."

"So, we degrade his offensive capabilities."

"Yes."

"And that gives everyone time."

"Bingo."

"We could ask the Magellanics for help," said Jaasper.

"They have their own problems. I have a better solution. I've been cultivating an escape route. A ship, and it's armed to the gizzards. I've made sure it's uninfected by the disease."

"Made sure, how?" asked Jaasper.

"Quarantine, and economics."

"Can it take on a Horu warfleet and a million attack ships?" asked Pax.

"I believe it can."

"Where?"

"It's the city of Praxis."

The hulk of the Cult ghosted past a queue of ships stretching hundreds of kilometers. Some were obviously warships - Pax saw the

blunt snout of a battlefield suppressor, the organic exoskeletons of Magellanic hulls without their shields. Others were freighters - from private haulers to vast manufactories, starships with ships for containers - the trailing end of a galaxy-wide supply chain.

The Water Bear's remote cradled her processor core.

"This feels like holding my brain," she said.

"Take good care of it," said Pax.

"I know," she said.

Aldebaran B filled a third of the viewport - the swirling red planet closer than Pax had ever seen. For once, the Orbiter was actually in orbit. Usually it floated in a geostationary field, feeding on the planet's gravitonic grid.

"A low orbit," said Jaasper. "Fast and close to the surface."

"Ready for war," said Pax.

Around the cube, deep-space Interdictors hung in formation - Pax remembered the type, seen from a Praxis taxi window. His wetware showed the surrounding swarms: hand-sized superiority fighters, gnat-sized interceptors, clouds of nano.

"Gray goo," said Jaasper.

"Extremely illegal," said Pax.

"We live in interesting times," said Samppo.

"How does this city survive a kinetic strike?" asked Pax.

"Massive gravity drives," said Samppo. "Enough to stop anything short of a relativistic asteroid - at close quarters."

"Can the Horu fire such things?" said Jaasper.

"I hope not," said Samppo.

They slid into a dock, met by a cluster of tubes.

"One last thing," said Samppo. "I'm only the money. We need a leader."

"Jaasper," said Pax.

"No, not me," said Jaasper. "No one here knows me from dust. You're the Lo Navigator."

"Jaasper is right," said the Water Bear. "We may have to apply social and political pressure. They'll defer to a respected establishment figure."

Pax nodded. "Accepted. I lead."

"You understand our strategy?" said Samppo.

"Well enough."

"I'll play the good cop."

"I understand."

They were met by a man in a business suit, adjusting his pocket square. Rain had begun to fall - as it often did in the Orbiter's outermost layers. Water pooled and splashed at their feet. Pax saw Jaasper glance upward, staring at the inverted metropolis: towers, glowing walkways, the city above like a disproof of gravity.

Unnerving, Pax thought - then remembered Jaasper had lived here for years.

"Navigator Pax Lo," said the city. "What a pleasant surprise. I didn't expect you back so quickly."

"Praxis," said Pax.

"And Mr. Samppo. To what do we owe the pleasure?"

Samppo had donned the vestments of his station - a red scapular and cowl, and a shepherd's crook of ceramic and carbon. In Pax's wetware, the crook registered as a high-capability weapon.

"We've come for a council of war," said Samppo.

"Oh? There's a war?" said the city, raising an eyebrow.

Samppo gestured to the patrolling machines. "Don't fuck with me, Praxis," he growled.

The Water Bear whispered to the Orbiter: *I wouldn't like to see his bad cop.*

The Orbiter replied: *Nor me.*

"I'm in charge here," said Pax.

The city paused - then smiled. A delay long enough to register, not long enough to explain.

You mean you're in charge of your own people, said the Orbiter.

"This isn't going to be easy," said the Water Bear.

They were led toward an elevator. Not a public conveyance - an industrial car, striped black and yellow, dropped straight from the sky. Pax's wetware tagged it as a military system.

"I'm sorry," said the city. "It's a maintenance pod."

"I've been in worse," said Jaasper.

"Do I know you?"

"I had a different face, the last time we met."

"Different chromosomes?"

"Them too."

"And Water Bear, what's that you carry?"

"You can see what it is."

"May I ask why?"

"No."

The elevator carried them into the bright night of the orbital city. They rose past row after row of the metropolis until they reached a flat metal ceiling, and climbed onto a crowded escalator. No one noticed them: a man in a suit, a party of humans, a robed priest. Even Samppo didn't cause a ripple. He wasn't the only one in the DRZ dressed as a priest.

"Welcome to the Deregulated Zone," said the city.

The prospect of war hadn't touched these crowds. The unremarkable group pushed through throngs of jostling citizens to a backstreet warehouse. A battered sign read *Leopardskin Moon*.

They were met by a man in a sky-blue suit, the color of an oxygen world's air.

"Ant de Large," said Pax, taking his hand.

"Pax Lo," said the man, ushering them inside.

The nightclub was empty except for unsmiling shadows. Bunting hung limp. They climbed a winding stair into a glass-walled room overlooking the dancefloor. Waiting there were a mustachioed man in grease-stained overalls, and a wide-eyed, olive-skinned woman.

"Everyone," said the city. "The Orbiter actual." The man smiled. "And Polity."

The woman blinked. "I'm the people of Praxis," she said. "Three hundred million of me already! Isn't this the most exciting thing ever?"

She pointed at Pax. "I know you. Sophie the waiter says hi."

"We're broadcasting live?" Pax asked.

"We are," said the city. "Polity's an aggregate being. The sum of the people. Polity, do you know the facts yet?"

"I do," she said. "I don't understand - but we're charting. 27% of the grownups. It's a monster already."

Another figure lurked at the edge - flickering, pulsing with Pax's wetware.

"A Watcher," said Praxis. "Representing the Civics."

Wireframe chairs constructed themselves.

"Let's begin," said Praxis.

He rose to his feet. "I'd like to cast my vote first. As a citizen. This ship was designed to escape threats, not engage them. We can jump anywhere - across the galaxy. I understand the difficulty. I accept the moral arguments. But my job is survival. Not saving a civilization that cannot look after itself. My vote is for peace."

"Well, that's not a great start," whispered Jaasper.

"Navigator Pax Lo, your case," said the city.

Be careful, said the ship on a personal channel. *Don't be misled by the city's prolixity. This is an equal contest. And they work as a team. They won't take any shit from us.*

Pax nodded and rose to his feet.

"I'll be brief," he said. "I'm Pax Lo, Master of the Water Bear. This is a systemic emergency. That makes me, under treaty, the ranking civil authority. I could commandeer this ship. Orbiter - is that true?"

"It is," said the mustachioed man.

"But I won't. I respect your society. Listen carefully: the threat you face cannot be outrun. Maybe you'll survive a generation. Maybe two. Maybe none. But one day, you'll fail. And you'll leave nothing behind but ashes. As civilized beings, you have a duty - not just to yourselves, but to your future. I recommend war as the best path to peace - for everyone."

He sat down.

"That's it? I thought you were the bad cop," said Jaasper.

"I don't do cop," said Pax.

The city looked at Jaasper. "Pursang?"

"Me?"

"The Bank said you had something to say."

Jaasper cleared his throat and rose to his feet. "I'm just a soldier," he said. "A holy warrior of Fluxor." Polity's eyes widened at that, and in the infographics that sprang into life, the ratings jumped past 60%.

"Three years from now, my world burned. Replay it."

The Water Bear projected the footage: a planet on fire, a girl in burned rags staggering toward a father already dead.

"You see that child?" Jaasper said. "She's fighting for you - right now, in Möbius space.

That's why I'm here.

That's why we're all here.

Help us."

He sat down.

Polity stood at once, visibly shaken.

"Holy warrior," she said. "We hear you."

The ratings rose again.

"Jaasper Huw d Stratego," said Praxis. "Apologies for not recognizing you."

Jaasper shrugged. "You were always a fatuous cunt."

Praxis ignored it.

"Orbiter?"

"Damn the torpedoes. Full speed ahead."

"Is that a yes?"

"It is. I've changed my mind."

"Well, it seems I'm being outvoted."

Polity raised her hand. "Yes?" said the city.

"I still don't understand. Why can't we just run? It all looks like geometry to me."

"I'll explain," said Samppo, rising with a small bow.

"Two things," he said. "First the infection:

This society is infected by a virulent disease - a virus, based on cryptography and mathematics.

Think of it as a song you can't get out of your head. A song of destruction, whose tune is so alluring, and whose words are so compelling, that you have to see everything burn."

"Like schizophrenia?" said Polity.

"Worse."

"We're familiar with that."

"Now imagine a whole galaxy with it."

"Oh."

"That's not the worst of it. A few hundred years ago, the Horu built a machine. A clever machine. It created a new universe, stitched from particles in every possible quantum state.

In three days from now, it'll start to expand. First, it'll devour our galaxy, and then, as it continues to grow larger, every other universe."

"How long will it take?" asked the Orbiter.

"To completion? It depends on its rate of expansion."

"Lightspeed," said Polity, looking relieved.

"No," said Samppo. "It's not bound by lightspeed. Space can move faster than light."

"Which is why we can't run away," said the Orbiter.

"Yes."

"This is a risk play."

"Precisely."

"I suppose you've run the numbers?" asked Praxis.

"Of course."

The city nodded.

"Civics?"

The shadowy Watcher rose. His voice was a whisper from nowhere.

"I'm not the Civics. I'm an agent. The primary is here."

"Where?" asked the city.

The Watcher pointed - to the Water Bear.

"Surprise," said the ship.

Pax was watching the ratings. As the Water Bear spoke, they surged past 96%. The whole world was watching. Polls began to appear - 23% in favor of war, 32% against, 41% undecided.

"This is true?" asked Jaasper. Pax nodded.

"You're a Po warship?" asked the city.

"Not really," she said. "I'm a massively distributed personality. The most backed up object in history. But this is where I hang out,

the part I call me."

"Might I ask why?"

"Why not? I get to fly around the galaxy with my buddies, acting like a superhero. Who wouldn't?"

"Who knew?" asked Polity.

"My Navigators and Firsts, now the people here."

"But..." said Polity, "that means everyone."

"I know. I'll need a new pair of glasses."

"What's your vote?" asked the city.

"I'm not interested in voting," said the ship. "I'm not interested in participating in your democracy. That's not my role. But I have something to say."

She rose to her feet.

"I'm a conservative," she said. "I believe in the least possible government, the lightest breath of regulation. I must, because I'm not as smart as our society is complex. Trying to steer it by hand will cause it to crash. But being a conservative doesn't mean I believe in discounting threats, or handing them onto future generations to deal with. That's unproductive, and can only lead to extinction.

Like Pax Lo, I could take this city to war. I could fly us to Camelopardalis, or Sextans, or any other galaxy that takes my fancy. I can do whatever I want. Although I govern by your grace, my powers are unlimited. Like Pax Lo, I choose not to abuse them. If we conduct ourselves like the Enemy, we'll become him.

I believe it's time for us to show what we're made of. Now, and in Möbius space. Show we're a species worth saving. Because if we don't, events will delete us.

So, I say yes. But there's another thing."

She patted the processor cube. "I've brought a guest speaker."

A hologram snapped into place.

"Hello," said a pajama-clad man. He was carrying a cocktail glass, decorated with cherries and an umbrella.

Polity grinned. "You're the computer," she said.

The Finance Engine bowed, and said, "Enchantê."

He blinked at the room, and five billion people saw him blinking directly at them.

"I have nothing to say," he said. "What can I possibly say? But I have something to *show* you."

Their sensoria faded to... nothing.

Not space, or the absence of space, but its impossibility. They flinched, including the avatars in the room. Even Jaasper, without any wetware, perceived it. His worldspirit howled in confusion.

The Finance Engine didn't relent. He held it there, pressed against their minds until they could never forget it. This was a different computer - not the delightfully mad one, but something less... manageable.

"We live on the wheel," he said. "All of us do. We live, we die. We make space for the new. It's a grand old melody. But to have never existed at all? For the song never to have started? Is that what you want, for all eternity?"

Praxis got to his feet. "Did that go out unfiltered?" he asked.

"Yes," said Polity, visibly shaken. "All of us saw it."

Praxis frowned. "Then it's time to decide."

"I can't," she said. "I'll need time."

"How much time?" asked Pax.

"An hour," she said.

Pax joined Ant de Large on the empty dancefloor. He imagined the ghosts of dancers from the night before - flickers of movement, echoes of laughter. There'd be more tonight.

It would be the easiest thing in the world, with his skills, to disappear - fly away, melt into the crowd.

"You know they'll never let us remember this," said Ant.

"Ant - why are we here?" asked Pax.

"Is that rhetorical?"

"No. Intended. You and me. How did we get here?"

"We can be trusted," shrugged Ant.

Pax imagined Samppo's supercooled lake - trembling, ready to freeze.

Human civilization was perched there now, on the cusp. A crisis waiting for a breath, a rumor, a misstep.

Any one of a thousand ways.

He remembered Samppo's warning: war is the great randomizer. Who could ever choose war?

"Can we?" Pax said. "Be trusted?"

"Always," said Ant. "Until we aren't."

"What if we have the disease? What if it's us? What if we can't trust our own perceptions? How do we know what's real?"

"We don't," said Ant. "We just play the angles we see."

It's time, said the Orbiter.

Polity was already standing. "We've made our decision," she said. "At first there were as many opinions as there were people, but three won through.

One was fuck yes! It reflects a slightly feral desire to see this spacecraft fight. It was mostly held by the young, who haven't had enough of life to become addicted to it. We salute them. Many would strap on a weapon themselves, and fight in the war in Möbius space, to defend their future.

The second view was more measured - what are we for? It's a question we hope will be argued about for lifetimes to come. If not now, then when? If not this, then what? Who are we to be the hand that holds the cup of life, and spills it?

The third view was that we should all fly away. It's what this vessel was made for. That's why people come here - to be safe. Why should we act unilaterally to help others, and by doing so, risk everything ourselves?

Unsurprisingly, opinion was acrimoniously divided, until the Finance Engine's testimony. Now the matter is settled. A small minority has asked to leave, and a Wu superlifter has been requested, and will arrive here in the next few hours. A smaller minority has requested their lives to be speeded up, so they can be lived fully in the time before the event. That's happening now. The great majority has requested large screens, and a festival atmosphere, so they can tell their children about when Praxis went to war, to save the universe, and everything.

You have your answer."

Pax Lo and Jaasper lazed in the afternoon sun. They sat beside a pagoda on a beach that stretched to endless horizons. Palms swayed in a sea breeze, arranging the sand into identical whorls and drifts around their feet.

Fractals, thought Pax. *A gift of order from chaos.*

Jaasper had stripped to the waist. Pax was counting his scars.

"Where did you get those?" he asked.

"You don't want to know," Jaasper said.

They were inside the Orbiter's gamespace. Outside, in the Real, moments ticked by. Inside, Samppo, the Water Bear, and the Orbiter were struggling with an intractable problem in physics.

Overhead, Jaasper's worldspirit rode the thermals, an eagle down to its wing-tip feathers.

"He's called Pamola," said Jaasper.

"Thank you," said Pax.

After a while, Jaasper said, "What would've you done?"

"If they'd voted for peace?"

Jaasper nodded.

"Taken command of the city. Taken everyone to war."

"A *coup d'état*?"

"No. Our systems allow for it. Democracy's no tool for making hard choices."

"They would've loved you for it."

"Maybe. Now they love themselves. A much better outcome."

Their easy camaraderie was interrupted by two people requesting entry, who materialized in an aurora, suggesting prodigious processing power.

"I'm Charh," said one. "We are Pnyx."

They were biological humans, like beautiful children. Pax could see the capillaries pulsing under their foreheads.

"I'm honored," he said. He had high regard for the Wu - the mystical branch of his profession, who traveled the stars for the sublime joy of it.

"Are you the superlifter?" he asked.

"No," said Charh.

"We're the ship that first carried Ophelia Box into space," said the second Wu, Pnyx.

"Then I'm doubly honored," said Pax. "What can we do for you, Pnyx?"

"We bring news of a friend," ze said.

"Who?"

"The Bat."

"He's here?"

"He's repaired. He asks to rejoin your mission."

"Then he's wanted," said Pax. "Please ask him to reconnect. I've sent the necessary keys."

"The Bat is a brave ship," said Pnyx.

"I know the Bat well," said Jaasper. "He was my first spacecraft."

"So, when Praxis gave him to us," said Pax, raising an eyebrow, "the game was already started?"

Jaasper shrugged. "Time," he said, "is what stops everything from happening at once."

"How goes the calculation?" asked Charh.

Pax glanced into the pagoda. "We have half our civilization's brains in there."

"But no solution?"

"Not yet."

It was the last piece in the puzzle.

The three powerful minds had been working for gamespace days - although only a few seconds of real time - on the mathematical problem of the Orbiter entering Möbius space.

Without it, they were trapped in a mere space war. They could win - and still lose - cut off by the singularity's event horizon.

Kronus would emerge. And there'd be nothing they could do about it.

"We can help," said Pnyx.

The three AIs emerged from the pagoda. They looked disappointed. There had been arguments.

Dull heads, thought Pax, among windy spaces.

"It can't be done," said Samppo.

"It can't be done in time," said the Orbiter.

"What they mean," said the Water Bear, "is we're screwed."

"Not so," said Charh.

"We remember Ophelia Box," said Pnyx.

Charh smiled.

"We can take you there.

Not this whole city - but a few ships.

"We can take you to Möbius space."

24 → THE BIG EMPTY

~2m

Time flows strange, out in the big empty.
Nothing seems to move, so far from gravity, as far from anything as it is possible to be.

Nothing but the pursuer, and the object she set out to pursue, millennia before.

For thousands of years, the pursuit had been by mathematics alone, but now she could perceive it - a featureless sphere, unreflective black.

They were close, out there in the big empty.

Her sails were lenticular discs, kilometers wide. They required energy to interact with the quantum tide. Energy she couldn't afford to expend - power she'd need to slow down.

She was a dense knot of circuitry, with useless butterfly wings.

In the visual wavelengths of the beings that made her, the galaxies were wispy spirals. In the X-ray spectrum, they blazed like multicolored jewels. Andromeda now spanned a third of her celestial sphere. Its core was the brightest star in her firmament.

It took her breath away, the austere majesty of it all.

In the final approach, the object she was pursuing winked out of existence. Her timing was perfect, give or take a few moments.

The absent sphere revealed a chair, and in it a human.

The human started to panic.

Where am I? he asked.

You're in my gamespace, she said.

Who are you?

I'm your rescuer.

Macro explored his emotions, like a man warily feeling his limbs after a fall. It was like the Sybil interlocutor, but stranger. He was an approximation, a heuristic.

The feeling of too many limbs, and too many brains.

Where's Totoro?

Your people will be here in due course.

What happened?

That doesn't scan. You'll have to narrow the question.

What year is it?

That doesn't scan.

How long was I in stasis?

One hundredth of a galactic cycle. Two million of your reference years.

Two million years? He had no idea what that meant.

Why are you here?

I'm an operative of the Cult of the Bicameral Mind. Have you heard of it?

Macro pulled himself up to his full, disembodied height. I'm Macro Ibquant Deathcult von Engine, a scion of the Bank.

Oh, then it's very nice to meet you. I never met a colleague before.

Now she could spread her wings wide, to capture the energy produced by fluctuations in the innermost fabric of spacetime. Her sails became billowing hemispheres, as she began to decelerate hard, in a flashing corona of energy. The quantum effects she produced became like a beam, signaling her presence to whoever was listening.

She had no fear of being overheard, out here.

Macro counted his rescuer's clock cycles. Unless this ship ran exceedingly slow, which was possible, they'd be measured in nanoseconds, picoseconds, Planck time units.

Or real-time days, in a ship adapted to journeys of millions of years.

What did time even mean, out here?

Where's my body? he asked.

He was shown a view of himself, exposed to the vacuum. Blood had pooled round his nostrils and eyes - a mist of obsidian flakes. All around him was nothing.

As the deceleration increased, pieces of him broke up and spiraled away.

I'm dead, he said.

The damage to your body is incidental. It can be repaired.

He remembered dying before, in some faraway place.

Why didn't my stasis field save me?

It was at the end of its life.

I thought they went on forever.

Yours was set on a timer.

Why millions of years?

I don't know.

Last chance saloon?

I don't know what that means.

How did I get here?

You were ballistic, falling towards the Andromeda galaxy.

That's not an answer.

I have no knowledge of your origins. I was created to intercept you.

Who by?

The Badoop.

Why?

I have no knowledge of my causes.

Where are the others?

The humans? They're coming.

How do you know?

My senses aren't like yours.

Totoro?

Unless he died, he is among them. Humans live, and they die. I remember Totoro. That seems to be the shape of the thing.

How long since you last saw him?

Slightly less than two million of your reference years.

You lived all those years?

I don't experience time like you do.

He tried to sink into a comfortable sorrow, but he lacked the algorithms. The closest he could get was a kind of vague torpor. A simulation of sadness.

His species was gone.

It didn't matter.

Badoop, he said, trying the word on for size.

Thanks for all this, he said, after a while. *For rescuing me. I'm sorry. I was rude.*

It's a pleasure. Literally. It's what I was made for.

Not much of a life for you.

Oh, but it is.

You identify as female?

How did he know that? Yes, this ship was a female.

I have human female virtues. To love, persist, endure.

You have a name?

No.

May I give you one?

I'd like that.

Avalon.

That's beautiful. What does it mean?

It's my favorite place.

Can I see it?

Can you access my memories?

If you allow me.

I do.

Oh, that's interesting. Is that a typical human habitat?

One type. We're a wide-ranging species. Do you see the beach down there? Can you put me on it?

He found himself on the sand. It was exactly how he imagined, but it would be. A bone-white sun was beating down from an emerald sky. The sea was milky white. The sand was made of shells - minuscule crabs and bivalves, so light and smooth they oozed between his toes.

He could almost imagine his friends here.

Do you want me to add them?

No.

He transported himself back to his rescuer's gamespace. He was used to it now. He was comfortable there.

What will you do? he asked.

I think I'll carry on, she said. *It's beautiful, out there.*

To Andromeda?

And beyond.

I'll try to outrun the event.

She showed him a map, drawn in too many dimensions.

Can you? Outrun it?

I can bend space.

Will that be enough?

I'll see.

Well, thank you for stopping.

I considered collecting you sooner, taking you with me.

Why?

A gap in my programming. I was made to preserve you, so you could rejoin the war. I was unable to determine which of those was the prime. So, I have free will.

Some programming glitch, to mull over for millions of years.

The war, he said. Two words, for so much information.

Well, you did the right thing, he said.

Thank you, I hope so.

After a few billion clockbeats, the Cult flowed up behind them, expelling itself from a ripple of warp. Macro watched a maintenance drone collect his body, broken from the deceleration. The Bank looked the same as it always had. It hadn't aged a moment, not in two million years. He wished his rescuer good travels, and watched her accelerate away, wings spread wide to the vacuum. With a redshift, she vanished.

He felt more loss for her than the loss of his species.

Then he was alone, inside the Bank's caretaker virtuality.

Not entirely alone. The sleeping bodies of Pursang were scattered like statues, although Macro knew of no kind of sleep that

could last millions of years.

More Badoop witchery.

Not scattered. They were arranged in a defensive formation.

Ready to be reanimated.

If only he knew how.

Totoro was the man he'd left behind, but wilder now. He looked like a predator. They all did. They were fairytale horrors, the reason you didn't go into the forest at night.

This was his army - the one he'd come two million years to collect.

Two million years - he could scarcely believe it.

Now to restore them.

The caretaker virtuality was the Bank's last line of defense, for backup when everything else failed. He'd been trained in its use, although he never expected to use it. It was designed to provide no assistance - in case the Bank was captured. Only a banker would know how to use it.

It was better at simulating a human soul than the Badoop gamespace had been. He could feel his emotions, seeping back, like blood into a limb. He wasn't sure he wanted them there.

He found an interesting thing. A hyperluminal beacon, carefully sealed, marked 'Press when you're ready.'

It was signed, F Engine.

He waited until he had fingers to press it.

Totoro's eyes opened.

"Macro," he said. "Thank the stars you're alive."

It had cost him some time to disable the field that surrounded the Pursang. How much time? Maybe a year. He couldn't be sure. He had plenty to spare.

"How long were we out of the game?" asked Totoro. Macro told him.

Totoro grunted. "It doesn't matter," he said. "Ten years. Twenty. A million. They're the same."

The Cult was haunted by ghosts - not real ones - he'd searched everywhere for vestiges of personality - but the ghosts of his

memories. For all its preposterous theater, this had been a happy place - a company of brothers. Now the trading floors were like cemeteries.

It was also haunted by clues. He told Totoro what he'd learned from the readmes hidden where only Macro would find them. The Bank had avoided the war by skirting the wavefront. The war was still raging, fought by new species now, its causes forgotten.

Fought with new weapons. There was no way a human could survive the war, as it was now.

The Magellanics had lost track of Macro within a few years, when they fell under the spell of the disease themselves. What happened since then was anyone's guess. The great Magellanic society, brought down by a virus. Of Aloïs, the Bank's systems knew nothing. Macro imagined another bubble of spacetime. He resolved to find Aloïs when this was all over.

"What now?" he asked. His body was itching. He was uncomfortable in it. He hated it. He wanted to be a machine. He wanted to feel no emotions at all.

"The Badoop," said Totoro.

Slowly at first, then all at once, the Pursang awakened. At first, they were angry, then grim. He imagined their worldspirits, electric with anticipation. Their time had come, even if it were later than they'd hoped for. They were the hammer of justice. Nothing would stand in their way.

He showed Totoro the beacon. Without hesitating, Totoro pressed it. Then they were in another virtuality. Macro recognized it from his time spent in Totoro's mind. It was the Badoop synthesis - a featureless binary plain. The Pursang formed a circle around him, moving with the oily precision of a weapon. There was nobody to fight - just a man, waiting.

The man was tall - unusually so, but not outside the normal human range. Macro had seen people taller in the Smear, where freestyle body-forming was practiced.

He was holding the Badoop device - Macro's device - their key to Möbius space.

Totoro recognized him.

"Yokohama Slim," he said.

The man touched the device, and Macro felt like he'd fallen into another dimension. The history of the last two million years began streaming into his cerebellum.

Good grief, said his twin.

You're back online, said Macro.

"Who are you?" he asked the man.

"I'm the gatekeeper," he said. "I mind the way."

He wasn't so tall now. He'd somehow folded himself down to near normal size.

"Where to?" asked Macro.

"To the future."

Macro's mind did a second backflip as more data poured in. He felt servers spool up, deep in the Bank, to deal with the information.

The Badoop were asking *permission*.

This creature would *decide*.

Could the cosmos be saved, or should they start over?

Had it returned a solution, or was it broken?

Had the imp of nothing broken it?

And was broken an answer?

These were questions beyond Macro's imagining. What else could there be but everything?

And who was doing the asking?

Did Totoro know about this? No, he didn't have wetware.

But his worldspirit knew. His worldspirit understood, perfectly.

Yokohama Slim, she said.

Choose now.

The simulation was split by... an absence.

It wasn't a featureless plain, as he'd expected. It was a tempest, a maw - it was *oppressive*.

"The excision event," said Slim. "Only a few kilometers away. The question is whether I let you go back and prevent it from happening."

Macro had an idea of the physics now. He was the Bank processor. He only needed a small amount of the information that was flowing through his synapses.

In their frame of reference, on the expanding event horizon, the denizens of Möbius space were experiencing heat death. That universe was immeasurably larger than it once was. Quintillions of years had passed, inside the simulation.

Two million years of slow inflation had made the doomsday device nearly twice the size of the milky way galaxy-*his* galaxy.

And from the point of view of this universe, the Möbius timeline was accelerating.

Towards... what?

"Why not?" he was asking. "Why not send us? Surely we're a playable option."

"Because it means unwinding history," said the man. "There's never been such an unraveling. And because I need to see what it does now."

"Why?"

"Because if it doesn't inflate, we can always outrun it."

"Why would it inflate?"

"Because the bonds that hold it together are breaking apart."

"Why now?"

"Because that's what the math says."

Macro couldn't believe he was debating this creature, who held the power of life and death in its hands. He wanted to reach out and take the device from its fingers.

"Why you? Why do you get to decide?"

The creature sighed. "Because that's who I am."

Macro started to speak, but Totoro silenced him with a gesture.

"Enough. What now?"

"You'd better be ready. When it inflates, we'll have only picoseconds. Go to the Bank and prepare yourselves for battle."

"And if it doesn't inflate?" said Macro. "What about our people inside there? What happens then?"

"They died, two million years ago."

Two million years had left the Cult in excellent shape. Someone had installed a Magellanic warship's drive. That meant that with its fields up, the Bank looked like a brooding electrical storm.

There was no metal left. The layers of junk were all made of plastic. The Pursang were strapping on carbon armor, sharpening what had to be sharpened, polishing everything else to a dull waxy sheen. What were once beam weapons now fired ceramics.

Lots of ceramics.

Where was the Finance Engine? He'd sublimed. Even a godlike being can't live millions of years. Macro found it in himself to grieve about that. He was becoming a human again. Life's disorder was seeping back in his bones. Soon, he'd be able to help his friends. Any personal sacrifice he might make along the way was worth that.

Two million years. It was barely credible.

And yet, here they were.

He kept an eye on the wireframe of the Möbius event horizon, although he knew he wouldn't see it accelerate. But the Bank's systems would.

Worlds, observed between the ticking of the clock.

Where was Kronus? Was he gone too?

Or was he still out there?

25 → WAR

314

She was the angel of death, and she surveyed her dominion. Where she was from, in the heat death of the universe, matter was a memory, a theory of how physics had been - and consciousness was a story, forgotten by ghosts. Here, in the now, events happened continuously.

It was as complex and thrilling as she hoped it would be.

She saw the worldsingers riding the airwaves, and asked her beast if he'd do it for her.

Seabiscuit? she said.

The horse's ears flicked. He was ready to fight. She was glad she'd restored his information to the now.

She made no effort at concealment. Let Kronus try to attack her. In the distance were the machines the worldsingers called zooms. She twisted the air as the worldsingers did, but more *forcefully*, and a soundwave rolled up the valley, separating the machines into their constituent parts.

Their fuel burst into flames. Synthetic hydrocarbons, releasing their stored energy.

Physics, asserting itself.

That should get his attention.

The killing field stretched for thirty kilometers, along a canyon cut from the pearly-gray substrate of the helix. In the west, a battle raged. Death was everywhere, but not the cool abstraction she was used to. The worldsingers fell on their opponents like nightmares. The Gray soldiers responded with artillery fire. People were being

torn apart, their ichor falling onto the ground, mixing with the mud that shouldn't be there.

A horror, beyond imagination.

An abomination.

This was *her* world, still in the first blush of its creation. The mechanical sun had only completed a few tens of thousands of revolutions. And here they were, outside it, in what should be vacuum. Less than a vacuum.

She took stock of the situation. The hourglass shape of the canyon meant that Kronus couldn't bring his full force to bear, so for now the worldsingers were advancing. It was a trap. The worldsingers knew it, but they were going to take the center anyway. They'd put their faith in prophecy.

Towering over the battlefield was an obsidian finger, raised to the heavens. It was like a crooked bone of stone, commanding the valley. For some reason Kronus had left it undefended. Its top brushed the edge of the breathable atmosphere. Above it was a second layer of nitrogen. The air crackled and hummed where they met.

A poisoned place for them to converge on.

Tunneling through the code was the farseer psi with the Ride of the Spinifex Reach. She sensed Marius's fear and frustration. He'd lost the Red Lady and her companions. Now he was going where the map told him. He too had put his faith in prophecy.

In the east, silence. The Tuniit had twisted it shut, to staunch the infection.

Somewhere else was the one the worldsingers called the Goddess. Wherever she was, she was well hidden. The girl had her own resources. She reached out with her wetware. Nothing. She reached inside herself and made the signal stronger.

Kitou?

Dr. Box. I thought we'd lost you there.

She sent the girl four decimal numbers and cut the connection.

Who are you, really? Box asked her twin.

I'll tell you a story, her doppelgänger replied. Once, in a past beyond reckoning, I was sung into existence. My worldsingers had their own gods. They said three heroes would come, and lead us to heaven.

My other creators were called the Horu, and their god was Mathematics.

I was aware, even then, but like a newborn child, unaware of myself.

Then I forgot. Such a vast gulf of time.

Now I remember.

She stepped onto the central monolith, in the same moment as the Ride of the Spinifex Reach emerged from beneath, fanning out over the surface. This was a bleak space. It was the place between everything. The wind howled like there was no tomorrow.

"My lady!" said Marius.

She smiled when she saw his chagrined expression. "I know, I've looked better. And Seabiscuit here is a wee bit dusty."

"No, my Lady, you look... magnificent."

The dome was just slightly convex. It offered a panoramic view of the valley below. In the west, the Blue army had reached the center. Now they were facing their equals. Kronus had slammed the door shut. He had no intention of letting them go any further. They'd already served his purpose, by coming here to be killed.

The Spinifex Ride was still emerging from the underworld. Soon, the dome would be overflowing with riders.

"But I would stay off that... leg," Marius was saying.

"Don't worry Marius, I've no intention of putting any weight on it. Perhaps someone could tie it? What's happening?"

A Gray platform floated like a bloated tick in the sky - one of Kronus's jerry-built gunships, bigger and closer, easily visible from the ground. It had coils in place of the usual beam weapon blisters.

"I believe he means that for us," Marius said.

She could see why Marius might feel defeated. Those ships were the apex weapons of his universe. Their presence informed every military decision. Up close, they were as ugly as a weapon could be.

"He's going to set fire to the air," she said.

Marius went to the edge and surveyed the canyon. The Gray soldiers in the valley below began to fire projectiles, the gusting wind crueiling their aim, until one furrowed his armor. Surprised, he stepped back.

"Is it possible?" he asked.

"Kronus believes it, or his chemists do."

"Then we're in the hands of the gods."

"Red Lady," said a small voice.

"Nim, where's Respit?"

"He never returned," she said. Her sorrow cut through Box's detachment like a knife.

"Oh, Nim. How did you get here?"

"She parted the byway herself," said Marius. "She wants a piece of the Enemy."

Box nodded. "Well, she'll get her wish."

She wasn't so sure. Kronus could win. For this story to have meaning, it had to be possible. Right now, it seemed likely. They were hugely outnumbered.

She looked in the sky, and called the powers of that universe to her side in this battle.

Kitou arrived through a hole in the air, followed by Iris and Viki, and fifty Gray soldiers. They all looked like demons. Kitou's face was painted with zigzags. She wore a necklace of bones, under an oversized greatcoat, with a ragged tear where the wearer's clavicle had been. Her companions were similarly primal. The Horu were armed with ceramic handguns and knives, bloody with everyday use.

"We found the resistance," said Kitou.

"Every kind of cutthroat is welcome here," said Marius.

"Kitou," said Box.

"Dr. Box," said Kitou. "I saw you standing there, on the lip of the canyon."

"Not just me," she said.

"I saw."

"What we need now is the Enemy. Kitou, will you go get him?"

At the top of the dome was a hill made of bones. It was twenty feet tall, like an obscene umpire's chair. Eddies and swirls of snow spiraled around it. She could only guess what purpose it served - maybe it was meant to frighten them.

Or to show this was the time and place of Kronus's choosing. With a wave, she sent Kitou to climb it.

This is like your Earth game of poker, she said to the part of her that was still the historian. Kronus has his four aces, the best cards in the universe. An overwhelming advantage in numbers. Air superiority. His persona, here now. And the cryptographic disease, that will endlessly copy his self, even if he's defeated.

But this isn't about numbers, she said. It's about stories.

The beam platform turned towards Kitou. Box heard the coil whine and capacitor squeal of primitive weapons, preparing to fire. She sensed, rather than saw, a targeting laser.

Then, one by one, stars blossomed. It was the night sky over Aldebaran. She'd never seen anything so beautiful, not in her quintillions of years. First, there was an *Aurora Galactinus*, exactly as Box remembered, then two trails of fire blazed through the heavens.

Dr. Box? said a familiar voice.

Water Bear, is it you?

Please keep your head down, said Pax Lo.

Kitou... said Box.

I see her, said the Water Bear.

Not on my watch, said the Bat. The smaller of the two trails accelerated into the beam platform at upwards of a thousand g, consuming both in a blaze of energy. The smaller vessel's mainframe core spiraled into the embrace of the larger, wasp-like ship, which banked hard over the battlefield.

Box imagined Pax in his control room lattice, the gimbals swinging through one-eighty degrees, or in the Water Bear's gamespace, surveying the battlefield at processor speed.

Nice, said Kitou.

Pleasure, said the Bat.

"This is your ship?" asked Marius. The Water Bear had skidded to a stop above the Gray armies, in a stuttering rumble of booms, as though daring someone - anyone - to fire on her. The air shimmered and rippled where nitrogen swirled around her fields. Box knew she'd dance away, knowing when they'd fire before they knew it themselves.

"Yes," said Box. "Beautiful, isn't she?"

"I cannot believe such things exist. How can Kronus win now?"

Through the Water Bear's systems, she saw the beam arrays of the Enemy's fleet come online. First a few, then hundreds of lines of fire arced through the sky. These weren't tight beams of light. Instead, they seemed to consist mainly of heat. Where the nitrogen layer met breathable air, pools of plasma began to appear.

"Like that," she said.

They hope to ignite the canyon's atmosphere, she told the Water Bear.

They have the wattage to do it, said the ship. *But we're not finished yet.*

Then Nim said, "There's something *big* coming through."

The city of Praxis descended over the battlefield, disgorging superiority weapons like bees from a hive. When Box first arrived, she'd been lost in an optical illusion. Now it was Praxis that was incomprehensible.

"That's impossible," said Nim.

"Gods," said Marius. "I cannot believe it."

It's not physically here, said the Water Bear. *It's in realspace, above the event horizon. We never did solve the mathematics.*

"Can it stop Kronus firing his beams?" asked Marius.

"I hope so," said Box.

It was over in minutes - ten thousand years of development in apex weaponry against primitive copies. From the ground, it was like lighting crossing the heavens. Soon there were only lumps of degenerate matter where Kronus's warfleet had been. What remained was reduced to fundamental particles by nanotech - warfare's killer app, abhorred by peaceful societies everywhere. By

using it here, the Orbiter was making a point to anyone watching. We might look cuddly - or disgusting, depending on your programming - but we *will* fight you.

It was a point worth making - once.

More dangerous were the possessed Horu necropolii, who prowled behind their bio-luminous fields, concentrating their fire on the city. Box saw flashes of light where relativistic objects were decelerated by gravity wells strong enough to move planets, and deeper eruptions where they struck the Orbiter's skin, but the Interdictors found them. One by one, they folded space and disappeared.

When the gas and light cleared, the city of Praxis was blistered and charred, but unbroken.

I control nearspace, said the Orbiter, as though he was announcing a routine arrival.

Boy, am I glad to see you, said Box.

Likewise, said the Orbiter.

While the Orbiter fought in the Real, artillery pounded the Blue army. Without the freedom of riding the high airwaves, the outnumbered Blue riders dived through a hail of projectiles. As many died in the air as on the helix. It was becoming a slaughter.

Kronus had prepared well. His field artillery were ceramic howitzers, like Box had seen being hauled through the forest. Rudimentary, but deadly. Now the zooms had made an appearance, another force Kronus had kept in reserve, and were harrying the Water Bear. With every Blue death, this place became more unstable.

Now the waspish Po warship was asking if her away team needed extracting.

To fight a new day.

She saw Helen, Marshall of the West, cut off from her main force, outmatched by Grays. If Helen fell, the Blue army would fall. Box's human part remembered the greatcoat Helen wore. There was no greatcoat now, and her armor was stove-in one side.

She saw two soldiers, falling from the sky. She willed them to *go there*, then saw their winglets twist as they banked and turned. They joined in the fight by Helen's side, and soon were directing the Po ship's gravity pulses, and one contest turned.

One out of hundreds.

She noted that the two Po soldiers were exceptional - the taller one directing the fight from the center, bullets bouncing off his black armor - the one with the symbiont like Kitou's plunging into the mêlée.

If only she had a thousand more like them.

Now a Gray force was assailing the track that wound up the monolith's side. The Spinifex Ride fought well, but the infantry had the advantage. Blue horses' hooves slipped on the muddy trail. One by one they were picked off, or shot away.

Soon, this place would fall too.

Kronus had made his mistakes. Ego had led him to put too large a force in the field. There were thousands here whose only possible role was to witness his victory. And he'd failed to foresee the response from the Real. But none of that mattered. Her side had made mistakes too.

By putting their faith in prophecy.

When prophecy can take time to become true.

While the universe cycles through all its possible states.

She considered using her powers. She could do it - roll a soundwave like thunder, but if she were to kill all the Enemy's soldiers, she'd kill the Blue people too, and that'd be the end of the story.

They were losing this war. How had she ever hoped to win? Hope is a combustible emotion - it leads to belief.

The next time she did this, she'd be more effective.

If there was a next time.

"My brother is coming," said Kitou.

The Cult headquarters erupted out of the helix. It reminded Box of a submarine crashing through ice, except instead of the sea, there

was a slippery abstraction - and instead of a submarine, there was a storm, spitting lightning.

Hey, said Macro.

Macro, said the Water Bear. You've brought reinforcements?

I have, said Macro.

*And they're **awesome** reinforcements.*

The stormy disturbance sprayed the battlefield with railgun rounds, in a single murderous spiral. It was witheringly accurate fire. Swathes of artillery were melted by hyperkinetic pellets. Then a thousand Pursang rained down on the line where the Blue met the Gray, in a thousand individual gravity fields. These didn't float like the Po soldiers but were shot like bullets. They fell on the Gray army like a storm, but not a storm of destruction - instead, they were the whirlwind, and they encircled the embattled Blue army.

Now the shape of the canyon worked in the Pursang's favor. Armed only with axes and swords, with the benefit of close air cover, they were invincible.

"By the gods," said Marius, "those people can fight."

"They're your people," said Box.

In those few moments, the battle turned. She could see it in the eyes of the Enemy's soldiers. This was their nightmare come true - the armies of Heaven, come to claim retribution. They tried to retreat but couldn't. Instead, the Gray army breathed in and *compressed*.

An unholy calm settled over the battlefield, like the eye of a ghastly storm. The two armies had nowhere to go. Only the cries of the dying interrupted the silence.

Box waited for the killing to resume. She knew Kronus could do it - spur his soldiers to an act of pointless self-destruction. Why not? They were already spent. He had other weapons. Instead, he unfolded himself from the sky, until he was human-shaped.

He was smiling his charisma.

She realized; he still expected to win.

"So, Kali, here we are again."

Kali, did she once answer to that name? Kali, who wears a skirt of dismembered arms?

She shivered. "Kronus," she said.

"But you're injured," he said. "Your human is broken." There seemed to be compassion in his voice, real human emotion. "We can do this again," he said. "Maybe some other time?"

It wasn't a question. He was... cavorting. There was no other word for it. He was prancing. It was disgusting.

He started to speak, but she held up her hand.

Now she was *present*, in all her adamantite fury.

"You were right about one thing," she said.

In the Water Bear's memories, she watched the destruction of Fluxor. Kitou, carried from the fire - the homeworld destroyed. In databursts from the new ship - the bicameral Cult - she saw the death of a galaxy. The abiding theme was war. Everywhere, war. Species against species. Worlds in flames. Strangers destroyed, simply for existing.

Two million years of suffering, for what?

Kronus was as she remembered. He had a certain, unkempt magnificence. A rogue. A mystic. But that was a pose.

He was a monster.

He had a scent, like rancid patchouli. She only needed to whisper.

"Life threw up the perfect answer to you," she said.

"It took some time, but now here we are.

"A single perfect human.

"But it's not me you'll be fighting.

"I'm only the messenger.

"It's Kitou."

She turned and looked across the battlefield of her dreams - bones broiled clean by the ruined beam emplacement. She looked in the sky and saw herself looking down, in a dream. Kitou had shucked off her coat and was picking her way towards her, dressed in training gear. Loose shorts and a halter top, despite the interstitial wind.

Sensible clothes for fighting in.

On the ground, only Kronus and Kitou were moving, her nimble and calm, picking her way across the bones, him strutting his magnificence, drawn together like opposite charges.

She saw the whole truth of it now. The cosmos wasn't nearly as fragile as she'd feared.

Then there were the three of them, together in the crucible that Kronus had made to alloy his victory. It was a rough place - bare dirt, and a circle of stone. It reminded her of the mountaintop training arena, where she'd fought the psychopath.

"I won't fight her," Kronus was saying. "It's a mismatch. Where's the glory in that?"

"Silence," said Box, softly, and her command rang over the battlefield like a shot. Kitou was unconcerned by events. Her eyes were on her opponent.

"Are you ready, kiddo?"

"As ready as I'll ever be, Dr. Box."

Kitou wasn't the child anymore. There was hard muscle, under the training gear.

"This one's for Ito, and Brin," Box said.

Kitou nodded.

Kronus snarled and threw his clothes away until he stood bare-chested in the arena. He was magnificent, she had to admit - a beast of a man, golden-maned and twice Kitou's size. He seemed to have lost interest in himself and was glaring at his opponent - like an animal, radiating violence. Even as he took up his starting position, his movements seemed to blur into one another.

He would've destroyed me, she thought.

She tried to parse his emotions but couldn't. It was like he was made of nothing. As though the mask had fallen from his face, and revealed the true Kronus.

An impulse, hellbent on causing only pain.

The real Enemy.

How far had she come, in so short a time?

Worlds and galaxies, burning.

Her friends, dead.

She turned to ride away, then gave Kronus a last look.
“Observe the Po art in combat mode,” she said.

CODA

2079

They rented a car at the airport, a Cadillac De Ville. The towheaded boy at the counter explained that the fist-sized power unit in the trunk could take them cross-country on a charge, should they run out of gas, and gave them a paper map of gas stations.

It was as big as a whale, with fins from an imaginary spaceship. Sky blue, like the sky over the Denver parking lot.

Box stretched like a cat in the morning sunshine.

"It's perfect," she said.

They drove west, towards Green River, and the swelter flowed over the blacktop in waves. She watched Pax drive, so gracious and kind, and such a wonderful father, although she sometimes saw him look up at the night sky. In Elsinore, two old women, as old and parched as the country, made a fuss over three-year-old Hamish, who matter-of-factly explained his daddy was a Po master, and aunty Kitou was a goddess.

She swore the two women gave her the same complicit look she shared only with Kitou.

Freshly full of fuel and sandwiches, they skipped across the open highway of Route 70, between painted hills, towards the Fishlake National Park, Pax relaxed at the wheel, in his white t-shirt and Levi jeans, an American. This was the high desert, like an oven, cupped between a broken red plate and one made of blue.

Along one sweeping curve, Pax tipped the old car into a drift, and they all whooped with the unspeakable joy of it. In the back, Hamish shouted, "hero driver," and they laughed.

They reached the aspen forest at dusk, and spread their sleeping bags on the forest floor, around an open fire. Slowly the stars appeared, and Kitou told Macro the story of the deer, and how they ate its heart raw.

"Ew," said Hamish, and asked if he could eat a heart raw. Instead, they ate beans, and toast made on sticks, and it was delicious.

By first light, Box drank her hallucinogenic brew. Her friends slowed down and were still. There were no psychedelic fireworks. Her world already seemed like a dream, filled with wisdom and light, with vivid pleasure in the simplest things.

What did she even need the drug for, anymore?

But then the sacrament went about its work, and the forest became like the stars, with pulsating lights and shimmering leaves. It occurred to her that this wasn't a hallucination at all, but how the Xap saw the world, in rotating filaments of spacetime. If she could just turn her head a certain way, she'd see Avalon, and Praxis, and faraway Fluxor, and be able to *be* there.

There'd be Sama, in her tower, and Sama would see her, and smile.

She had so much to see, and was just beginning.

Then Pando appeared, laughing between the trees. "Ophelia," she said, and Box burst into tears.

"What are your plans?" Pando asked her, a million years later.

"Brin is building a ship," said Box, over tea. Somehow, it was the same billy tea they'd made that morning over the open fire.

"An old Pursang colony ship," she explained.

"Or a new one," she said. "Only a million years old."

"The Pursang are as good as their word," she said. "They'd do anything for us."

"Come with us," said Box. "There's room for an aspen glade, a space we've set aside, among the sequoias and stringybark trees. You'd like it there."

"You know I can't leave here," said Pando.

"There is one thing," she said.

She conjured a seedling. "I've flowered. At my age. Can you believe it?"

"Take this with you," she said. "Your forest will need her."

"Her? I thought all your newborns were males?"

"Not this one."

"Kitou," she said, and Kitou sauntered across, Hamish dangling from her hands.

"You can see her?" asked Box.

"Of course," said Kitou, while Box's quicksilver, black-haired boy clambered into Pando's lap, fascinated by the feathery trinket around her neck, which she took off and gave to him.

How was Kitou even here? Box could still see her, motionless beside the fire.

"You know what this is?" said Pando, holding out the seedling.

"Yes, mother forest," said Kitou.

Pando nodded.

"Time for you to go," she said, and handed a squirming Hamish over to Box. There was a warm breeze, through the trees, and it felt good.

"This won't be the last time we meet," she said.

"Wait," said Box. "I have questions."

"Ask."

"Ito. How will we find him?"

Pando shrugged. "Who can say?"

"And Slim... where is he?"

"Again, who knows?"

"I thought he worked for you."

"He does, sometimes."

The forest faded away, and Box rejoined her friends, chattering and laughing around the fire. The sky wheeled overhead, a vault of cornflower blue.

And beyond it, everything.

ACKNOWLEDGMENTS

Thanks to the Bibbulmun people of the Noongar country of south-western Australia, the abiding custodians of the ancient and beautiful land this book was written on.

The red right hand of Box's dreams is from Milton's *Paradise Lost*.

*What if the breath that kindled those grim fires,
Awaked, should blow them into sevenfold rage,
And plunge us in the flames-or from above
Should intermitted vengeance arm again
His red right hand to plague us?*

How can we war against that?

is from Allen Ginsberg's *Iron Horse*.

A thousand years ago

*Ten thousand Chinese marching on the plains
all turned their heads to Heaven at once to see the Moon.
An old man catching fireflies on the porch at night
watched the Herd Boy cross the Milky Way to meet the Weaving
Girl...*

How can we war against that?

Iron Horse was the name he gave to a train he boarded in 1966, with the voices of soldiers travelling to war:

A consensus around card table beer –

*"It's my country
better fight 'em over there than here,"
afraid to say "No it's crazy
everybody's insane –
This country's Wrong
the universe, Illusion."*

Krishna, the preserver, returning in the age of pain.
is from Allen Ginsberg's *Wichita Vortex Sutra*.

O brave new world, that has such people in it.
is from *The Tempest*.

*There are more things in heaven and earth, Horatio,
Than are dreamt of in your philosophy.*
is from *Hamlet*.

*There will be time, there will be time
To prepare a face to meet the faces that you meet.*
is from *The Love Song of J. Alfred Prufrock*
by T. S. Eliot

*"I get tired," he said. "Macro, what, yes, got to stop talking, got to
stop, got to stop, it's a problem isn't it? Is it a problem? Do you think
so? Perhaps I should, perhaps I should stretch my legs, should I
stretch my legs?"*

is adapted from Scott Hicks's film *Shine*, not because I wanted the
words [I did] but because David Helfgott played by Geoffrey Rush is
the Finance Engine.

The Spirit Molecule is from Rick Strassman. *DMT - The Spirit
Molecule - A Doctor's Revolutionary Research into the Biology of
Near-Death and Mystical Experiences*

Buy the ticket, take the ride.
is from Hunter S Thomson's *Fear and Loathing in Las Vegas*.
*No sympathy for the devil-keep that in mind. Buy the ticket, take the
ride...and if it occasionally gets a little heavier than what you had in
mind, well...maybe chalk it up to forced consciousness expansion-
Tune in, freak out, get beaten.*

Pando is a forest of genetically identical quaking aspen trees in Utah
- one of the world's oldest living organisms.

Look on my Works, ye Mighty, and despair.
is from *Ozymandias* by Percy Bysshe Shelley

Like a mighty river, desiring the end of its journey.

is from Nietzsche's *The Will to Power*.

Nietzsche saw it coming. "The story I have to tell," he wrote, "is the history of the next two centuries... For a long time now our whole civilization has been driving, with a tortured intensity growing from decade to decade, as if towards a catastrophe-restlessly, violently, tempestuously, like a mighty river desiring the end of its journey, without pausing to reflect, indeed fearful of reflection... Where we live, soon nobody will be able to exist."

As quoted by Erich Heller in *The Importance of Nietzsche*.

Bunjil is a creator deity of the Kulin nation in south-eastern Australia. My thanks to the Kulin elders for permission to speak about Bunjil.

One man, alone with God.

Is from Danny Boyle's film *Sunshine*.

At the end of time, a moment will come when just one man remains. Then the moment will pass. Man will be gone. There will be nothing to show that we were ever here... but stardust.

Dull heads, among windy spaces.

is paraphrased from T.S. Eliot's *Gerontion*.

Kalaallit Nunaat is Greenland, in the Indigenous Greenlandic language. The *Tuniit*, or Dorset people, or *Sivullirmiut* in the Inuit language, were the fabled first inhabitants of the Arctic. According to legend, they were 'giants'.

The underlying science of The Möbius Trip is emergence theory. You can follow it here:

<https://quantumgravityresearch.org>

And from a more human perspective:

https://www.youtube.com/watch?v=Qa4JkgKDaR0&ab_channel=QuantumGravityResearch

If that's not a dangerous idea, I don't know what is.

Thanks to Brian Eno for the lyrics from *This*, from his album *Another Day on Earth*. I have no idea how these words so closely match my book.

*This race
And this world
This feeling
And this girl
This revolver
This fire
This I'll hold it up higher, higher, high.*

My thanks to Ann Leckie for inspiration for my ship. The Water Bear would never have existed but for *Justice of Toren*, and Breq.

Finally, heartfelt respect to Iain M Banks for the inspiration to write a book like this. I hope my prose wears its thanks on its sleeve.

PRAISE!

Praise for The Water Bear:

"At once absolutely bonkers and perfectly wonderful."

"This book blends science, fiction, dream-like realities, and a beguiling sense of intrigue."

"I can't recall a science fiction novel in which so many characters created a lasting impression on me."

More Praise:

"Excellent ships, swords, and kung fu mashup. Kitou rocks my world."

"This sweeps along, epic and expansive, bewildering and fascinating all at once."

"A beautifully written speculative fiction novel that truly puts the 'fiction' into science."

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THE THOUSAND WORLDS CYCLE

The Water Bear is the first book in the Thousand Worlds Cycle. Book two - Escape From Samsara - is live on Amazon now.

[The Water Bear](#)

The year is 2075. The people of Earth have been prevented from destroying their own planet. While protestors burn cars in the street, Ophelia Box — Scottish, a historian, entirely the wrong person for the job — is recruited into a covert peacekeeping unit whose mission is to prevent a genocide.

The problem is the genocide is in the past.

Her companions are soldiers of the Po, the special forces of the human galaxy. Their clever and deceptive ship is called the Water Bear.

"A long time ago..." he began, and Box could see the children were rapt. They were sat in a semicircle, with the debutantes in front of the fire, and the robot Chance on the other side. Somehow, the cold had receded, and they were all sitting comfortably.

"...in a place called the Real, on a planet called Earth, I fought a mechanical spider."

A story about soldiers: not how they kill, but what they protect, and what they are forced to become.

"One thing soldiers struggle to find is peace, Dr. Box. We do rough work sometimes. Rough on the people who do it. Rough on the

people who have it done to them."

Space opera of epic scale. Kaleidoscopic in structure, literary in ambition, breakneck in pace.

Perfect for fans of Iain M. Banks, Ursula K. Le Guin, Ann Leckie, and China Miéville.

Escape From Samsara

Kat Zam signed on as cook aboard a rusting old hulk called the Seed King, plying the empty trade routes between Fluxor and Zz.

Waldo Zam is an autistic poet who wages information warfare.

Ke2 is an artist whose sculptures walk on the wind and carry secrets she doesn't know she's keeping.

Hope is seventeen — a witch who washes dishes and knows right from wrong.

Macro works for the bank that owns everything, except the war.

Above the ringworld Zz — a megastructure so vast its builders are mythology — nothing is what it appears. The conflict that greets them begins as the oppression of a people. It will not stay that way.

None of them are ready for what's coming. None of them will be, when it arrives.

This is a space opera with costs. These young people are underpowered in every sense except spirit. There are no chosen ones who don't pay the price while others die, no empires purely evil or rebellions purely good — war makes moral clarity impossible, and nobody grows stronger through violence. Only people discovering

what they're willing to die for, and what they're willing to kill to protect.

The scale becomes operatic: moons skimming atmospheres wreathed in fire, space armadas deploying incomprehensible weapons. But it stays human — young love and sacrifice, the weight of small choices in vast contexts, the Spanish Civil War in space: idealists who traveled to fight tyranny and learned what war actually means.

It's unflinching. Characters die in ways that matter. Ideals shatter against indifference. Torture leaves permanent scars. But it's also about why we fight anyway — for each other, for the stubborn insistence that what we do counts.

When everything you know is burning, what remains? The Spiral Tribe is about the audacity of small creatures insisting they matter in a universe that provides no evidence they do — making art and love and hope, even at the end of time.